

Fermi Notes on Physics

Volume Six

Introductory Undergraduate
Lectures on Oscillations,
Waves, and Acoustics

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Simple, Damped, and Driven Oscillations

1.1 Simple Harmonic Motion Revisited

Simple harmonic motion was the subject of Sections 4.1 and 4.2 of Vol. 1, which treated it at greater length than this section does. We begin by stating the results found there, then draw the motion in the phase plane, then find the mean kinetic and potential energies, and finally set up the complex notation that the rest of the course uses, with two examples of oscillators whose restoring force is supplied by gravity acting on a liquid.

The Equation of Simple Harmonic Motion

Let a body of mass m be attached to a spring of spring constant κ (Vol. 1 writes it k), and let x be the displacement of the body from its equilibrium position. The force exerted by the spring is $-\kappa x$, by Hooke's law, and Newton's second law gives $m\ddot{x} = -\kappa x$, which can be rewritten as

$$\ddot{x} = -\omega_0^2 x, \quad \omega_0 = \sqrt{\frac{\kappa}{m}}. \quad (1.1)$$

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This is the equation of simple harmonic motion, and ω_0 is called the natural angular frequency of the oscillator. Its general solution is

$$x = A \cos(\omega_0 t + \phi), \quad (1.2)$$

where the amplitude A and the phase constant ϕ are fixed by the position and the velocity of the body at one instant. The motion repeats after a period $P = 2\pi/\omega_0$, which does not depend on the amplitude, and its frequency is $f = 1/P$. The energy of the oscillator is the sum of the kinetic energy of the body and the potential energy stored in the spring,

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}\kappa x^2 = \frac{1}{2}\kappa A^2, \quad (1.3)$$

and it stays constant, since no force acting on the body dissipates energy.

Section 4.2 of Vol. 1 showed why harmonic motion is so common. Near a minimum of the potential energy, the first term of its Taylor series that varies with position is the quadratic one, so any system described by a single coordinate q , with kinetic energy $\frac{1}{2}m_{\text{eff}}\dot{q}^2$ for a constant m_{eff} and potential energy $U(q)$ with a minimum at q_0 , oscillates harmonically about q_0 with

$$\omega_0^2 = \frac{U''(q_0)}{m_{\text{eff}}}, \quad (1.4)$$

as long as the oscillations are small enough that the cubic and higher terms of the series can be neglected. The coefficient m_{eff} is called the effective mass; it is the mass for a particle, and the moment of inertia about the pivot for the pendulums of Vol. 1.

The Phase Plane

In Section 8.1 of Vol. 1 the state of an oscillator at one instant was represented by a point in the phase plane, called the representative point, with the displacement x along one axis and the momentum

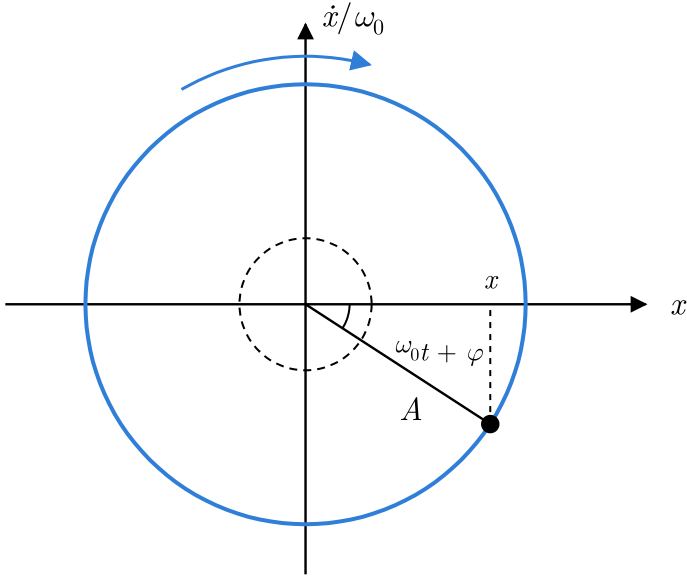


Figure 1. The phase plane of a simple harmonic oscillator, with the velocity divided by ω_0 on the vertical axis. The representative point moves clockwise round a circle of radius A at angular velocity ω_0 , and its horizontal coordinate is the displacement. The dashed circle is the trajectory of an oscillation of smaller energy, which takes the same time to go round.

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p along the other, and as time passes the point traces out a curve called the phase trajectory. For the harmonic oscillator each trajectory was an ellipse with semi-axes $\sqrt{2E/m\omega_0^2}$ and $\sqrt{2mE}$. For oscillations it is more convenient to plot the velocity divided by the natural angular frequency, $\dot{x}/\omega_0$, which has the dimensions of a length, on the vertical axis. Since $m = \kappa/\omega_0^2$, the energy can be written as

$$E = \frac{1}{2}\kappa \left[x^2 + \left(\frac{\dot{x}}{\omega_0} \right)^2 \right], \quad (1.5)$$

so that the energy is proportional to the square of the distance of the representative point from the origin. Conservation of energy then says that the point stays on a circle of radius $\sqrt{2E/\kappa} = A$, as shown in Figure 1. Substituting the solution gives the coordinates of the point as

$$x = A \cos(\omega_0 t + \phi), \quad \frac{\dot{x}}{\omega_0} = -A \sin(\omega_0 t + \phi), \quad (1.6)$$

so the radius from the origin to the point makes an angle $\omega_0 t + \phi$ with the positive x -axis, measured clockwise, and the point goes round the circle at the constant angular velocity ω_0 . The sense of rotation is clockwise because in the upper half of the plane $\dot{x}$ is positive, so the point moves towards larger x . The displacement is the projection of the point on the horizontal axis, which is the form, familiar from school, in which simple harmonic motion is the projection of uniform circular motion on a diameter.

Oscillations of different energies give concentric circles, and since each is traversed at the same angular velocity, every oscillation takes the same time to go round, which is the independence of the period from the amplitude. The position of the point on its circle also shows how the energy is divided between its kinetic and potential forms. The potential energy is $\frac{1}{2}\kappa x^2 = E \cos^2(\omega_0 t + \phi)$ and the kinetic energy is $\frac{1}{2}m\dot{x}^2 = E \sin^2(\omega_0 t + \phi)$, so all the energy is potential where the circle crosses the horizontal axis and all of it is kinetic where the circle crosses the vertical axis.

The Mean Kinetic and Potential Energies

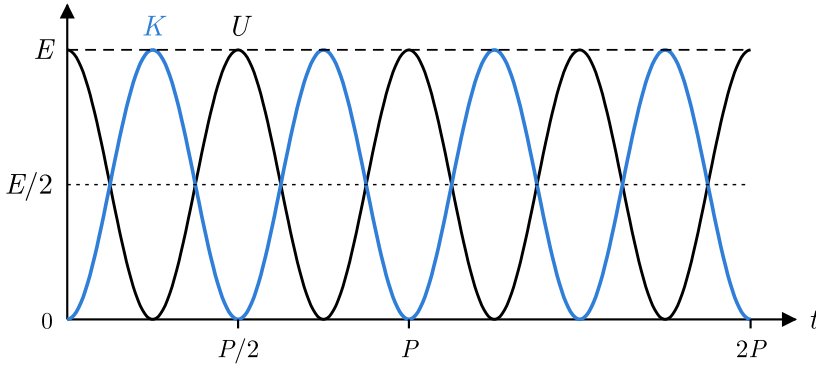


Figure 2. The kinetic energy K and the potential energy U of a simple harmonic oscillator over two periods. Each varies at twice the frequency of the displacement between zero and the total energy E , and their sum is constant. The mean of each over a period is $E/2$, shown by the dotted line.

The mean of a quantity $G(t)$ over one period is defined as

$$\langle G \rangle = \frac{1}{P} \int_0^P G(t) dt. \quad (1.7)$$

To find the means of the kinetic and the potential energies, we write the squares of the cosine and the sine with the identities $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$ and $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$, which follow from the formula $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ together with $\cos^2 \theta + \sin^2 \theta = 1$. The potential energy U and the kinetic energy K become

$$U = \frac{1}{2}E [1 + \cos 2(\omega_0 t + \phi)], \quad K = \frac{1}{2}E [1 - \cos 2(\omega_0 t + \phi)]. \quad (1.8)$$

Each energy oscillates about the value $E/2$ at the angular frequency $2\omega_0$, twice that of the displacement, as Figure 2 shows, and the kinetic and potential energies oscillate in antiphase, so that their sum is constant. The term $\cos 2(\omega_0 t + \phi)$ goes through two complete

cycles in one period P , and its integral over the period is therefore zero. The means of the two energies are then

$$\langle K \rangle = \langle U \rangle = \frac{1}{2}E. \quad (1.9)$$

The result, stated for the mass on a spring in Section 4.2 of Vol. 1, holds for small oscillations of any system of the kind described above, since the argument used only the fact that the kinetic energy is proportional to $\dot{q}^2$ and the potential energy to $(q - q_0)^2$.

The Complex Representation

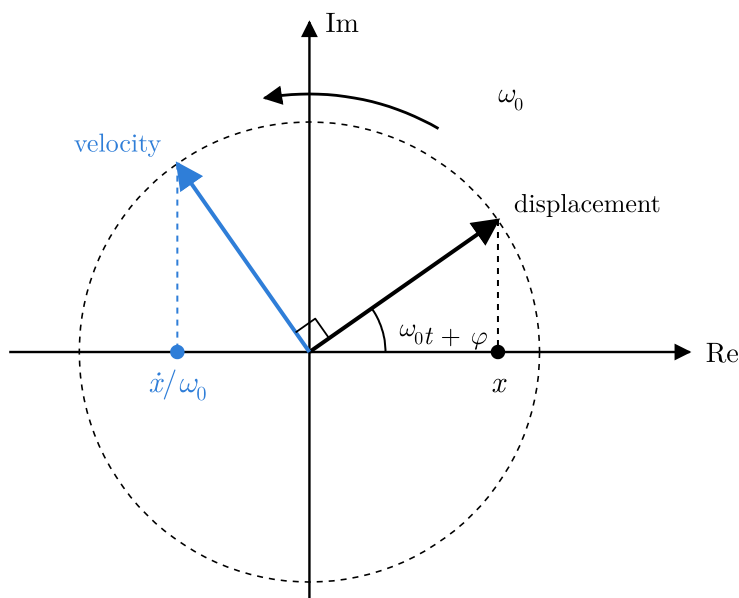


Figure 3. The phasor diagram of a simple harmonic oscillator. The complex number $\mathcal{A}e^{i\omega_0 t}$, labelled as the displacement, turns anticlockwise at angular velocity ω_0 , and its projection on the real axis is x . The phasor $i\mathcal{A}e^{i\omega_0 t}$, labelled as the velocity, is a quarter of a turn ahead, and its projection is $\dot{x}/\omega_0$.

The displacement can be written as the real part of a complex number that turns at a constant rate. Let

$$x = \operatorname{Re}(\mathcal{A}e^{i\omega_0 t}), \quad \mathcal{A} = Ae^{i\phi}, \quad (1.10)$$

so that Euler's formula returns $x = A \cos(\omega_0 t + \phi)$. The constant $\mathcal{A}$ is the complex amplitude of Section 1.2 of Vol. 4 of the Fermi Notes on Physics, where it was written C , and its modulus is the amplitude and its argument the phase constant. On the Argand diagram of Figure 3 the complex number $\mathcal{A}e^{i\omega_0 t}$, which is called a phasor, turns anticlockwise about the origin at angular velocity ω_0 , and the displacement is its projection on the real axis. Throughout this course every oscillation will be written with the time dependence $e^{i\omega t}$ in this way, as in Vol. 1 and in Section 1.2 of Vol. 4.

Differentiating the phasor with respect to time multiplies it by $i\omega_0$, so the velocity is

$$\dot{x} = \operatorname{Re}(i\omega_0 \mathcal{A}e^{i\omega_0 t}). \quad (1.11)$$

Since $i = e^{i\pi/2}$, multiplying by i turns a phasor anticlockwise through a right angle, and the velocity is therefore a quarter of a period ahead of the displacement. Differentiating a second time multiplies the phasor by $i\omega_0$ again, which gives $\ddot{x} = -\omega_0^2 x$, the equation of simple harmonic motion. The velocity divided by ω_0 is $\operatorname{Re}(i\mathcal{A}e^{i\omega_0 t}) = -\operatorname{Im}(\mathcal{A}e^{i\omega_0 t})$, so the representative point of the phase plane, with coordinates x and $\dot{x}/\omega_0$, is the reflection of the phasor in the real axis. The phase plane of Figure 1 is thus the mirror image of the Argand diagram of Figure 3, which is why the phasor turns anticlockwise while the representative point turns clockwise.

A product, such as the square of the velocity in the kinetic energy, cannot be formed directly from the complex form, since the real part of a product is not the product of the real parts. The mean of a product is found instead from the complex amplitudes, as you showed in Problem 7 of Section 1.2 of Vol. 4: for two oscillations $x_1 = \operatorname{Re}(\mathcal{A}_1 e^{i\omega t})$ and $x_2 = \operatorname{Re}(\mathcal{A}_2 e^{i\omega t})$ of the same angular frequency, with complex amplitudes $\mathcal{A}_1$ and $\mathcal{A}_2$,

$$\langle x_1 x_2 \rangle = \frac{1}{2} \operatorname{Re}(\mathcal{A}_1 \mathcal{A}_2^*). \quad (1.12)$$

Applied to the velocity, whose complex amplitude is $i\omega_0\mathcal{A}$, this gives $\langle \dot{x}^2 \rangle = \frac{1}{2}\omega_0^2|\mathcal{A}|^2$, and the mean kinetic energy is $\frac{1}{4}m\omega_0^2\mathcal{A}^2 = \frac{1}{2}E$, in agreement with the calculation above.

A Liquid in a U-Tube

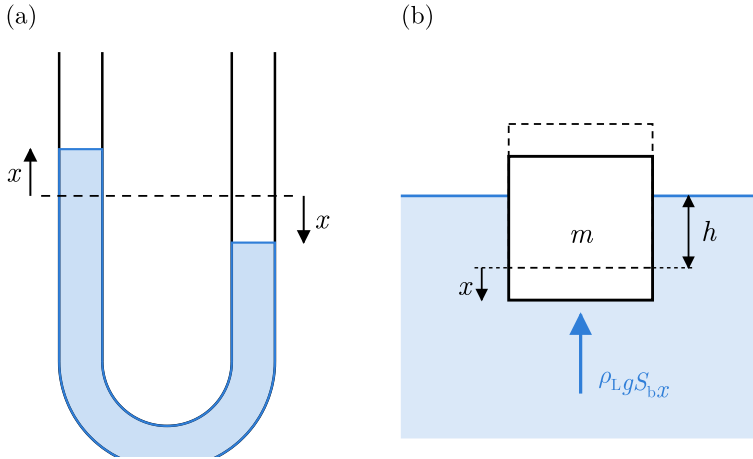


Figure 4. Two oscillators in which gravity supplies the restoring force. (a) A liquid in a U-tube, displaced so that its surface stands a distance x above the equilibrium level in one arm and x below it in the other. (b) A body of mass m floating in a liquid, pushed down a distance x below its equilibrium position, in which it is immersed to a depth h . The dashed outline marks the equilibrium position, and the extra buoyant force is $\rho_L g S_b x$.

As a first example, consider a liquid of density ρ_L in a U-tube of uniform cross-sectional area S with vertical arms, as in Figure 4(a), and let H be the total length of the column of liquid, measured along the tube. When the surface in one arm is raised by x , the surface in the other arm falls by x , because the liquid is incompressible and the tube is uniform. Every part of the column moves along the tube with the same speed $\dot{x}$, provided that the viscous drag of the walls is neglected and the liquid moves with the same speed at every point of a cross-section, so the kinetic energy is

$$K = \frac{1}{2}\rho_L S H \dot{x}^2. \quad (1.13)$$

To find the potential energy, we compare the displaced liquid with the liquid at rest. The only difference is that a column of liquid of length x has been taken from the top of the column in one arm, where its centre of mass was a distance $x/2$ below the equilibrium level, and placed on top of the column in the other arm, where its centre of mass is a distance $x/2$ above it. This column of liquid has mass $\rho_L Sx$ and has been raised through a height x , so

$$U = \rho_L g S x^2, \quad (1.14)$$

where g is the gravitational field strength. The effective mass is $\rho_L SH$ and $U''(0) = 2\rho_L gS$, so

$$\omega_0^2 = \frac{2\rho_L gS}{\rho_L SH} = \frac{2g}{H}. \quad (1.15)$$

The frequency depends neither on the density of the liquid nor on the area of the tube, and the period is that of a simple pendulum of length $H/2$. The same result follows from the forces. The liquid in the raised arm stands $2x$ higher than the liquid in the other arm, and the weight of this unbalanced column, $2\rho_L gSx$, accelerates the whole mass $\rho_L SH$ back towards equilibrium. The potential energy is exactly quadratic in x for as long as both surfaces remain in the vertical parts of the tube, so the oscillation of the liquid is harmonic for large displacements as well as small ones, unlike the swing of a pendulum.

A Floating Body

As a second example, consider a body of mass m floating upright in a liquid of density ρ_L , as in Figure 4(b). Let the body have vertical sides over the whole of the part that is immersed, so that its horizontal cross-section there has a constant area S_b , as a block or a vertical cylinder has, and let it be immersed to a depth h at equilibrium. By Archimedes' principle, the buoyant force on the body is the weight of the liquid it displaces, and at equilibrium this force balances the weight of the body, so that

$$m = \rho_L S_b h. \quad (1.16)$$

When the body is pushed down a further distance x , it displaces an extra volume $S_b x$ of liquid, and the buoyant force exceeds the weight by $\rho_L g S_b x$. The equation of motion is $m\ddot{x} = -\rho_L g S_b x$, which is the equation of simple harmonic motion with

$$\omega_0^2 = \frac{\rho_L g S_b}{m} = \frac{g}{h}, \quad (1.17)$$

where the second form uses the condition for equilibrium. The body bobs with the period of a simple pendulum whose length is the depth to which the body is immersed, so a body that floats high, with a small value of h , bobs quickly, and one that floats deep bobs slowly.

The calculation neglects the motion of the liquid around the body, which is not treated in this course. The liquid pushed aside by the body must also be set in motion, and its kinetic energy adds to the effective mass, which lowers the frequency somewhat, while the waves that the body sends out across the surface carry energy away and damp the motion.

1.2 Damped Oscillations and the Quality Factor

Every real oscillator loses energy, for example to friction, to the air around it, or to the material of its spring, and its oscillations die away. The damped oscillator was treated in Section 4.3 of Vol. 1, and in this section we first restate the three kinds of motion found there, then find the rate at which the oscillator loses energy, then relate the quality factor to the decay of the free oscillation, and finally compare the three kinds of motion in the phase plane.

Light, Heavy, and Critical Damping

Let the body of Section 1.1 be acted on, in addition to the force of the spring, by a resistive force $-b\dot{x}$ proportional to its velocity, where b is a positive constant called the damping coefficient. Such a force describes, for example, the drag on a body moving slowly

through a fluid, or the force on a piston attached to the moving body and pushed through a closed cylinder filled with oil, which resists the motion of the piston with a force proportional to its velocity. The equation of motion is $m\ddot{x} + b\dot{x} + \kappa x = 0$, and dividing by m gives

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0, \quad \gamma = \frac{b}{2m}, \quad (1.18)$$

where γ is called the damping rate. The equation is linear, with constant coefficients, and Section 4.3 of Vol. 1 solved it by trying $x = e^{\beta t}$, with β a constant to be determined. Substituting and cancelling the exponential gives $\beta^2 + 2\gamma\beta + \omega_0^2 = 0$, whose roots are

$$\beta_{\pm} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}. \quad (1.19)$$

The form of the motion depends on the sign of $\gamma^2 - \omega_0^2$, and there are three cases:

1. light damping, $\gamma < \omega_0$, for which the roots are complex and $x = Ae^{-\gamma t} \cos(\omega_d t + \phi)$, with $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$; 2. heavy damping, $\gamma > \omega_0$, for which both roots are real and negative and $x = D_+ e^{\beta_+ t} + D_- e^{\beta_- t}$; 3. critical damping, $\gamma = \omega_0$, for which the roots coincide at $-\gamma$ and $x = (D_1 + D_2 t)e^{-\gamma t}$.

In each case the two constants, A and ϕ for light damping, D_+ and D_- for heavy damping, and D_1 and D_2 for critical damping, are fixed by the initial position and velocity. A lightly damped oscillator goes on oscillating, at an angular frequency ω_d a little below ω_0 , inside an envelope $Ae^{-\gamma t}$ that shrinks with time. A heavily damped oscillator does not oscillate, and returns to equilibrium slowly, at a rate that is approximately $\omega_0^2/2\gamma$ when $\gamma \gg \omega_0$. A critically damped oscillator released from rest returns to equilibrium without overshooting, and more quickly than a heavily damped one released in the same way; like the heavily damped oscillator, it can pass through equilibrium at most once, which happens only when it is started with a large enough velocity towards equilibrium.

The Rate of Loss of Energy

The energy of the damped oscillator is defined as for the undamped one, $E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}\kappa x^2$. Differentiating it with respect to time, by the chain rule, and using the equation of motion in the form $m\ddot{x} + \kappa x = -b\dot{x}$, gives

$$\frac{dE}{dt} = m\dot{x}\ddot{x} + \kappa x\dot{x} = \dot{x}(m\ddot{x} + \kappa x) = -b\dot{x}^2. \quad (1.20)$$

The right-hand side is the power delivered to the body by the resistive force, which is the force $-b\dot{x}$ multiplied by the velocity $\dot{x}$. The result is exact, and holds for all three kinds of damping. The energy never increases, and it decreases at every instant except those at which the body is momentarily at rest.

For light damping we can go further by averaging over a cycle. When $\gamma \ll \omega_0$ the envelope $Ae^{-\gamma t}$ changes by only a small fraction, about $\gamma P = 2\pi\gamma/\omega_0$, in one period, and over a single cycle the motion is very nearly simple harmonic motion with an amplitude that is constant for that cycle. The mean kinetic energy over the cycle is then half the energy, as found in Section 1.1, so that $\langle \dot{x}^2 \rangle = 2\langle K \rangle/m = E/m$, and the mean rate of change of the energy is

$$\frac{dE}{dt} = -\frac{b}{m}E = -2\gamma E, \quad (1.21)$$

where E now stands for the energy averaged over a cycle. The solution of this equation is

$$E = E_0 e^{-2\gamma t}, \quad (1.22)$$

where E_0 is the energy at $t = 0$. The energy decays twice as fast as the amplitude, which agrees with Section 4.3 of Vol. 1, where the same result was obtained from the fact that the energy goes as the square of the amplitude. The energy falls by a factor of e in a time $1/2\gamma$.

The quality factor was defined in Section 4.3 of Vol. 1 as $Q = \omega_0/2\gamma$. The mean power dissipated by the resistive force is $2\gamma E$,

and multiplying the numerator and the denominator of $\omega_0/2\gamma$ by E gives

$$Q = \frac{\omega_0 E}{2\gamma E} = \omega_0 \frac{\text{energy stored}}{\text{mean power dissipated}}. \quad (1.23)$$

The energy dissipated in one period is the mean power multiplied by $P = 2\pi/\omega_0$, which gives the form found in Vol. 1,

$$Q = 2\pi \frac{\text{energy stored}}{\text{energy dissipated per cycle}}, \quad (1.24)$$

in which Q is the ratio of the energy stored in the oscillator to the energy it loses as its phase advances by one radian. An oscillator of high Q , such as a tuning fork or a pendulum swinging in air, loses a small fraction of its energy in each cycle and rings for many cycles, while an oscillator of low Q , such as the suspension of a car, is stopped within a cycle or two.

The Logarithmic Decrement

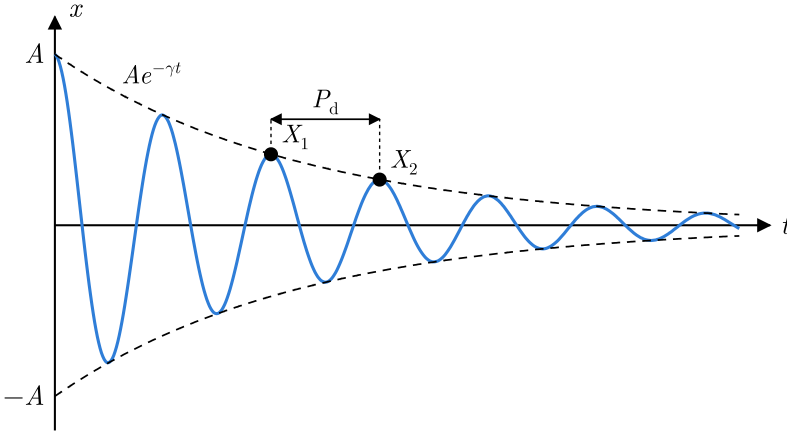


Figure 5. The displacement $x = Ae^{-\gamma t} \cos \omega_d t$ of a lightly damped oscillator, with the envelopes $\pm Ae^{-\gamma t}$ shown dashed. Successive maxima such as X_1 and X_2 are separated by the period $P_d = 2\pi/\omega_d$, and their ratio is the same for every pair.

The quality factor can be measured by watching the free oscillation die away. Let $P_d = 2\pi/\omega_d$ be the period of the lightly damped oscillator, the interval after which the cosine factor in $x = Ae^{-\gamma t} \cos(\omega_d t + \phi)$ repeats. At time $t + P_d$ the cosine has returned to its value at t , while the exponential has been multiplied by $e^{-\gamma P_d}$, so that

$$x(t + P_d) = e^{-\gamma P_d} x(t) \quad (1.25)$$

for every t . One period later, the whole curve of Figure 5 is a copy of itself reduced in the ratio $e^{-\gamma P_d}$. In particular, the maxima of the displacement are separated by P_d , as you showed in Problem 1 of Section 4.3 of Vol. 1, and if X_1 and X_2 are two successive maxima, then

$$\Lambda = \ln \frac{X_1}{X_2} = \gamma P_d = \frac{2\pi\gamma}{\omega_d}, \quad (1.26)$$

where Λ is called the logarithmic decrement. Since $\omega_d = \omega_0 \sqrt{1 - \gamma^2/\omega_0^2}$ and $\gamma/\omega_0 = 1/2Q$, the logarithmic decrement is related to the quality factor by

$$\Lambda = \frac{\pi}{Q} \frac{1}{\sqrt{1 - 1/4Q^2}} \approx \frac{\pi}{Q}, \quad (1.27)$$

where the approximation holds for light damping, $Q \gg 1$.

A maximum that comes n cycles after X_1 is smaller than X_1 by the factor $e^{-n\Lambda}$, because each cycle reduces the displacement by $e^{-\Lambda}$. If the amplitude falls by a factor of e in n_e cycles, then $n_e \Lambda = 1$, and

$$Q \approx \pi n_e, \quad (1.28)$$

so the quality factor of a lightly damped oscillator is π times the number of cycles in which its amplitude falls by a factor of e , which is the statement in Section 4.3 of Vol. 1 that such an oscillator

rings for about Q/π periods. It may be more convenient to count the cycles $n_{1/2}$ in which the amplitude falls to half, which gives $Q \approx \pi n_{1/2} / \ln 2$, as you will show in Problem 2. Counting cycles in this way, with the ratio of two maxima many cycles apart, is one way of measuring the quality factor of a lightly damped oscillator such as a pendulum, and it has the advantage that the error in measuring either maximum is spread over many cycles. The width of the resonance curve of a driven oscillator, which Section 4.3 of Vol. 1 related to Q , gives another way.

The Damped Oscillator in the Phase Plane

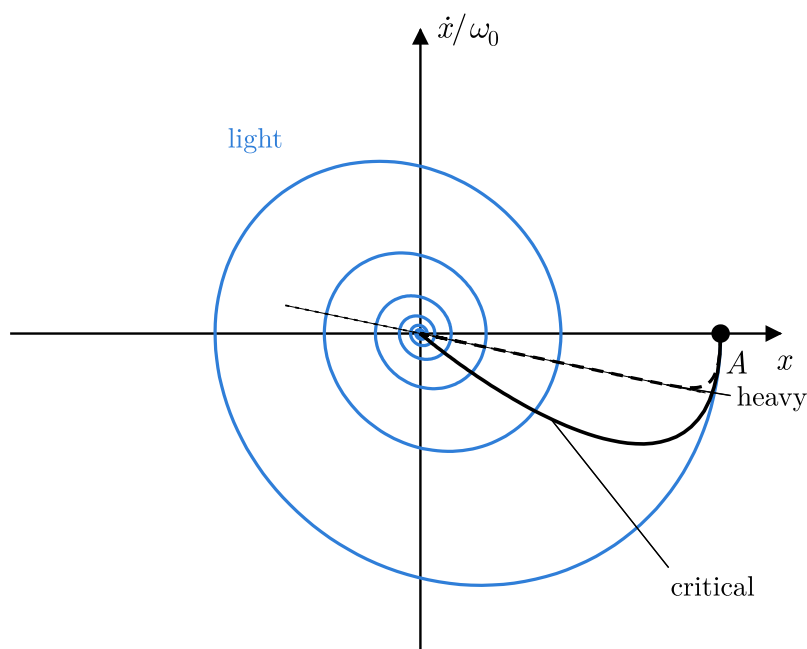


Figure 6. Phase trajectories of damped oscillators released from rest at displacement A , with the velocity divided by ω_0 on the vertical axis. The lightly damped oscillator spirals clockwise into the origin. The critically damped and heavily damped oscillators approach the origin without circling it, the heavily damped one along the dotted line $\dot{x} = \beta_+ x$, which is the trajectory of the slower exponential.

We now draw the damped motion in the phase plane of Section 1.1, with x along the horizontal axis and $\dot{x}/\omega_0$ along the vertical axis.

The energy is still $E = \frac{1}{2}\kappa[x^2 + (\dot{x}/\omega_0)^2]$, so it is still proportional to the square of the distance of the representative point from the origin. Since $dE/dt = -b\dot{x}^2$, the point moves steadily inwards, crossing the circles of constant energy that the undamped oscillator would follow, except at the instants when it crosses the horizontal axis, where $\dot{x} = 0$ and the point is moving along the circle through it.

For light damping the point goes round the origin clockwise, as the undamped oscillator does, and the radius $\sqrt{2E/\kappa}$ shrinks as it goes. Each complete turn takes a time P_d and reduces the radius by exactly the factor $e^{-\Lambda}$, since $\dot{x}$, like x , is multiplied by $e^{-\gamma P_d}$ in one period, and the trajectory is a spiral that winds into the origin, turning round it an unlimited number of times, as Figure 6 shows. A curve whose distance from a fixed point is multiplied by the same factor for every equal angle turned through about that point, so that the distance falls exponentially with the angle, is called a logarithmic spiral, and the trajectory of a lightly damped oscillator is approximately one.

For heavy damping the motion is $x = D_+e^{\beta_+t} + D_-e^{\beta_-t}$, with $\beta_- < \beta_+ < 0$. The motions in which one of the two constants is zero are pure exponentials, for which $\dot{x} = \beta_+x$ or $\dot{x} = \beta_-x$, and in the phase plane each of these is a straight line through the origin, with slope β_+/ω_0 or β_-/ω_0 in the plane of x and $\dot{x}/\omega_0$. A point on one of these lines moves along it into the origin. Every other motion is a mixture of the two exponentials, and since the term in e^{β_-t} decays faster, the point approaches the origin along the line $\dot{x} = \beta_+x$ of the slower exponential. The displacement can pass through zero at most once, because $x = 0$ requires $e^{(\beta_+ - \beta_-)t} = -D_-/D_+$, and the left-hand side increases steadily with time, so it can take the value on the right at no more than one instant. The trajectory can therefore cross the vertical axis of the phase plane at most once, and it never circles the origin. You will find in Problem 3 which starting points lead to a crossing.

For critical damping the two roots coincide at $\beta = -\omega_0$, and the two straight lines of the heavily damped oscillator merge into the single line $\dot{x} = -\omega_0x$, which makes an angle of $\pi/4$ with the axes of the phase plane. Every critically damped trajectory approaches the origin along this line, since the motion $x = (D_1 + D_2t)e^{-\omega_0t}$

gives $\dot{x}/x = -\omega_0 + D_2/(D_1 + D_2t)$, which tends to $-\omega_0$ as t grows.

The three cases are therefore distinguished in the phase plane by the straight-line trajectories. A heavily damped oscillator has two such lines, a critically damped one has one, and a lightly damped one has none, since its roots $\beta_{\pm}$ are complex, and its trajectory winds round the origin instead of approaching it along a line. Figure 6 shows the three kinds of trajectory for an oscillator released from rest at the same displacement.

1.3 Forced Oscillation, Resonance, and Impedance

The oscillator driven by a sinusoidal force was treated in Section 4.3 of Vol. 1, which found its steady motion and the transient that accompanies it. This section begins by restating the steady state, then follows the complex amplitude on the Argand diagram, introduces the mechanical impedance, and finds the mean power absorbed from the driving force, and finally returns to the transient and to an oscillator driven through the motion of its support.

The Steady State

Let the damped oscillator of Section 1.2 be driven by a force $F_0 \cos \omega t$, where the driving frequency ω can be chosen freely and is a different quantity from the natural frequency ω_0 . The equation of motion is

$$m\ddot{x} + b\dot{x} + \kappa x = F_0 \cos \omega t, \quad (1.29)$$

and Section 4.3 of Vol. 1 found its steady solution by replacing $F_0 \cos \omega t$ with $F_0 e^{i\omega t}$ and trying $x = \text{Re}(\mathcal{A}e^{i\omega t})$. Each differentiation multiplies $\mathcal{A}e^{i\omega t}$ by $i\omega$, and the complex amplitude is

$$\mathcal{A} = \frac{F_0/m}{\omega_0^2 - \omega^2 + 2i\gamma\omega}. \quad (1.30)$$

Writing $\mathcal{A} = Ae^{-i\delta}$, so that $x = A \cos(\omega t - \delta)$ and δ is the phase lag of the displacement behind the force, gives

$$A = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}}, \quad \tan \delta = \frac{2\gamma\omega}{\omega_0^2 - \omega^2}, \quad (1.31)$$

with δ between 0 and π . Slow driving gives $A = F_0/\kappa$ with the displacement in phase with the force; at $\omega = \omega_0$ the lag is $\pi/2$ and $A = F_0/b\omega_0 = QF_0/\kappa$; fast driving gives $A \approx F_0/m\omega^2$ with the displacement in antiphase. The amplitude is greatest at $\sqrt{\omega_0^2 - 2\gamma^2}$, a little below ω_0 for light damping, and this phenomenon is known as resonance. The general solution adds to the steady state one of the damped motions of Section 1.2, which is called the transient and dies away with time.

The Complex Amplitude on the Argand Diagram

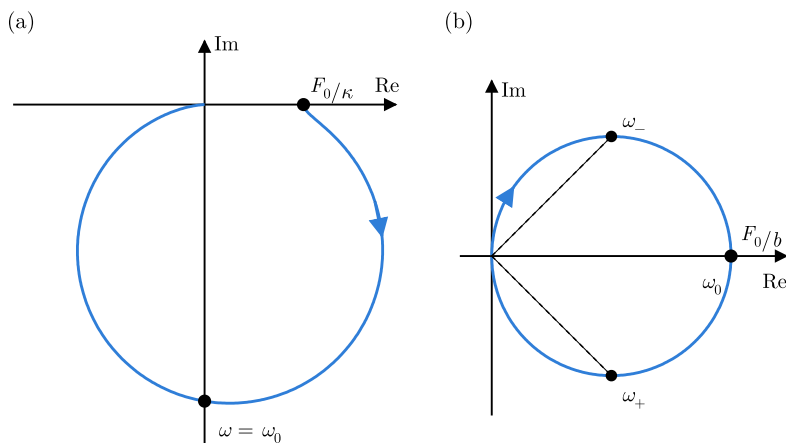


Figure 7. The steady response of a lightly damped driven oscillator on the Argand diagram as the driving frequency increases from zero. (a) The complex amplitude $\mathcal{A}$ of the displacement starts at F_0/κ , turns clockwise through the lower half of the plane, and ends at the origin. (b) The complex amplitude of the velocity, F_0/Z_m , goes round a circle of diameter F_0/b , reaching F_0/b at ω_0 and the points at angles of $\pi/4$ to the real axis at the half-power frequencies $\omega_{\pm}$.

As the driving frequency increases from zero, the complex amplitude traces out the curve in Figure 7(a). It starts on the positive

real axis at F_0/κ , where $\delta = 0$. As ω increases the lag δ grows, and since $\mathcal{A} = Ae^{-i\delta}$, the point turns clockwise about the origin through the lower half of the plane, where $\text{Im } \mathcal{A} = -A \sin \delta$ is negative. It crosses the negative imaginary axis at $-iQF_0/\kappa$ when $\omega = \omega_0$, and approaches the origin along the negative real axis as ω grows, since $\mathcal{A} \approx -F_0/m\omega^2$ there.

Multiplying out $\mathcal{A}e^{i\omega t}$ with Euler's formula gives $x = \text{Re}(\mathcal{A}) \cos \omega t - \text{Im}(\mathcal{A}) \sin \omega t$, so $\text{Re } \mathcal{A}$ is the amplitude of the part of the motion in phase with the force and $-\text{Im } \mathcal{A}$ that of the part a quarter of a period behind it, whose velocity is in phase with the force.

Mechanical Impedance

The velocity of the body is $\dot{x} = \text{Re}(i\omega \mathcal{A}e^{i\omega t})$, and the ratio of the complex force $F_0e^{i\omega t}$ to the complex velocity $i\omega \mathcal{A}e^{i\omega t}$ does not depend on time. This ratio,

$$Z_m = \frac{F_0}{i\omega \mathcal{A}} = \frac{\kappa - m\omega^2 + i b \omega}{i\omega} = b + i \left(m\omega - \frac{\kappa}{\omega} \right), \quad (1.32)$$

is called the mechanical impedance of the oscillator. Its real part b is called the mechanical resistance and its imaginary part $m\omega - \kappa/\omega$ the mechanical reactance. A force of complex amplitude F_0 gives a velocity of complex amplitude F_0/Z_m , so a large impedance means a small response. The reactance is the sum of a term $m\omega$ from the inertia of the body, which dominates well above ω_0 , and a term $-\kappa/\omega$ from the stiffness of the spring, which dominates well below it. At $\omega = \omega_0$ the reactance is zero, $Z_m = b$, and the velocity is in phase with the force.

As the frequency varies, the impedance moves along the vertical line $\text{Re } Z_m = b$ in the complex plane, from $-i\infty$ at low frequencies to $+i\infty$ at high ones, and the complex amplitude of the velocity, F_0/Z_m , moves round a circle. To show this, let $X_m = m\omega - \kappa/\omega$ be the reactance and subtract $F_0/2b$ from the velocity amplitude:

$$\frac{F_0}{b + iX_m} - \frac{F_0}{2b} = \frac{F_0(b - iX_m)}{2b(b + iX_m)}. \quad (1.33)$$

The numerator and the denominator of the fraction on the right contain complex numbers of the same modulus, $|b - iX_m| = |b + iX_m|$, so the modulus of the right-hand side is $F_0/2b$ for every value of X_m . The velocity amplitude therefore lies on the circle of radius $F_0/2b$ centred at $F_0/2b$, which passes through the origin and through the point F_0/b , as in Figure 7(b). Below resonance the reactance is negative, and the velocity amplitude lies in the upper half of the circle, ahead of the force in phase; above resonance it lies in the lower half, behind the force.

Power Absorbed and the Width of the Resonance

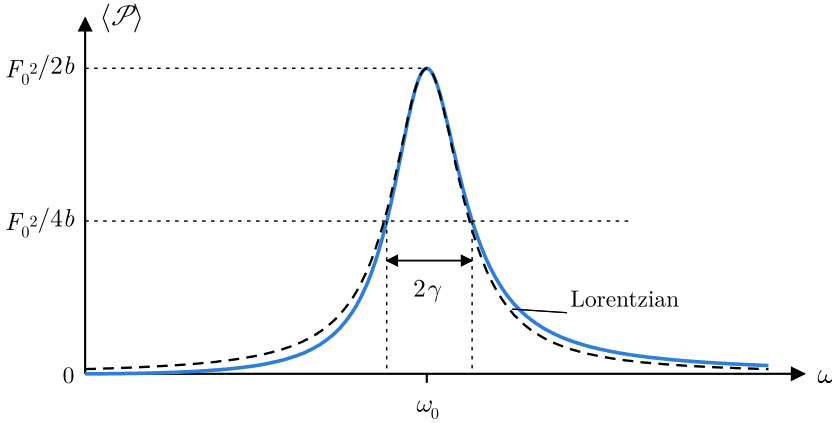


Figure 8. The mean power absorbed by a lightly damped oscillator from a driving force of amplitude F_0 , as a function of the driving frequency. The peak is at ω_0 , and the power falls to half its peak value at two frequencies separated by 2γ . The dashed curve is the Lorentzian approximation, which fits the exact curve closely near the peak.

The power delivered by the driving force is $\mathcal{P} = F_0 \cos \omega t \dot{x}$, the product of two oscillations of the same angular frequency, with complex amplitudes F_0 and F_0/Z_m . By the formula for the mean of a product in Section 1.1, the mean power over a cycle is

$$\langle \mathcal{P} \rangle = \frac{1}{2} \operatorname{Re} \left(\frac{F_0^2}{Z_m^*} \right) = \frac{F_0^2}{2} \frac{b}{|Z_m|^2} = \frac{F_0^2}{2} \frac{b}{b^2 + (m\omega - \kappa/\omega)^2}, \quad (1.34)$$

where the second form follows from $1/Z_m^* = Z_m/|Z_m|^2$, whose real part is $b/|Z_m|^2$. In the steady state the energy stored in the oscillator is the same at the end of each cycle as at its start, so this power is equal to the mean power dissipated by the resistive force, as you will check in Problem 1. The mean power is greatest when the reactance is zero, which is at $\omega = \omega_0$ exactly, and its greatest value is

$$\langle \mathcal{P} \rangle_{\max} = \frac{F_0^2}{2b}. \quad (1.35)$$

The power peaks at the natural frequency for any damping, while the displacement amplitude peaks below it, because the power goes as the square of the velocity amplitude, whose peak is at ω_0 , as you showed in Problem 3 of Section 4.3 of Vol. 1.

The mean power falls to half its peak value when the reactance is equal in magnitude to the resistance, $m\omega - \kappa/\omega = \pm b$. Multiplying by ω/m turns this condition into the quadratic equation

$$\omega^2 \mp 2\gamma\omega - \omega_0^2 = 0, \quad (1.36)$$

and by the quadratic formula the positive roots of the two equations, one for each sign, are

$$\omega_{\pm} = \sqrt{\omega_0^2 + \gamma^2} \pm \gamma. \quad (1.37)$$

The frequencies ω_+ and ω_- are called the half-power frequencies, and their difference is exactly

$$\Delta\omega = \omega_+ - \omega_- = 2\gamma = \frac{\omega_0}{Q}, \quad (1.38)$$

which is called the width of the resonance. For light damping, $\gamma \ll \omega_0$, the half-power frequencies are approximately $\omega_0 \pm \gamma$. The quality factor is therefore the ratio of the natural frequency to the width of the resonance, $Q = \omega_0/\Delta\omega$, a result that Section 4.3 of Vol. 1 obtained approximately from the square of the displacement

amplitude and that holds exactly for the power. At the half-power frequencies $\omega_{\pm}$ the impedance is $Z_m = b(1 \pm i)$, and the velocity amplitude lies at the two points of the circle in Figure 7(b) at angles of $\pi/4$ to the real axis.

The Lorentzian Shape of the Resonance

Near the peak the formula for the power takes a simpler form. The reactance can be written as

$$m\omega - \frac{\kappa}{\omega} = \frac{m}{\omega} (\omega^2 - \omega_0^2) = \frac{m}{\omega} (\omega - \omega_0) (\omega + \omega_0) , \quad (1.39)$$

and when $|\omega - \omega_0| \ll \omega_0$ the factor $(\omega + \omega_0)/\omega$ is approximately 2, so the reactance is approximately $2m(\omega - \omega_0)$. Substituting this into the mean power, and using $b = 2m\gamma$, gives $b^2 + (m\omega - \kappa/\omega)^2 \approx 4m^2[\gamma^2 + (\omega - \omega_0)^2]$, and so

$$\langle \mathcal{P} \rangle \approx \frac{F_0^2}{2b} \frac{\gamma^2}{(\omega - \omega_0)^2 + \gamma^2} . \quad (1.40)$$

A curve of this shape was called a Lorentzian in Section 6.2 of Vol. 4, where it appeared as the spectrum of a decaying exponential. It is symmetric about ω_0 , falls to half its peak at $\omega_0 \pm \gamma$, and far from the peak goes as $1/(\omega - \omega_0)^2$. The approximation holds across the entire peak, whose width is 2γ , provided that $\gamma \ll \omega_0$, which is to say $Q \gg 1$. In Figure 8 the exact curve lies below the Lorentzian well below ω_0 and above it well above, because the factor $(\omega + \omega_0)/\omega$ that the approximation replaced by 2 is greater than 2 below resonance and less than 2 above.

Starting from Rest

We now add the transient to the steady state for a lightly damped oscillator that is at rest at its equilibrium position when the force is switched on at $t = 0$. The general solution is

$$x = A \cos(\omega t - \delta) + e^{-\gamma t} (D_1 \cos \omega_d t + D_2 \sin \omega_d t) , \quad (1.41)$$

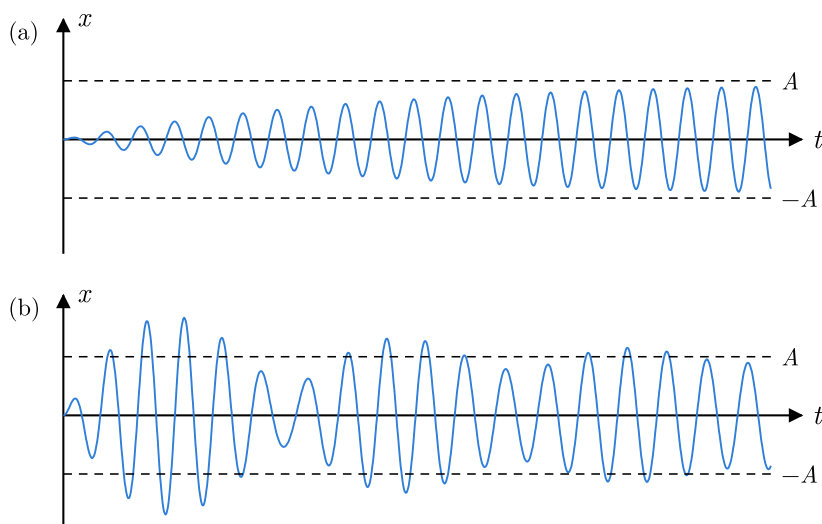


Figure 9. The displacement of a lightly damped oscillator started from rest by a sinusoidal force, with the steady amplitude $\pm A$ shown dashed. (a) Driven at ω_0 , the amplitude grows smoothly towards A as $A(1 - e^{-\gamma t})$. (b) Driven at a frequency somewhat below ω_0 , the oscillator beats at the difference frequency while the transient lasts, and the beats die away as the motion settles into the steady state.

and the conditions $x(0) = 0$ and $\dot{x}(0) = 0$ give two equations for the constants D_1 and D_2 . The first gives $A \cos \delta + D_1 = 0$, and the second, after differentiating by the product rule, gives $A\omega \sin \delta - \gamma D_1 + \omega_d D_2 = 0$, so that

$$D_1 = -A \cos \delta, \quad D_2 = -\frac{A}{\omega_d} (\omega \sin \delta + \gamma \cos \delta). \quad (1.42)$$

Let us take first an oscillator driven at its natural frequency, $\omega = \omega_0$. Then $\delta = \pi/2$, so that $D_1 = 0$ and $D_2 = -A\omega_0/\omega_d$, which is approximately $-A$ when the damping is light, and, replacing ω_d by ω_0 in the transient for the same reason, the motion is

$$x \approx A(1 - e^{-\gamma t}) \sin \omega_0 t. \quad (1.43)$$

The oscillation grows smoothly towards its steady amplitude over a time of order $1/\gamma$, which is Q/π periods, as in Figure 9(a).

Now let the driving frequency lie below ω_0 , differing from it by much more than γ while remaining close enough to it that ω and ω_d are nearly equal. Then $\tan \delta = 2\gamma\omega/(\omega_0^2 - \omega^2)$ is approximately $\gamma/(\omega_0 - \omega)$, which is small, so δ is small, and to a first approximation $D_1 = -A$ and $D_2 = 0$, since D_2 is of order $A\gamma/(\omega_0 - \omega)$. The motion is then

$$x \approx A \cos \omega t - A e^{-\gamma t} \cos \omega_d t, \quad (1.44)$$

a sum of two oscillations, one at the driving frequency with the steady amplitude and one at the frequency ω_d with the decaying amplitude $A e^{-\gamma t}$. Their phase difference, $(\omega - \omega_d)t$, changes slowly compared with either oscillation, so over any one cycle the sum is an oscillation whose amplitude follows from the addition of phasors in Section 1.2 of Vol. 4. The two oscillations are in antiphase at $t = 0$, and the amplitude of their sum is

$$A \sqrt{1 + e^{-2\gamma t} - 2e^{-\gamma t} \cos [(\omega - \omega_d)t]}. \quad (1.45)$$

This amplitude rises and falls with angular frequency $|\omega - \omega_d|$, between $A(1 - e^{-\gamma t})$ and $A(1 + e^{-\gamma t})$, which are beats at the difference between the driving frequency and the frequency ω_d of the free damped oscillation. As the transient decays, the beats die away and the amplitude settles at A , as in Figure 9(b). If the beat period $2\pi/|\omega - \omega_d|$ is short compared with $1/\gamma$, the displacement during the first beat approaches twice its steady amplitude.

Driving through the Support

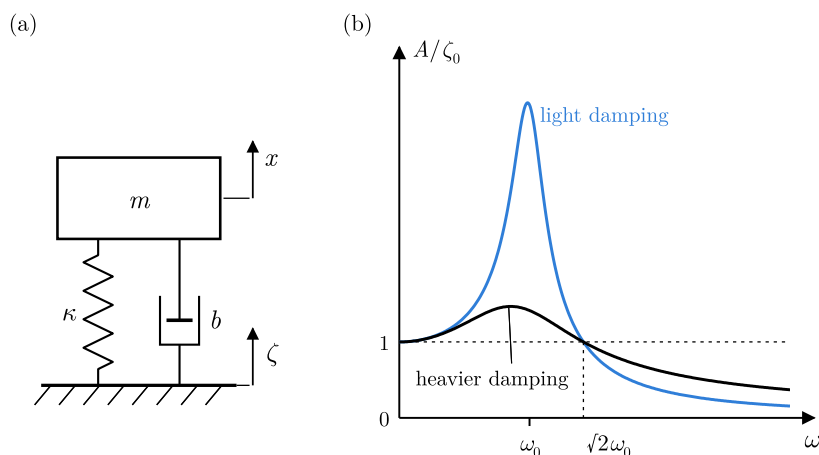


Figure 10. (a) A body of mass m joined to a moving support by a spring and, side by side with it, a piston in a cylinder of oil. The support moves with displacement ζ and the body with displacement x . (b) The ratio of the amplitude of the body to the amplitude of the support, for light damping and for heavier damping. Every curve passes through 1 at $\omega = \sqrt{2}\omega_0$, and above that frequency the body moves less than the support.

An oscillator can also be driven by moving the point to which its spring is attached, as the ground beneath a seismometer moves during an earthquake. A seismometer is an instrument that records the motion of the ground by measuring how a mass, hung from a spring inside a case fixed to the ground, moves relative to the case. Let a body of mass m be joined to a support by a spring of constant κ and, side by side with the spring, a piston in a cylinder of oil, as described in Section 1.2, with damping coefficient b , as in Figure

10(a), and let the support move with displacement $\zeta(t) = \zeta_0 \cos \omega t$. The displacement x of the body is measured from its equilibrium position when the support is at rest, and gravity, if the arrangement is vertical, only shifts that equilibrium position, as you showed in Problem 2 of Section 4.2 of Vol. 1 for a body hanging from a spring. The spring is stretched by $x - \zeta$, and the piston moves relative to its cylinder with velocity $\dot{x} - \dot{\zeta}$, so

$$m\ddot{x} = -\kappa(x - \zeta) - b(\dot{x} - \dot{\zeta}), \quad (1.46)$$

which can be rewritten as

$$m\ddot{x} + b\dot{x} + \kappa x = \kappa\zeta + b\dot{\zeta}. \quad (1.47)$$

The support drives the body through both the spring and the piston. With $\zeta = \text{Re}(\zeta_0 e^{i\omega t})$ and $x = \text{Re}(\mathcal{A} e^{i\omega t})$, each time derivative becomes a factor $i\omega$, and dividing by m gives

$$\frac{\mathcal{A}}{\zeta_0} = \frac{\omega_0^2 + 2i\gamma\omega}{\omega_0^2 - \omega^2 + 2i\gamma\omega}. \quad (1.48)$$

The ratio of the amplitude A of the body to the amplitude ζ_0 of the support is the modulus of this expression,

$$\frac{A}{\zeta_0} = \sqrt{\frac{\omega_0^4 + 4\gamma^2\omega^2}{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}}, \quad (1.49)$$

which is called the transmissibility and is plotted in Figure 10(b). When the support moves slowly, $\omega \ll \omega_0$, the ratio is close to 1 and the body follows the support. At $\omega = \omega_0$ the ratio is $\sqrt{1 + \omega_0^2/4\gamma^2} = \sqrt{1 + Q^2}$, approximately Q for light damping, and the motion of the support is amplified. The ratio is exactly 1 when the numerator and the denominator are equal, that is when $(\omega_0^2 - \omega^2)^2 = \omega_0^4$, which gives $\omega = \sqrt{2}\omega_0$ for any damping. Above this frequency the body moves less than the support. Far above it, the transmissibility is approximately $\sqrt{\omega_0^4 + 4\gamma^2\omega^2}/\omega^2$, which falls

as ω_0^2/ω^2 while $2\gamma\omega \ll \omega_0^2$ but only as $2\gamma/\omega$ at higher frequencies, so the damping that limits the amplification at resonance weakens the isolation at high frequencies.

A machine can therefore be isolated from vibrations of the floor by mounting it on springs soft enough that $\sqrt{2}\omega_0$ lies well below the angular frequency of the vibration. Such a machine passes through the resonance each time it is started, which is one reason designers of such mountings often accept a moderate amount of damping as a compromise between the height of the resonance and the isolation at the running speed. A seismometer records the displacement of the body relative to its frame, $x - \zeta$, whose complex amplitude is $\mathcal{A} - \zeta_0 = \zeta_0\omega^2/(\omega_0^2 - \omega^2 + 2i\gamma\omega)$. For $\omega \gg \omega_0$ this tends to $-\zeta_0$, so the body stays almost at rest while the frame moves with the ground, and the relative displacement reproduces the displacement of the ground. You will show in Problem 4 that at frequencies well below ω_0 the same instrument records the acceleration of the ground.

Problems

§1.1 Simple Harmonic Motion Revisited

1. Using the phase plane, find the displacements at which the kinetic and potential energies of a simple harmonic oscillator are equal, and show that the potential energy is greater than the kinetic energy for half of each period.
2. The arms of a U-tube of uniform cross-section make angles θ_1 and θ_2 with the horizontal, and the tube holds a column of liquid of total length H . Show that the angular frequency of small oscillations of the liquid is given by $\omega_0^2 = g(\sin\theta_1 + \sin\theta_2)/H$, and check that it reduces to the result for vertical arms.
3. A small body of mass m is fixed at the midpoint of a light string of length L , which is stretched to a tension τ between two fixed points. Neglecting gravity, find the angular frequency of small transverse oscillations of the body, assuming that the tension does not change as the body moves.
4. For an oscillator with $x = \text{Re}(\mathcal{A}e^{i\omega_0 t})$, use the formula for the

mean of a product of two oscillations to show that $\langle x\dot{x} \rangle = 0$. Explain this result by writing $x\dot{x}$ as the derivative of a periodic function, and interpret it in the phase plane.

5. A uniform solid cylinder of density ρ_b and length H_c floats upright in a liquid of greater density ρ_L . Find the depth to which it is immersed, and show that the angular frequency of its vertical oscillations is given by $\omega_0^2 = \rho_L g / (\rho_b H_c)$, neglecting the motion of the liquid.

§1.2 Damped Oscillations and the Quality Factor

1. Show that the logarithmic decrement is given exactly by $\Lambda = 2\pi / \sqrt{4Q^2 - 1}$, and that for large Q the estimate π / Λ differs from Q by a fraction of about $1/8Q^2$.
2. The amplitude of a lightly damped oscillator falls to half its initial value after $n_{1/2}$ cycles. Show that $Q \approx \pi n_{1/2} / \ln 2$, and find the number of cycles after which the energy has fallen to half its initial value.
3. A heavily damped oscillator starts at $x = A$, with $A > 0$, with velocity $\dot{x}_i$. Show that its displacement passes through zero at some later time if and only if $\dot{x}_i < \beta_- A$, and describe this condition in the phase plane.
4. A critically damped oscillator is released from rest at displacement A . Find its velocity as a function of time, and show that the greatest speed it reaches is $\omega_0 A / e$. Sketch its phase trajectory, and show that it approaches the line $\dot{x} = -\omega_0 x$ without meeting it.
5. Show that the energies of a lightly damped oscillator at two successive maxima of its displacement are exactly in the ratio $e^{2\Lambda}$, and hence that the fraction of the energy lost in one cycle is approximately $2\pi/Q$ when Q is large.

§1.3 Forced Oscillation, Resonance, and Impedance

1. Show that in the steady state the mean power dissipated by the resistive force, $\langle b\dot{x}^2 \rangle$, is equal to the mean power $\langle \mathcal{P} \rangle$ delivered by the driving force.

2. Show that the half-power frequencies satisfy $\omega_+\omega_- = \omega_0^2$, so that ω_0 is their geometric mean, and find the phase lag δ of the displacement behind the force at each of them.
3. Show that for an oscillator driven at ω_0 the total energy is constant in the steady state, and that the ratio of this energy to the energy dissipated in one cycle is $Q/2\pi$, as it is for the freely decaying oscillator of Section 1.2.
4. For the seismometer described in the text, show that when $\omega \ll \omega_0$ the relative displacement $x - \zeta$ is approximately $-\ddot{\zeta}/\omega_0^2$, so that it is proportional to the acceleration of the ground.
5. An undamped oscillator, initially at rest at equilibrium, is driven by the force $F_0 \cos \omega_0 t$ at its natural frequency. By taking the limit $\gamma \rightarrow 0$ of the result for driving at resonance, show that $x = (F_0 t / 2m\omega_0) \sin \omega_0 t$, and verify by substitution that this satisfies $m\ddot{x} + \kappa x = F_0 \cos \omega_0 t$.

Coupled Oscillators, Normal Modes, and Fourier Analysis

2.1 Two Coupled Oscillators

A pair of equal masses joined to each other and to two walls by springs is treated in Section 4.4 of Vol. 1 of the Fermi Notes on Physics, and again, as an eigenvalue problem, in Section 4.3 of Vol. 4 of the Fermi Notes on Physics. The normal modes, the normal coordinates, and the beats of such a pair are worked out in detail in those sections, so we recall them only briefly, in the notation of this course. The rest of the section takes them further. It first follows the energy as it passes between a pair of coupled pendulums, then shows why the passage is incomplete when the pendulums differ, and finally drives one of a pair of coupled masses and finds how each of them responds.

The Normal Modes Recalled

Let two equal masses m slide without friction along a line, each joined to the wall nearest it by a spring of constant κ and to the other mass by a coupling spring of constant κ_c , and let x_1 and x_2 be their displacements from equilibrium, both measured in the same direction. The coupling spring is stretched by $x_2 - x_1$, and Newton's second law for each mass gives

$$m\ddot{x}_1 = -\kappa x_1 + \kappa_c(x_2 - x_1), \quad m\ddot{x}_2 = -\kappa x_2 - \kappa_c(x_2 - x_1), \quad (2.1)$$

which are the equations of Section 4.4 of Vol. 1, with the spring constants written κ and κ_c in place of k and k_c . As in Section 4.3 of Vol. 4, we collect the displacements into a column $\mathbf{x}$, with components x_1 and x_2 , and write the pair of equations as a single matrix equation,

$$\ddot{\mathbf{x}} = -\mathbf{D} \mathbf{x}, \quad \mathbf{D} = \frac{1}{m} \begin{pmatrix} \kappa + \kappa_c & -\kappa_c \\ -\kappa_c & \kappa + \kappa_c \end{pmatrix}. \quad (2.2)$$

The matrix $\mathbf{D}$ is called the dynamical matrix of the system. A motion in which both masses oscillate with the same angular frequency ω and a fixed ratio of displacements has the form $\mathbf{x} = \mathbf{e} \cos(\omega t + \phi)$, where $\mathbf{e}$ is a constant column, and since its second derivative with respect to time is $-\omega^2 \mathbf{x}$, it satisfies the equation of motion when

$$\mathbf{D} \mathbf{e} = \omega^2 \mathbf{e}. \quad (2.3)$$

The squares of the normal frequencies are therefore the eigenvalues of $\mathbf{D}$, and the patterns of the normal modes are its eigenvectors. For the matrix above they were found in Problem 6 of Section 4.3 of Vol. 4, and they are

$$\omega_1^2 = \frac{\kappa}{m}, \quad \mathbf{e}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \omega_2^2 = \frac{\kappa + 2\kappa_c}{m}, \quad \mathbf{e}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \quad (2.4)$$

In the first mode the masses move together, the coupling spring keeps its natural length, and each mass oscillates at the natural frequency $\omega_0 = \sqrt{\kappa/m}$ of a mass on its outer spring alone, so that $\omega_1 = \omega_0$. In the second mode the masses move in opposite directions and the coupling spring adds to the restoring force on each. The normal coordinates $q_1 = \frac{1}{2}(x_1 + x_2)$ and $q_2 = \frac{1}{2}(x_1 - x_2)$ each obey the equation of a single oscillator, $\ddot{q}_1 = -\omega_1^2 q_1$ and $\ddot{q}_2 = -\omega_2^2 q_2$, and the general motion is a superposition of the two modes with four constants fixed by the initial positions and velocities of the masses.

Section 4.4 of Vol. 1 also worked out the case in which the coupling is weak, $\kappa_c \ll \kappa$, and the first mass is pulled aside a distance A and released from rest with the second at rest in equilibrium. Let

$$\bar{\omega} = \frac{1}{2}(\omega_1 + \omega_2), \quad \omega_b = \omega_2 - \omega_1 \approx \frac{\kappa_c}{m\omega_0}, \quad (2.5)$$

where the approximate value of ω_b follows from the binomial expansion $\omega_2 = \omega_0(1 + 2\kappa_c/\kappa)^{1/2} \approx \omega_0(1 + \kappa_c/\kappa)$. The displacements are then

$$x_1 = A \cos \frac{\omega_b t}{2} \cos \bar{\omega} t, \quad x_2 = A \sin \frac{\omega_b t}{2} \sin \bar{\omega} t. \quad (2.6)$$

Each mass oscillates at the mean frequency $\bar{\omega}$, with an amplitude that changes slowly, and the motion passes from the first mass to the second and back again. This slow alternation is called beating.

Energy Exchange between Coupled Pendulums

The beating of Section 4.4 of Vol. 1 is often demonstrated with a pair of pendulums. Let each pendulum have a bob of mass m on a light string of length L , and let the bobs be joined by a light spring of constant κ_c , as in Figure 11. When a bob is displaced a small horizontal distance x , its string makes an angle of about x/L with the vertical, and the tension, which is close to mg for small angles, has a horizontal component of about $-mgx/L$. Each pendulum, with the spring removed, therefore behaves as a mass on a spring of constant mg/L , and the equations of motion of the bobs are those above with $\kappa = mg/L$ and $\omega_0^2 = g/L$.

Let us follow the energy as it passes between the pendulums. Let the energy of the first pendulum be the sum of the kinetic energy of its bob and the potential energy of its restoring force, $E_1 = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\omega_0^2 x_1^2$, and define E_2 for the second pendulum in the same way. The coupling spring holds some energy as well, but you will show in Problem 1 that it is smaller than the energy of the pendulums by a factor of order $\kappa_c/m\omega_0^2$, which is small when the coupling is weak. Differentiating the expression for x_1 with the product rule gives

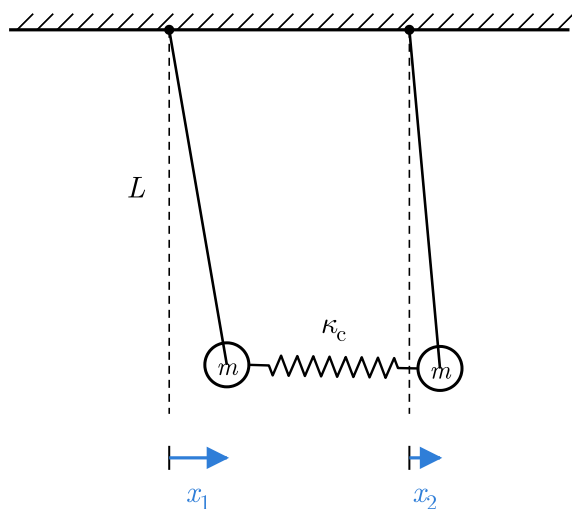


Figure 11. A pair of identical pendulums, with bobs of mass m on strings of length L , hung side by side and joined by a light spring of constant κ_c that has its natural length when both strings are vertical. The displacements x_1 and x_2 of the bobs are measured horizontally from their equilibrium positions, which lie on the dashed lines.

$$\dot{x}_1 = -A\bar{\omega} \cos \frac{\omega_b t}{2} \sin \bar{\omega} t - \frac{A\omega_b}{2} \sin \frac{\omega_b t}{2} \cos \bar{\omega} t. \quad (2.7)$$

The second term is smaller than the first by a factor of order $\omega_b/\bar{\omega}$, which is again of order $\kappa_c/m\omega_0^2$, and to the same accuracy $\bar{\omega}$ may be replaced by ω_0 . Keeping only the first term, and using $\sin^2 \bar{\omega} t + \cos^2 \bar{\omega} t = 1$, the energy of the first pendulum is

$$E_1 = \frac{1}{2}m\omega_0^2 A^2 \cos^2 \frac{\omega_b t}{2} = \frac{1}{2}E (1 + \cos \omega_b t), \quad (2.8)$$

where $E = \frac{1}{2}m\omega_0^2 A^2$ is the energy given to the first pendulum at the start, and the last step uses the identity $2 \cos^2 \theta = 1 + \cos 2\theta$. The same calculation for the second pendulum gives

$$E_2 = \frac{1}{2}m\omega_0^2 A^2 \sin^2 \frac{\omega_b t}{2} = \frac{1}{2}E (1 - \cos \omega_b t), \quad (2.9)$$

so that $E_1 + E_2 = E$ at every instant, to the accuracy of the approximation.

The energy therefore passes back and forth between the pendulums at the angular frequency ω_b , as Figure 12 shows. All of it has reached the second pendulum at the time found in Problem 4 of Section 4.4 of Vol. 1,

$$t = \frac{\pi}{\omega_b} \approx \frac{\pi m \omega_0}{\kappa_c}, \quad (2.10)$$

and all of it is back with the first at $t = 2\pi/\omega_b$. The period of a swing is $2\pi/\omega_0$, so the number of swings made during one complete transfer is about $\omega_0/2\omega_b = m\omega_0^2/2\kappa_c$, which is large when the coupling is weak. A weak coupling passes the energy slowly, and a spring of half the stiffness takes twice as long to pass it.

The energy is carried across by the coupling spring in the following way. The spring pushes the second bob with the force $\kappa_c(x_1 - x_2)$, and the rate at which this force does work on the second pendulum is $\kappa_c(x_1 - x_2)\dot{x}_2$. During the first half of the transfer, x_1 goes as

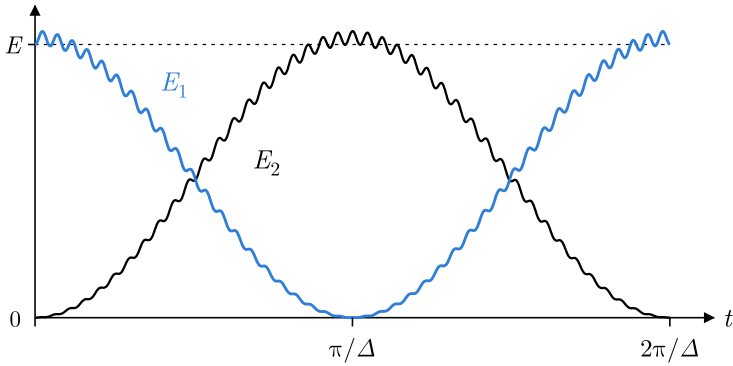


Figure 12. The energies E_1 (accent) and E_2 of a pair of weakly coupled pendulums, computed from the exact solution after the first pendulum is released from rest with the second at rest in equilibrium. The small ripples, at twice the frequency of the swings, come from the terms neglected in the approximate expressions for E_1 and E_2 , and E_1 can rise a little above E because the coupling spring, stretched by A at the start, holds a small energy of its own.

$\cos \bar{\omega}t$ and x_2 as $\sin \bar{\omega}t$, so the second pendulum is a quarter of a period behind the first, and its velocity $\dot{x}_2$, which goes as $\cos \bar{\omega}t$, is in phase with the force $\kappa_c x_1$ that the first pendulum exerts through the spring. This is the phase at which an oscillator absorbs power most readily from a driving force, as Section 1.3 showed for an oscillator driven at resonance, where the velocity is in phase with the force. Averaging the rate of working over one swing, and using the mean value $\frac{1}{2}$ of $\cos^2 \bar{\omega}t$ and the mean value zero of $\sin \bar{\omega}t \cos \bar{\omega}t$, gives the mean power

$$\langle \mathcal{P} \rangle = \frac{1}{2} \kappa_c A^2 \omega_0 \cos \frac{\omega_b t}{2} \sin \frac{\omega_b t}{2} = \frac{1}{4} \kappa_c A^2 \omega_0 \sin \omega_b t, \quad (2.11)$$

which agrees with the rate of change $dE_2/dt = \frac{1}{2} E \omega_b \sin \omega_b t$ found by differentiating the expression for E_2 , once $\omega_b = \kappa_c / m \omega_0$ is substituted. After the time π / ω_b the factor $\cos(\omega_b t / 2)$ in x_1 changes sign, the second pendulum is then a quarter of a period ahead of the first, and the power flows back.

Pendulums of Unequal Length

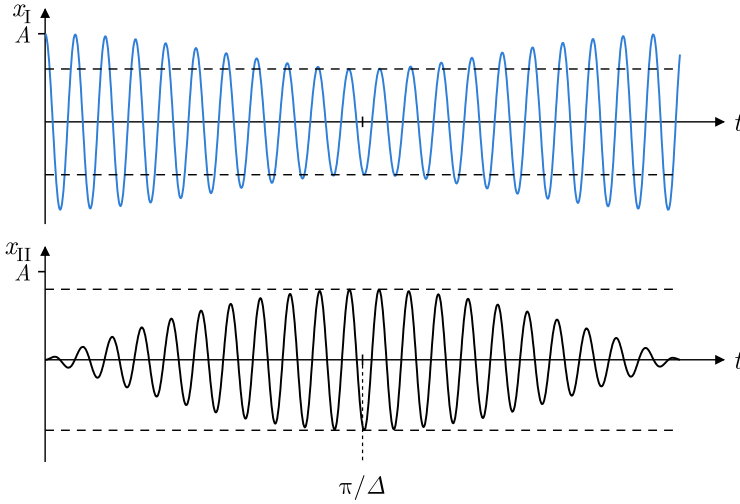


Figure 13. The displacements x_I (accent) and x_{II} of a pair of coupled pendulums of unequal length, computed from the normal modes after pendulum I is released from rest a distance A from equilibrium with pendulum II at rest in equilibrium. The amplitude of II never exceeds $A \sin 2\psi$, and that of I never falls below $A \cos 2\psi$, both marked by the dashed lines.

If the two pendulums have different lengths, the energy no longer passes completely from one to the other. Let the pendulums be labelled I and II, with strings of lengths L_I and L_{II} and bobs of the same mass m , and let

$$\omega_I^2 = \frac{g}{L_I} + \frac{\kappa_c}{m}, \quad \omega_{II}^2 = \frac{g}{L_{II}} + \frac{\kappa_c}{m}, \quad (2.12)$$

so that ω_I is the angular frequency with which pendulum I would swing if bob II were held fixed, and similarly for ω_{II} . The equations of motion are then $\ddot{x}_I = -\omega_I^2 x_I + (\kappa_c/m)x_{II}$ and $\ddot{x}_{II} = -\omega_{II}^2 x_{II} + (\kappa_c/m)x_I$, and the dynamical matrix is

$$D = \begin{pmatrix} \omega_I^2 & -\kappa_c/m \\ -\kappa_c/m & \omega_{II}^2 \end{pmatrix}. \quad (2.13)$$

Its characteristic equation is $(\omega_I^2 - \omega^2)(\omega_{II}^2 - \omega^2) - \kappa_c^2/m^2 = 0$, whose roots are the squared normal frequencies,

$$\omega_{1,2}^2 = \frac{1}{2} (\omega_I^2 + \omega_{II}^2) \mp \frac{1}{2} \sqrt{(\omega_I^2 - \omega_{II}^2)^2 + 4\kappa_c^2/m^2}. \quad (2.14)$$

The matrix is real and symmetric, so by Section 4.3 of Vol. 4 its normalised eigenvectors are perpendicular unit vectors. Any pair of perpendicular unit vectors in a plane can be obtained by turning the pair $(1, 0)$ and $(0, 1)$ through some angle ψ , and reversing one of them if necessary, so the eigenvectors can be written

$$\mathbf{e}_1 = \begin{pmatrix} \cos \psi \\ \sin \psi \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} -\sin \psi \\ \cos \psi \end{pmatrix}, \quad (2.15)$$

with $\mathbf{e}_1$ belonging to ω_1^2 and $\mathbf{e}_2$ to ω_2^2 . To find ψ , we use the result of Section 4.3 of Vol. 4 that the orthogonal matrix $\mathbf{S}$ with columns $\mathbf{e}_1$ and $\mathbf{e}_2$ satisfies $\mathbf{S}^T \mathbf{D} \mathbf{S} = \Lambda$, where Λ is diagonal with entries ω_1^2 and ω_2^2 . Multiplying on the left by $\mathbf{S}$ and on the right by $\mathbf{S}^T$ gives $\mathbf{D} = \mathbf{S} \Lambda \mathbf{S}^T$, and multiplying out the three matrices, with the double-angle formulas $2 \sin \psi \cos \psi = \sin 2\psi$ and $\cos^2 \psi - \sin^2 \psi = \cos 2\psi$, gives for the off-diagonal entry and for the difference of the diagonal entries

$$-\frac{\kappa_c}{m} = -\frac{1}{2} (\omega_2^2 - \omega_1^2) \sin 2\psi, \quad \omega_1^2 - \omega_2^2 = -(\omega_2^2 - \omega_1^2) \cos 2\psi. \quad (2.16)$$

Since $\omega_2^2 - \omega_1^2$ is the square root in the expression for the normal frequencies, the first of these relations gives

$$\sin 2\psi = \frac{2\kappa_c/m}{\sqrt{(\omega_I^2 - \omega_{II}^2)^2 + 4\kappa_c^2/m^2}}. \quad (2.17)$$

When the pendulums are equal, $\sin 2\psi = 1$, so $\psi = \pi/4$ and the eigenvectors are those of the equal masses, $(1, 1)/\sqrt{2}$ and $(-1, 1)/\sqrt{2}$.

Now let pendulum I be released from rest a distance A from equilibrium, with pendulum II at rest in equilibrium. The initial column $(1, 0)$ has the components $\cos \psi$ along $\mathbf{e}_1$ and $-\sin \psi$ along $\mathbf{e}_2$, found by taking its scalar product with each, and each mode then oscillates at its frequency, so

$$\begin{pmatrix} x_I \\ x_{II} \end{pmatrix} = A \cos \psi \mathbf{e}_1 \cos \omega_1 t - A \sin \psi \mathbf{e}_2 \cos \omega_2 t. \quad (2.18)$$

The lower component, with the identity $\cos \omega_1 t - \cos \omega_2 t = 2 \sin \bar{\omega} t \sin(\omega_b t/2)$, is

$$x_{II} = A \sin \psi \cos \psi (\cos \omega_1 t - \cos \omega_2 t) = A \sin 2\psi \sin \frac{\omega_b t}{2} \sin \bar{\omega} t, \quad (2.19)$$

with $\bar{\omega}$ and ω_b defined from ω_1 and ω_2 as before. The amplitude of pendulum II rises to $A \sin 2\psi$ at $t = \pi/\omega_b$ and falls back, and the amplitude of pendulum I, which you will find in Problem 2, falls to $A \cos 2\psi$ at the same instant and never reaches zero, as Figure 13 shows. When the coupling is weak and the pendulums are nearly equal, the energies of the two bobs at equal amplitudes are nearly equal, and the greatest fraction of the energy that reaches pendulum II is

$$\sin^2 2\psi = \frac{4\kappa_c^2/m^2}{(\omega_I^2 - \omega_{II}^2)^2 + 4\kappa_c^2/m^2}. \quad (2.20)$$

This fraction is close to one only if $|\omega_I^2 - \omega_{II}^2|$ is small compared with $2\kappa_c/m$, and it falls to one half when $|\omega_I^2 - \omega_{II}^2| = 2\kappa_c/m$. A weakly coupled pair of pendulums is therefore very sensitive to a small difference in their lengths. The reason is the one that governs the driven oscillator of Section 1.3. Pendulum A pushes on pendulum II through the spring at a frequency close to ω_I , which differs from the frequency ω_{II} at which B would swing if bob I were held fixed, and an oscillator driven away from its natural frequency falls out of step with the force, so that it takes energy during part of the time and returns it during the rest.

A Driven Pair of Oscillators

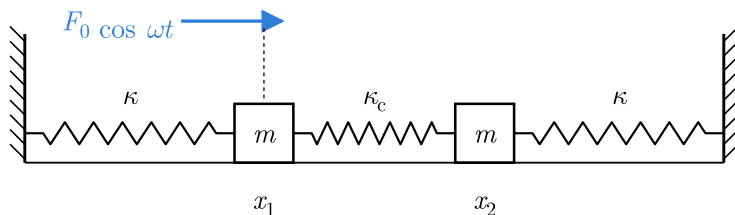


Figure 14. A pair of equal masses between two walls, joined to the walls by springs of constant κ and to each other by a spring of constant κ_c , with a driving force $F_0 \cos \omega t$ applied to the first mass only.

We now return to the two equal masses between walls and drive the first of them with a force $F_0 \cos \omega t$, as in Figure 14. The equations of motion become

$$\begin{aligned} m\ddot{x}_1 &= -\kappa x_1 + \kappa_c(x_2 - x_1) + F_0 \cos \omega t, \\ m\ddot{x}_2 &= -\kappa x_2 - \kappa_c(x_2 - x_1). \end{aligned} \quad (2.21)$$

The normal coordinates separate these equations as they separated the free ones. Adding the equations and dividing by two, and then subtracting them and dividing by two, gives

$$m\ddot{q}_1 = -\kappa q_1 + \frac{1}{2}F_0 \cos \omega t, \quad m\ddot{q}_2 = -(\kappa + 2\kappa_c) q_2 + \frac{1}{2}F_0 \cos \omega t, \quad (2.22)$$

so each mode is an undamped oscillator driven by half the applied force. The steady response of such an oscillator was found in Section 1.3. In complex form, with $q_1 = \text{Re}(\mathcal{A}e^{i\omega t})$ and the force written as $\text{Re}(F_0 e^{i\omega t})$, substituting into the first equation gives $m(\omega_1^2 - \omega^2)\mathcal{A} = \frac{1}{2}F_0$, and the second equation gives the same with ω_2 in place of ω_1 . Since $x_1 = q_1 + q_2$ and $x_2 = q_1 - q_2$, the complex amplitudes of the two masses are

$$\begin{aligned}\mathcal{A}_1 &= \frac{F_0}{2m} \left(\frac{1}{\omega_1^2 - \omega^2} + \frac{1}{\omega_2^2 - \omega^2} \right), \\ \mathcal{A}_2 &= \frac{F_0}{2m} \left(\frac{1}{\omega_1^2 - \omega^2} - \frac{1}{\omega_2^2 - \omega^2} \right).\end{aligned}\tag{2.23}$$

Both amplitudes are real, since there is no damping, and a negative amplitude means that the mass moves in antiphase with the force. Putting each pair of fractions over a common denominator, and using $\omega_1^2 + \omega_2^2 = 2(\kappa + \kappa_c)/m$ and $\omega_2^2 - \omega_1^2 = 2\kappa_c/m$, gives the equivalent forms

$$\mathcal{A}_1 = \frac{F_0 (\kappa + \kappa_c - m\omega^2)}{m^2 (\omega_1^2 - \omega^2) (\omega_2^2 - \omega^2)}, \quad \mathcal{A}_2 = \frac{F_0 \kappa_c}{m^2 (\omega_1^2 - \omega^2) (\omega_2^2 - \omega^2)}.\tag{2.24}$$

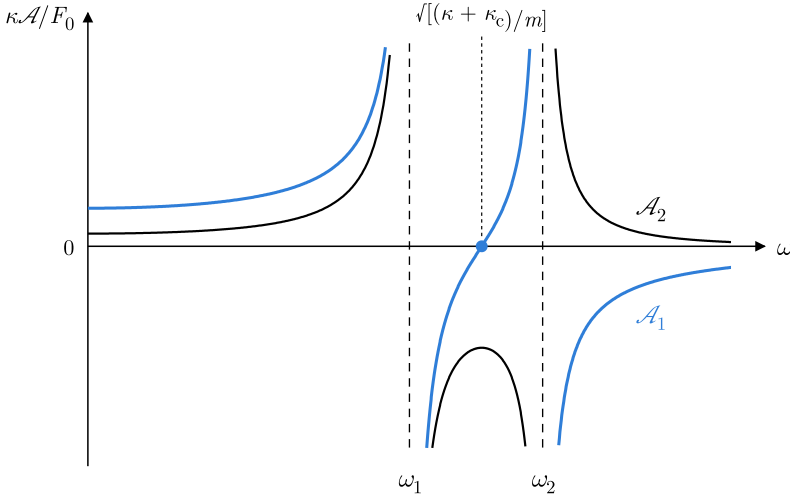


Figure 15. The steady amplitudes $\mathcal{A}_1$ (accent) and $\mathcal{A}_2$ of a pair of coupled masses when the first is driven at angular frequency ω , multiplied by κ/F_0 and cut off where they grow large. Both amplitudes grow without limit at the normal frequencies ω_1 and ω_2 , marked by the dashed lines, and $\mathcal{A}_1$ passes through zero at the frequency marked by the dot.

The two amplitudes are drawn in Figure 15. Both grow with-

out limit when the driving frequency approaches either normal frequency, since one of the modes is then driven at resonance. With a small resistive force $-b\dot{x}$ on each mass, each normal coordinate becomes a damped driven oscillator of the kind treated in Section 1.3, as you will show in Problem 4, and the infinite amplitudes are replaced by peaks of finite height. Below ω_1 both masses move in phase with the force, as a pair of masses pushed slowly would. Above ω_2 the driven mass moves in antiphase with the force, as a single mass driven far above its natural frequency does, and the other mass moves in antiphase with the driven one.

Between the two resonances there is a frequency at which the driven mass does not move. The numerator of $\mathcal{A}_1$ vanishes when

$$\omega^2 = \frac{\kappa + \kappa_c}{m} = \frac{1}{2}(\omega_1^2 + \omega_2^2), \quad (2.25)$$

and at this frequency $\omega_1^2 - \omega^2 = -\kappa_c/m$ and $\omega_2^2 - \omega^2 = \kappa_c/m$, so $\mathcal{A}_2 = -F_0/\kappa_c$. The second mass then oscillates in antiphase with the force, and the coupling spring, which is stretched by $x_2 - x_1 = x_2$, pulls on the first mass with the force $\kappa_c x_2 = -F_0 \cos \omega t$. This force cancels the driving force at every instant, so the first mass stays at rest, and the spring joining it to the wall exerts no force on it. A zero of the response at a frequency between two resonances is called an antiresonance. The frequency of the antiresonance can be interpreted as follows. With the first mass at rest, the second mass is held by its wall spring and by the coupling spring, whose far end is now fixed, and the natural frequency of a mass held by springs of constants κ and κ_c is $\sqrt{(\kappa + \kappa_c)/m}$. The second mass is therefore oscillating at the resonant frequency it has when the first mass is held fixed, with an amplitude just large enough to cancel the driving force on the first. Some devices that are attached to machines to stop them vibrating at a particular frequency work on this principle, and are called vibration absorbers; you will design one in Problem 3.

2.2 Many Coupled Oscillators and the Loaded String

A system of many oscillators, each coupled to its neighbours, is the next step towards a continuous medium. The simplest example is

a light string under tension carrying a row of equal beads, which is called a loaded string. This section first writes down the equations of motion of the beads, then finds all the normal modes by guessing their form and shows that the modes are orthogonal, and finally uses them to follow the motion that starts from any given displacements and velocities of the beads.

The Equations of Motion of the Beads

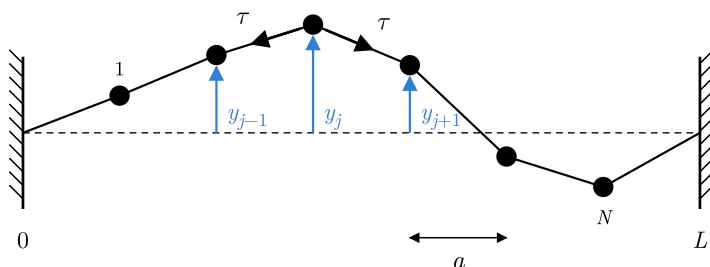


Figure 16. A loaded string, with N beads of mass m spaced a distance a apart on a light string of tension τ fixed at $x = 0$ and $x = L$. The beads are displaced transversely, and the segments of string between them are straight. The two arrows at bead j show the tension pulling it towards its neighbours.

Let a light string be stretched with tension τ between fixed points at $x = 0$ and $x = L$, and let it carry N beads, each of mass m , at the equally spaced positions $x = ja$ with $j = 1, 2, \dots, N$, so that $L = (N + 1)a$. Let y_j be the transverse displacement of bead j , as in Figure 16. The mass of the string is negligible compared with that of the beads, so each segment between neighbouring beads is straight. We assume that the displacements are small compared with the spacing a , so that every segment makes a small angle with the x -axis.

The segment to the right of bead j makes an angle whose tangent is $(y_{j+1} - y_j)/a$. For a small angle the sine and the tangent are nearly equal, so the transverse component of the tension that this segment

exerts on bead j is $\tau(y_{j+1} - y_j)/a$. The segment to the left pulls with a transverse component $-\tau(y_j - y_{j-1})/a$. The components along the string are τ multiplied by the cosines of the small angles, which are close to one, so they cancel, and the beads move only transversely. The tension itself stays equal to τ , since the length of a segment, $\sqrt{a^2 + (y_{j+1} - y_j)^2} \approx a + (y_{j+1} - y_j)^2/2a$ by the binomial expansion, differs from a only by a term of second order in the displacements. Newton's second law for bead j is therefore

$$m\ddot{y}_j = \frac{\tau}{a} (y_{j+1} - 2y_j + y_{j-1}), \quad j = 1, 2, \dots, N, \quad (2.26)$$

where the ends of the string are described by setting $y_0 = 0$ and $y_{N+1} = 0$. Let Ω be the frequency scale of the chain, with

$$\Omega^2 = \frac{\tau}{ma}. \quad (2.27)$$

The same equations describe a row of N masses m joined by $N + 1$ springs of constant κ between two walls and moving along the row, with y_j replaced by the displacement of mass j along the row and $\Omega^2 = \kappa/m$, since the springs on either side of mass j exert the forces $\kappa(y_{j+1} - y_j)$ and $-\kappa(y_j - y_{j-1})$. Everything in this section applies to both systems, and you will check the case of the springs in Problem 1.

In the notation of Section 2.1, the equations of motion are $\ddot{\mathbf{y}} = -\mathbf{D}\mathbf{y}$, where $\mathbf{y}$ is the column of the N displacements and the dynamical matrix is

$$\mathbf{D} = \Omega^2 \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \cdots & 0 \\ 0 & -1 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 2 \end{pmatrix}, \quad (2.28)$$

with $2\Omega^2$ along the diagonal, $-\Omega^2$ on the two lines next to it, and zeros elsewhere. The matrix is real and symmetric, so by Section 4.3 of Vol. 4 it has N real eigenvalues and N perpendicular eigenvectors,

and the system has N normal modes. For $N = 2$ the matrix is the one of the equal masses of Section 2.1 with $\kappa_c = \kappa$, and for a small number of beads the modes could be found by expanding the determinant. For large N that method is impractical, since the characteristic equation is a polynomial of degree N , and we look instead for the modes directly.

The Normal Modes

A normal mode is a motion in which every bead oscillates at the same frequency, $y_j = \text{Re}(\mathcal{A}_j e^{i\omega t})$, with complex amplitudes $\mathcal{A}_j$ that do not depend on time. Substituting into the equations of motion and cancelling the factor $e^{i\omega t}$ gives

$$(2\Omega^2 - \omega^2) \mathcal{A}_j = \Omega^2 (\mathcal{A}_{j+1} + \mathcal{A}_{j-1}) . \quad (2.29)$$

The relation has the same form for every bead, and a sine has the property that the sum of its values at two points equally spaced on either side of a third is a multiple of its value at the third. Since the string is fixed at $x = 0$, we therefore try amplitudes that vary as a sine of the position of the bead,

$$\mathcal{A}_j = \mathcal{A} \sin(kja) , \quad (2.30)$$

where $\mathcal{A}$ and k are constants. The function $\sin kx$ repeats itself each time x increases by $2\pi/k$, which is therefore its wavelength λ , and $k = 2\pi/\lambda$ is called the wave number. The compound-angle formulas $\sin(\theta_1 \pm \theta_2) = \sin \theta_1 \cos \theta_2 \pm \cos \theta_1 \sin \theta_2$, added together, give $\sin(\theta_1 + \theta_2) + \sin(\theta_1 - \theta_2) = 2 \sin \theta_1 \cos \theta_2$, and with $\theta_1 = kja$ and $\theta_2 = ka$ (and the same change in the identity $\sin \alpha - \sin \beta = \dots$ in the solution of Problem 2) the right-hand side of the relation becomes

$$\Omega^2 \mathcal{A} [\sin(kja + ka) + \sin(kja - ka)] = 2\Omega^2 \cos(ka) \mathcal{A} \sin(kja) . \quad (2.31)$$

Every term of the relation now contains the factor $\mathcal{A} \sin(kja)$, and the relation holds for every bead if $2\Omega^2 - \omega^2 = 2\Omega^2 \cos ka$. With

the identity $1 - \cos ka = 2 \sin^2(ka/2)$, this condition becomes

$$\omega = 2\Omega \sin \frac{ka}{2}. \quad (2.32)$$

The trial amplitudes satisfy the condition $y_0 = 0$ at the left end, since $\sin 0 = 0$. The condition $y_{N+1} = 0$ at the right end requires $\sin[k(N+1)a] = \sin kL = 0$, so kL must be a whole multiple of π , and the allowed wave numbers are $k_n = n\pi/L$. The normal modes of the loaded string are therefore

$$\begin{aligned} \mathcal{A}_j &= \mathcal{A} \sin \frac{jn\pi}{N+1}, \\ \omega_n &= 2\Omega \sin \frac{n\pi}{2(N+1)}, \\ n &= 1, 2, \dots, N. \end{aligned} \quad (2.33)$$

The whole number n need not be taken beyond N . For $n = 0$ and $n = N + 1$ the sine vanishes at every bead, which describes a string at rest. For $n' = 2(N + 1) - n$ the amplitudes are $\sin(2j\pi - jn\pi/(N + 1)) = -\sin(jn\pi/(N + 1))$, which is the pattern of mode n turned upside down, with the same frequency, and changing the sign of n , or adding a multiple of $2(N + 1)$ to it, gives nothing new in the same way. There are therefore exactly N distinct modes, as the N eigenvectors of $\mathbf{D}$ require. As a check, for $N = 2$ the formula gives $\omega_1 = 2\Omega \sin(\pi/6) = \Omega$ and $\omega_2 = 2\Omega \sin(\pi/3) = \sqrt{3}\Omega$, with the amplitudes in the ratios $(1, 1)$ and $(1, -1)$. These are the modes of the pair of masses held by three equal springs in Section 4.3 of Vol. 4, with frequencies $\sqrt{\kappa/m}$ and $\sqrt{3\kappa/m}$, once κ is replaced by τ/a .

The mode shapes are drawn in Figure 17 for five beads. In mode n the displacement of bead j is proportional to $\sin(n\pi x_j/L)$, where $x_j = ja$ is the position of the bead, so the beads lie on a sine curve with n half-wavelengths between the ends and wavelength $\lambda_n = 2L/n$. Mode n has $n - 1$ nodes between the ends, and in the modes of higher order the neighbouring beads move more nearly in opposite directions. In mode N each bead moves nearly opposite to its neighbours, and the string is bent sharply at every bead.

The frequencies are plotted against n in Figure 18. For the modes

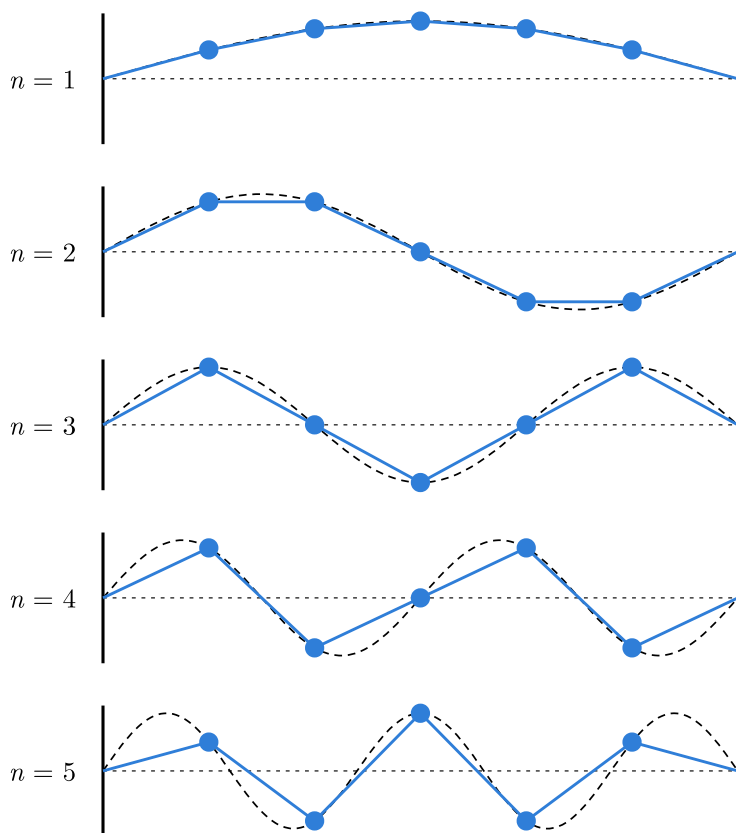


Figure 17. The normal modes of a loaded string with $N = 5$ beads, for $n = 1$ to 5 from top to bottom. The beads (accent) lie on the dashed curve $\sin(n\pi x/L)$, which has n half-wavelengths between the fixed ends, and the string between neighbouring beads is straight.

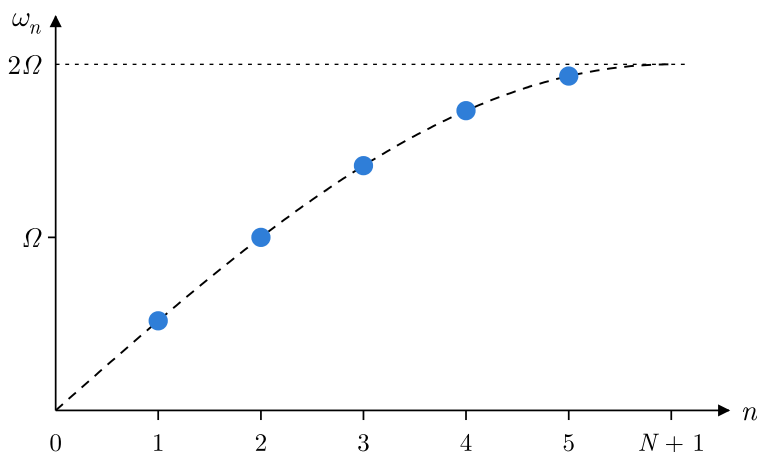


Figure 18. The normal frequencies $\omega_n = 2\Omega \sin[n\pi/2(N+1)]$ of a loaded string with $N = 5$ beads (accent), and the dashed curve that the formula gives when n is allowed to vary continuously. The frequencies are close to proportional to n for the lowest modes and approach 2Ω for the highest.

of low order, with $n \ll N$, the argument of the sine is small, and the approximation $\sin \theta \approx \theta$ for small θ gives

$$\omega_n \approx \frac{n\pi\Omega}{N+1}, \quad (2.34)$$

so the lowest frequencies are nearly whole multiples of the lowest, as are the harmonics of a stretched string met at school. For the higher modes the frequencies grow more slowly than n , and they crowd together below the value 2Ω , which none of them reaches. The reason for this limit can be seen from the forces. In a normal mode every bead has the same ratio of restoring force to displacement, $m\omega^2$. Let bead j be the bead with the largest displacement. The restoring force on it has the magnitude $(\tau/a)|2y_j - y_{j+1} - y_{j-1}|$, which is at most $(\tau/a)(2|y_j| + |y_{j+1}| + |y_{j-1}|)$, and since neither neighbour is displaced further than bead j , this is at most $4\tau|y_j|/a$. Therefore $m\omega^2 \leq 4\tau/a$, or $\omega \leq 2\Omega$. The limit would be reached only if both neighbours of the bead were displaced by the same amount

in the opposite direction, and then their neighbours in turn, which would require the alternating pattern $y_j \propto (-1)^j$ along the whole string. This pattern does not vanish at the fixed ends, so no mode of the loaded string reaches 2Ω .

Orthogonality of the Modes

The normal frequencies are all different, since $\sin \theta$ increases steadily over the range $0 < \theta < \pi/2$, and D is real and symmetric. By Section 4.3 of Vol. 4, eigenvectors of such a matrix that belong to different eigenvalues are orthogonal, so for $n \neq n'$

$$\sum_{j=1}^N \sin \frac{jn\pi}{N+1} \sin \frac{jn'\pi}{N+1} = 0. \quad (2.35)$$

We also need the value of the sum when $n' = n$. With the identity $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$ it becomes

$$\sum_{j=1}^N \sin^2 \frac{jn\pi}{N+1} = \frac{N}{2} - \frac{1}{2} \sum_{j=1}^N \cos \frac{2jn\pi}{N+1}. \quad (2.36)$$

The sum of cosines is the real part of $\sum_{j=1}^N \zeta^j$, where $\zeta = e^{2\pi in/(N+1)}$. Including the term $j = 0$, which is one, the sum is a geometric series, and the formula for its sum gives

$$\sum_{j=0}^N \zeta^j = \frac{1 - \zeta^{N+1}}{1 - \zeta} = 0, \quad (2.37)$$

since $\zeta^{N+1} = e^{2\pi in} = 1$ while $\zeta \neq 1$ for $1 \leq n \leq N$. The sum from $j = 1$ is therefore -1 , its real part is -1 , and so

$$\sum_{j=1}^N \sin \frac{jn\pi}{N+1} \sin \frac{jn'\pi}{N+1} = \frac{N+1}{2} \quad \text{if } n' = n, \quad (2.38)$$

and zero otherwise.

The General Motion

Since the equations of motion are linear, any superposition of the normal modes is also a motion of the string. Writing each mode with a real amplitude and a phase constant, the general motion is

$$y_j(t) = \sum_{n=1}^N A_n \sin \frac{jn\pi}{N+1} \cos(\omega_n t + \phi_n), \quad (2.39)$$

which contains $2N$ constants A_n and ϕ_n , as many as there are initial displacements and velocities of the beads. To find the constants, we set $t = 0$ in this expression and in its derivative with respect to time,

$$\begin{aligned} y_j(0) &= \sum_{n=1}^N A_n \cos \phi_n \sin \frac{jn\pi}{N+1}, \\ \dot{y}_j(0) &= - \sum_{n=1}^N \omega_n A_n \sin \phi_n \sin \frac{jn\pi}{N+1}. \end{aligned} \quad (2.40)$$

We then multiply each equation by $\sin[jn'\pi/(N+1)]$ and sum over j . By the orthogonality relation, only the term with $n = n'$ remains on the right, multiplied by $(N+1)/2$, and renaming n' as n gives

$$\begin{aligned} A_n \cos \phi_n &= \frac{2}{N+1} \sum_{j=1}^N y_j(0) \sin \frac{jn\pi}{N+1}, \\ A_n \sin \phi_n &= - \frac{2}{(N+1)\omega_n} \sum_{j=1}^N \dot{y}_j(0) \sin \frac{jn\pi}{N+1}. \end{aligned} \quad (2.41)$$

The amount of each mode in the motion is found by projecting the initial displacements and velocities onto the pattern of that mode, in the same way that the components of a vector are found by taking its scalar products with a set of perpendicular unit vectors. Once the constants are known, each mode oscillates at its frequency ω_n , independently of the others.

A String Carrying Three Beads

As an example, consider a string carrying three beads, with the middle bead pulled aside a distance η and all three released from rest. With $N = 3$ the mode patterns are $\sin(jn\pi/4)$, which for $j = 1, 2, 3$ are

$$\begin{aligned} n = 1 &: \left(\frac{1}{\sqrt{2}}, 1, \frac{1}{\sqrt{2}} \right), \\ n = 2 &: (1, 0, -1), \\ n = 3 &: \left(\frac{1}{\sqrt{2}}, -1, \frac{1}{\sqrt{2}} \right), \end{aligned} \tag{2.42}$$

with the frequencies $\omega_1 = 2\Omega \sin(\pi/8)$, $\omega_2 = \sqrt{2}\Omega$, and $\omega_3 = 2\Omega \sin(3\pi/8)$. The initial velocities are zero, so every $\sin \phi_n$ is zero and we may take every $\phi_n = 0$. Only the middle bead is displaced, so each sum over j has the single term $j = 2$, and

$$A_n = \frac{1}{2} \eta \sin \frac{n\pi}{2}, \tag{2.43}$$

which gives $A_1 = \eta/2$, $A_2 = 0$, and $A_3 = -\eta/2$. The second mode is absent because the middle bead is at rest in that mode, so a displacement of the middle bead alone has no component along it. The displacements are

$$y_2 = \frac{\eta}{2} (\cos \omega_1 t + \cos \omega_3 t), \quad y_1 = y_3 = \frac{\eta}{2\sqrt{2}} (\cos \omega_1 t - \cos \omega_3 t), \tag{2.44}$$

which satisfy the initial conditions $y_2(0) = \eta$ and $y_1(0) = y_3(0) = 0$.

The motion is drawn in Figure 19. Since $\sin(3\pi/8) = \cos(\pi/8)$, the ratio of the frequencies is $\omega_3/\omega_1 = \cot(\pi/8)$, and the half-angle formula $\cot(\theta/2) = (1 + \cos \theta)/\sin \theta$ with $\theta = \pi/4$ gives $\omega_3/\omega_1 = 1 + \sqrt{2}$. This is not a ratio of whole numbers, so the motion never repeats exactly.

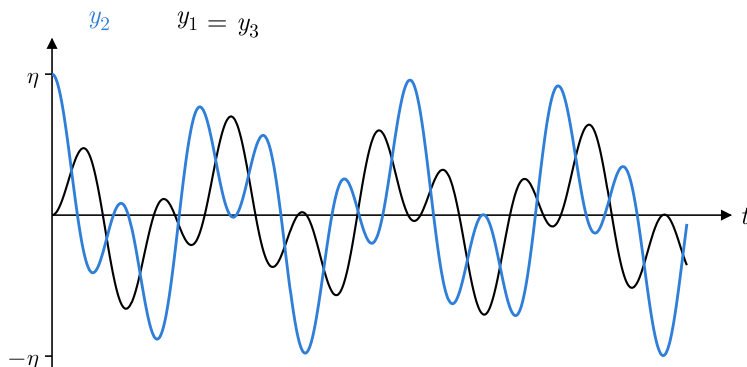


Figure 19. The motion of a string carrying three beads after the middle bead is pulled aside a distance η and all three are released from rest. The middle bead (accent) and the outer beads each move as a superposition of the first and third modes, and the motion does not repeat.

2.3 Symmetry and Normal Modes

The modes of the loaded string were found in Section 2.2 by guessing their form, and the guess worked because the beads are all alike and each is joined in the same way to its neighbours. This section turns that observation into a method. It first shows that a symmetry of a system of oscillators can be described by a matrix that commutes with the dynamical matrix, and that the modes can be found among the eigenvectors of the symmetry matrix. Then it uses a reflection to sort the modes of three coupled masses into two kinds before any determinant is expanded, and finally it uses a rotation to find the modes of a ring of masses, which leads to the relation between the frequency and the wave number of a long chain.

Symmetries and the Dynamical Matrix

Let a system of oscillators have the equations of motion $\ddot{\mathbf{x}} = -\mathbf{D}\mathbf{x}$, where $\mathbf{x}$ is the column of displacements and $\mathbf{D}$ the dynamical matrix, as in Sections 2.1 and 2.2. A symmetry of the system is an operation, such as exchanging the labels of two identical masses, that carries every motion of the system into another possible motion. Let the operation be described by a constant square matrix

$\mathbf{G}$, which carries the column $\mathbf{x}$ into $\mathbf{Gx}$. If $\mathbf{x}(t)$ is a motion, the second derivative of $\mathbf{Gx}$ with respect to time is $\mathbf{G}\ddot{\mathbf{x}} = -\mathbf{GD}\mathbf{x}$, and for $\mathbf{Gx}$ to be a motion as well this must equal $-\mathbf{DGx}$. Since the displacements at any one instant can be chosen freely, $\mathbf{G}$ describes a symmetry exactly when

$$\mathbf{GD} = \mathbf{DG}, \quad (2.45)$$

that is, when $\mathbf{G}$ and $\mathbf{D}$ commute.

The use of this relation rests on the following result. Let $\mathbf{G}$ commute with $\mathbf{D}$, and let $\mathbf{e}$ be an eigenvector of $\mathbf{G}$ with eigenvalue β , so that

$$\mathbf{Ge} = \beta \mathbf{e}. \quad (2.46)$$

Multiplying the column $\mathbf{De}$ by $\mathbf{G}$ and using the commutation relation and then the eigenvalue equation gives

$$\mathbf{G}(\mathbf{De}) = \mathbf{DGe} = \beta(\mathbf{De}), \quad (2.47)$$

so $\mathbf{De}$ is either zero or an eigenvector of $\mathbf{G}$ with the same eigenvalue β . Suppose first that β is not a repeated eigenvalue, so that every eigenvector of $\mathbf{G}$ with eigenvalue β is a multiple of $\mathbf{e}$. Then $\mathbf{De}$ is a multiple of $\mathbf{e}$, which we write $\omega^2 \mathbf{e}$, and $\mathbf{e}$ is the pattern of a normal mode with frequency ω , found without solving the eigenvalue problem of $\mathbf{D}$. Suppose next that β is repeated, with several independent eigenvectors. Then $\mathbf{D}$ carries each of them into a combination of the same eigenvectors, and the modes are found by solving a smaller eigenvalue problem among those combinations. In either case the symmetry sorts the modes into classes labelled by the eigenvalues of $\mathbf{G}$, and the modes can be chosen so that each belongs to a single class.

Reflection Symmetry of Three Masses

As an example of a reflection symmetry, consider three equal masses m in a row between two walls, with each of the outer masses joined to its wall by a spring of constant κ and neighbouring masses joined

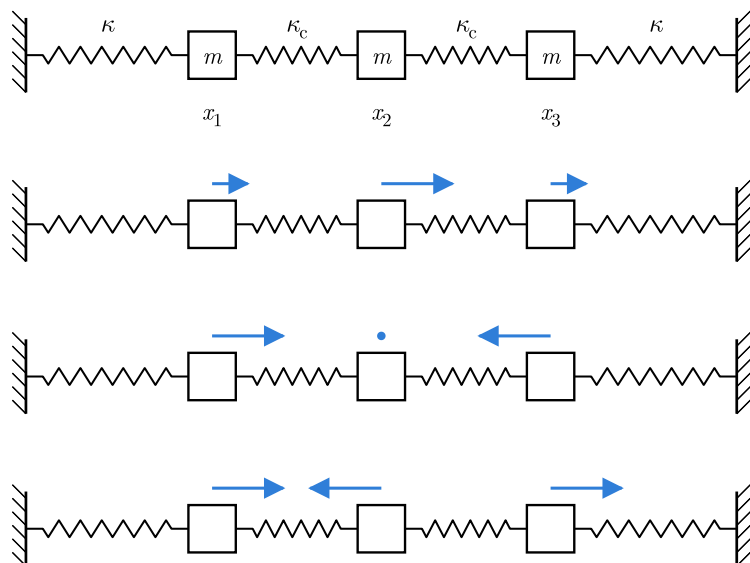


Figure 20. A row of three equal masses between two walls, joined to the walls by springs of constant κ and to each other by springs of constant κ_c (top). The three rows below show the normal modes in order of increasing frequency, drawn for coupling springs weaker than the outer springs: the slower even mode, the odd mode with the middle mass at rest (dot), and the faster even mode. Each arrow shows the displacement of a mass at an instant when all the displacements are greatest.

by springs of constant κ_c , as in Figure 20. Let x_1 , x_2 , and x_3 be the displacements of the masses along the row, all measured in the same direction. The equations of motion are

$$\begin{aligned} m\ddot{x}_1 &= -\kappa x_1 - \kappa_c(x_1 - x_2), \\ m\ddot{x}_2 &= -\kappa_c(x_2 - x_1) - \kappa_c(x_2 - x_3), \\ m\ddot{x}_3 &= -\kappa x_3 - \kappa_c(x_3 - x_2), \end{aligned} \quad (2.48)$$

and the dynamical matrix is

$$D = \frac{1}{m} \begin{pmatrix} \kappa + \kappa_c & -\kappa_c & 0 \\ -\kappa_c & 2\kappa_c & -\kappa_c \\ 0 & -\kappa_c & \kappa + \kappa_c \end{pmatrix}. \quad (2.49)$$

The outer masses have the same mass and are held by the same springs, so exchanging x_1 and x_3 turns the first equation into the third, the third into the first, and the second into itself. This exchange is described by the matrix

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad (2.50)$$

and multiplying out the products confirms that $PD = DP$. Exchanging the outer masses twice returns every displacement to its place, so P^2 is the identity matrix, and if $P\mathbf{e} = \beta\mathbf{e}$ then $\mathbf{e} = P^2\mathbf{e} = \beta^2\mathbf{e}$, and therefore $\beta^2 = 1$ and $\beta = \pm 1$. The eigenvectors with $\beta = 1$ are the columns unchanged by the exchange, which have the form (e_1, e_2, e_1) for any numbers e_1 and e_2 , and a mode of this form is called even. The eigenvectors with $\beta = -1$ are the columns whose sign is reversed by the exchange, $(e_1, 0, -e_1)$, and a mode of this form is called odd. A mirror placed at the middle mass also reverses the direction in which the displacements are measured, so it carries (x_1, x_2, x_3) into $(-x_3, -x_2, -x_1)$, but its matrix, $-P$, has the same eigenvectors and sorts the modes in the same way.

The eigenvalue -1 is not repeated, since every odd column is a multiple of $(1, 0, -1)$, so by the result above this column is a mode.

Multiplying it by $\mathbf{D}$ gives

$$\mathbf{D} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \frac{\kappa + \kappa_c}{m} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad (2.51)$$

so the odd mode has $\omega^2 = (\kappa + \kappa_c)/m$. In this mode the outer masses move in opposite directions and the middle mass stays at rest, since the two coupling springs pull on it with equal and opposite forces. Each outer mass is then held by its wall spring and by a coupling spring whose far end is fixed, which accounts for the frequency.

The eigenvalue $+1$ is repeated, with the two independent eigenvectors $(1, 0, 1)$ and $(0, 1, 0)$, and the even modes are combinations of them. Putting $\mathbf{e} = (e_1, e_2, e_1)$ into $\mathbf{D}\mathbf{e} = \omega^2\mathbf{e}$, the first and third rows give the same equation, and the first two rows give

$$(\kappa + \kappa_c - m\omega^2)e_1 - \kappa_c e_2 = 0, \quad -2\kappa_c e_1 + (2\kappa_c - m\omega^2)e_2 = 0. \quad (2.52)$$

These equations have a solution other than $e_1 = e_2 = 0$ when their determinant vanishes, which gives the quadratic equation

$$(m\omega^2)^2 - (\kappa + 3\kappa_c)m\omega^2 + 2\kappa\kappa_c = 0, \quad (2.53)$$

whose roots are

$$m\omega^2 = \frac{1}{2} \left[\kappa + 3\kappa_c \mp \sqrt{(\kappa + 3\kappa_c)^2 - 8\kappa\kappa_c} \right]. \quad (2.54)$$

The quantity under the square root can be rewritten as $(\kappa - \kappa_c)^2 + 8\kappa_c^2$, which is positive, so both roots are real. Without the symmetry the three frequencies would have been the roots of a cubic equation, and the reflection has split that equation into a linear one and a quadratic one. Substituting $m\omega^2 = \kappa + \kappa_c$ into the left-hand side of the quadratic gives $-2\kappa_c^2$, which is negative, so the frequency of the odd mode lies between the two even frequencies, and as the frequency rises the modes run even, odd, even, as Figure 20

shows. In the slower even mode the middle mass moves further than the outer ones and in the same direction, and in the faster even mode it moves opposite to them. As a check, when $\kappa_c = \kappa$ the system is a row of three masses and four equal springs, the roots are $m\omega^2 = (2 \mp \sqrt{2})\kappa$ and the odd mode has $m\omega^2 = 2\kappa$, and these agree with $\omega_n = 2\Omega \sin(n\pi/8)$ from Section 2.2, with $N = 3$ and $\Omega^2 = \kappa/m$, since $4\sin^2(\pi/8) = 2 - \sqrt{2}$ by the half-angle formula.

A Ring of Masses

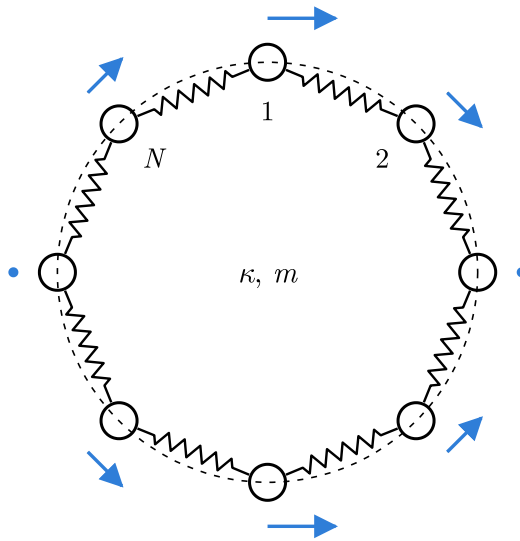


Figure 21. A ring of N equal masses on a smooth circular track, drawn for $N = 8$, with neighbouring masses joined by springs of constant κ that lie along the track. The accent arrows show the displacements along the track, measured in the direction of increasing j , in the standing wave $x_j \propto \cos[2\pi(j-1)/N] \cos \omega t$, which combines the modes $n = 1$ and $n = N - 1$, at an instant when the displacements are greatest. The dots mark the masses that stay at rest.

A translation symmetry is most easily described for a ring of masses. Let N equal masses m slide without friction round a circular track, with each mass joined to its two neighbours by springs of constant κ that lie along the track, and with mass N joined to mass 1, as

in Figure 21. Let a be the distance between neighbouring masses along the track, and let x_j be the displacement of mass j along the track, measured in the direction of increasing j . The equations of motion are those of Section 2.2,

$$m\ddot{x}_j = \kappa(x_{j+1} - 2x_j + x_{j-1}) , \quad (2.55)$$

with the difference that there are no walls, and the labels run round the ring, so that x_{N+1} means x_1 and x_0 means x_N . Let $\Omega^2 = \kappa/m$ as before.

Every mass is joined to its neighbours in the same way, so moving the whole pattern of displacements one place round the ring carries a motion into another motion. Let $\mathbf{R}$ be the matrix of this operation, which replaces each displacement x_j by x_{j+1} . Moving the pattern N places brings every displacement back to where it started, so $\mathbf{R}^N$ is the identity matrix, and an eigenvalue β of $\mathbf{R}$ must satisfy $\beta^N = 1$, by the argument used for $\mathbf{P}$ above. The N roots of this equation are $\beta = e^{-2\pi in/N}$ with $n = 0, 1, \dots, N-1$. For each of them the column with components

$$e_j = e^{-2\pi inj/N} \quad (2.56)$$

is an eigenvector, since moving it one place gives $e_{j+1} = e^{-2\pi in/N} e_j$. These are N different eigenvalues, each with a single eigenvector, and by the result of the first subsection each eigenvector is the pattern of a normal mode. The modes of the ring are therefore known without an eigenvalue problem being solved.

It is convenient to write the pattern in terms of a wave number. Let

$$k = \frac{2\pi n}{Na} , \quad (2.57)$$

so that $e_j = e^{-ijk a}$, and the mode is $x_j = \text{Re} [\mathcal{A} e^{i(\omega t - jka)}]$. Substituting this into the equation of motion and cancelling the common factor $\mathcal{A} e^{i(\omega t - jka)}$ gives

$$-m\omega^2 = \kappa (e^{-ika} - 2 + e^{ika}) = -2\kappa (1 - \cos ka) = -4\kappa \sin^2 \frac{ka}{2}, \quad (2.58)$$

by Euler's formula and the half-angle identity used in Section 2.2, and therefore

$$\omega = 2\Omega \left| \sin \frac{ka}{2} \right|. \quad (2.59)$$

With $\mathcal{A} = Ae^{i\phi}$, the real displacement is $x_j = A \cos(\omega t - jka + \phi)$. The phase of mass j is the same as that of mass $j + 1$ a time ka/ω later, so a crest of the pattern moves round the ring in the direction of increasing j at the speed ω/k , and the mode is a wave travelling round the ring, of wavelength $2\pi/k = Na/n$.

The frequencies are shared between modes. The mode with $N - n$ in place of n has $e_j = e^{-2\pi i(N-n)j/N} = e^{2\pi i n j/N}$, since $e^{-2\pi i j} = 1$, and this is a wave with k replaced by $-k$, travelling round the ring in the opposite direction. Its frequency is $2\Omega |\sin(\pi(N - n)/N)| = 2\Omega |\sin(\pi n/N)|$, the same as that of mode n . Each frequency therefore belongs to two modes, waves of the same wavelength running in opposite directions, with two exceptions for even N . The mode $n = 0$ has $\omega = 0$ and every mass displaced by the same amount; it is a rotation of the whole ring along the track, which, like the uniform translation of the pair of masses joined by a single spring in Problem 1 of Section 4.4 of Vol. 1, has no restoring force and moves at constant speed. The mode $n = N/2$ has $e_j = e^{-i\pi j} = (-1)^j$, each mass moving opposite to its neighbours, and the highest frequency, 2Ω ; a wave of this pattern looks the same whichever way it travels. For odd N only the mode $n = 0$ is unshared. Any combination of two modes with the same frequency is also a mode, and adding and subtracting the waves with k and $-k$ gives displacements proportional to $\cos(jka)$ and $\sin(jka)$, which are standing waves on the ring, like the modes of the loaded string. A standing wave of this kind with $n = 1$ is drawn in Figure 21.

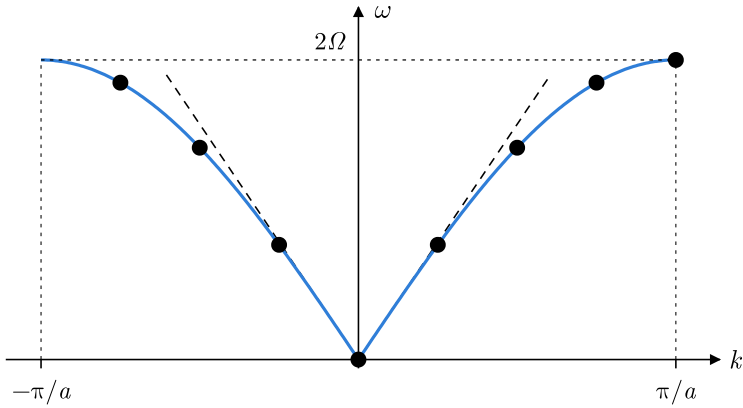


Figure 22. The dispersion relation $\omega = 2\Omega|\sin(ka/2)|$ of a chain of equal masses and springs (accent), for wave numbers between $-\pi/a$ and π/a . The dots mark the modes of a ring of eight masses, which lie in pairs at $\pm k$ with a single mode at $k = 0$ and one at $k = \pi/a$. The dashed lines show the approximation $\omega \approx \Omega a|k|$ for long wavelengths.

The Long Chain

The relation between the frequency and the wave number found for the ring is the same as the relation $\omega = 2\Omega \sin(ka/2)$ found for the loaded string in Section 2.2, with $\Omega^2 = \kappa/m$ in place of τ/ma . The two chains differ only in the wave numbers that their ends allow, $k = 2\pi n/Na$ for the ring and $k = n\pi/L$ for the string with fixed ends. In a long chain, with N large, these allowed wave numbers are closely spaced, and the frequency may be regarded as a function of a continuously variable k ,

$$\omega = 2\Omega \left| \sin \frac{ka}{2} \right|. \quad (2.60)$$

A relation of this kind between the frequency of a wave and its wave number is called a dispersion relation. It is drawn in Figure 22, with the modes of a ring of eight masses marked on it.

Only the range $-\pi/a < k \leq \pi/a$ needs to be considered. Replacing k by $k + 2\pi/a$ multiplies $e^{-ijk a}$ by $e^{-2\pi i j} = 1$, so the wave numbers

k and $k + 2\pi/a$ describe the same displacements of every mass. A wave whose wavelength is shorter than $2a$ gives the masses the same displacements as a wave of longer wavelength, because the masses sample the wave only at the points $x = ja$, and the pattern between them has no meaning. The greatest frequency of the chain is reached at $k = \pi/a$, where the wavelength is $2a$ and each mass moves opposite to its neighbours, and it is

$$\omega_{\max} = 2\Omega. \quad (2.61)$$

This value can be checked from the forces. When mass j is displaced by x_j and both its neighbours by $-x_j$, each of the springs on either side of it changes length by $2x_j$, and together they exert the force $-4\kappa x_j$, which gives $\omega^2 = 4\kappa/m$. The argument of Section 2.2, applied to the mass with the largest displacement, shows that no mode of a chain can have a higher frequency, so 2Ω is the highest frequency at which the chain can oscillate.

At the other end of the range, when the wavelength is long compared with the spacing, $ka \ll 1$ and the approximation $\sin \theta \approx \theta$ gives

$$\omega \approx \Omega a |k|. \quad (2.62)$$

The frequency is then proportional to the wave number, and waves of every long wavelength travel at the same speed, $\omega/k = \Omega a$. As the wavelength shortens, the curve of Figure 22 falls below the straight line, and the speed ω/k falls, to $2\Omega a/\pi$ at $k = \pi/a$. Waves on a chain therefore travel at a speed that depends on their wavelength, and only those whose wavelength is long compared with the spacing of the masses travel at a common speed.

2.4 The Continuum Limit

A real string, of steel or nylon, carries no beads, and its mass is spread evenly along its length. In this section we let the number of beads on the loaded string of Section 2.2 grow and their spacing shrink while the mass per unit length stays fixed. We first find

the equation that the displacement obeys in this limit, then the modes of the continuous string and its general motion, and finally its energy and the way in which the energy is divided among the modes when the string is plucked or struck.

From Beads to a Continuous String

As in Section 2.2, let N beads, each of mass m , be fixed to a string stretched with tension τ between two walls, with the beads a distance a apart and the walls at $x = 0$ and $x = L = (N + 1)a$. Bead j sits at $x_j = ja$ when the string is straight, and we let $y_j(t)$ be its transverse displacement, with $y_0 = y_{N+1} = 0$ at the walls. The segment between beads j and $j + 1$ has the slope $(y_{j+1} - y_j)/a$ and pulls on bead j with the transverse force $\tau(y_{j+1} - y_j)/a$ when the slopes are small, and adding the pull of the segment on the other side gives the equation of motion of Section 2.2,

$$m \frac{d^2 y_j}{dt^2} = \frac{\tau}{a} (y_{j+1} - 2y_j + y_{j-1}) . \quad (2.63)$$

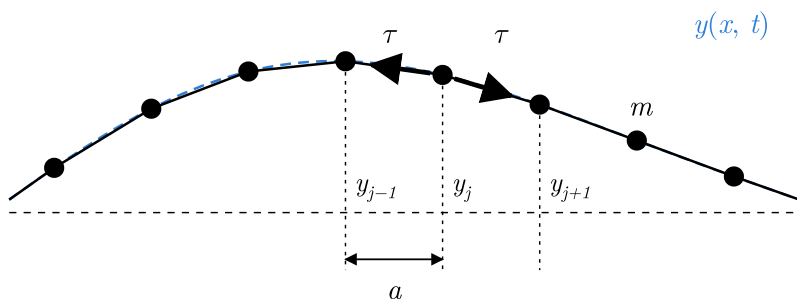


Figure 23. Part of the loaded string of Section 2.2, with beads of mass m a distance a apart. The tension τ pulls on bead j along the two segments that meet there. When the displacements y_j change little from one bead to the next, they are the values at $x = ja$ of a smooth function $y(x, t)$, drawn dashed in the accent colour.

We now let $N \rightarrow \infty$ and $a \rightarrow 0$, keeping the length L fixed, and let the mass of each bead shrink in proportion to the spacing, so that

the mass per unit length $\mu = m/a$ stays the same. Suppose that the displacements change little from one bead to the next, so that, as Figure 23 shows, they are the values $y_j(t) = y(x_j, t)$ of a smooth function $y(x, t)$ at the positions of the beads. The displacements of the neighbouring beads are then $y(x \pm a, t)$, and expanding each in a Taylor series about $x = x_j$ gives

$$y(x \pm a, t) = y \pm a \frac{\partial y}{\partial x} + \frac{a^2}{2} \frac{\partial^2 y}{\partial x^2} \pm \frac{a^3}{6} \frac{\partial^3 y}{\partial x^3} + \frac{a^4}{24} \frac{\partial^4 y}{\partial x^4} \pm \dots, \quad (2.64)$$

where y and its derivatives on the right-hand side are evaluated at x . When the two expansions are added, the terms with odd powers of a cancel. Dividing the equation of motion by a and writing $m/a = \mu$ therefore gives

$$\begin{aligned} \mu \frac{\partial^2 y}{\partial t^2} &= \tau \frac{y(x+a, t) - 2y(x, t) + y(x-a, t)}{a^2} \\ &= \tau \frac{\partial^2 y}{\partial x^2} + \frac{\tau a^2}{12} \frac{\partial^4 y}{\partial x^4} + \dots \end{aligned} \quad (2.65)$$

As $a \rightarrow 0$ the terms that carry powers of a vanish, and the displacement of the string satisfies

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}, \quad v = \sqrt{\frac{\tau}{\mu}}, \quad (2.66)$$

which is the wave equation for a stretched string. Section 7.1 of Vol. 4 derived the same equation directly, from the forces exerted on a short piece of a continuous string by the rest of the string (it writes the displacement as u and the mass per unit length as ρ_ℓ), and showed with d'Alembert's solution that v is the speed at which a disturbance of any shape travels along the string.

The limit also tells us when the wave equation holds. If the displacement varies along the string as $\sin kx$, then $\partial^4 y / \partial x^4 = -k^2 \partial^2 y / \partial x^2$, and the first term that was dropped is smaller than the term that was kept by the factor $k^2 a^2 / 12$. The wave equation is therefore a valid approximation when $ka \ll 1$, that is when the wavelength is much longer than the spacing of the beads. A real string is made

of atoms, and for it too the continuum limit is valid as long as the wavelength is much longer than the distance between neighbouring atoms. In the same limit the dispersion relation $\omega = 2\Omega|\sin(ka/2)|$ of the long chain of Section 2.3 becomes $\omega = \Omega ka = vk$, since $\Omega^2 = \tau/ma = \tau/\mu a^2$ for the loaded string.

The Modes in the Continuum Limit

In Section 2.2 we found that the loaded string has N modes, labelled by $n = 1, 2, \dots, N$, in which the displacement of bead j is proportional to $\sin[jn\pi/(N+1)]$ and the angular frequency is $\omega_n = 2\Omega\sin[n\pi/2(N+1)]$. With $\Omega = v/a$ and $(N+1)a = L$, the frequencies can be rewritten as

$$\omega_n = \frac{2v}{a} \sin \frac{n\pi a}{2L}. \quad (2.67)$$

For the modes of low order, with $n \ll N$, the argument of the sine is small, and replacing $\sin \theta$ by θ gives $\omega_n = n\pi v/L$. The mode shapes need no approximation, since $jn\pi/(N+1) = n\pi x_j/L$, so in every mode the beads sit on the curve $\sin(n\pi x/L)$. In the continuum limit, mode n of the string is

$$\begin{aligned} y(x, t) &= \sin \frac{n\pi x}{L} \cos(\omega_n t + \phi_n), \\ \omega_n &= \frac{n\pi v}{L} = vk_n, \\ k_n &= \frac{n\pi}{L}, \end{aligned} \quad (2.68)$$

where ϕ_n is a phase constant. These are the normal modes that Section 7.2 of Vol. 4 found by separating the variables in the wave equation. Mode n fits n half-wavelengths into the string, and its frequency is n times that of the lowest mode. Frequencies that are whole multiples of a lowest frequency are called harmonics, and the lowest is called the fundamental.

Figure 24 compares the two sets of frequencies. Keeping the next term of the Taylor series of the sine, $\sin \theta \approx \theta - \theta^3/6$, gives

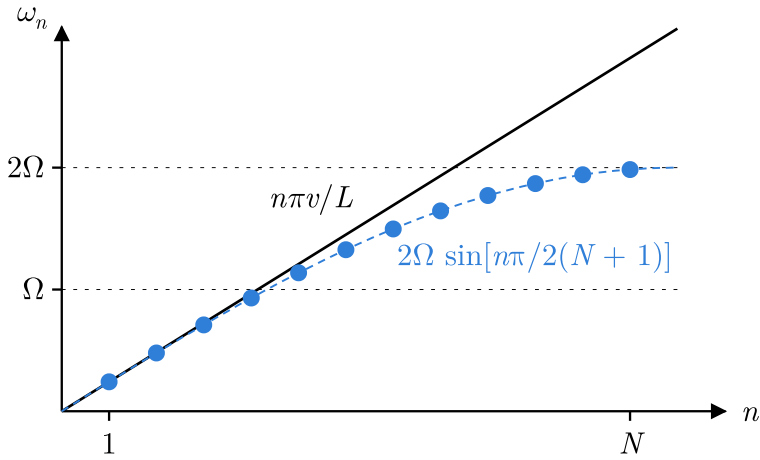


Figure 24. The angular frequencies $\omega_n = 2\Omega \sin[n\pi/2(N+1)]$ of the loaded string with N beads (dots, with the same formula drawn dashed for values of n between whole numbers), and the angular frequencies $n\pi v/L$ of the continuous string with the same length and the same mass per unit length (solid line). The two sets of frequencies agree for $n \ll N$ and differ more and more as n approaches N .

$$\omega_n \approx \frac{n\pi v}{L} \left(1 - \frac{n^2 \pi^2 a^2}{24L^2} \right), \quad (2.69)$$

so the loaded string has lower frequencies than the continuous string, by a fraction that is small as long as $na \ll L$, which is the condition of the previous subsection, since the wavelength of mode n is $2L/n$.

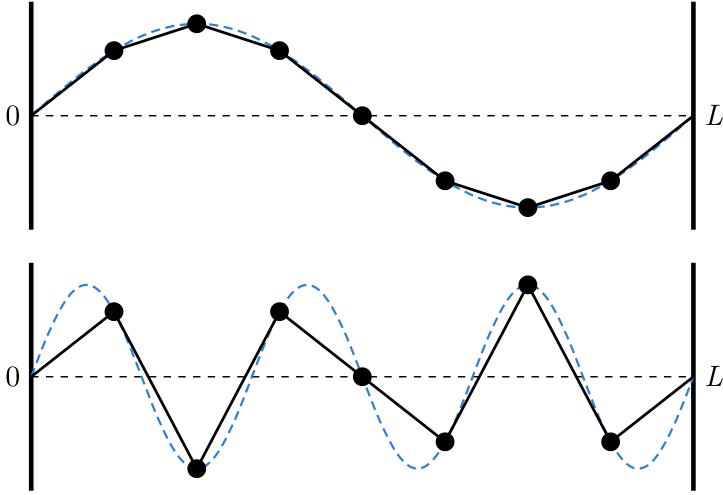


Figure 25. A mode of low order (upper panel) and a mode whose number n is close to the number of beads N (lower panel) of a loaded string with fixed ends. The beads lie on the curve $\sin(n\pi x/L)$, drawn dashed in the accent colour, in both modes, while the string between neighbouring beads is straight. In the lower panel neighbouring beads move almost in opposite directions, and the curve through them does not describe the shape of the string.

The modes of high order behave differently, as Figure 25 shows. For $n = N$ the displacement of bead j is $\sin[j\pi - j\pi/(N+1)] = -(-1)^j \sin[j\pi/(N+1)]$, by the formula for the sine of a difference, so the displacements change sign from one bead to the next. The string between the beads is then a zigzag of straight segments, and no smooth function describes it. The loaded string also has only N modes, one for each of the coordinates $y_1, \dots, y_N$. A value $n = N+1$ makes every $\sin[jn\pi/(N+1)]$ zero, and replacing n by $2(N+1) - n$ only changes the sign of every displacement, so larger values of n

repeat the modes already found. The continuous string has a mode for every positive whole number n .

The General Motion of the String

In Section 2.2 the general motion of the beads was written as a superposition of the modes, with the coefficient of each mode found by multiplying the initial displacements $y_j(0)$ by $\sin[jn\pi/(N+1)]$ and summing over the beads. In the continuum limit the superposition becomes an infinite series,

$$y(x, t) = \sum_{n=1}^{\infty} \sin \frac{n\pi x}{L} (a_n \cos \omega_n t + b_n \sin \omega_n t), \quad (2.70)$$

in which the amplitude and the phase of each mode are combined into the coefficients a_n and b_n . Let $y_i(x)$ and $\dot{y}_i(x)$ be the displacement and the transverse velocity of the string at $t = 0$. Setting $t = 0$ in the series and in its derivative with respect to t shows that y_i and $\dot{y}_i$ are half-range sine series, whose coefficients were found in Section 7.2 of Vol. 4 from the orthogonality of the sines established in Section 4.2 of that volume:

$$a_n = \frac{2}{L} \int_0^L y_i(x) \sin \frac{n\pi x}{L} dx, \quad b_n = \frac{2}{\omega_n L} \int_0^L \dot{y}_i(x) \sin \frac{n\pi x}{L} dx. \quad (2.71)$$

The integrals are the continuum limits of the sums over beads, since a sum over j multiplied by the spacing a becomes an integral over x . Since every ω_n is a whole multiple of $\omega_1 = \pi v/L$, increasing t by $2L/v$ leaves every term of the series unchanged, and every motion of the string repeats itself after that time, as you showed in Problem 3 of Section 7.2 of Vol. 4. The frequencies of the loaded string are not whole multiples of ω_1 , as Figure 24 shows, and a motion of the beads that contains several modes does not in general repeat itself exactly.

A String Plucked at a Point

As an example, consider a string that is pulled aside at the point $x = x_p$ to the displacement y_p , so that it forms two straight segments, and released from rest. Every b_n is zero, since $\dot{y}_i = 0$, and the coefficients a_n were worked out in Section 7.2 of Vol. 4, for a string plucked at its midpoint in the text and for a general point in Problem 2 of that section:

$$a_n = \frac{2y_p L^2}{n^2 \pi^2 x_p (L - x_p)} \sin \frac{n\pi x_p}{L}. \quad (2.72)$$

The modes that have a node at x_p are absent, and the coefficients go as $1/n^2$, the rate found in Section 6.1 of Vol. 4 for a function whose slope jumps while its value does not. When the string is plucked close to one end, so that $x_p \ll L$, the sine can be replaced by its argument for all n much smaller than L/x_p , and then $a_n \approx 2y_p L/[n\pi(L - x_p)]$. The coefficients of these modes fall only as $1/n$, so the high harmonics are relatively stronger for a string plucked near its end than for one plucked near its middle, which is one reason a guitarist might pluck close to the bridge to obtain a brighter tone.

The shape of the string, computed by summing the series, is drawn in Figure 26 at $t = 0, L/8v, L/2v$, and L/v , and it always consists of straight segments. The reason is clear from d'Alembert's solution of Section 7.1 of Vol. 4, which for a string released from rest is $y(x, t) = \frac{1}{2}[y_i(x - vt) + y_i(x + vt)]$, with y_i extended beyond the ends of the string as an odd function about each fixed end. Each of the two terms is a zigzag of straight lines whose corners travel at the speed v , one to the right and one to the left, and a sum of two straight lines is a straight line, so the string is straight everywhere except at the moving corners. At $t = L/v$ the oddness of the extension gives $y(x, L/v) = -y_i(L - x)$, which is the shape in the bottom panel.

The Energy of the String

The energy of the loaded string is the kinetic energy of the beads together with the potential energy stored in the stretched segments.

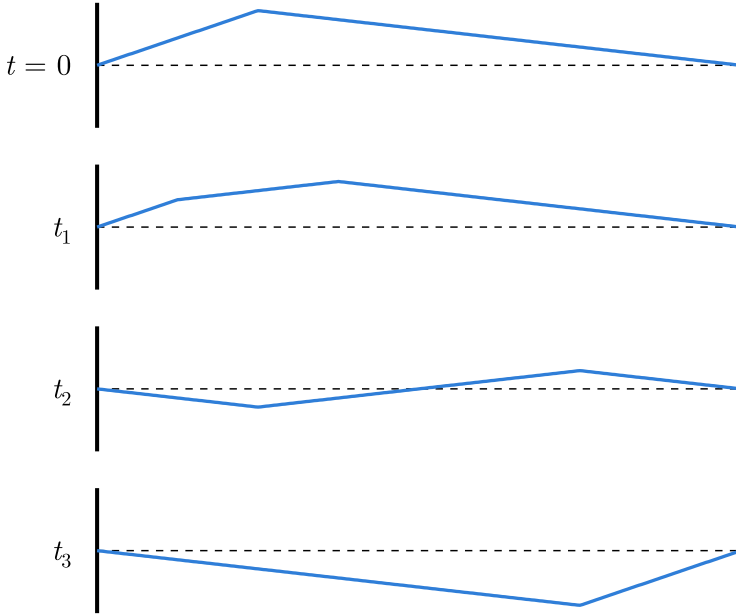


Figure 26. A string plucked at the point x_p and released from rest, computed from the series in the text at four times, which increase from the top panel to the bottom. The corner at the plucked point divides into two corners, which travel in opposite directions at the speed v and are inverted on reflection at the fixed ends. In the bottom panel, half a period after release, the string has the shape it started with, inverted and reflected end to end.

The segment between beads j and $j + 1$ has the length

$$\sqrt{a^2 + (y_{j+1} - y_j)^2} \approx a + \frac{(y_{j+1} - y_j)^2}{2a}, \quad (2.73)$$

by the binomial expansion $\sqrt{1 + (y_{j+1} - y_j)^2/a^2} \approx 1 + (y_{j+1} - y_j)^2/2a^2$, which holds when the slopes are small. The segment is lengthened by the second term against the tension τ , which to this order is the same in every segment, and the work done, $\tau(y_{j+1} - y_j)^2/2a$, is stored as potential energy. In the continuum limit $(y_{j+1} - y_j)/a$ becomes $\partial y/\partial x$, and the sums over beads of $\frac{1}{2}m\dot{y}_j^2 = \frac{1}{2}\mu a \dot{y}_j^2$ and of $\frac{1}{2}\tau a[(y_{j+1} - y_j)/a]^2$ become integrals, so the energy of the string is

$$E = \frac{1}{2} \int_0^L \left[\mu \left(\frac{\partial y}{\partial t} \right)^2 + \tau \left(\frac{\partial y}{\partial x} \right)^2 \right] dx. \quad (2.74)$$

To express this energy in terms of the modes, let $q_n(t) = a_n \cos \omega_n t + b_n \sin \omega_n t$, so that $y = \sum_n q_n \sin k_n x$. Differentiating term by term,

$$\frac{\partial y}{\partial t} = \sum_{n=1}^{\infty} \dot{q}_n \sin k_n x, \quad \frac{\partial y}{\partial x} = \sum_{n=1}^{\infty} k_n q_n \cos k_n x, \quad (2.75)$$

which are a half-range sine series and a half-range cosine series with no constant term. Parseval's theorem of Section 6.1 of Vol. 4, applied to the odd or the even extension f of such a series to $-L < x < L$, states that $(1/L) \int_{-L}^L f^2 dx$ is the sum of the squares of the coefficients of the series. The square of either extension is even, so its integral over $-L < x < L$ is twice its integral over $0 < x < L$, and therefore $\int_0^L (\partial y/\partial t)^2 dx = (L/2) \sum_n \dot{q}_n^2$ and $\int_0^L (\partial y/\partial x)^2 dx = (L/2) \sum_n k_n^2 q_n^2$. Since $\tau k_n^2 = \mu v^2 k_n^2 = \mu \omega_n^2$, the energy of the string is

$$E = \sum_{n=1}^{\infty} \left[\frac{1}{2} \frac{\mu L}{2} \dot{q}_n^2 + \frac{1}{2} \frac{\mu L}{2} \omega_n^2 q_n^2 \right]. \quad (2.76)$$

Each term is the energy of a simple harmonic oscillator of mass $\mu L/2$, angular frequency ω_n , and displacement q_n , the mass being half that of the string because the mean value of $\sin^2 k_n x$ along the string is one half. No term contains the coordinates of two different modes, so the energy of the string is the sum of the energies of its modes, and each mode keeps the energy it was given at the start,

$$E_n = \frac{\mu L}{4} \omega_n^2 (a_n^2 + b_n^2). \quad (2.77)$$

In the ideal string no energy passes from one mode to another, while a real string loses energy to the air and to its supports at rates that differ from one mode to another.

Plucked and Struck Strings

For the plucked string, substituting the coefficients a_n found above, with $\mu v^2 = \tau$, gives

$$E_n = \frac{\tau y_p^2 L^3}{\pi^2 x_p^2 (L - x_p)^2} \frac{1}{n^2} \sin^2 \frac{n\pi x_p}{L}, \quad (2.78)$$

You will check in Problem 2 that these energies add up to the work done in pulling the string aside.

As another example, consider a string that is straight at $t = 0$ and is struck by a hammer, which gives the short length σ of string centred on x_s the transverse velocity $\dot{y}_h$, while the rest of the string is at rest. Every a_n is zero, and

$$\begin{aligned} b_n &= \frac{2\dot{y}_h}{\omega_n L} \int_{x_s - \sigma/2}^{x_s + \sigma/2} \sin k_n x \, dx \\ &= \frac{2\dot{y}_h}{\omega_n k_n L} \left[\cos k_n \left(x_s - \frac{\sigma}{2} \right) - \cos k_n \left(x_s + \frac{\sigma}{2} \right) \right] \\ &= \frac{4\dot{y}_h}{\omega_n k_n L} \sin k_n x_s \sin \frac{k_n \sigma}{2}, \end{aligned} \quad (2.79)$$

where the last step uses the identity $\cos \theta_1 - \cos \theta_2 = 2 \sin[(\theta_1 +$

$\theta_2)/2] \sin[(\theta_2 - \theta_1)/2]$, which follows from the formulas for the cosine of a sum and of a difference. The energy of mode n is therefore

$$E_n = \frac{\mu L}{4} \omega_n^2 b_n^2 = \frac{4\mu \dot{y}_h^2}{L k_n^2} \sin^2 k_n x_s \sin^2 \frac{k_n \sigma}{2}. \quad (2.80)$$

For the modes with $k_n \sigma \ll 1$, that is with $n \ll L/\sigma$, the last sine can be replaced by its argument, and

$$E_n \approx \frac{\mu \dot{y}_h^2 \sigma^2}{L} \sin^2 \frac{n\pi x_s}{L}, \quad (2.81)$$

which does not fall with n . Apart from the factor $\sin^2(n\pi x_s/L)$, a narrow hammer divides its energy roughly equally among the modes up to n of order L/σ , beyond which the energies fall as $1/n^2$, so the narrower the hammer, the higher the harmonics it excites.

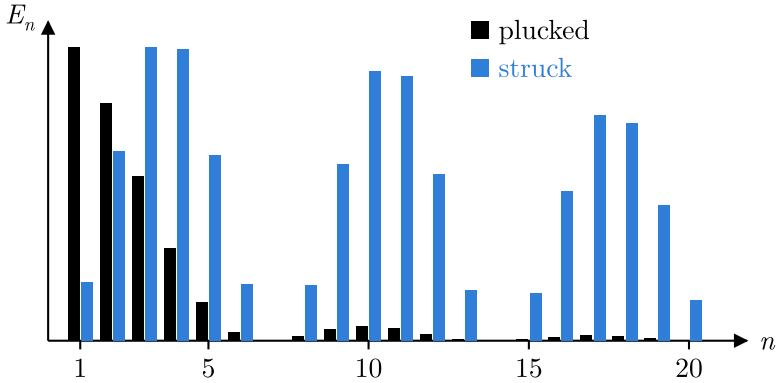


Figure 27. The energies E_n of the modes of a string plucked, and of the same string struck by a narrow hammer (accent), at the same point near one end, each set scaled so that its largest energy is drawn at the same height. The energies of the plucked string fall as $1/n^2$, while those of the struck string do not fall over the range of modes drawn. The modes with a node at the plucked or struck point are absent from both.

Figure 27 compares the two strings when $x_s = x_p$. The factor $\sin^2(n\pi x_p/L)$ is common to both, so the modes with a node at that

point receive no energy in either case, while the energies of the other modes fall quickly with n for the plucked string and stay large for the struck one. In this model a string struck by a narrow hammer is therefore much richer in high harmonics than the same string plucked at the same point.

2.5 Fourier Analysis of Signals

A quantity that varies in time and carries information about its source is called a signal, such as the displacement of one point of the string of Section 2.4, which is a sum of harmonics. We now describe signals by the frequencies they contain, first for signals that are periodic and then for signals that are not, and use this description to find how an oscillator responds to a force that is not sinusoidal.

The Spectrum of a Periodic Signal

Let $F(t)$ be a force that repeats itself with the period P , and let $\omega_f = 2\pi/P$ be its fundamental angular frequency (with ω_f replacing the fundamental ω in the series, the spectrum, Figure 28 and Figure 29 and their captions, the square wave, $\omega_0 = n_0\omega_f$, the swing, and Problems 1 and 2 and their solutions, while $\chi(\omega)$ and the transforms keep ω). The complex Fourier series of Section 6.1 of Vol. 4, with the position x replaced by the time t and the period $2L$ by P , so that π/L becomes ω , gives

$$F(t) = \sum_{n=-\infty}^{\infty} C_n e^{in\omega t}, \quad C_n = \frac{1}{P} \int_0^P F(t) e^{-in\omega t} dt. \quad (2.82)$$

Vol. 4 integrates over the interval from $-P/2$ to $P/2$, and the integral over any interval of one period gives the same result, since the integrand repeats itself with the period P . The force is real, so $C_{-n} = C_n^*$, and the terms with n and $-n$ combine as

$$C_n e^{in\omega t} + C_n^* e^{-in\omega t} = 2 \operatorname{Re} (C_n e^{in\omega t}) = 2|C_n| \cos(n\omega t + \arg C_n), \quad (2.83)$$

which is a harmonic oscillation of angular frequency $n\omega$ and real amplitude $2|C_n|$, while C_0 is the mean value of the force. The set of amplitudes $2|C_n|$, drawn against the angular frequencies $n\omega$, is called the spectrum of the signal. The spectrum leaves out the phases $\arg C_n$, which are needed to rebuild the signal but matter less for many purposes; the ear, for example, is to a good approximation insensitive to the relative phases of the harmonics of a steady note. Parseval's theorem of Section 6.1 of Vol. 4 gives the mean square of the force over a period as

$$\frac{1}{P} \int_0^P F^2 dt = \sum_{n=-\infty}^{\infty} |C_n|^2 = C_0^2 + \sum_{n=1}^{\infty} \frac{1}{2} (2|C_n|)^2, \quad (2.84)$$

and since the mean square of a harmonic oscillation is half the square of its amplitude, the mean square of the signal is the sum of the mean squares of its harmonics.

As an example, consider the square wave of Figure 28, which equals F_0 for $0 < t < P/2$ and $-F_0$ for $P/2 < t < P$. Its series was found in Section 6.1 of Vol. 4, and with $L = P/2$ it is

$$F(t) = \frac{4F_0}{\pi} \left(\sin \omega t + \frac{1}{3} \sin 3\omega t + \frac{1}{5} \sin 5\omega t + \cdots \right). \quad (2.85)$$

The even harmonics are missing because $F(t+P/2) = -F(t)$. Shifting the time by half a period multiplies $e^{in\omega t}$ by $e^{in\pi} = (-1)^n$, so a harmonic with even n is unchanged by the shift while one with odd n changes sign, and a signal that changes sign when shifted by half a period can contain only harmonics with odd n .

An Oscillator Driven by a Periodic Force

Consider a damped oscillator of mass m , spring constant κ , and damping coefficient b , driven by the force $F(t)$, so that

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = \frac{F(t)}{m}. \quad (2.86)$$

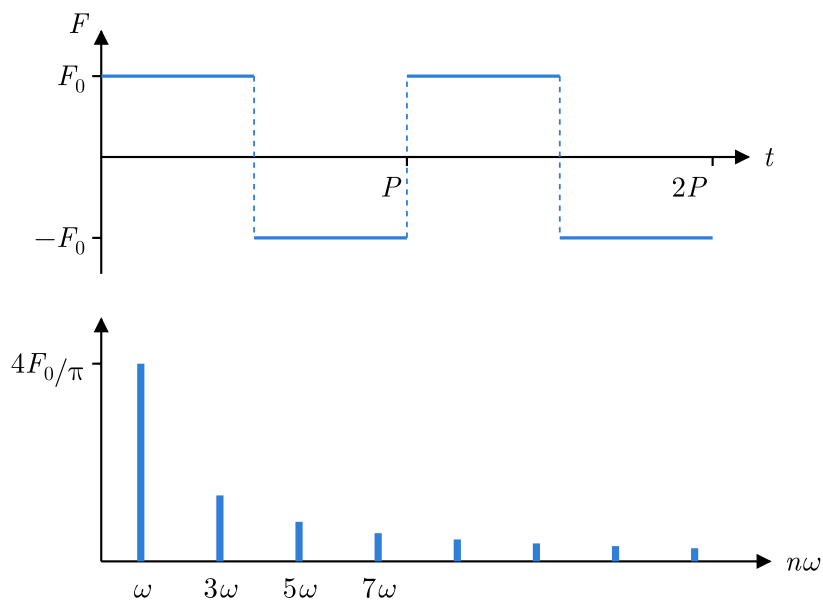


Figure 28. A square wave of amplitude F_0 and period P (upper panel), and its spectrum (lower panel), in which the harmonic of angular frequency $n\omega$ has the amplitude $4F_0/n\pi$ for odd n . The even harmonics are absent.

Section 1.3 found the steady response to the force $F_0 \cos \omega t = \operatorname{Re}(F_0 e^{i\omega t})$ to be $x = \operatorname{Re}(\mathcal{A} e^{i\omega t})$, with the complex amplitude $\mathcal{A} = \chi(\omega) F_0$, where

$$\chi(\omega) = \frac{1}{m(\omega_0^2 - \omega^2 + 2i\gamma\omega)}, \quad (2.87)$$

which is called the response function of the oscillator. Substituting $x = \chi(\omega) C e^{i\omega t}$ into the equation of motion shows that it is the steady response to the complex force $C e^{i\omega t}$, for any complex constant C and any real ω , positive or negative. The equation of motion is linear, so the response to a sum of forces is the sum of the responses to each of them, and the steady response to the periodic force is

$$x(t) = \sum_{n=-\infty}^{\infty} \chi(n\omega) C_n e^{in\omega t}. \quad (2.88)$$

The harmonic of angular frequency $n\omega$ in the response has the amplitude $2|\chi(n\omega)||C_n|$, so the spectrum of the response is the spectrum of the force multiplied, harmonic by harmonic, by

$$|\chi(\omega)| = \frac{1}{m\sqrt{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}}. \quad (2.89)$$

Let the oscillator be lightly damped, with $Q = \omega_0/2\gamma \gg 1$, and let its natural frequency be a whole multiple of the fundamental, $\omega_0 = n_0\omega$. The harmonic $n = n_0$ drives the oscillator at resonance, where $|\chi(\omega_0)| = 1/2m\gamma\omega_0 = Q/m\omega_0^2 = Q/\kappa$. For any other harmonic $|\omega_0^2 - n^2\omega^2|$ is at least of order ω^2 , while $2\gamma n\omega$ is smaller than this by a factor of order n_0/Q , so the damping term can be neglected and

$$|\chi(n\omega)| \approx \frac{1}{m|\omega_0^2 - n^2\omega^2|} = \frac{1}{\kappa|1 - n^2/n_0^2|}, \quad n \neq n_0. \quad (2.90)$$

A force applied very slowly would displace the oscillator by its amplitude divided by κ . Compared with this static displacement, the resonant harmonic is amplified by the factor Q , and the others by factors of order one that do not depend on Q .

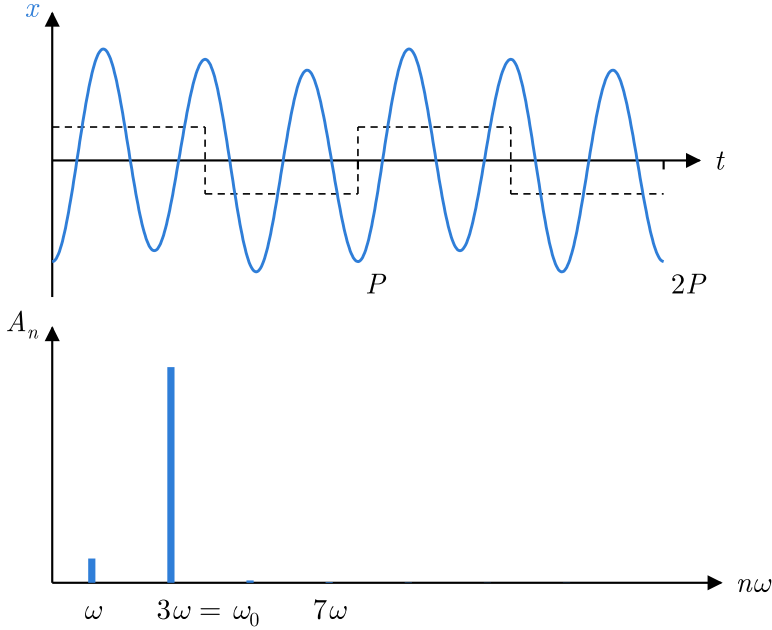


Figure 29. The steady response $x(t)$ (accent) of a lightly damped oscillator whose natural frequency is the third harmonic of a square-wave driving force, which is drawn dashed at a reduced scale (upper panel), and the amplitudes A_n of the harmonics of the response (lower panel). The response is almost a sinusoidal oscillation at ω_0 , with a smaller admixture of the fundamental.

For the square wave, with n_0 odd, the amplitude of the response at ω_0 is $(4F_0/n_0\pi)(Q/\kappa)$, and the amplitude of the response at the fundamental is $(4F_0/\pi)/[\kappa(1 - 1/n_0^2)]$, so the ratio of the first to the second is

$$\frac{Q(n_0^2 - 1)}{n_0^3}, \quad (2.91)$$

which is $8Q/27$ for $n_0 = 3$. Figure 29 shows the response for this

case. The force jumps between two values, and the oscillator responds with a nearly sinusoidal oscillation at three times the frequency of the force, since the lightly damped oscillator picks out the harmonic nearest its natural frequency. The ratio is large when $Q \gg n_0$, which is the condition that the width $\omega_0/Q = n_0\omega/Q$ of the resonance of Section 1.3 be much smaller than the spacing ω between neighbouring harmonics, so that only one harmonic falls inside the resonance.

As an example, consider a swing, which behaves as a lightly damped oscillator of natural angular frequency ω_0 , pushed briefly once in every second period of its oscillation. The pushes form a periodic force whose period is $4\pi/\omega_0$, so its fundamental is $\omega_0/2$ and its second harmonic is at ω_0 . A train of short pushes contains many harmonics of nearly equal amplitude, as you will show in Problem 1, and the second harmonic drives the swing at resonance. This is why a swing can be kept going by a push on every second swing, or on every third.

Signals That Are Not Periodic

A signal that does not repeat itself, such as a single pulse or a note that dies away, can be described by its Fourier transform. With the time t in place of the position x and the angular frequency ω in place of the wave number k , the Fourier transform of Section 6.2 of Vol. 4 and its inversion formula are

$$\tilde{h}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} h(t) e^{-i\omega t} dt, \quad h(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{h}(\omega) e^{i\omega t} d\omega, \quad (2.92)$$

where $h(t)$ is the signal. The inversion formula writes the signal as a superposition of oscillations $e^{i\omega t}$ with a continuous range of angular frequencies. The sum over the harmonics $n\omega$ of a periodic signal has become an integral over ω , and the quantity $\tilde{h}(\omega) d\omega/\sqrt{2\pi}$, the amplitude contributed by the angular frequencies between ω and $\omega + d\omega$, has taken the place of the coefficient C_n , the amplitude of the single oscillation $e^{in\omega t}$. By Parseval's theorem for Fourier transforms, from Section 6.2 of Vol. 4, $\int h^2 dt = \int |\tilde{h}|^2 d\omega$, so $|\tilde{h}(\omega)|^2 d\omega$ measures the part of the signal carried by the angular frequencies

between ω and $\omega + d\omega$, and $|\tilde{h}(\omega)|^2$ is often called the power spectrum of the signal. For a real signal $\tilde{h}(-\omega) = \tilde{h}^*(\omega)$, so the negative angular frequencies carry the same information as the positive ones.

For a force that acts for a finite time, each oscillation $e^{i\omega t}$ in the force produces the response $\chi(\omega)e^{i\omega t}$, and adding the responses gives

$$\tilde{x}(\omega) = \chi(\omega) \tilde{F}(\omega), \quad (2.93)$$

as you showed in Problem 2 of Section 6.2 of Vol. 4 with the derivative property of the Fourier transform.

The Spectrum of a Freely Decaying Oscillation

Consider a lightly damped oscillator that is set in motion at $t = 0$ and then left to itself. As in Section 1.2, its displacement decays as

$$x(t) = \begin{cases} 0, & t < 0, \\ Ae^{-\gamma t} \cos \omega_d t, & t > 0, \end{cases} \quad (2.94)$$

where the phase constant has been chosen to be zero. Writing the cosine as $\frac{1}{2}(e^{i\omega_d t} + e^{-i\omega_d t})$ turns the signal for $t > 0$ into the sum of two exponentials, $\frac{1}{2}Ae^{-(\gamma - i\omega_d)t}$ and $\frac{1}{2}Ae^{-(\gamma + i\omega_d)t}$. Each is the decaying exponential $e^{-\Gamma t}$ of Section 6.2 of Vol. 4 (which writes λ for Γ), whose Fourier transform is $1/[\sqrt{2\pi}(\Gamma + i\omega)]$, multiplied by $\frac{1}{2}A$ and with Γ now complex. The calculation of Vol. 4 holds for complex Γ as long as its real part is positive, because the exponential still vanishes at the upper limit of the integral, where its modulus is $e^{-\gamma t}$. Therefore

$$\tilde{x}(\omega) = \frac{A}{2\sqrt{2\pi}} \left[\frac{1}{\gamma + i(\omega - \omega_d)} + \frac{1}{\gamma + i(\omega + \omega_d)} \right]. \quad (2.95)$$

For positive ω near ω_d , the first term is of order $1/\gamma$, while the second is of order $1/\omega_d$, which is much smaller when $\gamma \ll \omega_d$. Neglecting the second term gives

$$|\tilde{x}(\omega)|^2 \approx \frac{A^2}{8\pi} \frac{1}{(\omega - \omega_d)^2 + \gamma^2} . \quad (2.96)$$

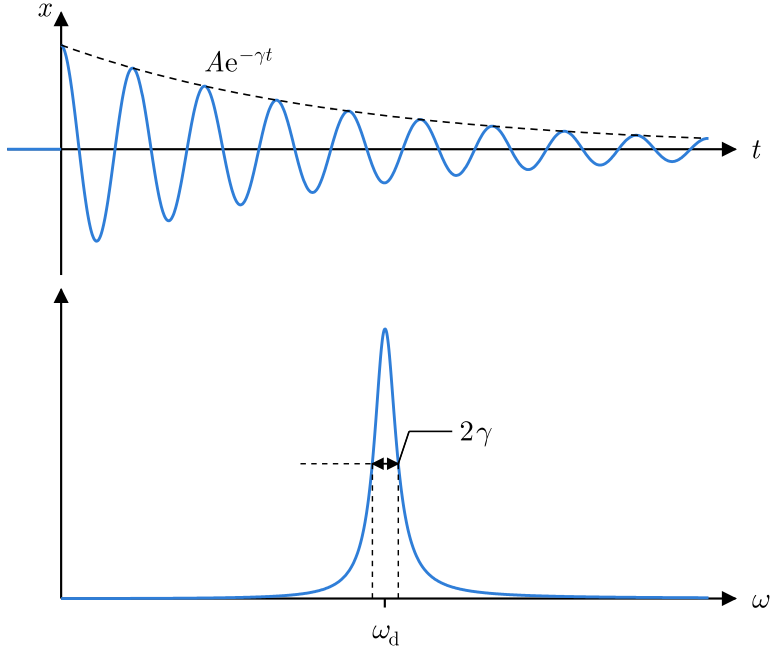


Figure 30. A freely decaying oscillation that begins at $t = 0$, with its envelope $Ae^{-\gamma t}$ dashed (upper panel), and the squared modulus of its Fourier transform (lower panel). The spectrum is a Lorentzian centred on ω_d , which falls to half its greatest value at a distance γ on either side of ω_d , so that its full width at half maximum is 2γ .

This curve is a Lorentzian, the shape met in Section 6.2 of Vol. 4, centred on ω_d and falling to half its greatest value at $\omega = \omega_d \pm \gamma$, as Figure 30 shows, so its full width at half maximum is $2\gamma = \omega_0/Q$. For light damping ω_d differs from ω_0 by a fraction of order $1/Q^2$, so the spectrum of the free oscillation has the width and the shape of the resonance of Section 1.3, where the mean power absorbed from the driving force was found to be a Lorentzian that falls to half its peak value at $\omega_0 \pm \gamma$.

The relation $\tilde{x} = \chi \tilde{F}$ explains why the spectrum of the free oscillation and the resonance have the same width. Suppose that the

oscillator, at rest, receives a sharp blow, a force that acts for a time much shorter than $1/\omega_0$ and whose integral over time is the impulse J . For angular frequencies up to well beyond ω_0 , the factor $e^{-i\omega t}$ in the Fourier transform of the force hardly changes while the force acts, so $\tilde{F}(\omega) \approx J/\sqrt{2\pi}$, which is an instance of the reciprocal relation of Section 6.2 of Vol. 4. After the blow the oscillator oscillates freely, and its spectrum is

$$|\tilde{x}(\omega)|^2 = \frac{J^2}{2\pi} |\chi(\omega)|^2. \quad (2.97)$$

Near ω_0 we can write $\omega_0^2 - \omega^2 = (\omega_0 + \omega)(\omega_0 - \omega) \approx 2\omega_0(\omega_0 - \omega)$ and $4\gamma^2\omega^2 \approx 4\gamma^2\omega_0^2$, so that $|\chi|^2 \approx 1/\{4m^2\omega_0^2[(\omega - \omega_0)^2 + \gamma^2]\}$, the same Lorentzian. The blow supplies every frequency near ω_0 equally, and the oscillator responds to each in proportion to $|\chi|$, so the free oscillation contains the frequencies that lie within the resonance. The longer the oscillator rings, the narrower its spectrum, and the product of the decay time $1/\gamma$ of the amplitude and the full width 2γ of the spectrum is 2 whenever the damping is light enough for the second term of $\tilde{x}(\omega)$ to be neglected. Among the ways of measuring the quality factor of a lightly damped oscillator, therefore, the following three agree with one another:

1. counting the cycles of its free decay, as in Section 1.2; 2. measuring the width of its resonance when it is driven, as in Section 1.3;
3. measuring the width of the spectrum of its free oscillation.

The third may be the most convenient when the oscillator is easily struck but not easily driven, and the width of a line in the spectrum of the sound of a struck bell, for example, can be used to find the quality factor of the corresponding mode of the bell.

A Finite Train of Oscillations

As a last example, consider an oscillation of angular frequency ω_s that lasts for a time Δt ,

$$h(t) = \begin{cases} \cos \omega_s t, & |t| < \Delta t/2, \\ 0, & |t| > \Delta t/2, \end{cases} \quad (2.98)$$

which could, for example, describe the displacement of a loud-speaker cone driven by a signal generator that is switched on for a time Δt . Section 6.2 of Vol. 4 found the Fourier transform of this signal, which it wrote with $T = \Delta t/2$ and with ω_0 in place of ω_s , to be

$$\tilde{h}(\omega) = \frac{1}{\sqrt{2\pi}} \left[\frac{\sin[(\omega - \omega_s)\Delta t/2]}{\omega - \omega_s} + \frac{\sin[(\omega + \omega_s)\Delta t/2]}{\omega + \omega_s} \right]. \quad (2.99)$$

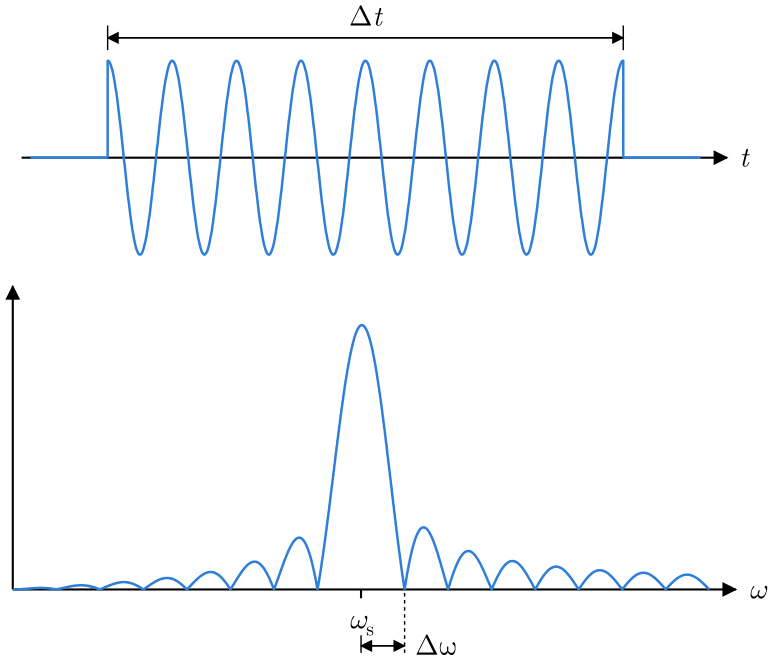


Figure 31. A train of oscillations of angular frequency ω_s lasting for the time Δt (upper panel), and the modulus of its Fourier transform for positive ω (lower panel). The spectrum has its peak at ω_s , and the first zeros on either side of the peak lie at the distance $\Delta\omega = 2\pi/\Delta t$ from it.

When the train contains many cycles, so that $\omega_s\Delta t \gg 1$, the peaks of the two terms at $\omega = \pm\omega_s$ hardly overlap, and near ω_s only the first term matters. It is largest at $\omega = \omega_s$, where its value is $\Delta t/2\sqrt{2\pi}$, and it first falls to zero where $(\omega - \omega_s)\Delta t/2 = \pm\pi$, as

Figure 31 shows. Let $\Delta\omega$ be the distance from the peak to the first zero on either side, which is a measure of the spread of the angular frequencies in the train. Then

$$\Delta\omega \Delta t = 2\pi. \quad (2.100)$$

(delete the sentence) The train contains $\omega_s \Delta t / 2\pi$ cycles, so the fractional spread of its frequencies is

$$\frac{\Delta\omega}{\omega_s} = \frac{2\pi}{\omega_s \Delta t}, \quad (2.101)$$

which is one divided by the number of cycles. A signal must therefore contain many cycles for its frequency to be sharply defined, and a note that lasts only a few cycles tends to be heard as a click rather than as a note of definite pitch.

The same relation holds, with widths defined loosely, for any signal. A signal that lasts for a time of order Δt has a spectrum spread over a range of angular frequencies of order $\Delta\omega$, where

$$\Delta\omega \Delta t \approx 2\pi. \quad (2.102)$$

The constant on the right-hand side depends on how the two widths are defined. For the freely decaying oscillation, with the decay time $1/\gamma$ as its duration and the half-width γ of its Lorentzian as the width of its spectrum, the product is one rather than 2π . A precise form of the relation, with the widths defined as standard deviations, is not treated in this course. The relation implies that an oscillation has to be observed for a time of at least about $2\pi/\Delta\omega$ for its frequency to be measured to within $\Delta\omega$, and you will see in Problem 4 that the same time is needed to tell apart two oscillations whose angular frequencies differ by $\Delta\omega$.

Problems

§2.1 Two Coupled Oscillators

1. For the pair of equal pendulums of Figure 11, released as in the text, show that the extension of the coupling spring is $x_2 - x_1 = -A \cos \omega_2 t$ exactly. Hence show that the mean energy stored in the coupling spring over one swing is smaller than E by the factor $\kappa_c/2m\omega_0^2$.
2. For the pendulums of unequal length of Figure 13, show that $x_1 = A (\cos^2 \psi \cos \omega_1 t + \sin^2 \psi \cos \omega_2 t)$, and that at $t = \pi/\omega_b$ this becomes $A \cos 2\psi \cos \omega_1 t$. For weak coupling, express the condition $\sin^2 2\psi = \frac{1}{2}$ in terms of g , L_I , L_{II} , κ_c , and m .
3. A machine of mass m rests on a mounting that acts as a spring of constant κ , and is shaken by a force $F_0 \cos \omega t$. A small mass m_a is attached to the machine by a spring of constant κ_a , and to nothing else. Show that the machine stays at rest if $\kappa_a/m_a = \omega^2$, and find the amplitude of the small mass.
4. Each of the two masses of Figure 14 is also acted on by a resistive force $-b\dot{x}$, with $\gamma = b/2m$. The normal coordinates still obey separate equations, as you showed in Problem 5 of Section 4.4 of Vol. 1; find $\mathcal{A}_1$. Show that at $\omega^2 = (\kappa + \kappa_c)/m$ the driven mass is no longer exactly at rest, and find $|\mathcal{A}_1|$ there when $2\gamma\omega \ll \kappa_c/m$.

§2.2 Many Coupled Oscillators and the Loaded String

1. A row of N equal masses m is joined by $N + 1$ equal springs of constant κ , with the outer springs fixed to walls, and the masses move along the row. Show that the equations of motion have the form of those of the loaded string with $\Omega^2 = \kappa/m$, and check that for $N = 2$ the frequencies agree with those found in Section 2.1 with $\kappa_c = \kappa$.
2. The right-hand end of a loaded string is tied to a light ring that slides without friction on a smooth rod perpendicular to the string at $x = L$, so that the last segment of string stays parallel to the x -axis. Show that the end condition becomes $y_{N+1} = y_N$, and that the normal frequencies are $\omega_n = 2\Omega \sin[(2n-1)\pi/2(2N+1)]$ for $n = 1, 2, \dots, N$. Check the result

for $N = 1$ from the forces on the bead.

3. Show that the potential energy of the loaded string, the work done against the tension in stretching the segments, is $(\tau/2a) \sum_{j=0}^N (y_{j+1} - y_j)^2$. Show also that the total energy of the string in mode n alone, with amplitude A_n , is $\frac{1}{4}(N+1)m\omega_n^2 A_n^2$.
4. A string carrying three beads is released from rest with the first bead displaced a distance η and the other two in equilibrium. Find the amplitudes of the three modes, and the displacement of the middle bead as a function of time.

§2.3 Symmetry and Normal Modes

1. Find the normal modes of the pair of equal masses of Section 2.1 from the symmetry that exchanges x_1 and x_2 , without expanding a determinant.
2. For the loaded string of Section 2.2, let $\mathbf{P}$ be the matrix that exchanges y_j and y_{N+1-j} for every j . Show that $\mathbf{P}$ commutes with $\mathbf{D}$, and that mode n is multiplied by $(-1)^{n+1}$ when $\mathbf{P}$ is applied to it, so that the modes are alternately even and odd.
3. Find the normal modes and frequencies of a ring of three equal masses joined by three equal springs. Describe the motion in a real mode of the highest frequency.
4. In the system of Figure 20, the middle mass is replaced by one of mass m_2 , the other masses and the springs being unchanged. Show that the exchange of x_1 and x_3 is still a symmetry, that the odd mode and its frequency are unchanged, and find the equation for the frequencies of the even modes.
5. For a ring of N masses, show that the real displacements $x_j = A \cos(jka) \cos \omega t$ and $x_j = A \sin(jka) \cos \omega t$, with $k = 2\pi n/Na$ and $\omega = 2\Omega |\sin(ka/2)|$, are motions of the ring, and that their patterns are orthogonal for $0 < n < N/2$.

§2.4 The Continuum Limit

1. Using $\sin \theta \approx \theta - \theta^3/6$, show that the angular frequencies of the loaded string are $\omega_n \approx (n\pi v/L)[1 - n^2\pi^2 a^2/(24L^2)]$ for $n \ll N$. Find the largest mode number n for which ω_n differs from $n\pi v/L$.

by less than a fraction ν of $n\pi v/L$, where $\nu \ll 1$ (and β replaced by ν in the last sentence of the problem and throughout its solution), and express it in terms of N and β .

2. A string is pulled aside at $x = x_p$ to the displacement y_p , as in the text. Show that the potential energy of the string before it is released is $\frac{1}{2}\tau y_p^2 L/[x_p(L - x_p)]$. By equating this energy to the sum of the energies E_n of the modes, show that $\sum_{n=1}^{\infty} \sin^2(n\theta)/n^2 = \frac{1}{2}\theta(\pi - \theta)$ for $0 \leq \theta \leq \pi$.
3. A string is plucked at its midpoint to the displacement y_p and released from rest. Show that the displacement of the midpoint is $(8y_p/\pi^2) \sum_{n \text{ odd}} \cos(n\pi vt/L)/n^2$. Using the half-range cosine series of $|x|$ from Section 6.1 of Vol. 4, show that for $0 \leq t \leq L/v$ the midpoint moves with the constant speed $2y_p v/L$ from y_p to $-y_p$, and explain this motion with the moving corners of Figure 26.
4. Show that in mode n of a string, with $q_n = a_n \cos \omega_n t + b_n \sin \omega_n t$, the kinetic energy and the potential energy are each equal to $\frac{1}{2}E_n$ when averaged over a period $2\pi/\omega_n$. Hence show that for any motion of the string, the kinetic energy averaged over the time $2L/v$ is half the total energy.

§2.5 Fourier Analysis of Signals

1. A force equals F_0 for $0 < t < t_1$ and zero for $t_1 < t < P$, and repeats itself with the period P , where $0 < t_1 < P$. Show that its complex Fourier coefficients are $C_n = (F_0/n\pi) \sin(n\omega t_1/2) e^{-in\omega t_1/2}$ for $n \neq 0$, with $\omega = 2\pi/P$, and find C_0 . Show that when the pulses are short, the harmonics with $n \ll P/\pi t_1$ all have nearly the same amplitude, twice the mean force, and that the harmonics for which nt_1/P is a whole number are absent.
2. The lightly damped oscillator of this section, with $Q \gg 1$, is driven by the square wave of Figure 28, with $\omega_0 = 2\omega$. Explain why no harmonic of the force drives the oscillator at resonance, and show that the amplitude of the response at 3ω is one fifth of the amplitude of the response at ω .
3. Using the approximate form of $|\tilde{x}(\omega)|^2$ near ω_d and the corre-

sponding form near $-\omega_d$, show that $\int_{-\infty}^{\infty} |\tilde{x}|^2 d\omega \approx A^2/4\gamma$ for the freely decaying oscillation. Show also that $\int_0^{\infty} x^2 dt = A^2/4\gamma + A^2\gamma/4(\gamma^2 + \omega_d^2)$, and check that the two integrals agree, as Parseval's theorem for Fourier transforms requires, when $\gamma \ll \omega_d$.

4. A signal is the sum of two trains of the kind treated in this section, each lasting for the time Δt , with angular frequencies ω_{s1} and $\omega_{s2} > \omega_{s1}$, both much larger than $2\pi/\Delta t$. Show that when $\omega_{s2} - \omega_{s1} = 2\pi/\Delta t$ the peak of the spectrum of each train lies at the first zero of the spectrum of the other. Explain why the two peaks can be told apart in the spectrum of the sum only if $\omega_{s2} - \omega_{s1}$ is at least of order $2\pi/\Delta t$.

Wave Motion, the Wave Equation, and Wave Energy

3.1 Waves at School

A wave is a disturbance that travels through a medium and carries energy from one place to another, while each part of the medium only oscillates about its equilibrium position, as the water at each point of a pond bobs up and down while a ripple crosses it. The school treatment of waves describes them by a handful of quantities and a few relations between them, which this section collects.

Transverse and Longitudinal Waves

In a transverse wave the medium moves perpendicular to the direction in which the wave travels, as a stretched string does when one end is shaken from side to side. In a longitudinal wave the medium moves along the direction of travel, as the air does when a sound passes through it, and the wave then consists of alternate regions in which the medium is squeezed together and spread apart, called compressions and rarefactions. Both kinds are shown in Figure 32.

A wave in which every part of the medium oscillates with simple harmonic motion is called a harmonic wave. Its amplitude A is the greatest displacement of the medium from equilibrium, and its

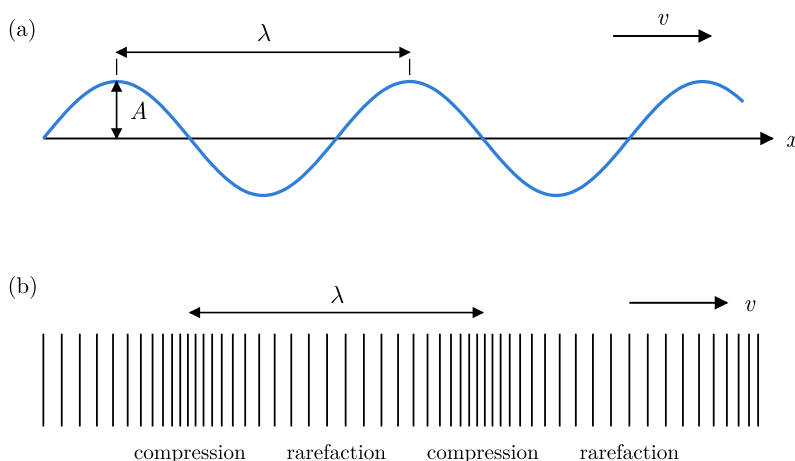


Figure 32. (a) A transverse wave on a string at one instant, with its wavelength λ and its amplitude A . (b) A longitudinal wave at one instant, drawn as the displaced positions of layers of the medium that are equally spaced when the medium is undisturbed. The wavelength is the distance between successive compressions. Both waves travel with speed v in the direction of the arrow.

wavelength λ is the distance between successive points that are in the same phase of their oscillation, such as two neighbouring crests. Its period P is the time taken by one oscillation of a point of the medium, and its frequency is $f = 1/P$. In one period the wave advances by one wavelength, so its speed is

$$v = f\lambda. \quad (3.1)$$

The frequency is fixed by the source and the speed by the medium, so when a wave passes from one medium into another its frequency stays the same and its wavelength changes in proportion to its speed.

The displacement of a harmonic wave travelling in the direction of increasing x is often written at school as

$$y = A \sin \left[2\pi \left(\frac{t}{P} - \frac{x}{\lambda} \right) \right], \quad (3.2)$$

which repeats itself when t increases by a period or x by a wavelength. Points of the medium a distance Δx apart along the direction of travel oscillate with a phase difference $2\pi \Delta x/\lambda$, so points a whole number of wavelengths apart move in step, and points half a wavelength apart move in opposite directions.

Wavefronts and Rays

A wavefront is a surface on which every point of the medium is in the same phase of its oscillation, such as the surface through the crests of a wave on water. A ray is a line perpendicular to the wavefronts, which points in the direction in which the wave travels. Waves spreading from a small source have spherical wavefronts centred on the source, and their rays point radially outwards. Far from the source, or near a large flat source such as a vibrating panel, the wavefronts are for all practical purposes planes and the rays are parallel, and the wave is called a plane wave.

The Speed of a Wave on a String

A transverse wave on a string under tension τ , with mass per unit length μ , travels with speed

$$v = \sqrt{\frac{\tau}{\mu}}. \quad (3.3)$$

A tighter or lighter string therefore carries waves faster, and the speed does not depend on the amplitude or on the frequency of the wave, provided the amplitude is small.

Intensity and the Inverse-Square Law

The intensity I of a wave is the power it carries across unit area of a surface perpendicular to its direction of travel, and it is measured in watts per square metre. For a given wave in a given medium, the intensity is proportional to the square of the amplitude. A small source that radiates a power $\mathcal{P}$ equally in all directions, in a medium that absorbs none of it, sends the whole of that power across every sphere centred on the source, and a sphere of radius r has area $4\pi r^2$, so the intensity at a distance r is

$$I = \frac{\mathcal{P}}{4\pi r^2}, \quad (3.4)$$

which is called the inverse-square law. Since the intensity goes as the square of the amplitude, the amplitude of the wave falls as $1/r$.

The Doppler Effect

When a source of sound moves towards a listener, the frequency heard is higher than the frequency emitted, and when it moves away the frequency heard is lower, which is why the pitch of an ambulance siren drops as the ambulance passes. This change of frequency is called the Doppler effect. For sound travelling with speed c through still air, a source of frequency f moving with speed V_s directly towards a listener at rest is heard at the frequency

$$f' = \frac{c}{c - V_s} f, \quad (3.5)$$

and a source moving directly away is heard at $cf/(c + V_s)$. When the source is at rest and the listener moves towards it with speed V_o , the frequency heard is

$$f' = \frac{c + V_o}{c} f, \quad (3.6)$$

and when the listener moves away it is $(c - V_o)f/c$.

3.2 Travelling Waves on a String

The wave equation for a stretched string was obtained in Section 2.4 as the limit of a string loaded with many beads, and the equation and d'Alembert's general solution of it, with the reflection of a pulse at a fixed end, are worked out in detail in Section 7.1 of Vol. 4 of the Fermi Notes on Physics. This section takes the equation and its general solution as its starting point. It first describes harmonic travelling waves in real and complex form, then distinguishes the velocity of the string from the speed of the wave, and finally solves for

the motion of an infinite string released with a given displacement and a given velocity.

The Wave Equation for a String

In Section 2.4 the wave equation was obtained as the limit of the loaded string of Section 2.2, in which the number of beads grows and their spacing shrinks while the mass per unit length is held fixed. Section 7.1 of Vol. 4 derived the same equation directly, by applying Newton's second law to the transverse motion of a short piece of string. Let $y(x, t)$ be the transverse displacement at time t of the point of the string whose undisturbed position is x , and let the string have tension τ and mass per unit length μ . The displacement satisfies

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}, \quad v = \sqrt{\frac{\tau}{\mu}}, \quad (3.7)$$

which Vol. 4 wrote with ρ_ℓ for the mass per unit length and a different letter for the displacement. The derivation rests on the following assumptions:

1. each point of the string moves only perpendicular to the undisturbed string;
2. the weight of the string is negligible compared with the tension;
3. the slope $\partial y / \partial x$ is small compared with one everywhere.

The third assumption enters the derivation in two places. The tension is the same all along the string because the angle ψ between the string and the x -axis is small, so that $\cos \psi$ differs from one only in the second order of ψ , and the transverse component of the tension, $\tau \sin \psi$, is replaced by $\tau \partial y / \partial x$ because $\sin \psi$ and $\tan \psi$ are equal to first order in ψ . When the slope is not small the length of the string, and with it the tension, changes appreciably as the string moves, and the equation of motion becomes nonlinear. The motion of a string of large slope is not treated in this course.

The wave equation is linear and homogeneous, so, as Section 7.1 of Vol. 4 showed, any sum of solutions with constant coefficients is again a solution. This is the principle of superposition, and it allows a complicated wave to be built up as a sum of simpler ones.

D'Alembert's Solution

In Section 7.1 of Vol. 4 the wave equation on an infinite line was solved by changing variables to $x - vt$ and $x + vt$ with the chain rule for several variables (Section 2.2 of Vol. 4), which gave the general solution

$$y(x, t) = f_1(x - vt) + f_2(x + vt), \quad (3.8)$$

where f_1 and f_2 are any functions that can be differentiated twice. A profile with a corner, such as the triangle of a plucked string, is treated as the limit of smooth profiles whose corners are rounded off, and the solution holds for it in the same form. The value of $f_1(x - vt)$ at the position x and the time t is the value of f_1 at the position $x - vt$ at time zero, so the profile described by f_1 moves in the direction of increasing x at speed v without changing its shape, as Figure 33 shows, and the profile described by f_2 moves in the direction of decreasing x in the same way. A disturbance of either kind was called a travelling wave in Vol. 4, and the constant v is the speed of the wave.

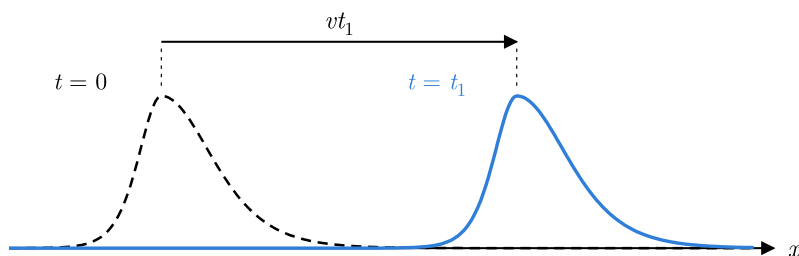


Figure 33. A pulse $y = f_1(x - vt)$ on a string at $t = 0$ (dashed) and at a later time t_1 (solid). The profile has moved a distance vt_1 in the direction of increasing x and has kept its shape, even though the pulse is not symmetric.

A relation that we will use repeatedly follows from the chain rule. For a wave $y = f_1(x - vt)$ travelling in the direction of increasing x , the derivatives with respect to position and time are $\partial y / \partial x =$

$f_1'(x - vt)$ and $\partial y/\partial t = -vf_1'(x - vt)$, where the prime denotes the derivative of f_1 with respect to its argument, so

$$\frac{\partial y}{\partial t} = -v \frac{\partial y}{\partial x}. \quad (3.9)$$

For a wave $y = f_2(x + vt)$ travelling the other way, the same argument gives $\partial y/\partial t = +v \partial y/\partial x$.

Harmonic Travelling Waves

A travelling wave whose profile is a sinusoid is called a harmonic travelling wave. Consider the displacement

$$y(x, t) = A \cos(\omega t - kx + \phi), \quad (3.10)$$

in which A is the amplitude, ϕ is the phase constant, and ω and k are positive constants. Differentiating twice gives $\partial^2 y/\partial x^2 = -k^2 y$ and $\partial^2 y/\partial t^2 = -\omega^2 y$, so the wave equation is satisfied when

$$\omega = vk. \quad (3.11)$$

At a fixed time the displacement repeats itself whenever x increases by $2\pi/k$, so the wavelength is $\lambda = 2\pi/k$, and k is called the wave number. At a fixed position the displacement repeats itself whenever t increases by $2\pi/\omega$, so the period is $P = 2\pi/\omega$ and ω is the angular frequency. Figure 34 shows the displacement against position at a fixed time and against time at a fixed position. Since $\omega/k = (2\pi f)(\lambda/2\pi) = f\lambda$, the condition $\omega = vk$ is the relation $v = f\lambda$ of Section 3.1.

The argument $\omega t - kx + \phi$ of the cosine is called the phase of the wave. A point of fixed phase, such as a crest, moves so that $\omega t - kx$ stays constant, and differentiating this condition with respect to time gives $dx/dt = \omega/k = v$, so the crests move in the direction of increasing x at the speed of the wave. We can also see this by writing the phase as $-k(x - vt) + \phi$, which shows that the harmonic wave is a function of $x - vt$ alone, as a wave travelling in the direction

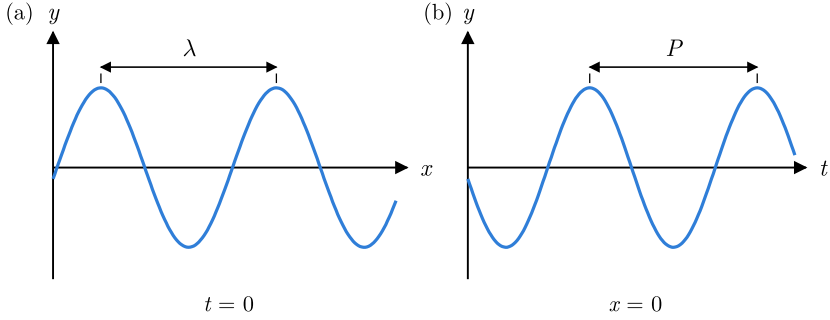


Figure 34. A harmonic travelling wave $y = A \cos(\omega t - kx + \phi)$. (a) The displacement against position at $t = 0$, which repeats after a wavelength $\lambda = 2\pi/k$. (b) The displacement against time at $x = 0$, which repeats after a period $P = 2\pi/\omega$.

of increasing x must be. A harmonic wave travelling in the direction of decreasing x is $A \cos(\omega t + kx + \phi)$.

As with the oscillator of Section 1.1, calculations with harmonic waves are simpler in complex form. With the time dependence $e^{i\omega t}$ used throughout this course, the wave travelling in the direction of increasing x is

$$y(x, t) = \text{Re} [\mathcal{A} e^{i(\omega t - kx)}] , \quad \mathcal{A} = A e^{i\phi} , \quad (3.12)$$

and the wave travelling in the direction of decreasing x is $\text{Re} [\mathcal{A} e^{i(\omega t + kx)}]$. The real part of $A e^{i\phi} e^{i(\omega t - kx)}$ is $A \cos(\omega t - kx + \phi)$, so the complex form describes the same wave as the real one. Differentiating the complex exponential with respect to t multiplies it by $i\omega$, and differentiating it with respect to x multiplies it by $-ik$, so substituting it into the wave equation gives $(-ik)^2 = (i\omega)^2/v^2$, which is the condition $\omega = vk$ again. Because the wave equation is linear and its coefficients are real, the real part of a complex solution is itself a solution, and we may carry out the calculation with the complex form and take the real part at the end.

The Velocity of the String and the Speed of the Wave

A wave on a string involves two quite different velocities. The speed v of the wave is the speed at which the profile, and with it the energy of the wave, moves along the string. The transverse velocity $\partial y/\partial t$ is the velocity with which a piece of the string moves perpendicular to the string. For the harmonic wave the transverse velocity is

$$\frac{\partial y}{\partial t} = -A\omega \sin(\omega t - kx + \phi), \quad (3.13)$$

whose greatest value is $A\omega$. The string itself does not move along the x -axis; what travels along it is the shape of the displacement.

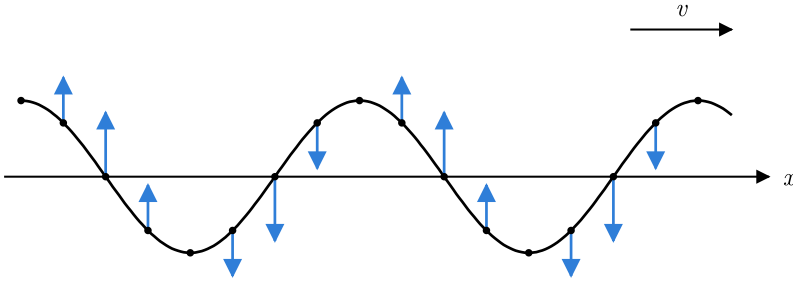


Figure 35. A harmonic wave travelling in the direction of increasing x , with the transverse velocity $\partial y/\partial t = -v \partial y/\partial x$ of the string drawn as arrows at equally spaced points. The string is rising ahead of each crest and falling behind it, and it is momentarily at rest at the crests and the troughs, where the slope is zero.

The relation $\partial y/\partial t = -v \partial y/\partial x$ tells us how the two velocities are linked. Figure 35 shows the transverse velocity along a harmonic wave. The part of the string just ahead of a crest is rising, because the crest is about to arrive there, and the part just behind it is falling, and the string moves fastest where it is steepest. The ratio of the greatest transverse speed to the speed of the wave is

$$\frac{A\omega}{v} = kA = \frac{2\pi A}{\lambda}, \quad (3.14)$$

which is also the greatest slope of the string. The condition of small slopes under which the wave equation holds is therefore the condition $A \ll \lambda/2\pi$ that the amplitude be small compared with the wavelength, and when it is met each piece of the string moves slowly compared with the wave.

A String Released from Rest

As a first example, consider an infinite string held in the shape $y_0(x)$ and released from rest at $t = 0$. The initial conditions are $y(x, 0) = y_0(x)$ and $\partial y / \partial t(x, 0) = 0$. Substituting d'Alembert's solution into them, and using the prime for the derivative of each function with respect to its argument, gives

$$f_1(x) + f_2(x) = y_0(x), \quad -vf_1'(x) + vf_2'(x) = 0. \quad (3.15)$$

The second condition says that $f_2 - f_1$ has zero derivative, so it is a constant, and since adding a constant to f_2 and subtracting it from f_1 leaves y unchanged, we may take the constant to be zero. Then $f_1 = f_2 = \frac{1}{2}y_0$, and

$$y(x, t) = \frac{1}{2}y_0(x - vt) + \frac{1}{2}y_0(x + vt). \quad (3.16)$$

The initial shape therefore divides into two pulses of the same shape and half the height, one travelling in each direction. The division is equal because a string released from rest has no velocity to favour one direction over the other.

Figure 36 shows the motion for the pulse $y_0(x) = y_m e^{-x^2/\sigma^2}$, of height y_m and width of order σ . While vt is smaller than about σ , the two halves overlap and their sum is a single hump that spreads and flattens, and once vt is several times σ the string carries two separate pulses, each of height $\frac{1}{2}y_m$, with the string flat and at rest between them.

A Pulse Launched in One Direction

As a second example, suppose that at $t = 0$ the string is given the displacement $y_0(x)$ and also a transverse velocity $q(x)$, so that

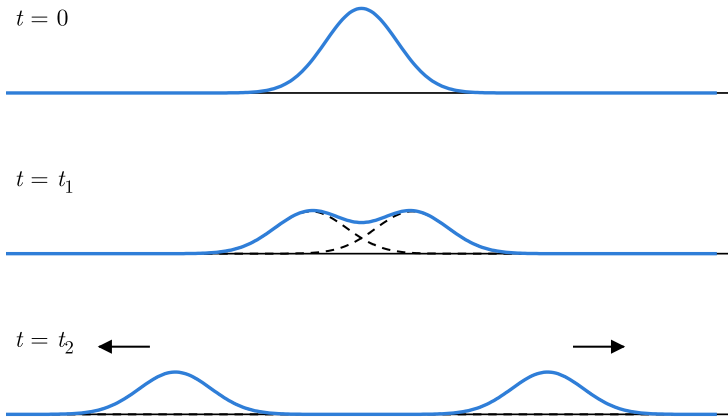


Figure 36. A string released from rest with the displacement $y_0(x) = y_m e^{-x^2/\sigma^2}$, shown at $t = 0$ and at two later times $t_1 < t_2$. The solid curve is the displacement and the dashed curves are the two halves, $\frac{1}{2}y_0(x - vt)$ and $\frac{1}{2}y_0(x + vt)$. At t_1 the halves still overlap, and by t_2 they have separated into two pulses of height $\frac{1}{2}y_m$ travelling apart at speed v .

$y(x, 0) = y_0(x)$ and $\partial y / \partial t(x, 0) = q(x)$. D'Alembert's solution now requires

$$f_1(x) + f_2(x) = y_0(x), \quad -vf_1'(x) + vf_2'(x) = q(x). \quad (3.17)$$

Integrating the second condition from 0 to x gives $f_2(x) - f_1(x) = v^{-1} \int_0^x q(x') dx'$, once more up to a constant that we may take to be zero, and adding this equation to the first condition and subtracting it from the first condition gives

$$\begin{aligned} f_2(x) &= \frac{1}{2} \left[y_0(x) + \frac{1}{v} \int_0^x q(x') dx' \right], \\ f_1(x) &= \frac{1}{2} \left[y_0(x) - \frac{1}{v} \int_0^x q(x') dx' \right]. \end{aligned} \quad (3.18)$$

Substituting these functions into $y = f_1(x - vt) + f_2(x + vt)$ gives d'Alembert's formula of Section 7.1 of Vol. 4,

$$y(x, t) = \frac{1}{2} [y_0(x - vt) + y_0(x + vt)] + \frac{1}{2v} \int_{x-vt}^{x+vt} q(x') dx', \quad (3.19)$$

and the two expressions above show how any initial state divides into a part travelling in each direction.

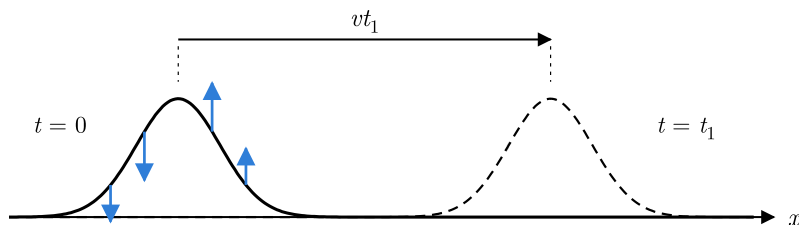


Figure 37. A pulse $y_0(x)$ given the initial velocity $-v y'_0(x)$, drawn as arrows, at $t = 0$ (solid). The front of the pulse is rising and the back is falling, as in a wave travelling in the direction of increasing x , and at a later time t_1 the whole pulse (dashed) has moved a distance vt_1 without dividing.

Let us choose the initial velocity to be $q(x) = -v y'_0(x)$, which is the velocity that the relation $\partial y / \partial t = -v \partial y / \partial x$ requires of a wave travelling in the direction of increasing x . The integral is then $v^{-1} \int_0^x q dx' = -[y_0(x) - y_0(0)]$, so f_2 is the constant $\frac{1}{2} y_0(0)$ and $f_1(x) = y_0(x) - \frac{1}{2} y_0(0)$, and the constants cancel in the sum, leaving

$$y(x, t) = y_0(x - vt). \quad (3.20)$$

The whole pulse travels in the direction of increasing x without dividing, as Figure 37 shows. The front of the pulse, where the slope is negative, is given an upward velocity, and the back of the pulse a downward one, which is the pattern of velocities in Figure 35. A pulse that is given the opposite velocity, $q = +v y'_0$, travels in the direction of decreasing x , as you will show in Problem 3, and a pulse given no velocity divides equally between the two directions, as in the first example.

3.3 Longitudinal Waves in Bars and Fluids

The string of Section 3.2 carries transverse waves. A solid bar, a column of fluid, or a spring can carry waves in which the medium moves along the direction in which the wave travels, and the compressions and rarefactions of Figure 32(b) are then set up in it. In this section we derive the wave equation for these longitudinal waves, first for a thin bar, then for a fluid in a narrow tube, and finally for a stretched spring, and we then compare the speeds of transverse and longitudinal waves on the same wire.

Strain and Young's Modulus

Consider a thin, straight bar lying along the x -axis, with cross-sectional area S and density ρ . When a longitudinal wave passes, each thin slice of the bar moves along the axis, and we let $\xi(x, t)$ be the displacement at time t of the slice whose undisturbed position is x , so that the slice is at $x + \xi(x, t)$.

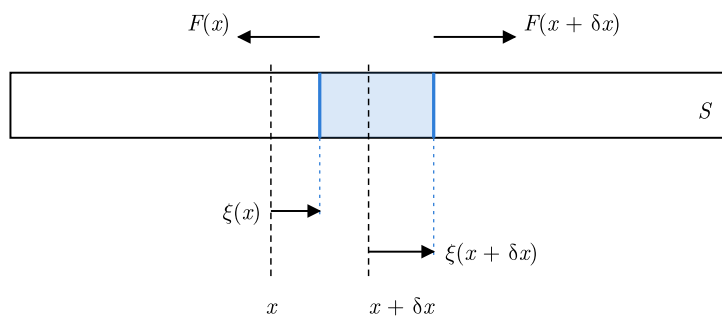


Figure 38. A thin bar of cross-sectional area S carrying a longitudinal wave. The slice of material that lies between x and $x + \delta x$ when the bar is undisturbed (dashed lines) is displaced to lie between $x + \xi(x)$ and $x + \delta x + \xi(x + \delta x)$ (shaded). The material on either side pulls on the slice with the forces $F(x)$ and $F(x + \delta x)$.

Figure 38 shows the slice that lies between x and $x + \delta x$ when the bar is undisturbed. When the bar is disturbed, the faces of the slice move to $x + \xi(x, t)$ and $x + \delta x + \xi(x + \delta x, t)$, so its length becomes

$$\delta x + \xi(x + \delta x, t) - \xi(x, t) = \left(1 + \frac{\partial \xi}{\partial x}\right) \delta x, \quad (3.21)$$

where we have expanded $\xi(x + \delta x, t)$ as a Taylor series in δx and kept the first two terms. The fractional change in the length of the slice, $\partial \xi / \partial x$, is called the strain. It is positive where the material is stretched and negative where it is compressed, and it is zero wherever the displacement is the same for neighbouring slices, since a uniform displacement moves the bar without deforming it.

The force per unit area that the material on one side of a cross-section exerts on the material on the other is called the stress. When a wire or a rod is stretched by a small amount, experiment shows that the stress is proportional to the strain, which is Hooke's law for a solid, and the ratio

$$Y = \frac{\text{stress}}{\text{strain}} \quad (3.22)$$

is called Young's modulus. It is a property of the material, measured, for example, by hanging known loads from a long wire and measuring its extension, and Hooke's law holds as long as the strain is small enough that the material returns to its original length when the load is removed. Let $F(x, t)$ be the force that the material to the right of the cross-section at x exerts on the material to its left, counted positive when it is a pull in the direction of increasing x . Hooke's law gives

$$F = YS \frac{\partial \xi}{\partial x}, \quad (3.23)$$

which is a tension where the bar is stretched and a thrust, with $F < 0$, where it is compressed. By Newton's third law, the material to the left of the cross-section exerts the force $-F$ on the material to its right. A bar also becomes slightly narrower when it is stretched, and the sideways motion that this requires is neglected here, which is a valid approximation provided the width of the bar is small compared with the wavelength.

The Wave Equation for a Bar

The slice between x and $x + \delta x$ has mass $\rho S \delta x$. The material to its right pulls it with the force $F(x + \delta x, t)$ in the direction of increasing x , and the material to its left pulls it with $F(x, t)$ in the opposite direction, so Newton's second law gives

$$\begin{aligned} \rho S \delta x \frac{\partial^2 \xi}{\partial t^2} &= F(x + \delta x, t) - F(x, t) \\ &= YS \left[\frac{\partial \xi}{\partial x}(x + \delta x, t) - \frac{\partial \xi}{\partial x}(x, t) \right]. \end{aligned} \quad (3.24)$$

Dividing both sides by $\rho S \delta x$ and letting δx tend to zero turns the bracket divided by δx into $\partial^2 \xi / \partial x^2$, by the definition of the derivative, and gives

$$\frac{\partial^2 \xi}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \xi}{\partial t^2}, \quad v = \sqrt{\frac{Y}{\rho}}. \quad (3.25)$$

This is the wave equation of Section 3.2, with the longitudinal displacement ξ in place of the transverse displacement y , with YS in place of the tension τ , and with ρS , the mass per unit length of the bar, in place of μ . The product YS is the force per unit strain, so it measures how strongly the bar resists being stretched, as the tension measures how strongly the string resists being bent away from the axis. The results of Section 3.2 therefore carry over, including d'Alembert's solution and the harmonic waves $\xi = A \cos(\omega t - kx + \phi)$ with $\omega = vk$.

The speed of the wave depends only on the material, through its stiffness and its density, and not on the cross-section or on any tension applied to the bar. Stiff, light materials carry longitudinal waves fastest, which is one reason these waves travel much faster in steel than in rubber, whose Young's modulus is smaller by a very large factor while its density is smaller by only a modest one.

A Column of Fluid

Consider next a fluid filling a narrow tube of cross-sectional area S lying along the x -axis, with the fluid at rest at the pressure p_0 and

the density ρ_0 . When a longitudinal wave passes, the fluid in each thin slice moves along the tube, and we let $\xi(x, t)$ be its displacement as before. The pressure at the slice becomes $p_0 + p$, where p is called the excess pressure. A fluid resists being compressed, and a small increase Δp in its pressure produces a fractional decrease in its volume that is proportional to Δp , so that

$$\Delta p = -B \frac{\Delta V_f}{V_f}, \quad (3.26)$$

where V_f is the volume of a given amount of fluid and ΔV_f its change. The positive constant B is called the bulk modulus of the fluid. It is large for a liquid such as water, which is hard to compress, and much smaller for a gas. For a gas the value of B also depends on whether heat flows into or out of the gas while it is compressed, so in this section B is treated as a parameter to be measured under the conditions of the wave.

The slice of fluid between x and $x + \delta x$ has volume $S \delta x$ when the fluid is at rest, and, as for the bar, its length becomes $(1 + \partial \xi / \partial x) \delta x$ when the fluid is disturbed. The fractional change in its volume is therefore $\partial \xi / \partial x$, and the excess pressure is

$$p = -B \frac{\partial \xi}{\partial x}, \quad (3.27)$$

which is positive where the fluid is compressed. The fluid on the left of the slice pushes on it with the force $(p_0 + p(x, t))S$ in the direction of increasing x , and the fluid on the right pushes on it with $(p_0 + p(x + \delta x, t))S$ in the opposite direction. The ambient pressure p_0 contributes equally to the two forces and cancels, so Newton's second law for the slice, whose mass is $\rho_0 S \delta x$, gives

$$\begin{aligned} \rho_0 S \delta x \frac{\partial^2 \xi}{\partial t^2} &= -S [p(x + \delta x, t) - p(x, t)] \\ &= BS \left[\frac{\partial \xi}{\partial x}(x + \delta x, t) - \frac{\partial \xi}{\partial x}(x, t) \right]. \end{aligned} \quad (3.28)$$

Dividing by $\rho_0 S \delta x$ and letting δx tend to zero gives the wave equation for ξ with the speed

$$c = \sqrt{\frac{B}{\rho_0}}. \quad (3.29)$$

The bulk modulus takes the place of Young's modulus, and the pressure the place of the tension, with a change of sign because a fluid pushes where a bar pulls. The derivation assumes that the fluid moves only along the tube, that the friction between the fluid and the walls can be neglected, and that the disturbance is small, $|\partial\xi/\partial x| \ll 1$, so that the excess pressure is small compared with B . Differentiating the wave equation for ξ with respect to x and multiplying by $-B$ shows that the excess pressure satisfies the same wave equation.

Compressions, Rarefactions, and Pressure

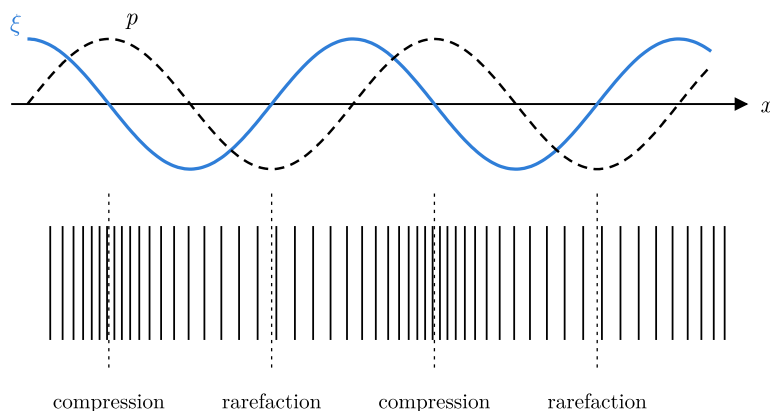


Figure 39. A harmonic longitudinal wave at one instant. The solid curve is the displacement ξ and the dashed curve is the excess pressure $p = -B \partial\xi/\partial x$, drawn to different scales. Below them, vertical lines mark the displaced positions of layers of the medium that are equally spaced when it is undisturbed. The layers crowd together where $\partial\xi/\partial x < 0$, which is where the excess pressure is greatest and the displacement is zero.

Consider the harmonic wave $\xi = A \cos(\omega t - kx)$ travelling along the tube in the direction of increasing x , with $\omega = ck$. The excess pressure is

$$p = -B \frac{\partial \xi}{\partial x} = -BkA \sin(\omega t - kx), \quad (3.30)$$

so the pressure varies harmonically with the amplitude BkA , which can be rewritten as $\rho_0 c \omega A$ since $B = \rho_0 c^2$ and $ck = \omega$. The pressure is a quarter of a wavelength out of step with the displacement. Figure 39 shows the displacement, the pressure, and the positions of layers of fluid at $t = 0$. The fluid is most compressed where the displacement is zero and decreases with x , because there the fluid on the left of the point is displaced towards it and the fluid on the right is displaced towards it as well, and the fluid is most rarefied where the displacement is zero and increases with x . Where the displacement is greatest, the neighbouring layers are displaced by nearly the same amount, and the fluid there is at its ambient pressure. The same picture holds for the bar, with the stress F/S in place of $-p$.

A Stretched Spring

As a third example, consider a spring of natural length L_0 , spring constant κ , and mass m , stretched to a length L between two fixed supports, as in Figure 40. The spring is then under the tension $\tau = \kappa(L - L_0)$, and we assume that its coils are equally spaced along it, so that its mass per unit length is $\mu = m/L$. Let x be the distance of a coil from one support when the spring is at rest, and let $\xi(x, t)$ be its displacement along the spring.

To find the tension at each point, we need the spring constant of a short piece of the spring. When a number of springs are joined end to end, the same tension acts in each of them and their extensions add, so a spring made of N identical pieces extends N times as much as one piece under the same tension, and each piece has a spring constant N times that of the whole spring. The piece of the stretched spring between x and $x + \delta x$ is a fraction $\delta x/L$ of the spring, and its spring constant is therefore $\kappa L/\delta x$. When the spring

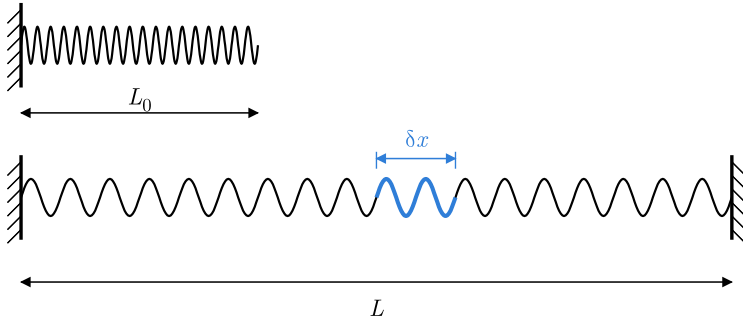


Figure 40. A spring of natural length L_0 (top) stretched to a length L between fixed supports (bottom). The piece of equilibrium length δx , drawn in the accent colour, is a fraction $\delta x/L$ of the spring and has the spring constant $\kappa L/\delta x$.

is disturbed, the extra extension of this piece is $\xi(x+\delta x, t) - \xi(x, t) = (\partial \xi / \partial x) \delta x$, and by Hooke's law the tension in the piece becomes

$$\tau + \frac{\kappa L}{\delta x} \frac{\partial \xi}{\partial x} \delta x = \tau + \kappa L \frac{\partial \xi}{\partial x}. \quad (3.31)$$

This has the form of the force in the bar, with κL in place of YS , together with the constant tension τ , which pulls equally on the two ends of any piece and so drops out of its equation of motion. Newton's second law for the piece, whose mass is $\mu \delta x$, therefore gives the wave equation in the same way as for the bar, and longitudinal waves travel along the spring with the speed

$$v_1 = \sqrt{\frac{\kappa L}{\mu}} = L \sqrt{\frac{\kappa}{m}}. \quad (3.32)$$

The same spring carries transverse waves as a string does, with the tension τ and the mass per unit length μ , at the speed $v_t = \sqrt{\tau/\mu} = \sqrt{\kappa(L - L_0)L/m}$. The ratio of the two speeds is

$$\frac{v_t}{v_l} = \sqrt{\frac{L - L_0}{L}}, \quad (3.33)$$

which is always less than one, and approaches one when the spring is stretched to many times its natural length. On a spring of negligible natural length, transverse and longitudinal pulses travel at the same speed.

Transverse and Longitudinal Waves on a Wire

A wire of cross-sectional area S , density ρ , and Young's modulus Y , stretched with the tension τ , can carry both transverse and longitudinal waves. Its mass per unit length is ρS , so transverse waves travel at $v_t = \sqrt{\tau/\rho S}$, and longitudinal waves travel at $v_l = \sqrt{Y/\rho}$, the speed found for the bar. The ratio of their squares is

$$\frac{v_t^2}{v_l^2} = \frac{\tau}{YS}, \quad (3.34)$$

which, by Hooke's law, is the strain produced in the wire by the tension. A metal wire that returns to its original length when the load is removed has extended by only a small fraction of its length, so transverse waves on a wire are much slower than longitudinal ones. The result has the same form as for the spring, where the square of the ratio of the speeds is also the extension produced by the tension divided by the stretched length of the spring. The spring can be stretched to many times its natural length because its coils bend and twist while the metal of the wire in the coils is strained only slightly, and this is why the two speeds can be comparable on a spring and not on a straight wire.

3.4 Energy and Power in Waves

The school statement that the intensity of a wave is proportional to the square of its amplitude was quoted in Section 3.1. This section derives the corresponding statement for the power carried by a wave on a string, by finding where the energy of the wave resides and how fast it moves. The results lead to the characteristic impedance of

a string, which describes how a string responds to a force applied to it, and they are then carried over to the longitudinal waves of Section 3.3.

Kinetic and Potential Energy of a String

The piece of string between x and $x + \delta x$ has mass $\mu \delta x$ and moves with the transverse velocity $\partial y / \partial t$, so its kinetic energy is $\frac{1}{2} \mu \delta x (\partial y / \partial t)^2$. The kinetic energy per unit length of the string is therefore $\frac{1}{2} \mu (\partial y / \partial t)^2$.

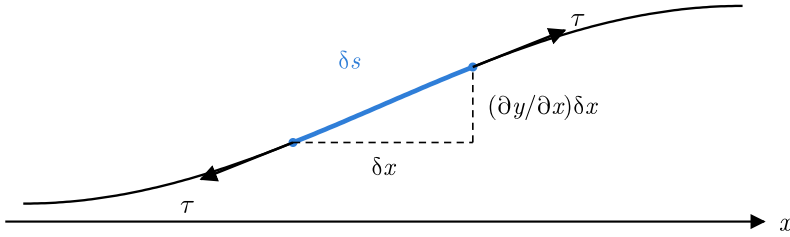


Figure 41. A piece of a displaced string. The piece that occupies the length δx of the x -axis has the length δL , which is longer than δx because the string is inclined with the slope $\partial y / \partial x$. (And δL for δs in the figure label, in the display for the length of the piece, and in 'lengthening the piece from δx to δs '.) The rest of the string pulls on each end of the piece with the tension τ along the string. The slope is drawn much larger than the derivation assumes.

The potential energy of the string comes from the work done against the tension in lengthening the string as it is displaced. Figure 41 shows that the piece of string above the interval δx is the hypotenuse of a right-angled triangle whose other sides are δx and $(\partial y / \partial x) \delta x$, so its length is

$$\delta s = \delta x \sqrt{1 + \left(\frac{\partial y}{\partial x} \right)^2} \approx \delta x \left[1 + \frac{1}{2} \left(\frac{\partial y}{\partial x} \right)^2 \right], \quad (3.35)$$

where the square root has been expanded by the binomial expansion, which is a valid approximation because the slope is small. In Section 3.2 we saw that the tension has the same value τ everywhere when the slope is small, so the work done against it in lengthening the piece from δx to δs is $\tau(\delta s - \delta x) = \frac{1}{2}\tau(\partial y/\partial x)^2 \delta x$. Taking the potential energy of the undisplaced string to be zero, the energy per unit length of the string is

$$\varepsilon = \frac{1}{2}\mu \left(\frac{\partial y}{\partial t} \right)^2 + \frac{1}{2}\tau \left(\frac{\partial y}{\partial x} \right)^2, \quad (3.36)$$

in which the first term is the kinetic energy and the second the potential energy per unit length.

Energy in a Travelling Wave

For a wave $y = f(x - vt)$ travelling in the direction of increasing x , the chain rule gives $\partial y/\partial t = -vf'$ and $\partial y/\partial x = f'$. The kinetic energy per unit length is then $\frac{1}{2}\mu v^2 f'^2$, and since $\mu v^2 = \tau$ this is equal to the potential energy per unit length, $\frac{1}{2}\tau f'^2$. In a travelling wave the kinetic and potential energies are therefore equal at every point and at every instant, and

$$\varepsilon = \tau f'^2 = \mu v^2 f'^2. \quad (3.37)$$

The same holds for a wave travelling in the direction of decreasing x . The oscillator of Section 1.1 behaves quite differently, since its energy passes back and forth between kinetic and potential forms twice in every cycle, and the two energies are equal only on average. In the wave, each piece of string gains kinetic and potential energy together, because the pieces that are moving fastest are those where the string is steepest and so most stretched.

For the harmonic wave $y = A \cos(\omega t - kx + \phi)$, the derivative is $f' = kA \sin(\omega t - kx + \phi)$, and the energy per unit length is

$$\varepsilon = \tau k^2 A^2 \sin^2(\omega t - kx + \phi) = \mu \omega^2 A^2 \sin^2(\omega t - kx + \phi), \quad (3.38)$$

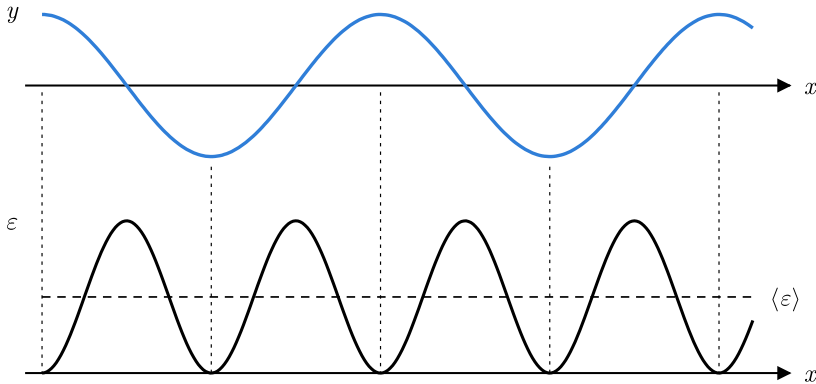


Figure 42. A harmonic wave travelling along a string (top) and its energy per unit length at the same instant (bottom). The energy is greatest where the string crosses the axis, where it is steepest and moving fastest, and it is zero at the crests and the troughs, where the string is momentarily at rest and horizontal. The dashed line is the mean energy per unit length.

since $\tau k^2 = \mu v^2 k^2 = \mu \omega^2$. The energy is concentrated where the string crosses its equilibrium position, and it is zero at the crests and the troughs, as Figure 42 shows. The identity $\sin^2 \chi = \frac{1}{2}(1 - \cos 2\chi)$ shows that the mean of $\sin^2$ over a period, or over a wavelength, is one half, because the mean of the cosine is zero. The mean energy per unit length is therefore

$$\langle \varepsilon \rangle = \frac{1}{2} \mu \omega^2 A^2, \quad (3.39)$$

which is proportional to the square of the amplitude and to the square of the frequency.

The Power Carried by a Wave

The energy of a wave moves along the string because each piece of string does work on the next. The part of the string to the left of the point x pulls on the part to the right with a force of magnitude τ directed along the string towards the left, and the transverse component of this force is $-\tau \sin \psi \approx -\tau \partial y / \partial x$, where

ψ is the angle between the string and the x -axis. The component of the force along the axis does no work, since the string does not move along the axis. The rate at which the left part does work on the right part, which is the power carried past x in the direction of increasing x , is the transverse force multiplied by the transverse velocity,

$$\mathcal{P}(x, t) = -\tau \frac{\partial y}{\partial x} \frac{\partial y}{\partial t}. \quad (3.40)$$

For a wave travelling in the direction of increasing x , substituting $\partial y/\partial x = f'$ and $\partial y/\partial t = -vf'$ gives $\mathcal{P} = \tau v f'^2 = v\varepsilon$. The power at each point is the energy per unit length there multiplied by the speed of the wave, so the energy is carried along with the profile at speed v . For the harmonic wave the mean power is

$$\langle \mathcal{P} \rangle = v \langle \varepsilon \rangle = \frac{1}{2} \mu v \omega^2 A^2. \quad (3.41)$$

For a wave travelling in the direction of decreasing x , the power is $-v\varepsilon$, which is negative because the energy flows towards decreasing x . The energy per unit length and the power are related for any motion of the string by $\partial \varepsilon/\partial t + \partial \mathcal{P}/\partial x = 0$, as you will show in Problem 1. Integrated between two points x_1 and x_2 , this relation says that the energy of the string between the two points changes only by the difference between the power flowing in at x_1 and the power flowing out at x_2 , which is the conservation of energy.

The Characteristic Impedance

In Section 1.3 the mechanical impedance of an oscillator was defined as $Z_m = F/\dot{x}$, the ratio of the driving force to the velocity it produces. The corresponding quantity for a string is the ratio of the transverse force that the part of the string to the left of a point exerts on the part to the right, $-\tau \partial y/\partial x$, to the transverse velocity at that point. For a wave travelling in the direction of increasing x , the velocity is $\partial y/\partial t = -v \partial y/\partial x$, so the ratio is

$$Z = \frac{-\tau \partial y / \partial x}{\partial y / \partial t} = \frac{\tau}{v} = \sqrt{\tau \mu} = \mu v, \quad (3.42)$$

which is called the characteristic impedance of the string. It is a property of the string alone and does not depend on the shape or the frequency of the wave. It is also real, so the force is in phase with the velocity, and it has the units of a damping coefficient, a force divided by a velocity. The power carried by the wave can be written as $\mathcal{P} = Z(\partial y / \partial t)^2$, and the mean power of a harmonic wave as $\langle \mathcal{P} \rangle = \frac{1}{2} Z \omega^2 A^2$. For a wave travelling in the direction of decreasing x the same ratio is $-Z$.

A String Driven at One End

As an example, consider a long string lying along $x > 0$, whose end is attached to a light ring that slides without friction on a smooth rod perpendicular to the string, as in Figure 43(a). A transverse force $F_e(t)$ is applied to the ring. We take the string to be long enough that no wave reflected from its far end returns during the time we consider, so the only wave on the string is the one travelling away from the ring, $y = f(x - vt)$.

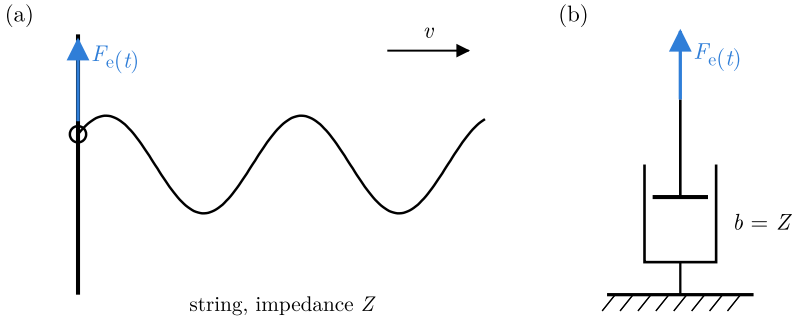


Figure 43. (a) A long string whose end is attached to a light ring that slides on a smooth rod, driven by a transverse force $F_e(t)$ applied to the ring. (b) A piston in a cylinder of oil, with resistance b , driven by the same force. When $b = Z$, the ring and the piston move with the same velocity.

The ring has no mass, so the forces on it add to zero. The string

pulls on the ring with the transverse force $+\tau \partial y / \partial x$ at $x = 0$, which is minus the force that the ring exerts on the string, so

$$F_e(t) = -\tau \frac{\partial y}{\partial x}(0, t) = Z \frac{\partial y}{\partial t}(0, t). \quad (3.43)$$

The force needed to move the end of the string with velocity $\partial y / \partial t$ is therefore Z times that velocity. A piston in a cylinder of oil, which exerts the resistive force $-b\dot{x}$ of Section 1.2, needs the force $b\dot{x}$ to move with velocity $\dot{x}$, and its mechanical impedance is $Z_m = b$. To the driving force, therefore, the long string is equivalent to a piston in oil of resistance Z , shown in Figure 43(b). The piston and the string differ in what happens to the energy: the oil turns the work done on the piston into heat, while the string carries it away as a wave. The equivalence holds for any force $F_e(t)$, harmonic or not, so a long string presents the same resistance at every frequency.

For the harmonic force $F_e = F_0 \cos \omega t$, the velocity of the end is $(F_0/Z) \cos \omega t$, and the mean power absorbed from the force is

$$\langle \mathcal{P} \rangle = \frac{F_0^2}{Z} \langle \cos^2 \omega t \rangle = \frac{F_0^2}{2Z}. \quad (3.44)$$

The wave produced has the amplitude $A = F_0/(Z\omega)$, since the greatest velocity of the end is ωA , and substituting this amplitude into $\frac{1}{2}Z\omega^2 A^2$ gives the same mean power, as the conservation of energy requires. A heavy string, with a large impedance, moves little under a given force and absorbs little power, and a light string moves more, and absorbs more, for the same force.

Energy in Longitudinal Waves

The results for the string carry over to the longitudinal waves of Section 3.3. In a bar, the slice between x and $x + \delta x$ has mass $\rho S \delta x$ and velocity $\partial \xi / \partial t$, so the kinetic energy per unit volume is $\frac{1}{2}\rho(\partial \xi / \partial t)^2$. When the slice is stretched by $(\partial \xi / \partial x) \delta x$, the force needed grows in proportion to the extension, from zero to $YS \partial \xi / \partial x$, so the work done is half the final force times the extension, as for a spring, and the potential energy per unit volume is $\frac{1}{2}Y(\partial \xi / \partial x)^2$. The factor of one half arises differently in the two media. On the

string the tension is constant and the extension goes as the square of the slope, and in the bar there is no force in the undisturbed state and the force grows with the extension.

For a wave travelling in the direction of increasing x , the kinetic and potential energies per unit volume are again equal, since $Y = \rho v^2$. The material to the left of a cross-section pushes on the material to the right with the force $-F = -YS \partial \xi / \partial x$ in the direction of increasing x , and the power carried across the cross-section is

$$\mathcal{P} = -YS \frac{\partial \xi}{\partial x} \frac{\partial \xi}{\partial t}. \quad (3.45)$$

The ratio of the force $-F$ to the velocity $\partial \xi / \partial t$ in a wave travelling in the direction of increasing x is YS/v , as you showed in Problem 1 of Section 3.3, so the characteristic impedance of the bar is

$$Z = \frac{YS}{v} = \rho v S = S \sqrt{\rho Y}, \quad (3.46)$$

and the mean power of the harmonic wave $\xi = A \cos(\omega t - kx + \phi)$ is $\frac{1}{2} \rho v S \omega^2 A^2$. For longitudinal waves it is useful to divide the mean power by the area of the cross-section, since this quotient does not depend on the size of the bar. The mean power per unit area is called the intensity, and for the harmonic wave it is

$$I = \frac{\langle \mathcal{P} \rangle}{S} = \frac{1}{2} \rho v \omega^2 A^2. \quad (3.47)$$

For the fluid in a narrow tube of Section 3.3, the same argument, with ρ_0 in place of ρ , c in place of v , and the excess pressure p in place of the stress $-F/S$, gives the power $\mathcal{P} = pS \partial \xi / \partial t$ and the intensity $I = \frac{1}{2} \rho_0 c \omega^2 A^2$ of a harmonic wave. In each medium the intensity is proportional to the square of the amplitude, which is the school statement of Section 3.1.

3.5 The Doppler Effect and Shock Waves

The Doppler formulas quoted in Section 3.1 can be derived by counting the wavefronts that leave the source and reach the observer. We do so for sound travelling with speed c through air at rest, although the argument holds for any waves whose speed is fixed by a medium at rest. We first treat a moving source and a moving observer on the line joining them, then a source moving at an angle to the line of sight, and finally a source moving faster than the waves it emits.

A Moving Source

Consider a source that emits sound of frequency f , so that it sends out a crest every period $P = 1/f$, and that moves with speed V_s , less than c , directly towards an observer at rest. Once a crest has left the source, it travels through the air at the speed c , independently of the motion of the source that emitted it, because the speed of sound is fixed by the properties of the air, its bulk modulus and its density, as Section 3.3 showed. During one period the crest emitted at the start of the period travels a distance cP towards the observer, and the source follows it a distance V_sP before it emits the next crest. The distance between successive crests ahead of the source is therefore

$$\lambda' = (c - V_s)P. \quad (3.48)$$

The crests reach the observer at the speed c , so they arrive at the rate c/λ' , and the frequency heard is

$$f' = \frac{c}{c - V_s} f. \quad (3.49)$$

Behind the source the crests are spaced by $(c + V_s)P$, and an observer there hears the lower frequency $cf/(c + V_s)$. Figure 44 shows the crests crowded together ahead of the source and spread apart behind it.

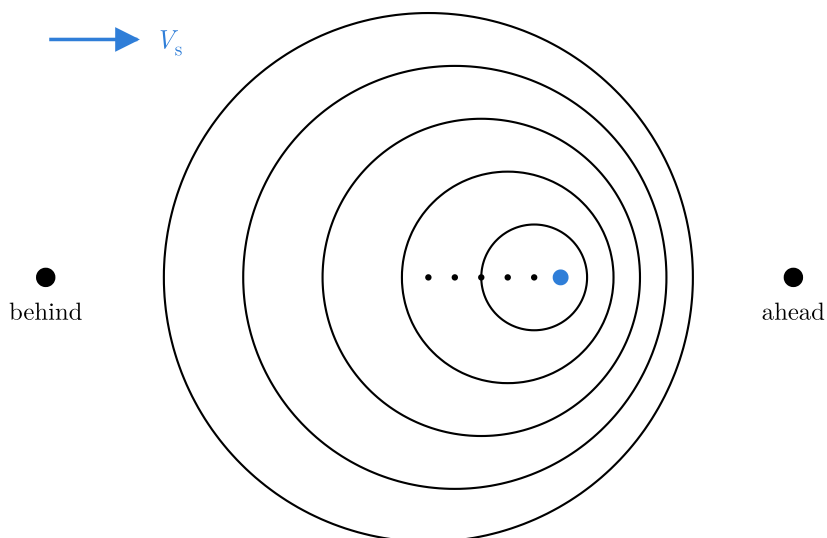


Figure 44. Wavefronts emitted by a source moving through still air with a speed V_s less than c . Each circle is a crest emitted one period after the next larger one, centred on the point, marked with a dot, where the source was at the moment of emission. The crests are crowded together ahead of the source, towards the observer marked ahead, and spread apart behind it, towards the observer marked behind.

A Moving Observer

Consider now a source at rest and an observer moving directly towards it with speed V_o . The crests are spaced by the wavelength $\lambda = c/f$ that the source produces in the air. They travel towards the observer at the speed c relative to the air, and the observer travels towards them at V_o , so the observer meets them at the rate $(c + V_o)/\lambda$ and hears the frequency

$$f' = \frac{c + V_o}{c} f. \quad (3.50)$$

An observer moving away from the source hears $(c - V_o)f/c$. When both the source and the observer are moving along the line joining them, the source fixes the spacing of the crests in the air, $(c - V_s)/f$, and the observer meets them at the relative speed $c + V_o$, so the frequency heard is

$$f' = \frac{c + V_o}{c - V_s} f, \quad (3.51)$$

where V_s and V_o are counted positive when the source and the observer move towards each other.

The Source Shift and the Observer Shift Compared

Let us compare a source approaching a stationary observer with speed V_s and an observer approaching a stationary source with the same speed, $V_o = V_s$, so that the relative velocity of source and observer is the same in the two cases. The frequencies heard are $f/(1 - V/c)$ and $f(1 + V/c)$, and Figure 45 shows that they are different. When V is small compared with c , the first can be expanded as a geometric series,

$$\frac{1}{1 - V_s/c} = 1 + \frac{V_s}{c} + \frac{V_s^2}{c^2} + \dots, \quad (3.52)$$

which converges because $V_s/c < 1$. The two frequencies therefore agree to first order in V_s/c and differ in the second order. The

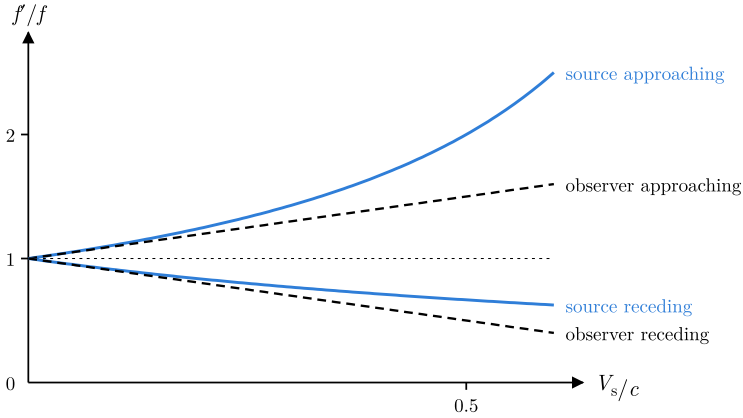


Figure 45. The ratio of the frequency heard to the frequency emitted, plotted against V_s/c , for a source approaching an observer at rest, or receding from it, with speed V_s (solid), and for an observer approaching a source at rest, or receding from it, with the same speed, $V_o = V_s$ (dashed). The curves agree when V_s is small compared with c and separate as V_s grows.

reason they differ is that the two situations are physically different. The air has a rest frame, in which the sound travels at the same speed c in every direction, and in one case the source moves through the air while in the other the observer does. The frequency heard depends on the velocities of the source and of the observer relative to the air, and not only on their velocity relative to each other. The difference is clearest near $V_s = c$: an observer approaching a stationary source at the speed of sound hears twice the emitted frequency, while a source approaching at the speed of sound keeps pace with the crests it emits, which pile up in front of it, and the formula gives an infinite frequency.

A Source Moving at an Angle

When the source does not move along the line joining it to the observer, only the component of its velocity along that line matters. Let the direction from the source to the observer make the angle θ with the velocity of the source, as in Figure 46(a), and let the observer be far away compared with the distance $V_s P$ that the source moves in one period, so that the lines from successive positions of

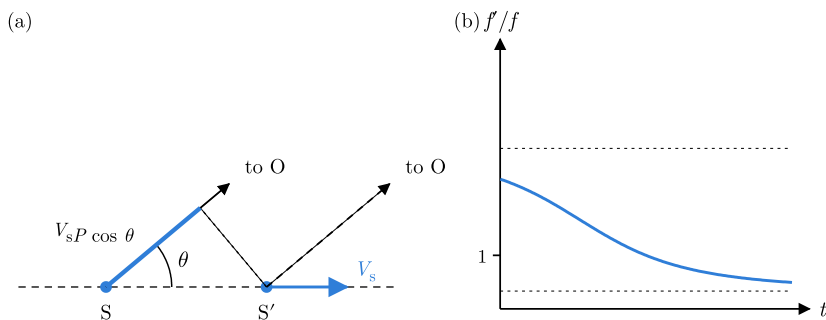


Figure 46. (a) A source moving with velocity V_s emits one crest at S and the next, a period P later, at S'. The direction from the source to a distant observer O makes the angle θ with the velocity, and the second crest has a shorter distance to travel by $V_s P \cos \theta$, the length of the thick segment. (b) The frequency heard by an observer at rest as a source moving in a straight line passes at a closest distance D , plotted against the time of arrival of the sound. The dashed lines are the limits $f/(1 - V_s/c)$, approached long before the source passes, and $f/(1 + V_s/c)$, approached long after.

the source to the observer are for all practical purposes parallel. A crest emitted at S at time zero and the next crest, emitted at S' at time P , travel towards the observer along parallel lines, and since S' lies a distance $V_s P$ from S along the velocity, the second crest has a distance shorter by $V_s P \cos \theta$ to travel. It therefore arrives a time $P - V_s P \cos \theta / c$ after the first, and the frequency heard is

$$f' = \frac{f}{1 - (V_s/c) \cos \theta}. \quad (3.53)$$

The formula reduces to the results for a source approaching the observer when $\theta = 0$ and receding when $\theta = \pi$, and it gives no shift when the source moves perpendicular to the line of sight. The angle θ is the angle at the moment of emission, which for a distant source is earlier than the moment at which the sound is heard.

As an example, consider a source moving in a straight line with constant speed V_s , which passes an observer at rest at a closest distance D . Long before it passes, the angle θ is nearly zero and

the observer hears $f/(1 - V_s/c)$. The angle then grows through $\pi/2$, at which the source is at its closest point and the emitted frequency is heard unchanged, a time D/c later, and long after the source has passed the angle is nearly π and the observer hears $f/(1 + V_s/c)$. Figure 46(b), computed from the formula with the position of the source and the travel time of the sound worked out at each instant, shows that the frequency falls smoothly from one limit to the other over a time of about D/V_s , in which the angle changes appreciably. The closer the observer is to the path, the more abrupt the fall.

The Mach Cone

A source that moves faster than the waves it emits, with $V_s > c$, is always ahead of the wavefronts it emits. Consider the crest emitted a time t before the present moment. It is a sphere of radius ct centred on the point where the source was then, which lies a distance $V_s t$ behind the present position of the source. Figure 47 shows these spheres, cut by a plane through the path of the source, for several earlier times.

A line from the present position of the source that touches the sphere emitted a time t ago is perpendicular to the radius at the point of contact, so the line, the radius ct , and the distance $V_s t$ back along the path form a right-angled triangle, drawn for one circle in Figure 47. The angle θ_M between the line and the path therefore satisfies

$$\sin \theta_M = \frac{ct}{V_s t} = \frac{c}{V_s}, \quad (3.54)$$

which does not depend on t . Every wavefront emitted in the past therefore touches the same cone, with its apex at the source and its axis along the path, which is called the Mach cone. The ratio V_s/c is called the Mach number of the source, and the faster the source, the narrower the cone. Outside the cone no sound from the source has yet arrived, and on the cone the disturbances emitted at all earlier times arrive together, so the wave there has a large amplitude. A disturbance of this kind is called a shock wave. The formation of the shock and its structure depend on the behaviour

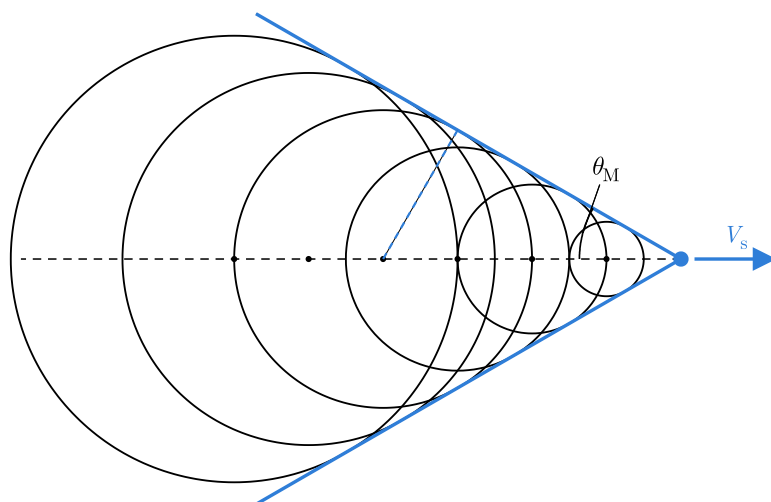


Figure 47. Wavefronts emitted by a source moving with a speed V_s greater than c , cut by a plane through its path. Each circle is a crest emitted at an earlier time, centred on the point, marked with a dot, where the source was at the moment of emission. The circles all touch two lines through the present position of the source, which make the angle θ_M with the path. The dashed radius of one circle meets one of the lines at a right angle.

of sound waves of large amplitude, which obey nonlinear equations and are not treated in this course.

The Sonic Boom

An aircraft flying level at a speed greater than the speed of sound carries its Mach cone with it, and the cone sweeps over the ground behind the aircraft along the whole of its path. An observer on the ground hears nothing of the aircraft until the cone arrives, and then hears the sudden change of pressure on the cone as a loud bang, which is called a sonic boom. The boom is heard by every observer along the path, some time after the aircraft has passed overhead, and not only at the moment when the aircraft first exceeds the speed of sound. In practice the aircraft forms a cone at its nose and another at its tail, and the boom is often heard as two bangs in quick succession. These statements assume air of uniform temperature, so that c is the same everywhere; in the real atmosphere the speed of sound varies with height, and the cone is distorted. You will find in Problem 3 how long after the aircraft passes overhead the boom is heard.

The Bow Wave of a Boat

As a further example of the Mach cone, consider a boat moving across water of depth h . Waves on the surface of water whose depth is much less than their wavelength travel with the speed $v_w = \sqrt{gh}$, where g is the acceleration due to gravity, independently of their wavelength. The derivation of this speed needs the dynamics of a fluid with a free surface, which is beyond the scope of this course. A boat moving faster than v_w outruns the waves it makes, and the construction of Figure 47, with circles on the surface in place of spheres, shows that the crests form a V-shaped bow wave with its apex at the bow and a half-angle θ_M given by

$$\sin \theta_M = \frac{v_w}{V_s}, \quad (3.55)$$

where V_s is the speed of the boat. Measuring the half-angle of the bow wave in water of known depth therefore gives the speed of the boat. In deep water the speed of surface waves depends on their

wavelength, and the wake of a boat has a different form, which is not treated in this course.

Problems

§3.1 Waves at School

1. A harmonic wave travels along a string and passes onto a second string that has a quarter of the mass per unit length of the first and is under the same tension. Find the ratios of the speed, the frequency, and the wavelength on the second string to those on the first.
2. A small source radiates sound equally in all directions, and the intensity at a distance r_1 is I_1 . Find the distance at which the intensity is $I_1/9$, and the ratio of the amplitudes at the two distances.
3. A source emitting sound of frequency f moves in a straight line at speed V_s through still air. A listener directly ahead of the source hears the frequency f_a , and one directly behind it hears f_b . Show that $V_s = c(f_a - f_b)/(f_a + f_b)$.
4. A listener moves at speed V_o directly towards a source of sound at rest, passes it, and moves directly away at the same speed. Find the ratio of the frequencies heard before and after passing the source.

§3.2 Travelling Waves on a String

1. A harmonic wave travels along a string in the direction of decreasing x , and the displacement at $x = 0$ is $A \cos \omega t$. Write down $y(x, t)$ in real and in complex form, and find the transverse velocity of the point $x = \lambda/4$ at $t = 0$.
2. An infinite string is released from rest with the triangular displacement $y_0(x) = y_m(1 - |x|/x_0)$ for $|x| < x_0$ and $y_0 = 0$ elsewhere. Find the displacement of the point $x = 0$ as a function of time.
3. Show that a string given the displacement $y_0(x)$ and the velocity $q(x) = +v y'_0(x)$ at $t = 0$ has the displacement $y_0(x + vt)$ at later

times.

4. At $t = 0$ an infinite string has the displacement $y_0(x)$ and the velocity $q(x)$, both of which are zero for $|x| > x_0$. Show that the point at a distance $x > x_0$ from the origin is at rest until $t = (x - x_0)/v$, and that after $t = (x + x_0)/v$ it is at rest again with the displacement $(2v)^{-1} \int_{-x_0}^{x_0} q(x') dx'$.
5. The pulse $y = y_m e^{-(x-vt)^2/\sigma^2}$ travels along a string. Find the greatest transverse speed of the string, and show that it is small compared with v provided $y_m \ll \sigma$.

§3.3 Longitudinal Waves in Bars and Fluids

1. Show that for a longitudinal wave travelling along a bar in the direction of increasing x , the strain is $-v^{-1} \partial \xi / \partial t$, and that the force $-F$ exerted across a cross-section by the material on the left on the material on the right is $\rho v S \partial \xi / \partial t$.
2. Let ρ' be the excess density of the fluid in the tube, so that its density is $\rho_0 + \rho'$. Using the conservation of the mass of a slice, show that $\rho' = -\rho_0 \partial \xi / \partial x$ to first order in small quantities, and hence that $p = c^2 \rho'$.
3. Show that a longitudinal pulse takes the time $\sqrt{m/\kappa}$ to travel the length of the stretched spring of the text, for every length L to which the spring is stretched, and find the time taken by a transverse pulse.
4. A harmonic wave of displacement amplitude A and angular frequency ω travels along a column of fluid. Show that the greatest strain in the fluid is $\omega A/c$, and hence that the condition of small disturbances is that the displacement amplitude be small compared with $\lambda/2\pi$.

§3.4 Energy and Power in Waves

1. Show that for any solution of the wave equation for a string, the energy per unit length and the power satisfy $\partial \epsilon / \partial t + \partial \mathcal{P} / \partial x = 0$.
2. A string carries the harmonic wave $A_1 \cos(\omega t - kx)$ travelling in the direction of increasing x and the wave $A_2 \cos(\omega t + kx)$

- travelling in the direction of decreasing x . Show that the mean power carried in the direction of increasing x is $\frac{1}{2}Z\omega^2(A_1^2 - A_2^2)$.
3. A mass m slides without friction on a smooth rod perpendicular to a long string, is pulled towards the position $y = 0$ by a spring of spring constant κ , and is attached to the end of the string, which lies along $x > 0$. Show that, as long as no wave returns from the far end of the string, the displacement of the mass satisfies $m\ddot{y} + Z\dot{y} + \kappa y = 0$, and find the quality factor of the oscillator in terms of Z , m , and κ .
 4. The pulse $y = y_m e^{-(x-vt)^2/\sigma^2}$ travels along a string. Show that its energy per unit length is greatest at the two points $x - vt = \pm\sigma/\sqrt{2}$, and zero at its peak.
 5. A long bar lying along $x > 0$ is driven at its end by a force $F_e(t)$ along its length. Show that the velocity of the end is $F_e/(\rho v S)$ as long as no wave returns from the far end of the bar.

§3.5 The Doppler Effect and Shock Waves

1. A wind blows at speed V_a from a source of sound towards an observer, both at rest relative to the ground. Working in the frame in which the air is at rest, show that the observer hears the emitted frequency f , and find the wavelength of the sound measured by an observer at rest relative to the ground.
2. A car moving at speed V_s towards a flat wall sounds its horn at frequency f . Show that the driver hears the echo from the wall at the frequency $f(c + V_s)/(c - V_s)$, and find the beat frequency between the echo and the horn heard directly.
3. An aircraft flies level at height H with speed $V_s > c$ in air of uniform temperature. Show that an observer on the ground directly below its path hears the sonic boom a time $(H/c)\sqrt{1 - c^2/V_s^2}$ after the aircraft passes overhead.
4. An observer moves with speed V_o in a direction that makes the angle θ with the direction from the observer to a distant source at rest. Show that the observer hears the frequency $f(1 + V_o \cos \theta/c)$.
5. A boat crossing water of depth h , shallow compared with the

wavelengths of the waves it makes, leaves a bow wave of half-angle θ_M . Find the speed of the boat, and explain why a boat moving more slowly than $\sqrt{gh}$ makes no such bow wave.

Superposition, Boundaries, and Interference

4.1 Superposition at School

In this part we study what happens when waves are added together and when a wave reaches a boundary at which part of it is reflected, such as the junction of two strings with different masses per unit length, or the end of a pipe. We begin by reviewing the school treatment of superposition, then add oscillations of the same and of different frequencies, then follow waves on a string across a junction and find how their reflection can be prevented, and finally treat standing waves and the interference of waves from several sources.

The Principle of Superposition

The material of this section will be familiar from school, and we review it briefly because the rest of the part builds on it. When two waves pass through the same region of a medium, the displacement at each point is the sum of the displacements that the two waves would produce there separately. This statement is called the principle of superposition. It holds for waves on a stretched string and for sound in air as long as the displacements are small, since the wave equation of Section 3.2 is then linear. When two pulses are sent towards each other from the ends of a long string, for ex-

ample, they pass through each other and emerge with their shapes unchanged, and while they overlap the shape of the string is the sum of their two shapes. The adding of waves in this way is called interference. Where the displacements of the two waves have the same sign the interference is called constructive, and where they have opposite signs it is called destructive.

Standing Waves on a String

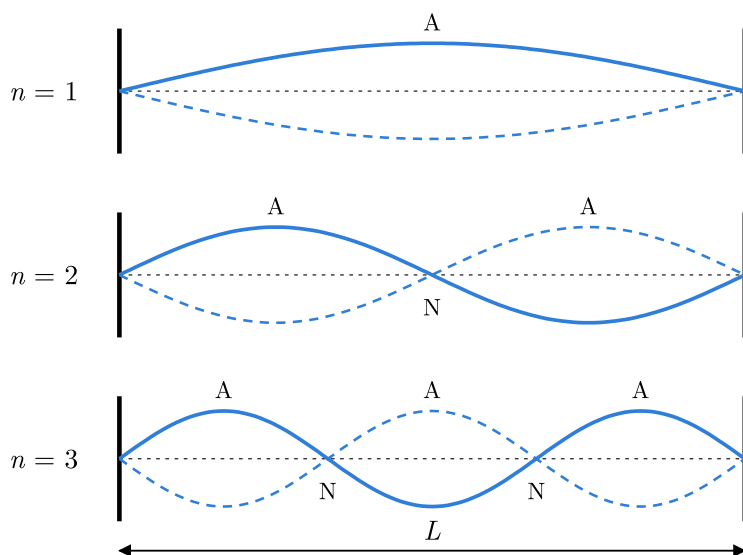


Figure 48. The first three standing waves on a string of length L fixed at both ends, each drawn at the two instants of greatest displacement (solid and dashed). The nodes are marked N and the antinodes A.

A string of length L fixed at both ends can vibrate in a pattern that does not travel along it. Waves travelling in the two directions are reflected at the ends and superpose, and at certain frequencies the result is a standing wave, in which some points of the string, called nodes, do not move, and the points halfway between neighbouring nodes, called antinodes, move with the largest amplitude. The ends are nodes, so a whole number of half-wavelengths must fit into the length of the string, $L = n\lambda/2$, and with $v = f\lambda$ the frequencies of the standing waves are

$$f_n = \frac{nv}{2L}, \quad n = 1, 2, 3, \dots, \quad (4.1)$$

where $v = \sqrt{\tau/\mu}$ is the speed of waves on the string. The lowest frequency f_1 is called the fundamental, and f_n is called the n th harmonic. Figure 48 shows the first three standing waves. Neighbouring nodes are half a wavelength apart, and each antinode lies a quarter of a wavelength from the nodes on either side of it.

Standing Waves in Pipes

The air in a pipe has standing waves as well, although sound in air is a longitudinal wave, so that the air moves along the pipe. At a closed end the air cannot move, and the end is a node of the displacement. At an open end the air is free to move, and the end is, to a good approximation, an antinode of the displacement. A pipe open at both ends has antinodes at both ends, and a whole number of half-wavelengths fits into its length, as on the string, so its frequencies are $nc/2L$, where c is the speed of sound, and every harmonic is present. A pipe closed at one end has a node at one end and an antinode at the other, and an odd number of quarter-wavelengths must fit into its length, $L = (2n - 1)\lambda/4$, so that

$$f_n = \frac{(2n - 1)c}{4L}, \quad n = 1, 2, 3, \dots. \quad (4.2)$$

Only the odd harmonics of the fundamental $c/4L$ are present. The fundamental of a closed pipe is half that of an open pipe of the same length, an octave lower, and the absence of the even harmonics is one reason why the two pipes sound different.

Interference from Two Sources

Consider two sources that emit waves of the same wavelength λ , in phase with each other, such as two loudspeakers driven from the same signal generator. Let r_1 and r_2 be the distances from the two sources to a point. The waves arrive there in phase, and interfere constructively, when the difference between the distances is a whole

number of wavelengths, and they arrive in antiphase, and interfere destructively, when it is an odd number of half-wavelengths:

$$\begin{aligned} r_2 - r_1 &= q\lambda \quad (\text{constructive}), \\ r_2 - r_1 &= \left(q + \frac{1}{2}\right)\lambda \quad (\text{destructive}), \end{aligned} \tag{4.3}$$

where $q = 0, \pm 1, \pm 2, \dots$. As a listener walks along a line in front of the two loudspeakers, the sound therefore becomes alternately loud and quiet.

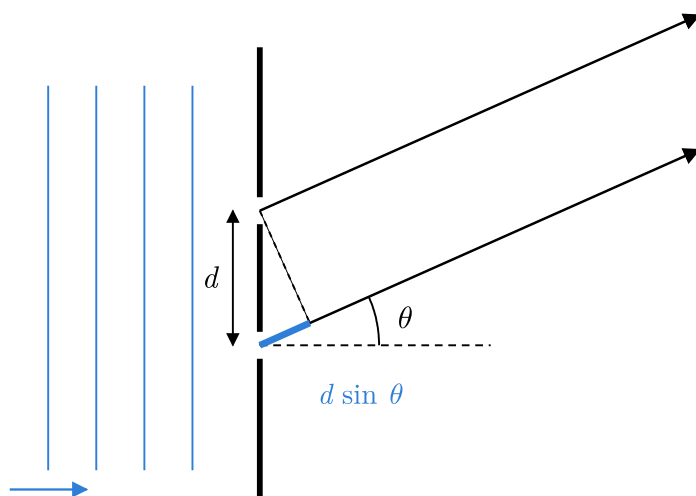


Figure 49. Plane waves arriving from the left at a pair of slits a distance d apart. For a distant point in the direction θ , the path from the lower slit is longer than the path from the upper slit by $d \sin \theta$ (accent). The dashed line from the upper slit is perpendicular to the two rays.

In Young's double-slit experiment, light from a single source falls on two narrow parallel slits a distance d apart, and the light from the slits falls on a screen a distance D away, with D much larger than d . For a point on the screen in the direction θ from the normal to the slits, the path difference is $d \sin \theta$, as Figure 49 shows, and bright fringes appear where

$$d \sin \theta = q\lambda. \tag{4.4}$$

For small angles $\sin \theta \approx \theta$, and the bright fringes are equally spaced on the screen, a distance $\lambda D/d$ apart. Young used this experiment at the beginning of the nineteenth century to show that light is a wave and to measure its wavelength.

Gratings and a Single Slit

A diffraction grating is a large number of equally spaced parallel slits. With a slit spacing d , the waves from neighbouring slits reach a distant point with the path difference $d \sin \theta$, and the condition $d \sin \theta = q\lambda$ again gives the directions of the bright maxima, where the waves from all the slits arrive in phase. The integer q is called the order of the maximum. With many slits the maxima are much sharper and brighter than those of two slits, which is why a grating is often used to measure wavelengths.

A single slit of width w also produces a pattern, since each part of the slit acts as a source. The central maximum is bright and wide, and the first minima on either side of it lie in the directions given by

$$\sin \theta = \frac{\lambda}{w}, \quad (4.5)$$

so a narrower slit gives a wider pattern. The spreading of a wave after it passes through an opening or around an obstacle is called diffraction.

Coherence

A steady pattern of interference is seen only if the sources keep a constant phase difference, and sources that do so are said to be coherent. Loudspeakers driven from the same signal are coherent, whereas separate lamps are not, since the light from each is emitted in short bursts whose phases vary at random, so that the pattern they would produce changes far too quickly to be seen. Young obtained two coherent sources by passing the light from a single source through the two slits.

4.2 Superposition of Oscillations

The principle of superposition applies to the oscillations at a single point as well as to waves spread through space. When two sounds reach the ear, for example, the excess pressure at the eardrum is the sum of the excess pressures that the two sounds would produce separately. In this section we first add oscillations along the same line, at the same frequency and then at different frequencies, then add oscillations along perpendicular lines, and finally treat an oscillation whose amplitude varies periodically. The addition of two oscillations of the same frequency with phasors is worked out in Section 1.2 of Vol. 4 of the Fermi Notes on Physics, and we begin by quoting its results.

Oscillations of the Same Frequency

Let two oscillations of the same angular frequency be

$$x_a = A_a \cos(\omega t + \phi_a), \quad x_b = A_b \cos(\omega t + \phi_b), \quad (4.6)$$

with the complex amplitudes $\mathcal{A}_a = A_a e^{i\phi_a}$ and $\mathcal{A}_b = A_b e^{i\phi_b}$, so that $x_a = \text{Re}(\mathcal{A}_a e^{i\omega t})$ and $x_b = \text{Re}(\mathcal{A}_b e^{i\omega t})$. Since the real part of a sum is the sum of the real parts,

$$x_a + x_b = \text{Re}[(\mathcal{A}_a + \mathcal{A}_b)e^{i\omega t}], \quad (4.7)$$

and the sum is an oscillation of the same frequency, whose complex amplitude $\mathcal{A} = \mathcal{A}_a + \mathcal{A}_b$ is found by adding the two phasors as vectors on the Argand diagram.

Figure 50 shows the construction. Let $\Delta\phi = \phi_b - \phi_a$ be the phase difference between the two oscillations. As Section 1.2 of Vol. 4 shows, with the cosine rule for the triangle of the figure, the amplitude $A = |\mathcal{A}|$ of the sum is given by

$$A^2 = A_a^2 + A_b^2 + 2A_a A_b \cos \Delta\phi. \quad (4.8)$$

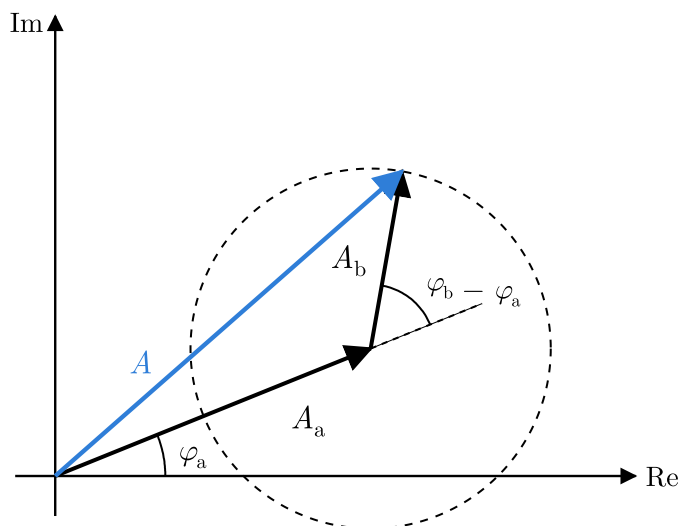


Figure 50. Adding two phasors. The complex amplitudes of lengths A_a and A_b are added head to tail, and the resultant (accent) has length A . If the two oscillations have different frequencies, the second phasor turns relative to the first, and the tip of the resultant moves round the dashed circle.

Oscillations in phase give $A = A_a + A_b$, oscillations in antiphase give $|A_a - A_b|$, and two oscillations of equal amplitude A_a give $A = 2A_a|\cos(\Delta\phi/2)|$, which vanishes when they are in antiphase.

The energy of an oscillator is proportional to the square of its amplitude, and the mean of x^2 over a cycle is $\frac{1}{2}A^2$, as you found in Problem 7 of Section 1.2 of Vol. 4. The mean square of the sum is therefore

$$\langle (x_a + x_b)^2 \rangle = \frac{1}{2}A_a^2 + \frac{1}{2}A_b^2 + A_a A_b \cos \Delta\phi, \quad (4.9)$$

where the angle brackets denote the mean over a cycle. The last term, which can be positive or negative, is called the interference term, and it is responsible for the loud and quiet places in the pattern of two sources reviewed in Section 4.1. If the phase difference changes at random during the time over which the mean is taken, the cosine takes positive and negative values equally often and the interference term averages to zero, so that the mean squares of the two oscillations add. This is why two sources that are not coherent produce no steady pattern.

Beats

Let us now add two oscillations whose angular frequencies ω_a and ω_b are different. Their phase difference $(\omega_b - \omega_a)t + \phi_b - \phi_a$ runs through every value as time passes, and we may choose the origin of time at an instant at which it is zero, so that the two phase constants are equal. A common phase constant only shifts the phase of the rapid oscillation found below, and we drop it. For equal amplitudes the sum is

$$x = A \cos \omega_a t + A \cos \omega_b t = 2A \cos \left(\frac{\omega_a - \omega_b}{2} t \right) \cos \left(\frac{\omega_a + \omega_b}{2} t \right), \quad (4.10)$$

where we have used the identity $\cos \chi_1 + \cos \chi_2 = 2 \cos \frac{1}{2}(\chi_1 - \chi_2) \cos \frac{1}{2}(\chi_1 + \chi_2)$, which follows from adding the formulas for the cosine of the sum and of the difference of the angles $\frac{1}{2}(\chi_1 + \chi_2)$ and $\frac{1}{2}(\chi_1 - \chi_2)$. When the two frequencies are close, so that $|\omega_a - \omega_b|$ is

much smaller than ω_a , the first cosine changes little during a period of the second. The sum is then an oscillation at the mean frequency whose amplitude, $2A|\cos \frac{1}{2}(\omega_a - \omega_b)t|$, rises and falls slowly. This amplitude returns to its greatest value each time the argument of the cosine increases by π , since the modulus of a cosine has half the period of the cosine itself, and so it does so at intervals of $2\pi/|\omega_a - \omega_b|$. When tuning forks of nearly equal frequency are sounded together, they are heard as a single tone whose loudness swells and dies away $|f_a - f_b|$ times a second. This phenomenon is known as beating, and $|f_a - f_b|$ is called the beat frequency. The slow transfer of energy between two weakly coupled pendulums in Section 2.1 is an instance of it, with the two normal frequencies as ω_a and ω_b . Since the beats slow down as the frequencies approach each other and stop when they are equal, a piano tuner will often tune a string by listening to its beats against a reference tone.

When the amplitudes are unequal the phasors give the result most easily. Taking out the factor $e^{i\omega_a t}$,

$$x = \text{Re} \left[e^{i\omega_a t} (A_a + A_b e^{-i(\omega_a - \omega_b)t}) \right] . \quad (4.11)$$

The quantity in the round brackets is a complex amplitude that changes slowly. On the Argand diagram of Figure 50 it is the sum of a fixed phasor of length A_a and a phasor of length A_b that turns relative to it with angular velocity $\omega_b - \omega_a$, so that its tip moves round the dashed circle. The formula for the amplitude found above, with $\Delta\phi$ replaced by the slowly changing phase difference $-(\omega_a - \omega_b)t$, gives

$$A(t)^2 = A_a^2 + A_b^2 + 2A_a A_b \cos [(\omega_a - \omega_b)t] , \quad (4.12)$$

since the cosine is an even function.

The amplitude now varies between $A_a + A_b$, when the two phasors point the same way, and $|A_a - A_b|$, when they point in opposite directions, and it never falls to zero unless the amplitudes are equal. Figure 51 shows the sum with its envelope, which is the curve traced by the maxima of the rapid oscillation. The beat frequency is again $|f_a - f_b|$. The phase of the sum also varies slightly during each beat,

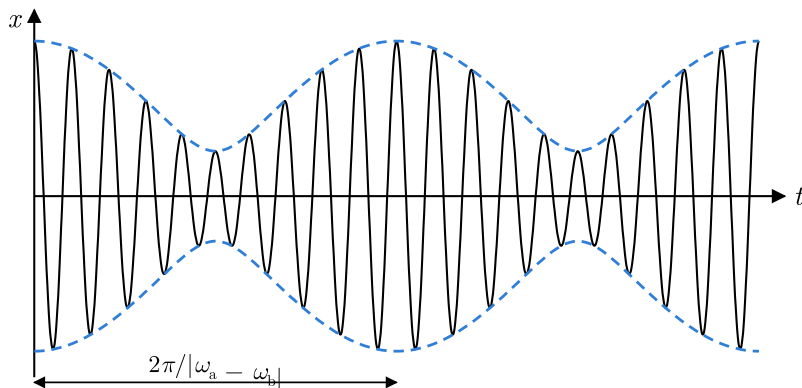


Figure 51. The sum of two oscillations of different amplitudes and nearby frequencies (solid), with the envelopes $\pm A(t)$ (accent, dashed). The amplitude varies between $A_a + A_b$ and $|A_a - A_b|$, and the time between neighbouring maxima is $2\pi/|\omega_a - \omega_b|$.

because the resultant phasor swings to either side of the phasor of length A_a , and you will find the size of this swing in Problem 2. The description of the sum as an oscillation with a slowly varying amplitude is a valid approximation as long as $|\omega_a - \omega_b|$ is much smaller than ω_a .

Perpendicular Oscillations of the Same Frequency

A point can also oscillate along two perpendicular lines simultaneously. A point of a stretched string, for example, can vibrate in both of the directions perpendicular to the string, and if the string vibrates only in its fundamental mode, the two motions have the same frequency. Let y and z be the displacements of the point in the two directions, and let

$$y = A_y \cos \omega t, \quad z = A_z \cos(\omega t + \Delta\phi), \quad (4.13)$$

where the origin of time has been chosen so that y has no phase constant. To find the path of the point, we eliminate the time. Expanding the cosine of a sum gives $z/A_z = \cos \omega t \cos \Delta\phi - \sin \omega t \sin \Delta\phi$, and substituting $\cos \omega t = y/A_y$ gives

$$\frac{z}{A_z} - \frac{y}{A_y} \cos \Delta\phi = -\sin \omega t \sin \Delta\phi. \quad (4.14)$$

Squaring both sides, and using $\sin^2 \omega t = 1 - y^2/A_y^2$, we find

$$\frac{y^2}{A_y^2} + \frac{z^2}{A_z^2} - \frac{2yz}{A_y A_z} \cos \Delta\phi = \sin^2 \Delta\phi. \quad (4.15)$$

This is a quadratic equation in y and z whose curve is closed and lies within the rectangle $|y| \leq A_y$, $|z| \leq A_z$, and a closed curve with a quadratic equation is an ellipse, or, in the cases $\Delta\phi = 0$ and $\Delta\phi = \pi$ treated below, a line segment. The ellipse touches each side of the rectangle once, since y reaches A_y once in each period, and similarly for $-A_y$ and for $\pm A_z$.

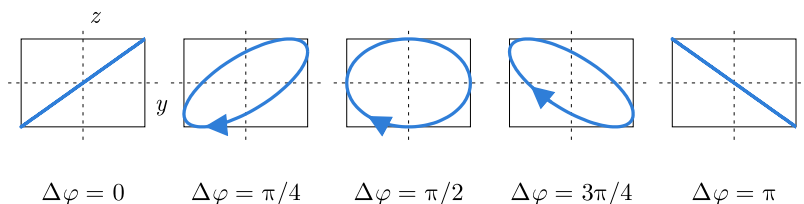


Figure 52. The path of a point whose perpendicular displacements y (horizontal) and z (vertical) oscillate with the same frequency, for several values of the phase difference $\Delta\phi$. Each path lies in the rectangle drawn around it, and the arrows show the direction in which the point moves.

Figure 52 shows the path for several values of $\Delta\phi$. When $\Delta\phi = 0$ the equation becomes $(y/A_y - z/A_z)^2 = 0$, and the point moves back and forth along the diagonal $z/A_z = y/A_y$; when $\Delta\phi = \pi$ it moves along the other diagonal. Motion along a line of this kind is called linear. When $\Delta\phi = \pm\pi/2$ the cross term vanishes, and the path is an ellipse with its axes along the y and z directions, which becomes a circle when also $A_y = A_z$. Uniform motion round a circle is therefore the sum of two perpendicular simple harmonic motions of equal amplitude, a quarter of a period apart. For the other values the axes of the ellipse are tilted, and you will find the

angle of tilt in Problem 3. The direction in which the point goes round the ellipse follows from the velocity at $t = 0$, when the point is at $y = A_y$ and $\dot{z} = -\omega A_z \sin \Delta\phi$. For $0 < \Delta\phi < \pi$ the point is then moving towards negative z , so that it goes round clockwise when y is drawn to the right and z upwards, as in the figure, and for $-\pi < \Delta\phi < 0$ it goes round anticlockwise.

Lissajous Figures

Suppose now that the perpendicular oscillations have different frequencies,

$$y = A_y \cos \omega_y t, \quad z = A_z \cos(\omega_z t + \Delta\phi), \quad (4.16)$$

and that the ratio of the frequencies is a ratio of whole numbers, $\omega_y/\omega_z = n_y/n_z$, with n_y and n_z having no common factor. The two oscillations are then harmonics of the same fundamental angular frequency $\omega_y/n_y = \omega_z/n_z$, and after one period of that fundamental, $2\pi n_y/\omega_y$, both y and z have returned to their starting values. The path is therefore a closed curve, traced over and over again. Curves of this kind are called Lissajous figures, after Lissajous, who displayed them in 1857 by fixing a small mirror to each of two tuning forks, vibrating in perpendicular planes, and reflecting a narrow beam of light from one mirror to the other and onto a screen, where the spot of light moved horizontally with one fork and vertically with the other.

During one period of the fundamental, y reaches its greatest value n_y times and z reaches its greatest value n_z times. For most values of the phase constant these maxima are separate points, and the figure touches each vertical side of its rectangle n_y times and each horizontal side n_z times, so the ratio of the frequencies can be read off from the figure, as Figure 53 shows. You will show in Problem 4 that for the ratio 1 : 2 with $\Delta\phi = 0$ the figure closes up into an arc of a parabola traced back and forth, which is one of the special values of the phase constant for which the counting fails.

If the ratio of the frequencies is not a ratio of whole numbers, the curve never closes, and over a long time it fills the rectangle. If the ratio is close to n_y/n_z , we may write the phase of z as $\omega_z t + \Delta\phi =$

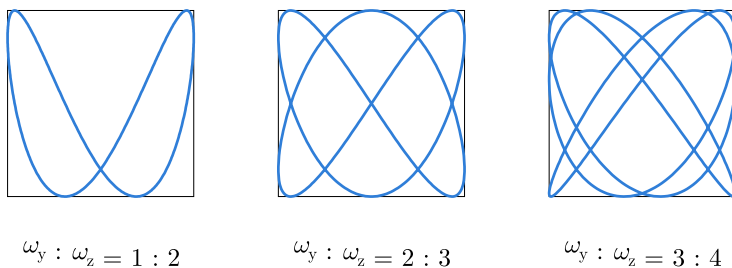


Figure 53. Lissajous figures for three ratios of the frequencies, drawn with the same phase constant $\Delta\phi$ in each. A figure with frequency ratio $\omega_y : \omega_z = n_y : n_z$ touches each vertical side of its rectangle n_y times and each horizontal side n_z times.

$(n_z/n_y)\omega_y t + \Delta\phi'$, where $\Delta\phi' = \Delta\phi + (\omega_z - n_z\omega_y/n_y)t$ changes slowly. The figure is then the closed figure for the ratio n_y/n_z with a phase constant that drifts, and it changes shape slowly and returns to its starting shape after the time $2\pi/|\omega_z - n_z\omega_y/n_y|$, as nearby frequencies on the same line give beats. The slower the drift of the figure, the closer the ratio of the frequencies is to n_y/n_z , and this allowed Lissajous to compare the frequencies of tuning forks very precisely.

Amplitude Modulation

The amplitude of an oscillation can itself be made to vary periodically. In a vibraphone, for example, discs driven by a motor open and close the tops of the tubes that resonate below the bars, so that the loudness of a note rises and falls several times a second. Let

$$x = A(1 + h \cos \omega_m t) \cos \omega t, \quad (4.17)$$

with $0 \leq h \leq 1$ and ω_m much smaller than ω . The oscillation at the frequency ω is called the carrier, ω_m is called the modulation frequency, and h is called the depth of modulation. The amplitude $A(1 + h \cos \omega_m t)$ varies between $A(1 + h)$ and $A(1 - h)$, and it is never negative, since $h \leq 1$. An oscillation of this kind is called amplitude-modulated.

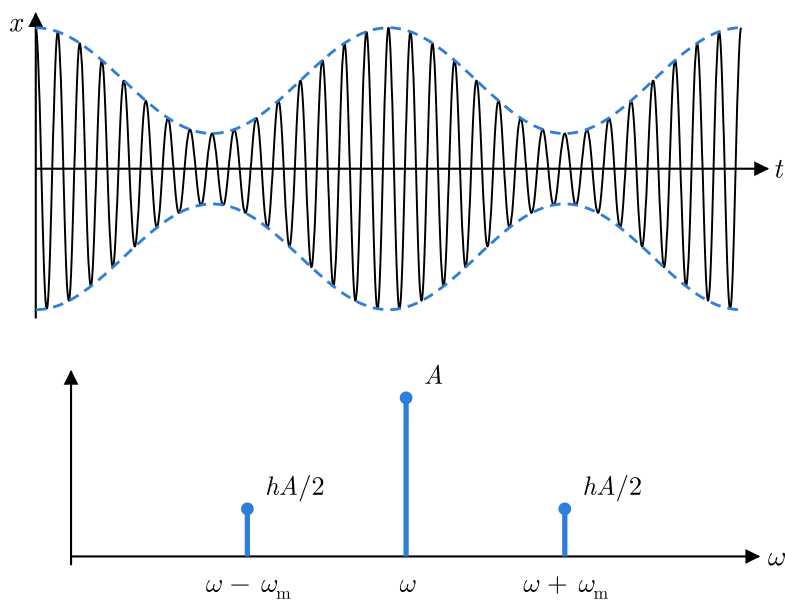


Figure 54. An amplitude-modulated oscillation (upper), with its envelope $\pm A(1 + h \cos \omega_m t)$ (accent, dashed), and its spectrum (lower), which consists of the carrier and two sidebands.

To find the spectrum of this oscillation, in the sense of Section 2.5, we use the identity $\cos \chi_1 \cos \chi_2 = \frac{1}{2}[\cos(\chi_1 + \chi_2) + \cos(\chi_1 - \chi_2)]$, which also follows from adding the formulas for the cosine of a sum and of a difference. With $\chi_1 = \omega t$ and $\chi_2 = \omega_m t$ it gives

$$x = A \cos \omega t + \frac{hA}{2} \cos(\omega + \omega_m)t + \frac{hA}{2} \cos(\omega - \omega_m)t. \quad (4.18)$$

The amplitude-modulated oscillation is therefore the sum of three oscillations of constant amplitude: the carrier, and two oscillations with frequencies $\omega + \omega_m$ and $\omega - \omega_m$, which are called the upper and the lower sidebands. Its spectrum, drawn in Figure 54, consists of three lines. An oscillation with a varying amplitude and a set of oscillations with constant amplitudes are thus two equivalent descriptions of the same motion, and each of them is useful. The equal-amplitude beats above are an example, since their sum is an oscillation at the mean frequency $\frac{1}{2}(\omega_a + \omega_b)$ whose amplitude factor $2A \cos \frac{1}{2}(\omega_a - \omega_b)t$ varies with the angular frequency $\frac{1}{2}|\omega_a - \omega_b|$. That amplitude factor passes through zero and changes sign, and the spectrum of the beats has the two lines at ω_a and ω_b and no line at the mean frequency. The amplitude factor $1 + h \cos \omega_m t$ of the modulated oscillation contains a constant term, and this term produces the line of the carrier.

If the amplitude is varied in proportion to 1 plus a periodic signal made up of several harmonics, each harmonic of the signal, of angular frequency ω' , produces a pair of sidebands at $\omega \pm \omega'$. The spectrum then consists of the carrier with a band of lines on either side of it, and each band is as wide as the highest frequency in the signal. A signal that changes over a time Δt therefore spreads the spectrum of the carrier over a range of frequencies of order $2\pi/\Delta t$, as the reciprocal relation $\Delta\omega \Delta t \approx 2\pi$ of Section 2.5 leads us to expect.

4.3 Reflection and Transmission at a Boundary

A wave travelling along a uniform string keeps its shape, as d'Alembert's solution of Section 7.1 of Vol. 4 shows. When it reaches a place where the properties of the string change, part of it is reflected, and Section 7.1 of Vol. 4 worked out the extreme

case of a string fixed at one end, where the whole of the wave is reflected and turned upside down. We now join two strings of different mass per unit length and find the amplitudes of the reflected and transmitted waves. We then check that energy is conserved at the junction, follow a pulse through it, and see how a bead fixed at the junction makes the reflection depend on the frequency. The section ends with the junction of two bars, which is treated in the same way.

Two Strings Joined at a Point

Let a string of mass per unit length μ_1 occupy the region $x < 0$, and let it be joined at $x = 0$ to a string of mass per unit length μ_2 that occupies $x > 0$. The knot that joins them has negligible mass, so the forces on it must balance, and as long as the slopes of the strings are small the components of these forces along the x -axis balance only if the tension τ is the same on both sides. The speeds of waves on the two strings are $v_1 = \sqrt{\tau/\mu_1}$ and $v_2 = \sqrt{\tau/\mu_2}$, and their characteristic impedances, as defined in Section 3.4, are $Z_1 = \sqrt{\tau\mu_1}$ and $Z_2 = \sqrt{\tau\mu_2}$.

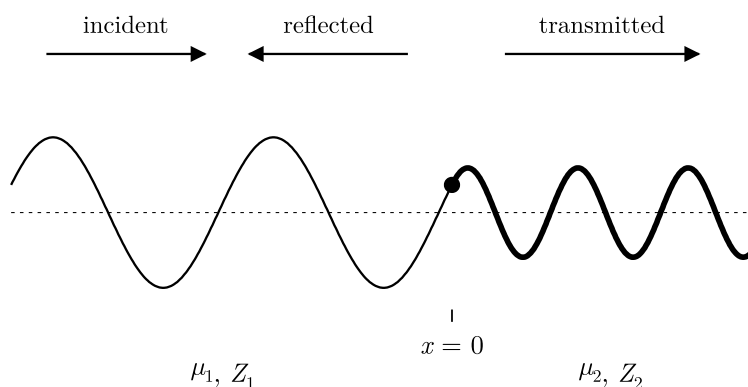


Figure 55. A harmonic wave arriving from the left at the junction of a light string and a heavier one (thicker line), drawn at one instant. The displacement and the slope of the string are continuous at the junction, and the transmitted wave has a shorter wavelength than the incident one.

A harmonic wave of angular frequency ω arrives from the left, as in

Figure 55. Part of it is reflected, and part is transmitted into the second string. The junction moves up and down at the frequency ω and drives the second string at that frequency, so all three waves have the same frequency, and we write them as

$$\begin{aligned} y_i &= \text{Re} [\mathcal{A} e^{i(\omega t - k_1 x)}] , \\ y_r &= \text{Re} [\mathcal{A}_r e^{i(\omega t + k_1 x)}] , \\ y_t &= \text{Re} [\mathcal{A}_t e^{i(\omega t - k_2 x)}] , \end{aligned} \quad (4.19)$$

with $k_1 = \omega/v_1$ and $k_2 = \omega/v_2$. The displacement of the first string is $y_i + y_r$ and that of the second is y_t . The incident amplitude $\mathcal{A}$ is given, and the reflected and transmitted amplitudes $\mathcal{A}_r$ and $\mathcal{A}_t$ are to be found from two conditions at the junction.

The first condition is that the two strings are joined, so their displacements at $x = 0$ are equal at every instant. The second concerns the transverse forces on the knot. The tension in a string acts along the string, and where the slope is small its transverse component is τ times the slope. The string on the left pulls the knot towards negative x , with the transverse force $-\tau \partial y / \partial x$ evaluated just to the left of the junction, and the string on the right pulls it towards positive x , with the transverse force $+\tau \partial y / \partial x$ evaluated just to the right. A resultant transverse force on a knot of negligible mass would give it an unlimited acceleration, so the two forces cancel and the slope is the same on both sides. Substituting the three waves into these conditions at $x = 0$ and cancelling the common factor $e^{i\omega t}$, and in the second condition also the factor $-i$, gives

$$\mathcal{A} + \mathcal{A}_r = \mathcal{A}_t , \quad k_1 (\mathcal{A} - \mathcal{A}_r) = k_2 \mathcal{A}_t . \quad (4.20)$$

Multiplying the second equation by τ/ω turns each wave number into an impedance, since $\tau k / \omega = \tau / v = \sqrt{\tau \mu} = Z$, and it becomes $Z_1 (\mathcal{A} - \mathcal{A}_r) = Z_2 \mathcal{A}_t$. This form of the condition can also be seen from Section 3.4, where the transverse force at a point of a wave travelling towards positive x was found to be Z times the transverse velocity there; for a wave travelling towards negative x the force is $-Z$ times the velocity. Substituting $\mathcal{A}_t$ from the first equation into

the second and solving for $\mathcal{A}_r$, and then using the first equation again, gives

$$\mathcal{R} = \frac{\mathcal{A}_r}{\mathcal{A}} = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad \mathcal{T} = \frac{\mathcal{A}_t}{\mathcal{A}} = \frac{2Z_1}{Z_1 + Z_2}. \quad (4.21)$$

The ratios $\mathcal{R}$ and $\mathcal{T}$ are called the amplitude reflection and transmission coefficients. They are real, so the transmitted wave is in phase with the incident wave at the junction, and the reflected wave is either in phase with it or in antiphase, according to the sign of $Z_1 - Z_2$. They do not depend on the frequency, and the two strings enter them only through their impedances. The first condition at the junction appears in them as $1 + \mathcal{R} = \mathcal{T}$.

Limiting Cases

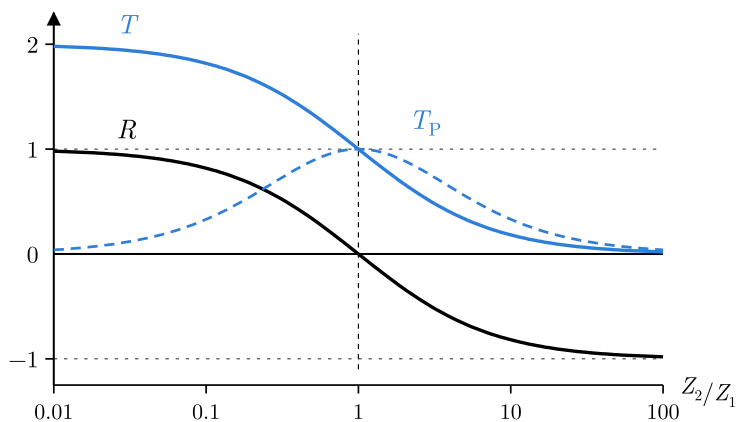


Figure 56. The amplitude reflection coefficient $\mathcal{R}$ (solid) and transmission coefficient $\mathcal{T}$ (accent), and the fraction $\mathcal{T}_P$ of the incident power that is transmitted (accent, dashed), plotted against the ratio Z_2/Z_1 of the impedances on a logarithmic scale.

Figure 56 shows how the coefficients depend on the ratio of the impedances. When the second string is much heavier than the first, $Z_2 \gg Z_1$, the coefficients tend to $\mathcal{R} = -1$ and $\mathcal{T} = 0$. The heavy string hardly moves, the junction behaves as a fixed end, and the

reflected wave has the same amplitude as the incident wave and is inverted, as Section 7.1 of Vol. 4 found by the method of an image pulse. When the second string is much lighter, $Z_2 \ll Z_1$, the coefficients tend to $\mathcal{R} = 1$ and $\mathcal{T} = 2$. The light string exerts almost no transverse force on the junction, which then behaves as a free end, and the wave is reflected without inversion. A free end can be made by tying the end of a string to a light ring that slides without friction on a smooth rod perpendicular to the string, and you will show in Problem 1 that the ring moves with twice the amplitude of the incident wave. When the impedances are equal there is no reflected wave. With the same tension in both strings this requires equal masses per unit length, so that the two strings are the same; for bars the corresponding condition can be met by different materials, as we find at the end of this section.

Energy at the Junction

In Section 3.4 the mean power carried by a harmonic wave of amplitude A on a string was found to be $\frac{1}{2}\mu v\omega^2 A^2$, which is $\frac{1}{2}Z\omega^2 A^2$. The fractions of the incident power that are reflected and transmitted are therefore

$$\mathcal{R}_P = \frac{Z_1 |\mathcal{A}_r|^2}{Z_1 |\mathcal{A}|^2} = \left(\frac{Z_1 - Z_2}{Z_1 + Z_2} \right)^2, \quad \mathcal{T}_P = \frac{Z_2 |\mathcal{A}_t|^2}{Z_1 |\mathcal{A}|^2} = \frac{4Z_1 Z_2}{(Z_1 + Z_2)^2}, \quad (4.22)$$

and their sum is one, since $(Z_1 - Z_2)^2 + 4Z_1 Z_2 = (Z_1 + Z_2)^2$. The power carried away from the junction by the reflected and transmitted waves is equal to the power brought to it by the incident wave. This had to be so, since the knot has negligible mass and stores no energy, so that the rate at which the first string does work on it is equal to the rate at which it does work on the second string.

The transmission coefficient $\mathcal{T}$ approaches 2 when the second string is very light, and yet the transmitted power is then small. The reason is that the power carried by a wave of a given amplitude is proportional to the impedance of the string, so that a wave of large amplitude on a light string carries little power. The fraction $\mathcal{T}_P$ is unchanged when Z_1 and Z_2 are interchanged, so a wave passing

from a heavy string to a light one loses the same fraction of its power by reflection as a wave passing the other way, and Figure 56 shows that $\mathcal{T}_P$ is largest, and equal to one, when the impedances are equal.

A Pulse Crossing the Junction

The coefficients $\mathcal{R}$ and $\mathcal{T}$ do not depend on the frequency, so every harmonic wave is reflected and transmitted in the same proportions, and we should expect a pulse of any shape to keep its shape. Let us show this directly from d'Alembert's solution, without resolving the pulse into harmonic waves. Let the incident pulse be $y = \eta_i(x - v_1 t)$, and let the reflected and transmitted pulses be $\eta_r(x + v_1 t)$ and $\eta_t(x - v_2 t)$, where η_i is given and η_r and η_t are to be found, and all three functions vanish before the incident pulse reaches the junction. The conditions at $x = 0$ are

$$\eta_i(-v_1 t) + \eta_r(v_1 t) = \eta_t(-v_2 t), \quad \eta'_i(-v_1 t) + \eta'_r(v_1 t) = \eta'_t(-v_2 t), \quad (4.23)$$

where the primes denote derivatives with respect to the argument. Differentiating the first condition with respect to time by the chain rule gives $-v_1 \eta'_i + v_1 \eta'_r = -v_2 \eta'_t$. Adding v_2 times the second condition to this equation removes η'_t , and leaves

$$\eta'_r(v_1 t) = \frac{v_1 - v_2}{v_1 + v_2} \eta'_i(-v_1 t) = -\mathcal{R} \eta'_i(-v_1 t), \quad (4.24)$$

where the second form follows from $v = \tau/Z$. The left-hand side is $v_1^{-1} d[\eta_r(v_1 t)]/dt$ and the right-hand side is $\mathcal{R} v_1^{-1} d[\eta_i(-v_1 t)]/dt$, by the chain rule, so integrating with respect to time, with both functions zero before the pulse arrives, gives $\eta_r(v_1 t) = \mathcal{R} \eta_i(-v_1 t)$. The first condition then gives $\eta_t(-v_2 t) = (1 + \mathcal{R}) \eta_i(-v_1 t) = \mathcal{T} \eta_i(-v_1 t)$. Writing these results for a general argument, the reflected and transmitted pulses are

$$y_r = \mathcal{R} \eta_i(-x - v_1 t), \quad y_t = \mathcal{T} \eta_i\left(\frac{v_1}{v_2} (x - v_2 t)\right). \quad (4.25)$$

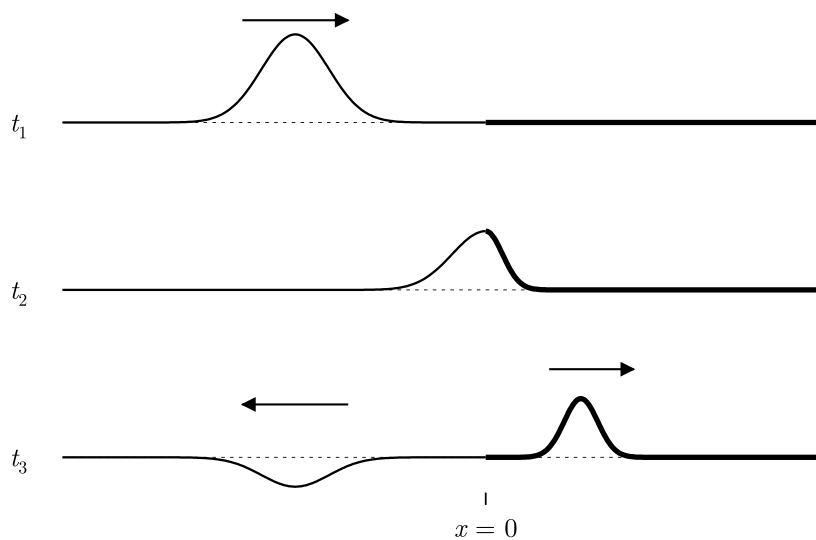


Figure 57. A pulse crossing the junction from a light string to a heavier one (thicker line), at three instants $t_1 < t_2 < t_3$. The reflected pulse is inverted, and the transmitted pulse is shorter than the incident one in the ratio v_2/v_1 .

The reflected pulse is the incident pulse reversed from left to right, as a pulse travelling in the opposite direction must be, and multiplied by $\mathcal{R}$. The transmitted pulse is multiplied by $\mathcal{T}$, and its length is changed in the ratio v_2/v_1 , because it takes the same time to cross the junction as the incident pulse did while it travels at the speed v_2 . Figure 57 shows a pulse passing from a light string to a heavier one, for which $\mathcal{R}$ is negative and $\mathcal{T}$ is less than one, and the transmitted pulse is shorter than the incident one. You will check in Problem 4 that the energies of the reflected and transmitted pulses add up to the energy of the incident pulse.

A Bead at the Junction

Let a bead of mass m now be fixed to the strings at the junction. The transverse forces on the bead no longer cancel, and Newton's second law for the bead reads

$$m \frac{\partial^2 y}{\partial t^2}(0, t) = \tau \frac{\partial y}{\partial x}(0^+, t) - \tau \frac{\partial y}{\partial x}(0^-, t), \quad (4.26)$$

where 0^+ and 0^- denote points just to the right and just to the left of the junction. The displacements are still continuous, so the first condition, $\mathcal{A} + \mathcal{A}_r = \mathcal{A}_t$, is unchanged, and the displacement of the bead is $\text{Re}(\mathcal{A}_t e^{i\omega t})$. Substituting the three waves, cancelling $e^{i\omega t}$, and dividing by $i\omega$ turns the law of motion into

$$Z_1 (\mathcal{A} - \mathcal{A}_r) = (Z_2 + i\omega m) \mathcal{A}_t. \quad (4.27)$$

The impedance Z_2 of the second string is replaced by $Z_2 + i\omega m$, which is the sum of Z_2 and the mechanical impedance of a free mass, found from the mechanical impedance $Z_m = b + i(m\omega - \kappa/\omega)$ of Section 1.3 by setting $b = \kappa = 0$. The bead and the second string are both driven by the motion of the junction, so their impedances add, as the forces needed to move them add. The coefficients become

$$\mathcal{R} = \frac{Z_1 - Z_2 - i\omega m}{Z_1 + Z_2 + i\omega m}, \quad \mathcal{T} = \frac{2Z_1}{Z_1 + Z_2 + i\omega m}, \quad (4.28)$$

which are complex, so that the reflected and transmitted waves are shifted in phase, and which depend on the frequency.

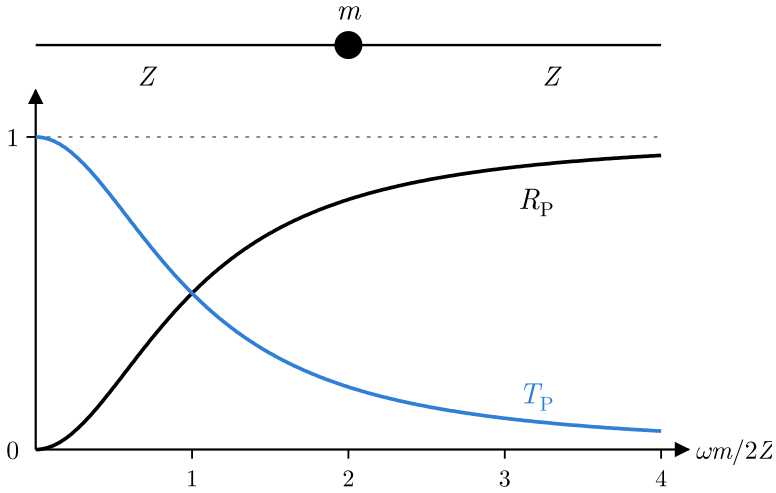


Figure 58. A bead of mass m at the junction of two identical strings of impedance Z (upper), and the fractions $\mathcal{R}_P$ (solid) and $\mathcal{T}_P$ (accent) of the incident power that are reflected and transmitted, plotted against $\omega m/2Z$ (lower).

The result is simplest for a bead on a uniform string, with $Z_1 = Z_2 = Z$. The transmission coefficient is then $\mathcal{T} = 1/(1 + i\omega m/2Z)$, and the fractions of the incident power that are transmitted and reflected are

$$\mathcal{T}_P = |\mathcal{T}|^2 = \frac{1}{1 + (\omega m/2Z)^2}, \quad \mathcal{R}_P = |\mathcal{R}|^2 = \frac{(\omega m/2Z)^2}{1 + (\omega m/2Z)^2}. \quad (4.29)$$

They are drawn in Figure 58, and they add to one, since the bead stores energy during part of each cycle and returns it during the rest, and so absorbs no power on average. At low frequencies the bead passes the wave on almost unchanged, because the force needed to accelerate it, $m\omega^2$ times its displacement, is small compared with the transverse forces of the strings on it. At high frequencies the bead hardly moves, and it reflects the wave almost completely, as

a fixed point would. The change from one behaviour to the other occurs near $\omega = 2Z/m$, where half the power is transmitted. A pulse contains a range of frequencies, which the bead reflects in different proportions, so the pulse no longer keeps its shape. A pulse whose duration is much longer than $m/2Z$ passes almost undisturbed, while a much shorter one is mostly reflected.

The same argument applies to any object fixed at the junction that exerts a force proportional to its displacement or to one of its derivatives, such as a bead held by a spring or by a piston in oil. The object enters the conditions only through its mechanical impedance Z_m , and Z_2 is replaced by $Z_2 + Z_m$. You will find in Problem 3 that a bead held by a spring transmits the whole of the incident power at its natural frequency ω_0 .

The Junction of Two Bars

The results carry over to the longitudinal waves of Section 3.3. Let two bars, of densities ρ_1 and ρ_2 , Young's moduli Y_1 and Y_2 , and cross-sections S_1 and S_2 , be bonded end to end at $x = 0$, and let $\xi(x, t)$ be the longitudinal displacement. As in Section 3.3, the force with which the material on the right of a cross-section pulls the material on the left is $YS \partial \xi / \partial x$, which is a tension when it is positive and a compression when it is negative. The bars are bonded, so the displacement is continuous at the joint. The joint itself has negligible mass, so the forces on it from the two bars cancel, and $Y_1 S_1 \partial \xi / \partial x$ on one side is equal to $Y_2 S_2 \partial \xi / \partial x$ on the other.

These conditions have the same form as those for the strings, with $\tau \partial y / \partial x$ replaced by $YS \partial \xi / \partial x$. For harmonic waves, the second condition becomes $Y_1 S_1 k_1 (\mathcal{A} - \mathcal{A}_r) = Y_2 S_2 k_2 \mathcal{A}_t$, and multiplying by $1/\omega$ turns each YSk into

$$\frac{YSk}{\omega} = \frac{YS}{v} = S\sqrt{Y\rho} = \rho v S, \quad (4.30)$$

using $v = \sqrt{Y/\rho}$. This is the impedance $Z = \rho v S$ of a bar found in Section 3.4. The coefficients $\mathcal{R}$ and $\mathcal{T}$, and the fractions of the power, are therefore given by the formulas for the strings, with

$Z_1 = \rho_1 v_1 S_1$ and $Z_2 = \rho_2 v_2 S_2$. A free end of a bar has $Z_2 = 0$ and a rigidly clamped end has $Z_2 \rightarrow \infty$. Unlike two strings under the same tension, two bars of different materials can have equal impedances, so that a wave passes from one to the other without reflection. For bars of equal cross-section this happens when the products ρv of the two materials are equal, and otherwise the cross-sections can be chosen so that $\rho_1 v_1 S_1 = \rho_2 v_2 S_2$.

At a free end the displacement is reflected without inversion, while the strain $\partial\xi/\partial x$, which measures the tension or compression, is reflected with its sign reversed, as you will show in Problem 5. A pulse of compression sent along a bar therefore returns from a free end as a pulse of tension, and a bar made of a material that is weak in tension, such as a brittle rock or concrete, can break near its free end when its other end is struck.

4.4 Impedance Matching and Evanescent Waves

Section 4.3 showed that part of a wave is reflected wherever the impedance of the medium changes. Reflection is often unwanted, for example when the power carried by a wave is to be delivered to a load at the end of a string or passed from one medium into another. We first show how the reflection at the end of a string can be removed, and then how it can be removed at the junction of two strings for waves of one frequency. We then treat a string held by springs along its length, on which waves below a certain frequency cannot travel, and finally show that a wave can nevertheless cross a short stretch of such a string. This last result is the classical counterpart of the tunnelling of a particle through a potential barrier, which is treated in Section 2.5 of Vol. 3 of the Fermi Notes on Physics.

A String Ended by a Piston in Oil

Let a string of tension τ and impedance Z occupy $x < 0$, and let its end be tied to a light ring that slides without friction on a smooth rod along the line $x = 0$, as in Figure 59. The ring is connected to a piston in a cylinder of oil, which exerts on it the resistive force $-b\dot{y}$, where $\dot{y}$ is the transverse velocity of the ring and b is the damping coefficient of the piston. The string exerts on the ring the

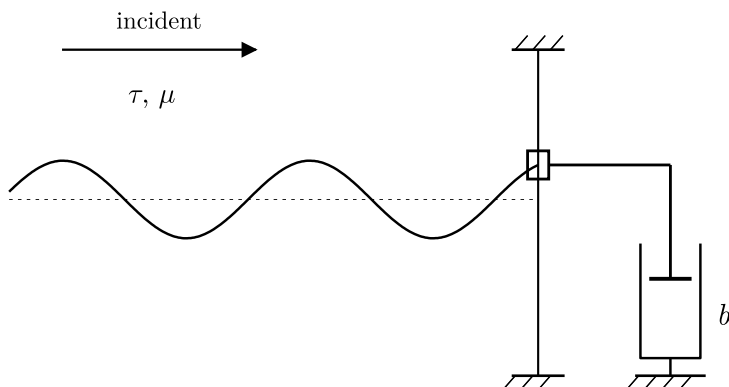


Figure 59. A string ended by a light ring that slides on a smooth rod perpendicular to the string. A piston in a cylinder of oil, with damping coefficient b , resists the motion of the ring.

transverse force $-\tau \partial y / \partial x$, as it did on the knot in Section 4.3, and since the ring has negligible mass, the two forces cancel:

$$-\tau \frac{\partial y}{\partial x} - b \frac{\partial y}{\partial t} = 0 \quad \text{at } x = 0. \quad (4.31)$$

On the string the displacement is $\text{Re}\{[\mathcal{A}e^{-ikx} + \mathcal{A}_r e^{ikx}]e^{i\omega t}\}$, the sum of an incident and a reflected wave. Substituting it into the condition at the end gives $ik\tau(\mathcal{A} - \mathcal{A}_r) = i\omega b(\mathcal{A} + \mathcal{A}_r)$, and dividing by $i\omega$ and using $\tau k / \omega = Z$ gives $Z(\mathcal{A} - \mathcal{A}_r) = b(\mathcal{A} + \mathcal{A}_r)$, so that

$$\mathcal{R} = \frac{Z - b}{Z + b}. \quad (4.32)$$

This is the result of Section 4.3 for a junction, with the piston in place of the second string and its damping coefficient b in place of Z_2 . If $b = Z$, nothing is reflected, and the whole of the incident power is absorbed by the piston. Since $\mathcal{R}$ does not depend on the frequency, this holds for every harmonic wave, and therefore for a pulse of any shape. The string cannot tell whether it ends on the piston or continues without end, since in both cases the force at $x = 0$ is Z times the velocity there. This is the converse of the

result of Section 3.4 that a long string, driven at one end, absorbs power as a piston in oil of resistance Z would.

The same argument applies to any device attached to the ring whose force on it is $-Z_m \dot{y}$ in complex notation, where Z_m is the mechanical impedance of the device, as in Section 1.3. Replacing b by Z_m gives

$$\mathcal{R} = \frac{Z - Z_m}{Z + Z_m}, \quad (4.33)$$

which vanishes when $Z_m = Z$. A load whose mechanical impedance is equal to the characteristic impedance of the string is said to be matched to the string, and choosing a load in this way is called impedance matching. A spring or a mass on the ring has a purely imaginary impedance, $-i\kappa/\omega$ or $i\omega m$, and you will show in Problem 2 that such a load reflects the whole of the incident power with a change of phase, since it stores energy during part of each cycle and returns it during the rest.

The Quarter-Wave Matching Section

Suppose now that power is to be passed from a string of impedance Z_1 into a string of impedance Z_3 , both under the tension τ . Joined directly, the strings reflect the fraction $[(Z_1 - Z_3)/(Z_1 + Z_3)]^2$ of the incident power. Let us instead insert between them a middle string, of impedance Z_2 and length L , occupying $0 < x < L$, with the first string, of impedance Z_1 , in $x < 0$ and the last string, of impedance Z_3 , in $x > L$, as in the upper part of Figure 60.

To find the reflection at $x = 0$, we need the relation between the transverse force and the transverse velocity there. Let the transverse force exerted at $x = 0$ by the first string on the rest be $F = -\tau \partial y / \partial x$, and let Z_{in} be the ratio of F to the velocity $\dot{y}$ at $x = 0$, which is called the input impedance of the rest of the string. The conditions at $x = 0$ involve only the displacement and the transverse force there, so the first string reflects in the same way as if it ended on a device of mechanical impedance Z_{in} , and the result for the end load gives

$$\mathcal{R} = \frac{Z_1 - Z_{\text{in}}}{Z_1 + Z_{\text{in}}} . \quad (4.34)$$

In the middle string the displacement is $\text{Re}\{[\mathcal{B}_+ e^{-ik_2 x} + \mathcal{B}_- e^{ik_2 x}]e^{i\omega t}\}$, the sum of a wave travelling towards positive x and a wave travelling towards negative x . The second of these is the first reflected at $x = L$, where the junction with the last string has the reflection coefficient $\mathcal{R}_{23} = (Z_2 - Z_3)/(Z_2 + Z_3)$ of Section 4.3, so that $\mathcal{B}_- e^{ik_2 L} = \mathcal{R}_{23} \mathcal{B}_+ e^{-ik_2 L}$. At $x = 0$ the displacement is proportional to $\mathcal{B}_+ + \mathcal{B}_-$, the velocity to $i\omega(\mathcal{B}_+ + \mathcal{B}_-)$, and the force $-\tau \partial y / \partial x$ to $i\tau k_2(\mathcal{B}_+ - \mathcal{B}_-)$, so that, with $\tau k_2 / \omega = Z_2$,

$$Z_{\text{in}} = Z_2 \frac{\mathcal{B}_+ - \mathcal{B}_-}{\mathcal{B}_+ + \mathcal{B}_-} = Z_2 \frac{1 - \mathcal{R}_{23} e^{-2ik_2 L}}{1 + \mathcal{R}_{23} e^{-2ik_2 L}} . \quad (4.35)$$

Let the middle string be a quarter of a wavelength long, so that $k_2 L = \pi/2$ and $e^{-2ik_2 L} = e^{-i\pi} = -1$. With $1 + \mathcal{R}_{23} = 2Z_2/(Z_2 + Z_3)$ and $1 - \mathcal{R}_{23} = 2Z_3/(Z_2 + Z_3)$, the input impedance becomes

$$Z_{\text{in}} = Z_2 \frac{1 + \mathcal{R}_{23}}{1 - \mathcal{R}_{23}} = \frac{Z_2^2}{Z_3} . \quad (4.36)$$

The reflection at $x = 0$ vanishes if $Z_{\text{in}} = Z_1$, that is if

$$Z_2 = \sqrt{Z_1 Z_3} . \quad (4.37)$$

A section a quarter of a wavelength long whose impedance is the geometric mean of the impedances on either side therefore transmits the whole of the incident power into the last string. Such a section is called a quarter-wave matching section. Since all the strings have the same tension, and $Z = \sqrt{\tau \mu}$, the condition requires the mass per unit length of the middle string to be the geometric mean $\sqrt{\mu_1 \mu_3}$.

The result can be understood by following the reflected waves. With $Z_2 = \sqrt{Z_1 Z_3}$, the reflection coefficients at $x = 0$ and at $x = L$ are equal, each being $(\sqrt{Z_1} - \sqrt{Z_3})/(\sqrt{Z_1} + \sqrt{Z_3})$. The wave reflected at $x = L$ travels half a wavelength further than the wave

reflected at $x = 0$, so the two arrive back at $x = 0$ in antiphase and nearly cancel, and the calculation above, which includes the waves reflected back and forth within the section, shows that the cancellation is exact.

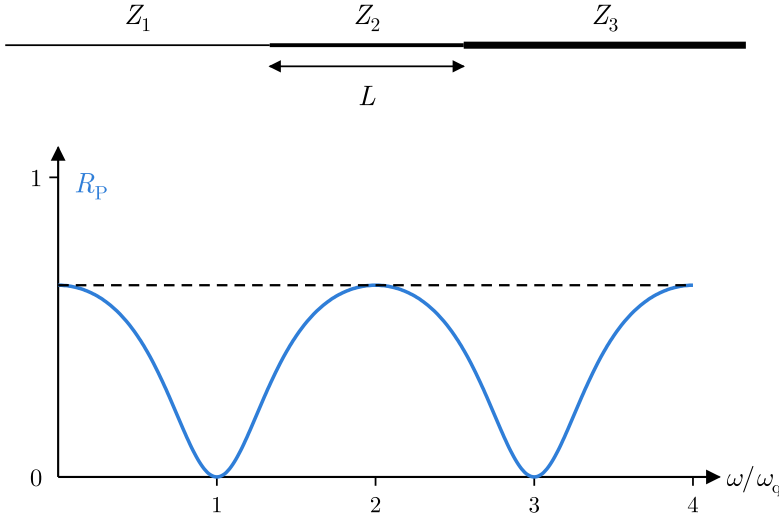


Figure 60. A quarter-wave matching section of impedance $Z_2 = \sqrt{Z_1 Z_3}$ and length L between strings of impedances Z_1 and Z_3 (upper). The fraction $\mathcal{R}_P$ of the incident power that is reflected (accent) is plotted against ω/ω_q , where ω_q is the frequency at which the section is a quarter of a wavelength long (lower). The dashed line is the reflected fraction when the first and last strings are joined directly.

The section is a quarter of a wavelength long at only one frequency, which we call ω_q . At other frequencies the input impedance is given by the general formula above, and the fraction $|\mathcal{R}|^2$ of the power that is reflected is drawn in the lower part of Figure 60. The reflection vanishes at ω_q and again at $3\omega_q, 5\omega_q, \dots$, where the section is an odd number of quarter wavelengths long, and it is small over a band of frequencies around each of these. At $2\omega_q$ the section is half a wavelength long, and you will show in Problem 3 that it then has no effect. Matching layers a quarter of a wavelength thick are often placed on the faces of the transducers used to send ultrasound into the body, for the same reason.

A String on an Elastic Foundation

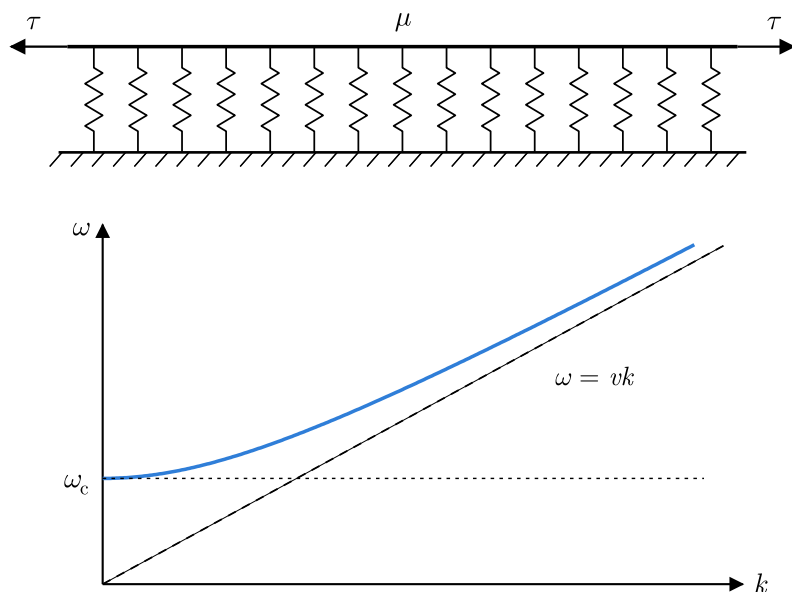


Figure 61. A string under tension τ held by springs along its length (upper), and its dispersion relation $\omega^2 = v^2k^2 + \omega_c^2$ (lower, accent). The dashed line is the relation $\omega = vk$ for the same string without the springs.

Let a string of tension τ and mass per unit length μ be attached along its whole length to a bed of springs, as in Figure 61, which pulls each element of the string back towards the axis with a force proportional to its displacement. Let κ_f be the stiffness of the bed per unit length, so that the springs exert the force $-\kappa_f y \delta x$ on an element of length δx . A support of this kind is called an elastic foundation; an example is a row of closely spaced pendulums that swing across a stretched string threaded through their bobs, since gravity then pulls each bob back towards the axis. Adding this force to the force from the tension, which gave the wave equation in Section 2.4, we find

$$\mu \frac{\partial^2 y}{\partial t^2} = \tau \frac{\partial^2 y}{\partial x^2} - \kappa_f y. \quad (4.38)$$

Substituting a harmonic wave $y = \text{Re}[\mathcal{A}e^{i(\omega t - kx)}]$ gives $-\mu\omega^2 = -\tau k^2 - \kappa_f$, and dividing by $-\mu$ gives

$$\omega^2 = v^2 k^2 + \omega_c^2, \quad \omega_c^2 = \frac{\kappa_f}{\mu}, \quad (4.39)$$

where $v = \sqrt{\tau/\mu}$ is the speed of waves on the same string without the springs. This is the dispersion relation of the string, in the sense of Section 2.3, and it is drawn in Figure 61.

As k tends to zero, ω tends to ω_c . The whole string then moves up and down together, the tension exerts no transverse force on a straight string, and each element oscillates on its springs as a mass on a spring does, with the angular frequency $\sqrt{\kappa_f/\mu}$. For large k , ω approaches vk , since the force from the tension, which grows as k^2 , is then much larger than the force from the springs. For every ω greater than ω_c a harmonic wave can travel, with the wave number $k = \sqrt{\omega^2 - \omega_c^2}/v$. The crests of this wave travel at the speed ω/k , which is greater than v and depends on the frequency, so that harmonic waves of different frequencies travel at different speeds on this string. For ω less than ω_c the dispersion relation has no real solution for k . The frequency ω_c is called the cutoff frequency of the string.

Evanescent Waves

Let the end $x = 0$ of a long string on an elastic foundation be moved up and down as $y(0, t) = A \cos \omega t$, with $\omega < \omega_c$. The dispersion relation now gives $k^2 = -(\omega_c^2 - \omega^2)/v^2$, which is negative, so k is imaginary. Let

$$\beta = \frac{\sqrt{\omega_c^2 - \omega^2}}{v}. \quad (4.40)$$

The solutions of the equation of motion of the form $e^{i\omega t}$ times a function of x are then $e^{-\beta x} e^{i\omega t}$ and $e^{\beta x} e^{i\omega t}$, as substituting them shows. The second grows without limit along the string, and it cannot describe a disturbance produced at the end of a long string, whose only source of energy is the motion of that end. The displacement is therefore

$$y(x, t) = Ae^{-\beta x} \cos \omega t. \quad (4.41)$$

Every point of the string moves in phase with the end, and the amplitude falls exponentially with the distance from the end, by a factor e in the distance $1/\beta$. There are no crests that travel along the string. A disturbance of this kind is called an evanescent wave. The upper part of Figure 62 shows it at several instants. The distance $1/\beta$ grows without limit as ω approaches ω_c from below, and it tends to $v/\omega_c = \sqrt{\tau/\kappa_f}$ at low frequencies.

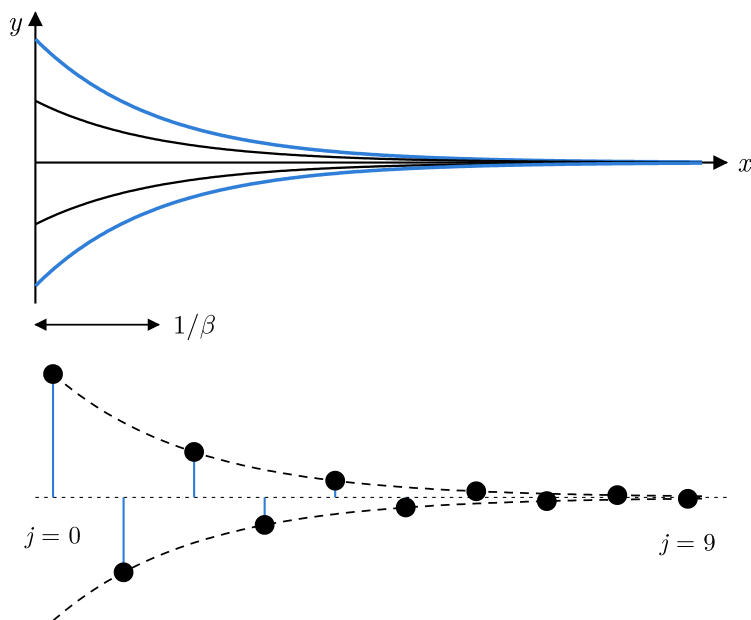


Figure 62. An evanescent wave on a string on an elastic foundation driven at its end below the cutoff frequency, drawn at four instants a sixth of a period apart (upper), and the displacements of the masses of a chain of masses and springs driven at its first mass above its highest frequency, with the envelope $\pm e^{-\beta a j}$ dashed (lower).

An evanescent wave carries no power along the string. The transverse force exerted by the string to the left of a point on the string to its right is $-\tau \partial y / \partial x = \tau \beta A e^{-\beta x} \cos \omega t$, which is in phase with

the displacement, while the velocity $\partial y/\partial t = -\omega A e^{-\beta x} \sin \omega t$ is a quarter of a period ahead of it. The rate at which work is done on the string beyond the point is the product of these, which is proportional to $\cos \omega t \sin \omega t = \frac{1}{2} \sin 2\omega t$ and averages to zero over a cycle. In complex notation, the ratio of the force to the velocity at every point is $\tau\beta/i\omega = -i\tau\beta/\omega$, which is the mechanical impedance $-i\kappa/\omega$ of a spring of constant $\tau\beta$. The end of the string therefore behaves as a spring, and the device that moves it does work on the string during part of each cycle and has the same amount of work returned to it during the rest.

The chain of masses and springs of Section 2.3 behaves in the same way above its highest frequency 2Ω . Let the masses m be joined by springs of constant κ and spaced a distance a apart, so that $\Omega^2 = \kappa/m$, and let the displacement of the j th mass be x_j . As in Section 2.3, its equation of motion is

$$m\ddot{x}_j = \kappa (x_{j+1} - 2x_j + x_{j-1}) . \quad (4.42)$$

Suppose that the first mass is driven at a frequency $\omega > 2\Omega$, and let us try $x_j = \text{Re}[\mathcal{A}(-1)^j e^{-\beta a j} e^{i\omega t}]$, in which neighbouring masses move in antiphase, as in the mode of highest frequency, and the amplitude falls by the factor $e^{-\beta a}$ from each mass to the next. Substituting and dividing by $\mathcal{A}(-1)^j e^{-\beta a j} e^{i\omega t}$ gives

$$-m\omega^2 = \kappa (-e^{-\beta a} - 2 - e^{\beta a}) = -2\kappa (1 + \cosh \beta a) , \quad (4.43)$$

and the identity $1 + \cosh \beta a = 2 \cosh^2(\beta a/2)$, which follows from the definition $\cosh u = \frac{1}{2}(e^u + e^{-u})$ in Section 1.2 of Vol. 4 by squaring $\cosh(\beta a/2)$, gives

$$\omega = 2\Omega \cosh \frac{\beta a}{2} . \quad (4.44)$$

Since $\cosh$ takes every value greater than one, this equation has a real, positive solution for β at every $\omega > 2\Omega$, and the disturbance is evanescent, as the lower part of Figure 62 shows. The higher the driving frequency, the faster the amplitude falls along the chain.

Above its highest frequency the chain cannot pass a wave along, and the motion stays near the driven end.

Transmission Through an Evanescent Region

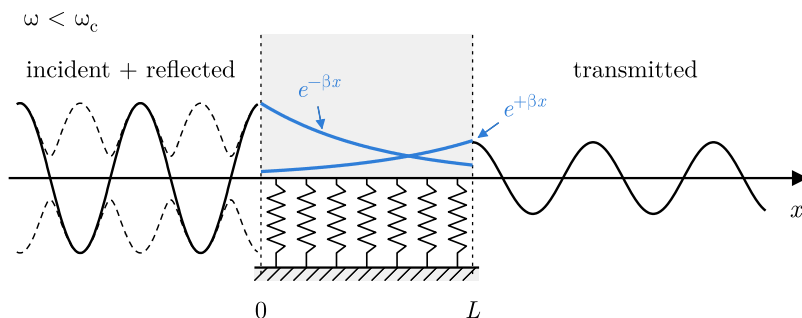


Figure 63. A string carrying an elastic foundation only over the stretch $0 < x < L$, with a harmonic wave of frequency below the cutoff arriving from the left. On the left the string carries the incident wave together with the reflected one, and the dashed curves are the envelope of their sum over a cycle. Within the stretch the displacement is a combination of a falling exponential (accent) and a rising one, the falling one dominating; beyond the stretch a wave of the same wavelength travels on, with a smaller amplitude than the incident wave.

Consider now a string of tension τ and mass per unit length μ that lies on an elastic foundation only in the stretch $0 < x < L$, and is free of springs on either side of it, as in Figure 63. A harmonic wave of frequency $\omega < \omega_c$ arrives from the left. On the left the displacement is the sum of the incident and the reflected waves, $\mathcal{A}e^{-ikx} + \mathcal{A}_r e^{ikx}$, with $k = \omega/v$, and on the right there is only a transmitted wave $\mathcal{A}_t e^{-ik(x-L)}$, where the factor $e^{i\omega t}$ is understood. In the stretch on the foundation the displacement is $\mathcal{B}_+ e^{-\beta x} + \mathcal{B}_- e^{\beta x}$, and here both exponentials are allowed, since the stretch has a finite length. At $x = 0$ and at $x = L$ the displacement is continuous, since the string is continuous, and so is the slope $\partial y / \partial x$, since the springs exert a force per unit length and so exert no finite force on a single point of the string, while the tension on either side of such a point is the same.

These equations have the same form as those of Section 2.5 of Vol. 3 for a particle of mass m and energy E meeting a rectangular barrier of height $V_0 > E$ and width w . With the factor $e^{i\omega t}$ removed, the displacement satisfies $d^2y/dx^2 = -(\omega^2/v^2)y$ off the foundation and $d^2y/dx^2 = -[(\omega^2 - \omega_c^2)/v^2]y$ on it. These are the equations for the wave function outside and inside the barrier, $d^2\psi/dx^2 = -(2mE/\hbar^2)\psi$ and $d^2\psi/dx^2 = -[2m(E - V_0)/\hbar^2]\psi$, with ω^2/v^2 in place of $2mE/\hbar^2$ and ω_c^2/v^2 in place of $2mV_0/\hbar^2$, and in both problems the function and its first derivative are continuous at the edges. The ratio E/V_0 therefore becomes ω^2/ω_c^2 , the width w becomes L , and the decay constant of the wave function inside the barrier becomes β . Section 2.5 of Vol. 3 found the probability of transmission to be the reciprocal of $1 + V_0^2 \sinh^2 \Theta / [4E(V_0 - E)]$, where Θ is the product of the width of the barrier and that decay constant. Since the strings on either side of the stretch are identical, the corresponding quantity here is $|\mathcal{A}_t/\mathcal{A}|^2$, the fraction of the incident power that is transmitted, and translating the result gives

$$\mathcal{T}_P = \left[1 + \frac{\omega_c^4 \sinh^2 \beta L}{4\omega^2(\omega_c^2 - \omega^2)} \right]^{-1}. \quad (4.45)$$

The fraction is not zero, so a wave crosses a stretch of string on which it cannot travel, and this is the classical counterpart of tunnelling. When the stretch is long compared with the decay length, so that $\beta L \gg 1$, we have $\sinh \beta L \approx \frac{1}{2}e^{\beta L}$, the second term in the bracket is much larger than one, and

$$\mathcal{T}_P \approx \frac{16\omega^2(\omega_c^2 - \omega^2)}{\omega_c^4} e^{-2\beta L}, \quad (4.46)$$

so the transmitted power falls exponentially with the length of the stretch. Above the cutoff frequency waves travel on the stretch, with the wave number $k' = \sqrt{\omega^2 - \omega_c^2}/v$, and the fraction transmitted is the result you found in Problem 4 of Section 2.5 of Vol. 3, translated in the same way. It is one when $k'L$ is a whole multiple of π , so that the stretch holds a whole number of half-wavelengths and the waves reflected from its two ends cancel. Figure 64 shows both ranges.

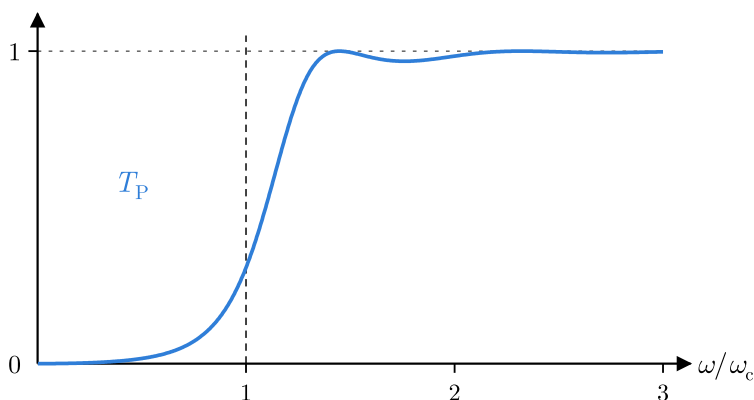


Figure 64. The fraction $\mathcal{T}_P$ of the incident power transmitted across a stretch of string on an elastic foundation, plotted against ω/ω_c . Below the cutoff frequency (dashed line) the wave crosses the stretch as an evanescent disturbance. Above it the transmission is complete only at the frequencies at which the stretch holds a whole number of half-wavelengths.

The evanescent disturbance of a long string carries no power, and yet power crosses the stretch, because on the stretch both exponentials are present. You will show in Problem 4 that the mean power carried along the stretch is $\tau\beta\omega \operatorname{Im}(\mathcal{B}_+\mathcal{B}_-^*)$, which is the same at every point of the stretch, and which is not zero when the decaying and the growing parts of the disturbance differ in phase.

4.5 Standing Waves in Strings, Pipes, and Cavities

The standing waves on a string and in a pipe were reviewed in Section 4.1 as they are met at school, with their nodes, antinodes, and harmonics. Here we derive them from the wave equation and the conditions at the ends, first for a string, then for a column of air in a pipe, and finally for the air in a closed box. The modes of a string with both ends fixed are worked out at length by separation of variables in Section 7.2 of Vol. 4, and they were found again in Section 2.4 as the limit of the loaded string, so we treat them only briefly here.

A Standing Wave as Two Travelling Waves

Let two harmonic waves of the same amplitude A and the same angular frequency ω travel along a stretched string in opposite directions. With the harmonic travelling waves of Section 3.2, the displacement of the string is

$$y(x, t) = A \cos(\omega t - kx) + A \cos(\omega t + kx). \quad (4.47)$$

Adding the formulas for the cosine of a sum and of a difference, $\cos(\omega t \mp kx) = \cos \omega t \cos kx \pm \sin \omega t \sin kx$, the terms in $\sin \omega t \sin kx$ cancel, and the sum becomes

$$y(x, t) = 2A \cos kx \cos \omega t. \quad (4.48)$$

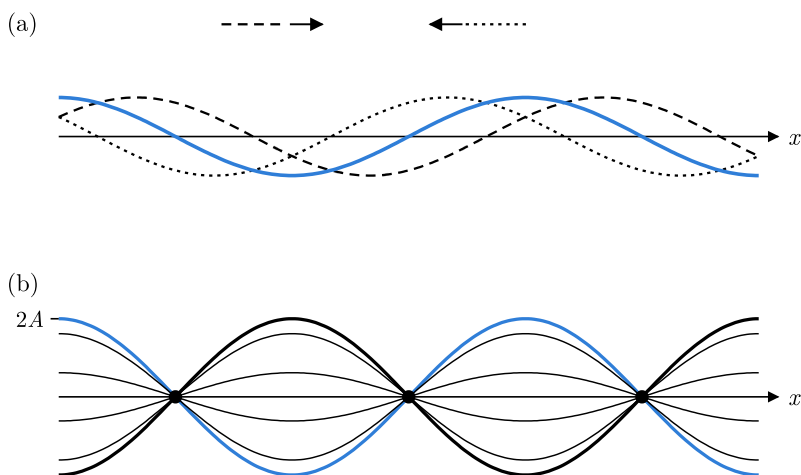


Figure 65. (a) Two harmonic waves of equal amplitude on a string, one travelling towards increasing x (dashed) and one towards decreasing x (dotted), and their sum (accent), all at one instant. (b) The sum at several instants during half a period, from an instant at which the displacement is greatest (accent) to half a period later (solid). The dots mark the nodes, where the string stays at rest.

Every point of the string now moves in simple harmonic motion with the angular frequency ω , and all the points move either in phase or in antiphase with one another, with an amplitude $2A|\cos kx|$ that depends on the position. The shape of the string grows and shrinks without moving along it. The points where $\cos kx = 0$, which are half a wavelength apart, stay at rest and are called nodes, and the points midway between them, where the amplitude is $2A$, are called antinodes. A disturbance of this form is called a standing wave, and Figure 65 shows how it is built from the two travelling waves.

A standing wave carries no power along the string, and this can be checked directly. The power carried past a point, found in Section 3.4 as the rate at which the string on the left of the point does work on the string on its right, is

$$\mathcal{P} = -\tau \frac{\partial y}{\partial x} \frac{\partial y}{\partial t}. \quad (4.49)$$

For the standing wave, $\partial y/\partial x = -2Ak \sin kx \cos \omega t$ and $\partial y/\partial t = -2A\omega \cos kx \sin \omega t$. Multiplying these derivatives, and using the double-angle formula $2 \sin \chi \cos \chi = \sin 2\chi$ for the factors in x and again for the factors in t , gives

$$\mathcal{P} = -\tau A^2 k \omega \sin 2kx \sin 2\omega t. \quad (4.50)$$

The mean of $\sin 2\omega t$ over a cycle is zero, so the mean power $\langle \mathcal{P} \rangle$ is zero at every point of the string. This agrees with Section 3.4, since the two travelling waves each carry the mean power $\frac{1}{2}\mu v \omega^2 A^2$ found there, in opposite directions. The power at a given instant is zero only at the nodes, where the string does not move, and at the antinodes, where its slope is zero, and between them the energy flows one way and then the other, twice in each period. With the energies per unit length of Section 3.4, the kinetic energy per unit length, $2\mu\omega^2 A^2 \cos^2 kx \sin^2 \omega t$, is greatest at the antinodes, and the potential energy per unit length, $2\tau k^2 A^2 \sin^2 kx \cos^2 \omega t$, is greatest at the nodes, where the string is stretched most. Since $\tau k^2 = \mu\omega^2$, because $k = \omega/v$ and $v^2 = \tau/\mu$, the energy passes from potential energy near the nodes to kinetic energy near the antinodes and

back. You will show in Problem 1 that the total energy between two neighbouring nodes does not change in time.

Strings with Fixed and Free Ends

A string of length L with both ends fixed can oscillate steadily only in standing waves that have nodes at $x = 0$ and $x = L$. Section 2.4 found these modes as the limit of the loaded string, and Section 7.2 of Vol. 4 found them by separation of variables, in the form

$$\begin{aligned} y_n(x, t) &= \sin \frac{n\pi x}{L} (A_n \cos \omega_n t + B_n \sin \omega_n t) , \\ \omega_n &= \frac{n\pi v}{L} , \\ n &= 1, 2, 3, \dots , \end{aligned} \tag{4.51}$$

where A_n and B_n are constants. The n th mode fits n half-wavelengths into the length of the string. The same condition follows from the reflections at the ends, each of which changes the phase of a wave by π , since a fixed end has the amplitude reflection coefficient $\mathcal{R} = -1$ (Section 4.3). After a round trip of length $2L$ a wave has fallen behind in phase by $2kL + 2\pi$, and it adds steadily to the wave still arriving only if this phase is a whole multiple of 2π , which requires $kL = n\pi$.

Let us now free one end. The end $x = L$ is attached to a ring of negligible mass that slides without friction on a smooth rod perpendicular to the string, as in Figure 66. The rod can exert a force on the ring only at right angles to the rod, that is, along the string, so the only transverse force on the ring is the force $-\tau \partial y / \partial x$ exerted by the string. A ring of negligible mass must have zero total force on it, since a finite force would give it an unlimited acceleration, and therefore

$$\frac{\partial y}{\partial x} = 0 \quad \text{at} \quad x = L , \tag{4.52}$$

so that the string always meets the rod at right angles. This is the free end of Section 4.3, which reflects a wave with $\mathcal{R} = +1$. A standing wave with a node at the fixed end $x = 0$ has the form $y =$

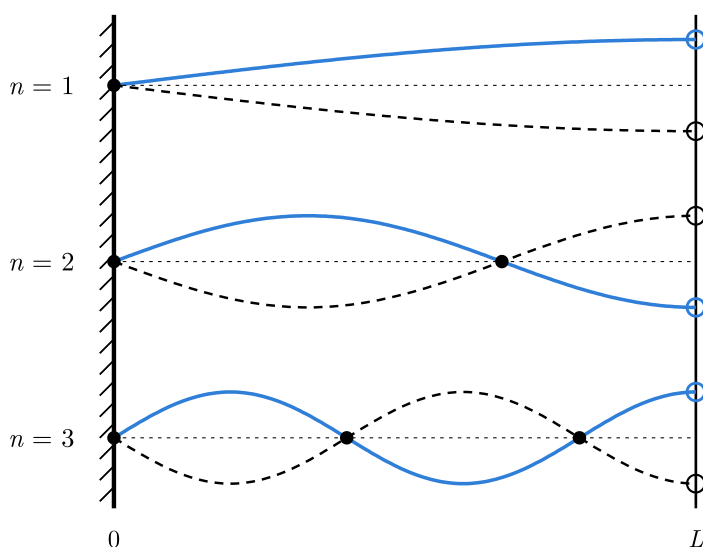


Figure 66. The three lowest modes of a string fixed at $x = 0$ and attached at $x = L$ to a ring of negligible mass that slides without friction on a smooth rod. The accent and dashed curves show the string at the two extremes of its oscillation, half a period apart, and the dots mark the nodes. The string meets the rod at right angles, and the free end is an antinode.

$A \sin kx \cos \omega t$, and the condition at the free end requires $k \cos kL = 0$. Therefore $kL = (n - \frac{1}{2})\pi$, and

$$\omega_n = \frac{(2n - 1)\pi v}{2L}, \quad n = 1, 2, 3, \dots \quad (4.53)$$

The lowest mode fits a quarter of a wavelength into the string, and its frequency is half that of the same string with both ends fixed. The frequencies of the modes are in the ratios 1 : 3 : 5 and so on, and the free end is an antinode of every mode. The round trip gives the same result, since the free end reflects without a change of phase, so that $2kL + \pi$ must be a whole multiple of 2π .

Standing Waves in Pipes

A column of air in a narrow pipe carries the longitudinal waves of Section 3.3. The displacement $\xi(x, t)$ of the air along the pipe satisfies the wave equation with the speed $c = \sqrt{B/\rho_0}$, and the excess pressure is

$$p = -B \frac{\partial \xi}{\partial x}. \quad (4.54)$$

The equation of motion of a thin slice of the air, from which Section 3.3 derived the wave equation, is $\rho_0 \partial^2 \xi / \partial t^2 = -\partial p / \partial x$, and it says that the air is accelerated towards lower pressure. These two relations fix the conditions at the ends of a pipe. At a closed end, a rigid cap stops the air from moving along the pipe, so $\xi = 0$ there at all times, and by the equation of motion $\partial p / \partial x = 0$ as well. A closed end is therefore a node of displacement and an antinode of pressure. At an open end the air in the pipe meets the air outside, which can move away in every direction and so cannot hold a pressure much different from the ambient pressure. The excess pressure is therefore approximately zero at an open end, and so is $\partial \xi / \partial x$, which makes an open end a node of pressure and an antinode of displacement. The condition $p = 0$ is a valid approximation when the width of the pipe is small compared with the wavelength. The air just outside the end takes part in the motion, so that the pipe behaves as though it were slightly longer than it is. The added length is called the end

correction, and its calculation, which comes from the radiation of sound out of the open end, is beyond the scope of this course.

A standing wave in the pipe has the form $\xi = (G_1 \cos kx + G_2 \sin kx) \cos \omega t$, with $\omega = ck$ and constants G_1 and G_2 , and the excess pressure that goes with it is

$$p = -B \frac{\partial \xi}{\partial x} = Bk (G_1 \sin kx - G_2 \cos kx) \cos \omega t. \quad (4.55)$$

Wherever the displacement varies as a cosine the pressure varies as a sine, and the other way round, so the nodes of pressure sit at the antinodes of displacement, and each is a quarter of a wavelength from the nearest node of displacement. The reason is that the air is compressed most where the air on either side of a point moves towards the point together, which is at a node of displacement, and it is not compressed at an antinode of displacement, where the neighbouring air moves as one piece.

The modes of a pipe of length L follow from these conditions, and they are drawn in Figure 67. There are three cases to consider:

1. In a pipe open at both ends, $\partial \xi / \partial x = 0$ at $x = 0$ and at $x = L$. The first condition gives $G_2 = 0$, so that $\xi \propto \cos kx$, and the second requires $\sin kL = 0$. The frequencies are $f_n = nc/2L$, with $n = 1, 2, 3, \dots$, and every harmonic of the lowest frequency is present.
2. In a pipe closed at both ends, $\xi = 0$ at both ends, so $\xi \propto \sin kx$ with $\sin kL = 0$. The frequencies are again $f_n = nc/2L$, but the patterns are different, since the excess pressure has antinodes at the ends of the closed pipe and nodes at the ends of the open one.
3. In a pipe closed at $x = 0$ and open at $x = L$, $\xi \propto \sin kx$ and $p \propto \cos kx$, and the open end requires $\cos kL = 0$. The frequencies are $f_n = (2n - 1)c/4L$, with $n = 1, 2, 3, \dots$, so only the odd harmonics of the lowest frequency $c/4L$ are present.

The pipe closed at one end has the modes of the string with one free end found above, and the correspondence is exact: the displacement ξ corresponds to the displacement y of the string, the excess pressure $-B \partial \xi / \partial x$ to the transverse force $-\tau \partial y / \partial x$, and the closed and open ends to the fixed and free ends of the string. A clarinet is approximately a cylindrical pipe closed at the mouthpiece by the

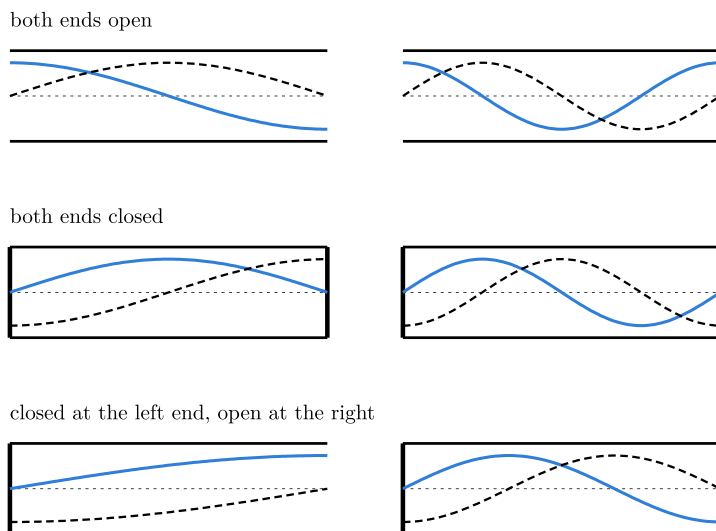


Figure 67. The two lowest modes of a pipe open at both ends, of a pipe closed at both ends, and of a pipe closed at the left end and open at the right, shown on the left and on the right of each row. The accent curve is the displacement ξ along the pipe and the dashed curve is the excess pressure p , both drawn at an instant at which they are greatest. Each node of pressure sits at an antinode of displacement, and a closed end is a node of displacement.

reed, and in its lowest notes the odd harmonics are much stronger than the even ones.

The Modes of a Rectangular Cavity

Consider now the air in a closed box with rigid walls, whose sides L_x , L_y , and L_z lie along the three axes. The pressure can now vary in all three directions, and we need the equations of Section 3.3 in three dimensions. Let the displacement of the air have the components ξ_x , ξ_y , and ξ_z . The argument that gave the equation of motion of a thin slice, applied to the faces of a small box of air normal to each axis in turn, gives

$$\rho_0 \frac{\partial^2 \xi_x}{\partial t^2} = -\frac{\partial p}{\partial x}, \quad \rho_0 \frac{\partial^2 \xi_y}{\partial t^2} = -\frac{\partial p}{\partial y}, \quad \rho_0 \frac{\partial^2 \xi_z}{\partial t^2} = -\frac{\partial p}{\partial z}. \quad (4.56)$$

The side of the small box along x is stretched by the fraction $\partial \xi_x / \partial x$, and similarly for the other sides, so its volume is multiplied by $(1 + \partial \xi_x / \partial x)(1 + \partial \xi_y / \partial y)(1 + \partial \xi_z / \partial z)$. To first order in the small derivatives the fractional change of volume is their sum, and since the bulk modulus relates the excess pressure to the fractional change of volume, as in Section 3.3,

$$p = -B \left(\frac{\partial \xi_x}{\partial x} + \frac{\partial \xi_y}{\partial y} + \frac{\partial \xi_z}{\partial z} \right). \quad (4.57)$$

We differentiate this relation twice with respect to t , exchange the order of the derivatives in each term, and substitute the three equations of motion, which gives

$$\frac{\partial^2 p}{\partial t^2} = c^2 \left(\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \frac{\partial^2 p}{\partial z^2} \right), \quad c^2 = \frac{B}{\rho_0}. \quad (4.58)$$

The bracket is the Laplacian of p , from Section 3.1 of Vol. 4, and when p depends on x alone the equation becomes the wave equation of Section 3.3. At a rigid wall normal to the x axis the air cannot move across the wall, so $\xi_x = 0$ there, and by the first equation of motion $\partial p / \partial x = 0$, as at the closed end of a pipe.

Following the method of separation of variables of Section 7.2 of Vol. 4, we look for modes that are products of standing waves along the three axes,

$$p = A \cos k_x x \cos k_y y \cos k_z z \cos \omega t, \quad (4.59)$$

which satisfy $\partial p / \partial x = 0$ on the wall $x = 0$ and the corresponding conditions on the walls $y = 0$ and $z = 0$. On the wall $x = L_x$ the condition requires $\sin k_x L_x = 0$, and similarly for the other two walls, so that

$$\begin{aligned} k_x &= \frac{n_x \pi}{L_x}, \\ k_y &= \frac{n_y \pi}{L_y}, \\ k_z &= \frac{n_z \pi}{L_z}, \\ n_x, n_y, n_z &= 0, 1, 2, \dots \end{aligned} \quad (4.60)$$

Substituting the product into the wave equation, each second derivative with respect to a coordinate multiplies it by $-k_x^2$, $-k_y^2$, or $-k_z^2$, and the second derivative with respect to time multiplies it by $-\omega^2$, so that $\omega^2 = c^2(k_x^2 + k_y^2 + k_z^2)$ and

$$\omega = \pi c \sqrt{\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2}}. \quad (4.61)$$

Each set of three integers, not all zero, is a mode. When two of the integers are zero the mode is a standing wave along one axis, with the frequencies of a pipe closed at both ends, and when one is zero the pressure does not vary along that axis. In a cube several sets of integers give the same frequency, as you will find in Problem 4, because the three axes of a cube are equivalent.

The number of modes with angular frequency below a given ω can be counted in the space of wave vectors, as Figure 68 shows for one plane of it. Each mode is a point (k_x, k_y, k_z) , and the points sit at the corners of every cell of a rectangular lattice whose cells

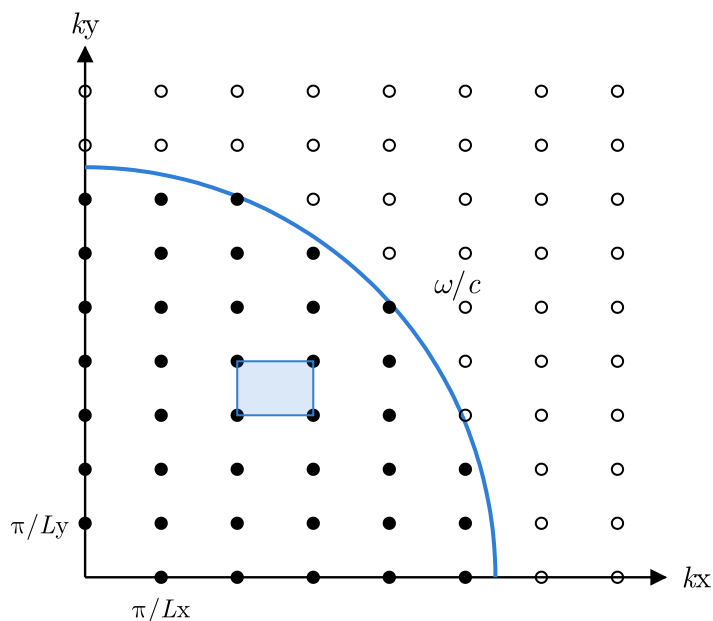


Figure 68. The modes of a rectangular cavity in the plane $k_z = 0$ of the space of wave vectors. Each point, at $(n_x\pi/L_x, n_y\pi/L_y)$, is one mode, and the shaded rectangle, with sides π/L_x and π/L_y , is one cell of the lattice, so that there is one point per cell. The filled points, inside the quarter circle of radius ω/c , are the modes with angular frequency less than ω .

have sides π/L_x , π/L_y , and π/L_z , so that there is one point per cell, and the volume of a cell is π^3/V , where $V = L_x L_y L_z$ is the volume of the cavity. The modes with angular frequency below ω lie inside the sphere of radius ω/c , in the eighth of it where all three components are positive or zero. Provided the wavelength $2\pi c/\omega$ is small compared with every side of the cavity, so that the sphere holds a great many cells, the number of these modes is the volume of the eighth of the sphere divided by the volume of a cell,

$$\mathcal{N}(\omega) \approx \frac{1}{8} \cdot \frac{4\pi}{3} \left(\frac{\omega}{c}\right)^3 \frac{V}{\pi^3} = \frac{V\omega^3}{6\pi^2 c^3}, \quad (4.62)$$

and differentiating with respect to ω gives the number of modes with angular frequency between ω and $\omega + d\omega$ as

$$\frac{d\mathcal{N}}{d\omega} d\omega = \frac{V\omega^2}{2\pi^2 c^3} d\omega. \quad (4.63)$$

The number depends on the volume of the cavity and not on its shape, as long as the wavelength is small compared with its sides. The same count was made in Section 5.6 of Vol. 2 of the Fermi Notes on Physics for light in a cube with conducting walls, and gave $V\omega^2 d\omega/\pi^2 c^3$, with c the speed of light, which is twice the result above because light has two independent polarisations for each set of three integers. In Section 5.5 of Vol. 2, where the modes of a crystal are counted, each set of integers carries three modes, one for each polarisation of sound in a solid. The standing waves of light have nodes at the walls, whereas the pressure in the cavity above has antinodes there, and this difference changes only whether the points with an integer equal to zero are counted, which are a small fraction of all the points when the wavelength is small compared with the sides.

A String Driven at One End

As an example of standing waves that are driven, consider a string of length L fixed at $x = L$, whose end at $x = 0$ is moved up and down by a driver so that $y(0, t) = A_0 \cos \omega t$, as in Figure 69. Once any transient motion has been removed by small losses in the string,

the string oscillates at the driving frequency. The wave sent out by the driver is reflected at the fixed end with $\mathcal{R} = -1$, so the two waves have equal amplitudes and add, as in the first subsection, to a standing wave with a node at the fixed end,

$$y(x, t) = G \sin k(L - x) \cos \omega t, \quad k = \frac{\omega}{v}, \quad (4.64)$$

where G is a constant to be determined. At $x = 0$ this must equal $A_0 \cos \omega t$, which gives $G = A_0 / \sin kL$.

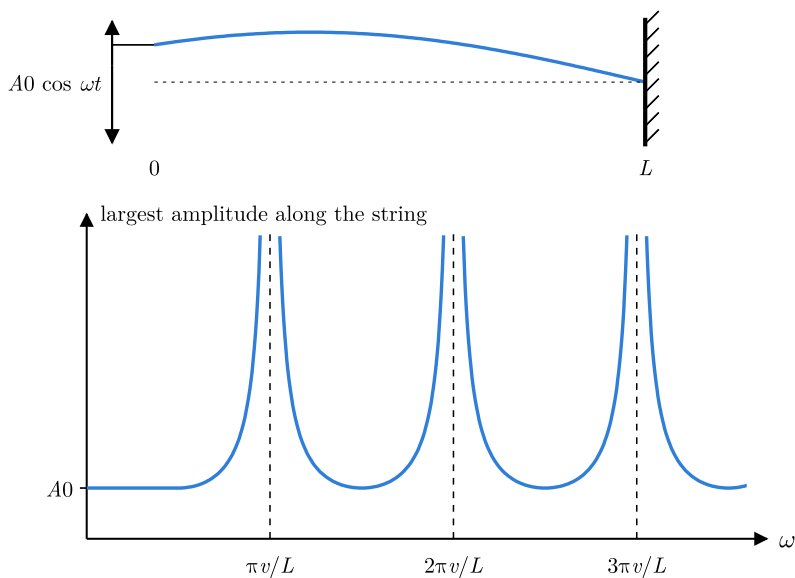


Figure 69. A string fixed at $x = L$ whose end at $x = 0$ is moved with the displacement $A_0 \cos \omega t$ (above), and the greatest amplitude of its steady oscillation along the string as a function of the driving frequency (below). The amplitude rises without limit at the frequencies $n\pi v/L$ of the modes of the string with both ends fixed, marked by the dashed lines, and falls to A_0 midway between them.

The amplitude at the antinodes of the standing wave, $A_0/|\sin kL|$, which is the largest amplitude on the string once kL exceeds $\pi/2$, grows without limit as the driving frequency approaches one of the frequencies $n\pi v/L$ of the modes of the string with both ends fixed, where $\sin kL = 0$, as Figure 69 shows. At the frequency of a mode

the driven end is itself a node of that mode, so an arbitrarily small motion of the end keeps up a large standing wave. In a real string the losses, from the resistance of the air and from friction inside the string, keep the amplitude finite, as the damping does at the resonance of Section 1.3.

The driver exerts on the end of the string the transverse force $F = -\tau \partial y / \partial x$ at $x = 0$, which is $\tau k A_0 \cot kL \cos \omega t$. Written with the time dependence $e^{i\omega t}$, the force is $\tau k A_0 \cot kL e^{i\omega t}$ and the velocity of the end is $i\omega A_0 e^{i\omega t}$, so the ratio of the force to the velocity, which is the impedance of Section 1.3 for the end of the string, is

$$\frac{F}{\dot{y}} = -i \frac{\tau k}{\omega} \cot kL = -iZ \cot kL, \quad (4.65)$$

where we have used $\tau k / \omega = \tau / v = Z$, the characteristic impedance of Section 3.4. The ratio is purely imaginary, so the force is a quarter of a period out of phase with the velocity and the driver supplies no mean power, as expected for a standing wave. You will show in Problem 2 that a driver that applies a force of given amplitude, in place of a displacement, produces the largest motion at a different set of frequencies.

4.6 Interference, Coherence, and Diffraction

The school treatment of interference was reviewed in Section 4.1. Two coherent sources give maxima where the path difference is a whole number of wavelengths and zeros where it is an odd number of half-wavelengths, and the grating and the single slit give the patterns whose maxima and minima are located there. We now add the waves with the phasors of Section 4.2 and find the intensity in every direction, first for two sources, then for a row of equally spaced sources, then for a slit of finite width, and finally we ask under what conditions two sources give a steady pattern. The acoustic example throughout is a pair of loudspeakers driven from one signal.

Two Coherent Sources

Let two small loudspeakers a distance d apart be driven from one amplifier with the same sinusoidal signal of angular frequency ω , so

that they vibrate with the same amplitude and in phase, and let P be a point at the distances r_1 and r_2 from them. By the principle of superposition, the excess pressure at P is the sum of the excess pressures of the two waves. Each wave falls behind in phase by k for every unit of distance it travels, as the harmonic travelling waves of Section 3.2 do. When P is far away compared with d , the distances r_1 and r_2 differ by a small fraction of either, and the two waves arrive with nearly equal amplitudes. The excess pressure at P is then

$$p = \operatorname{Re} [\mathcal{A}e^{i(\omega t - kr_1)} + \mathcal{A}e^{i(\omega t - kr_2)}] = \operatorname{Re} [\mathcal{A}e^{i(\omega t - kr_1)} (1 + e^{-i\Delta\phi})] , \quad (4.66)$$

where $\mathcal{A}$ is the complex amplitude of each wave at P and $\Delta\phi = k(r_2 - r_1)$ is the phase difference between them.

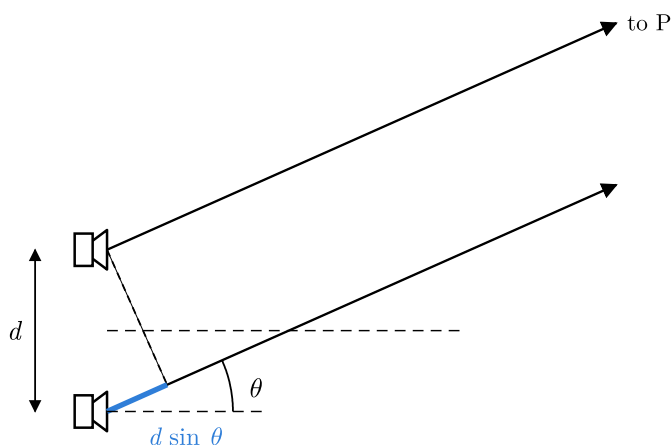


Figure 70. Two loudspeakers a distance d apart, driven from one signal, and the directions from them to a distant point P, which make the angle θ with the normal to the line joining the loudspeakers. The dashed line from the upper loudspeaker is perpendicular to the two paths, so the path from the lower loudspeaker is longer by $d \sin \theta$ (accent).

When P is far away, the paths from the two loudspeakers to P are nearly parallel, and they make the same angle θ with the normal to the line joining the loudspeakers, as in Figure 70. The dashed

line drawn from the upper loudspeaker perpendicular to the paths cuts off a small right-angled triangle whose hypotenuse is d . The dashed side makes the angle θ with the line joining the loudspeakers, since it is perpendicular to the paths, and the path from the lower loudspeaker is longer by the side of the triangle that lies along it, so that

$$r_2 - r_1 \approx d \sin \theta, \quad \Delta\phi = kd \sin \theta = \frac{2\pi d \sin \theta}{\lambda}, \quad (4.67)$$

an approximation that holds when the distance of P is much larger than d .

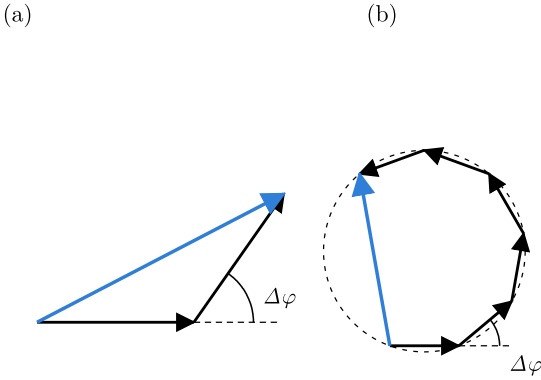


Figure 71. Phasor diagrams for the sum of waves from coherent sources. (a) Two waves of equal amplitude with the phase difference $\Delta\phi$, whose resultant (accent) has the length $2|\mathcal{A}|\cos(\Delta\phi/2)$. (b) The waves from a row of equally spaced sources, each phasor turned through $\Delta\phi$ from the one before, so that the phasors are chords of a circle (dashed), and the resultant (accent) is a chord of the same circle.

The two terms in the bracket $1 + e^{-i\Delta\phi}$ are phasors of equal length, one turned through the angle $\Delta\phi$ from the other, as in Figure 71(a). Taking out the factor $e^{-i\Delta\phi/2}$, which has modulus one, gives $1 + e^{-i\Delta\phi} = e^{-i\Delta\phi/2}(e^{i\Delta\phi/2} + e^{-i\Delta\phi/2}) = 2e^{-i\Delta\phi/2}\cos(\Delta\phi/2)$, so the resultant has the amplitude $2|\mathcal{A}|\cos(\Delta\phi/2)$. The intensity is proportional to the square of the amplitude, as Section 3.4 showed, and therefore

$$I = 4I_1 \cos^2 \frac{\Delta\phi}{2}, \quad (4.68)$$

where I_1 is the intensity at P of one loudspeaker alone. The intensity is $4I_1$ where $\Delta\phi$ is a whole multiple of 2π , that is where $d \sin \theta = q\lambda$, and zero where $d \sin \theta = (q + \frac{1}{2})\lambda$, which are the conditions quoted in Section 4.1. Since the mean of $\cos^2$ over a cycle is one half, the intensity averaged over a range of directions that includes many maxima, which requires d to be many wavelengths, is $2I_1$, the sum of the intensities of the two loudspeakers. The interference redistributes the power of the loudspeakers among the directions, making some louder and others quieter than either loudspeaker alone.

Since $|\sin \theta|$ cannot exceed one, the number of maxima is about $2d/\lambda$. When d is less than half a wavelength there is no direction in which the intensity is zero, and at low frequencies two loudspeakers close together radiate much as a single source would. At high frequencies the maxima are many and close together, and a listener walking across in front of the loudspeakers, in open air, hears the sound grow louder and quieter. In space the directions that make the same angle θ with the normal lie on a cone about the line joining the loudspeakers, so the loud and quiet regions are cones.

A Row of Sources

Let N identical loudspeakers be placed in a row with the spacing d , all driven from one signal. Far away, in the direction θ , the path from each loudspeaker is longer than the path from the one before it by $d \sin \theta$, so neighbouring waves differ in phase by $\Delta\phi = kd \sin \theta$, and the excess pressure is

$$p = \text{Re} \left[\mathcal{A} e^{i(\omega t - kr_1)} \Sigma \right], \quad \Sigma = \sum_{j=0}^{N-1} e^{-ij\Delta\phi}. \quad (4.69)$$

The phasors in the sum Σ each have length one and each is turned through $\Delta\phi$ from the one before, so they form part of a regular polygon, as in Figure 71(b). The sum is a geometric series, whose modulus you found in Problem 6 of Section 1.2 of Vol. 4, and we

repeat the short calculation in the present notation. Multiplying it by $e^{-i\Delta\phi}$ gives $\Sigma e^{-i\Delta\phi} = \sum_{j=1}^N e^{-ij\Delta\phi}$, and subtracting this from Σ leaves only the first term of one sum and the last term of the other, so that $\Sigma(1 - e^{-i\Delta\phi}) = 1 - e^{-iN\Delta\phi}$. For any real χ , taking out the factor $e^{-i\chi/2}$ gives $1 - e^{-i\chi} = e^{-i\chi/2}(e^{i\chi/2} - e^{-i\chi/2}) = 2ie^{-i\chi/2} \sin(\chi/2)$, and using this in the numerator and the denominator gives

$$|\Sigma| = \left| \frac{\sin(N\Delta\phi/2)}{\sin(\Delta\phi/2)} \right|, \quad I = I_1 \frac{\sin^2(N\Delta\phi/2)}{\sin^2(\Delta\phi/2)}, \quad (4.70)$$

where I_1 is again the intensity of one loudspeaker alone. For $N = 2$, the double-angle formula $\sin \Delta\phi = 2 \sin(\Delta\phi/2) \cos(\Delta\phi/2)$ turns this into the result for two loudspeakers.

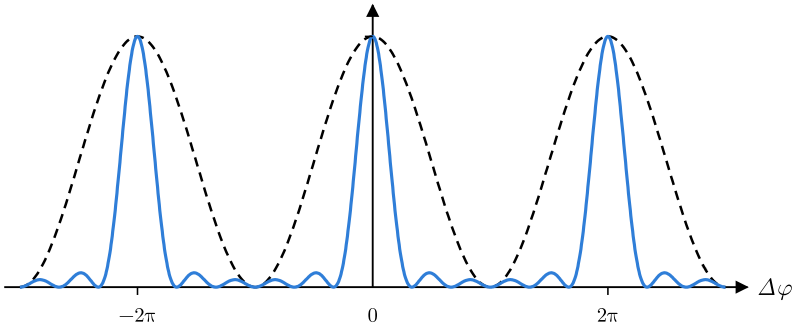


Figure 72. The intensity against the phase difference $\Delta\phi$ between neighbouring sources, for two sources (dashed) and for a row of N sources (accent), each divided by its greatest value. The principal maxima of the row are at the same places as the maxima for two sources and are much narrower, and between neighbouring principal maxima there are $N - 2$ subsidiary maxima.

Where $\Delta\phi$ is a whole multiple $2\pi q$ of 2π , all the phasors point the same way, and the numerator and the denominator of the intensity both vanish. Writing $\Delta\phi = 2\pi q + \delta\phi$ with a small $\delta\phi$, the ratio of the sines becomes $\sin(N\delta\phi/2)/\sin(\delta\phi/2)$ apart from its sign, and since the sine of a small angle is approximately the angle, the ratio tends to N as $\delta\phi$ tends to zero. The intensity there is $N^2 I_1$, and these peaks are called principal maxima. They occur where $d \sin \theta = q\lambda$,

which is the condition for the maxima of a grating given in Section 4.1, the same condition as for two sources. The intensity is zero where the numerator vanishes and the denominator does not, that is where $N\Delta\phi/2$ is a multiple of π that is not a multiple of $N\pi$. At these angles the phasors close into a complete polygon, or go round it several times, and their sum is zero. Between two neighbouring principal maxima there are therefore $N - 1$ zeros, and between the zeros lie $N - 2$ much weaker peaks, which are called subsidiary maxima. Figure 72 compares the pattern of a row with that of two sources.

The zeros nearest to the principal maximum of order q are at $\Delta\phi = 2\pi q \pm 2\pi/N$. Since $\Delta\phi = 2\pi d \sin\theta/\lambda$, the value of $\sin\theta$ changes by λ/Nd between the principal maximum and the first zero beside it, and since a small change of $\sin\theta$ is $\cos\theta$ times the change of θ , the angle between them is

$$\Delta\theta \approx \frac{\lambda}{Nd \cos\theta}. \quad (4.71)$$

The principal maxima become higher as N^2 and narrower as $1/N$, and you will show in Problem 2 that the intensity averaged over a period of $\Delta\phi$ is NI_1 , the sum of the intensities of the loudspeakers. When d is less than a wavelength only the principal maximum with $q = 0$ exists, since $\sin\theta = q\lambda/d$ would exceed one for any other q , and the row sends its sound out in a single beam perpendicular to the row, with the angular half-width λ/Nd , the wavelength divided by the length of the row. This is one reason a designer might stack loudspeakers in a tall column, less than a wavelength apart, to send sound in a beam that is narrow in the vertical direction. For light, a grating with a great many slits gives principal maxima so narrow that nearby wavelengths are separated, as you will show in Problem 3.

A Slit of Finite Width

Let a plane sound wave arrive at normal incidence on a wall with a long opening of width w in it, such as a doorway, and let us find the intensity in the directions that make the angle θ with the normal to the wall, in the plane that crosses the opening. As at school, we

treat the opening as a continuous line of sources that all vibrate in phase, which is a valid approximation for directions not too far from the normal. A long, flat loudspeaker whose face moves as one piece is described in the same way.

Let us divide the opening into N strips of width w/N and treat each strip as one of the sources of the previous subsection, with the spacing $d = w/N$, so that neighbouring strips differ in phase by $\Delta\phi = k(w/N) \sin \theta$. Let

$$\beta = \frac{N\Delta\phi}{2} = \frac{kw \sin \theta}{2} = \frac{\pi w \sin \theta}{\lambda}, \quad (4.72)$$

which does not depend on N . As N grows, $\Delta\phi/2 = \beta/N$ becomes small, and $\sin(\beta/N) \approx \beta/N$, so the intensity of the row becomes

$$I = I_1 \frac{\sin^2 \beta}{(\beta/N)^2} = I_0 \left(\frac{\sin \beta}{\beta} \right)^2, \quad (4.73)$$

where $I_0 = N^2 I_1$ is the intensity straight ahead, at $\theta = 0$. The intensity I_0 stays fixed as N grows, since the wave of each strip becomes weaker as the strips become narrower.

The phasor diagram shows the same result, as Figure 73(a) illustrates. As N grows, the polygon of Figure 71(b) becomes a smooth arc whose length is the amplitude $|\mathcal{A}_0|$ of the wave straight ahead, where all the phasors point the same way. Between the first strip and the last the direction of the phasors turns through $N\Delta\phi = 2\beta$, so the arc is part of a circle of radius $|\mathcal{A}_0|/2\beta$, and a chord that subtends the angle 2β at the centre of a circle has the length $2 \sin \beta$ times the radius. The resultant is therefore $|\mathcal{A}_0| \sin \beta/\beta$, and squaring it gives the intensity above.

The intensity is zero where $\sin \beta = 0$ with $\beta \neq 0$, that is where $w \sin \theta = q\lambda$ with $q = \pm 1, \pm 2, \dots$, and the first zero, where the arc closes into a full circle, is at $\sin \theta = \lambda/w$. At this angle each strip in one half of the opening can be paired with the strip half the width further on, whose path is longer by $\lambda/2$, so the waves of each pair cancel. Between the zeros the intensity has weaker maxima, near $\beta = (q + \frac{1}{2})\pi$, where $(\sin \beta/\beta)^2 \approx 1/(q + \frac{1}{2})^2 \pi^2$, so the first of

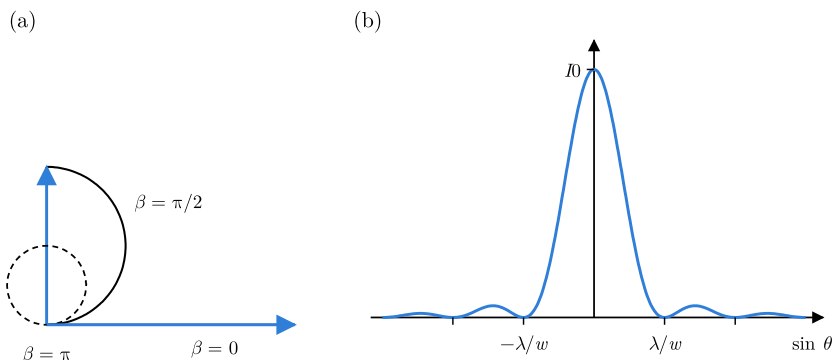


Figure 73. (a) Phasor diagrams for a slit of width w , with $\beta = \pi w \sin \theta / \lambda$. The chain of phasors from the strips of the slit becomes an arc of fixed length, which is straight when $\beta = 0$, a semicircle when $\beta = \pi/2$, and a closed circle (dashed) when $\beta = \pi$, and the resultant (accent) is the chord that joins its ends. (b) The intensity $I_0(\sin \beta/\beta)^2$ against $\sin \theta$, with its first zeros at $\sin \theta = \pm \lambda/w$.

them has about $4/9\pi^2 \approx 0.045$ of the intensity straight ahead. The central maximum is twice as wide as the others, as Figure 73(b) shows.

A wave passing through an opening of width w therefore spreads into directions within about $\sin \theta = \lambda/w$ of the normal. When the opening is narrower than a wavelength, λ/w exceeds one, the first zero is not reached, and the sound spreads into every direction in front of the wall. Sound of long wavelength is therefore heard well round the edges of a doorway, while sound of short wavelength passes through it in a narrower beam, and a loudspeaker whose face moves as one piece sends its high frequencies out in a narrower beam than its low ones.

The paths from the strips to P were taken to be parallel, and the condition for this can be found as follows. For a point P at the distance r straight ahead of the centre of the opening, the distance from a strip at the distance x' from the centre is $\sqrt{r^2 + x'^2} \approx r + x'^2/2r$, by the binomial expansion, so the path from an edge of the opening, where $x' = w/2$, is longer by about $w^2/8r$ than the parallel paths allow. The result above holds when this extra

path is small compared with a wavelength, that is when r is large compared with w^2/λ . The same condition, with Nd in place of w , applies to the row of sources.

Coherence

Let the two loudspeakers now be driven by two separate signals, with the amplitudes A_1 and A_2 at P and the phase difference $\Delta\phi$. The resultant of two phasors of lengths A_1 and A_2 with the angle $\Delta\phi$ between them has, by the cosine rule applied to the triangle they form with it, the squared length $A_1^2 + A_2^2 + 2A_1A_2 \cos \Delta\phi$, and therefore

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \Delta\phi, \quad (4.74)$$

where I_1 and I_2 are the intensities of the two loudspeakers alone. The last term is called the interference term. As an example, let the two signals come from two separate oscillators set to nearly the same angular frequencies ω and ω' . The phase difference then changes steadily at the rate $\omega' - \omega$, the pattern of loud and quiet directions sweeps past a listener at a fixed point, and the listener hears the beats of Section 4.2 at the difference frequency. The mean of $\cos \Delta\phi$ over a time long compared with $2\pi/|\omega' - \omega|$ is zero, so a measurement that lasts that long finds $I = I_1 + I_2$, and there is no steady pattern. The same holds whenever the phase difference wanders at random through all its values during the measurement. Two sources whose phase difference stays fixed during the time of a measurement are called coherent, and two whose phase difference wanders are called incoherent. For incoherent sources the intensities add.

A signal from a single source need not be coherent with itself at two different times. As a model, let the signal be made of sinusoidal trains of about the same duration Δt , each with a phase unrelated to that of the one before, as in Figure 74(a), which is approximately how the output of a noise generator, a circuit that produces a randomly varying voltage, behaves after it has passed through a filter that lets through only a narrow band of frequencies. By Section 2.5, a train of duration Δt has a spectrum of width $\Delta\omega \approx 2\pi/\Delta t$,

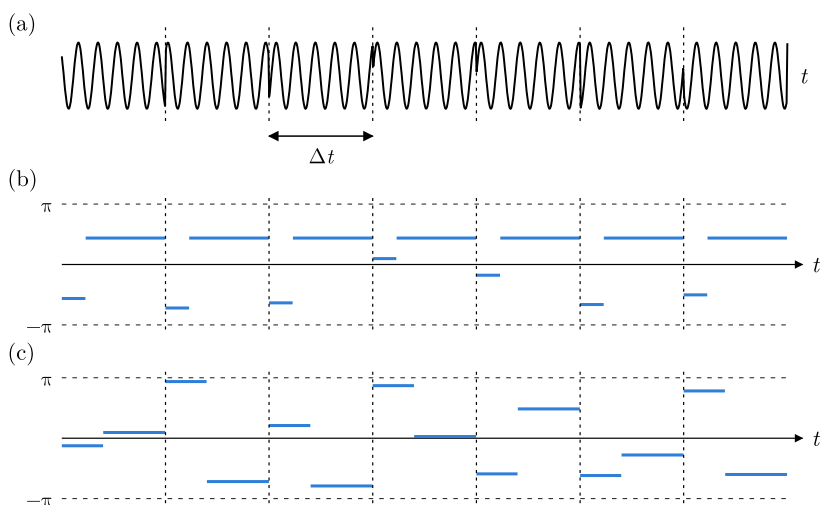


Figure 74. (a) A signal made of sinusoidal trains of duration Δt , drawn as equal for simplicity, whose phase jumps to an unrelated value at the start of each train (dotted lines). (b) The phase difference, reduced to the range from $-\pi$ to π , between the signal and a copy of it delayed by a time short compared with Δt ; it keeps one value except for short intervals after each jump. (c) The same for a delay longer than Δt , when the phase difference takes an unrelated value after every jump.

so the signal occupies a band of angular frequencies of this width.

Let this one signal drive both loudspeakers. The wave that reaches P from the farther loudspeaker left it a time $t_d = (r_2 - r_1)/c$ earlier than the wave from the nearer one, so the phase difference at P is ωt_d plus the difference between the phases of the signal at two times t_d apart. When t_d is short compared with Δt , both waves come from the same train for most of the time, and the phase difference keeps the value ωt_d except for the short intervals after each jump, as in Figure 74(b), so the pattern is steady. When t_d is longer than Δt , the two waves come from different trains, the phase difference takes an unrelated value after every jump, as in Figure 74(c), and the intensities add. The time over which the phase of a signal stays predictable is called its coherence time t_{coh} , and by the argument above

$$t_{\text{coh}} \approx \frac{2\pi}{\Delta\omega} = \frac{1}{\Delta f}, \quad (4.75)$$

where $\Delta f = \Delta\omega/2\pi$ is the width of the band in frequency. The distance $\Lambda_{\text{coh}} = c t_{\text{coh}}$ is called the coherence length, and a steady pattern is found only where the path difference is less than the coherence length. Near $\theta = 0$, where the paths are equal, the pattern is always present, and it fades out in the directions where $d \sin \theta$ approaches Λ_{coh} .

The same result follows from the spectrum of the signal. Each angular frequency in the band makes a separate pattern, with the phase difference ωt_d at P, and across the band this phase difference varies by $\Delta\omega t_d$. When $\Delta\omega t_d$ approaches 2π , the maxima of some frequencies fall on the zeros of others, and the sum of the patterns is uniform. This happens when $t_d \approx 2\pi/\Delta\omega$, which is the coherence time found above.

Problems

§4.1 Superposition at School

1. A string of length L fixed at both ends carries waves of speed v . Find the frequencies of the standing waves that have a node at

the point a distance $L/4$ from one end.

2. A pipe open at both ends and a pipe closed at one end have the same fundamental frequency. Find the ratio of their lengths, and the ratio of the frequencies of their second-lowest standing waves.
3. Loudspeakers a distance d apart are driven in phase at wavelength λ . A listener walks along a line parallel to the line joining the loudspeakers, at a distance D from it, with D much larger than d and than λ . Find the distance between neighbouring quiet positions near the point opposite the midpoint of the loudspeakers.
4. A grating of slit spacing d is illuminated with waves of the two wavelengths λ and λ' . Show that the maximum of order 2 for λ lies in the same direction as the maximum of order 3 for λ' if $2\lambda = 3\lambda'$. Show also that the highest order that can be seen with the wavelength λ is the largest integer less than d/λ .

§4.2 Superposition of Oscillations

1. Let three oscillations along the same line have the same amplitude A and angular frequency ω , and the phase constants 0 , $\Delta\phi$, and $2\Delta\phi$. Show that the amplitude of their sum is $A|1+2\cos\Delta\phi|$, and find the value of $\Delta\phi$ between 0 and π for which the sum vanishes. Draw the phasor diagram for this value.
2. Oscillations with amplitudes $A_a > A_b$ and nearby frequencies ω_a and ω_b are added. Show that the phase of the sum, measured relative to the phase $\omega_a t$ of the larger oscillation, swings between $-\sin^{-1}(A_b/A_a)$ and $\sin^{-1}(A_b/A_a)$ during each beat.
3. Show that the axes of the ellipse traced by $y = A_y \cos \omega t$, $z = A_z \cos(\omega t + \Delta\phi)$ make an angle θ_e with the y direction (and θ_e for ψ throughout the problem and its solution), where $\tan 2\psi = 2A_y A_z \cos \Delta\phi / (A_y^2 - A_z^2)$. Check the result against Figure 52.
4. Show that the Lissajous figure $y = A_y \cos \omega t$, $z = A_z \cos 2\omega t$ is the arc of the parabola $z/A_z = 2y^2/A_y^2 - 1$ between $y = -A_y$ and $y = A_y$, traced back and forth.
5. Show that the mean of x^2 over a time much longer than $2\pi/\omega_m$

for the amplitude-modulated oscillation of this section is $\frac{1}{2}A^2(1 + \frac{1}{2}h^2)$. Hence show that the fraction of the mean square that lies in the two sidebands is $h^2/(2 + h^2)$, which is at most one third.

§4.3 Reflection and Transmission at a Boundary

1. A string of impedance Z occupies $x < 0$, and its end is tied to a light ring that slides without friction on a smooth rod along the line $x = 0$. Show that $\partial y / \partial x = 0$ at the end, and hence that $\mathcal{R} = 1$ and that the ring oscillates with twice the amplitude of the incident wave.
2. A harmonic wave travels on the second string of this section towards the junction, from the right. Show that its amplitude reflection and transmission coefficients are $\mathcal{R}' = -\mathcal{R}$ and $\mathcal{T}' = 2Z_2/(Z_1 + Z_2)$, and that $\mathcal{T}\mathcal{T}' = 1 - \mathcal{R}^2$.
3. A bead of mass m at the junction of two identical strings of impedance Z is also attached to a spring of constant κ , whose other end is fixed. Show that the fraction of the incident power that is transmitted is $\mathcal{T}_P = 4Z^2/[4Z^2 + (m\omega - \kappa/\omega)^2]$. Show that the whole of the power is transmitted when $\omega = \omega_0$, and find the frequencies at which half of it is transmitted.
4. A pulse $y = \eta_i(x - v_1t)$ crosses the junction of this section. Using the result of Section 3.4 that the kinetic and potential energies of a travelling wave are equal, so that the energy per unit length is $\mu(\partial y / \partial t)^2$, show that the energy of the incident pulse is $\tau \int \eta_i'(s)^2 ds$, and that the energies of the reflected and transmitted pulses add up to this value.
5. A longitudinal pulse $\xi = \eta_i(x - vt)$ travels along a bar that occupies $x < 0$ towards its free end at $x = 0$, where the force $YS \partial \xi / \partial x$ must vanish. Show that the reflected pulse is $\eta_i(-x - vt)$, and that each part of the pulse returns from the end with its strain reversed in sign, so that a compression is reflected as a tension.

§4.4 Impedance Matching and Evanescent Waves

1. A string of impedance Z ends on a light ring attached to a piston in oil of damping coefficient b , as in Figure 59. Show that the

fraction of the incident power absorbed by the piston is $4bZ/(b+Z)^2$, and that it is largest when $b = Z$.

2. The ring at the end of the string is attached instead to a spring of constant κ , whose other end is fixed. Show that $|\mathcal{R}| = 1$, and find the change of phase on reflection. Show that the end behaves as a fixed end at low frequencies and as a free end at high frequencies.
3. Show that when the middle string of the quarter-wave matching section is half a wavelength long, at the frequency $2\omega_q$, its input impedance is Z_3 , so that the fraction of the power reflected is the same as if the first and last strings were joined directly. Show also that, at ω_q , a middle string of impedance Z_2 reflects the fraction $[(Z_1Z_3 - Z_2^2)/(Z_1Z_3 + Z_2^2)]^2$ of the incident power.
4. On the stretch of string on an elastic foundation the displacement is $\text{Re}[(\mathcal{B}_+e^{-\beta x} + \mathcal{B}_-e^{\beta x})e^{i\omega t}]$. Using the result of Problem 7 of Section 1.2 of Vol. 4 for the mean of a product of two oscillations, show that the mean power carried towards positive x is $\tau\beta\omega \text{Im}(\mathcal{B}_+\mathcal{B}_-^*)$, and that it is the same at every point of the stretch.
5. A long string on an elastic foundation, occupying $x > 0$, is held at rest with the displacement y_0 at $x = 0$. Show that its displacement is $y_0e^{-x\sqrt{\kappa_f/\tau}}$, and compare the distance $\sqrt{\tau/\kappa_f}$ with the decay length $1/\beta$ of an evanescent wave at low frequencies.

§4.5 Standing Waves in Strings, Pipes, and Cavities

1. For the standing wave $y = 2A \cos kx \cos \omega t$ on a string of mass per unit length μ , show that the total energy between two neighbouring nodes is $\frac{1}{2}\mu\omega^2 A^2 \lambda$ at every instant, and that this is the energy that two travelling waves of amplitude A carry, on average, in a length of half a wavelength.
2. The end $x = 0$ of a string fixed at $x = L$ is attached to a ring of negligible mass on a smooth rod, as in Figure 66, and a driver applies to the ring the transverse force $F_0 \cos \omega t$. Use the ratio $F/\dot{y} = -iZ \cot kL$ found in the text to find the amplitude of the velocity of the ring, and show that it is greatest at the frequencies of a string fixed at $x = L$ and free at $x = 0$. Show also that at

low frequencies the string acts on the ring as a spring with the spring constant τ/L .

3. In Kundt's experiment, a metal bar of length L_b is clamped at its midpoint and stroked along its length so that it vibrates longitudinally in its lowest mode. A light disc on one end of the bar drives the air in a glass tube closed at its far end, and a fine powder spread along the tube gathers in heaps at the nodes of displacement, which are a distance Δx apart (and in the statement and solution $c = \Delta x \sqrt{Y/\rho}/L_b$, the wavelength in the air $2\Delta x$, and $c = 2f\Delta x$). Using the condition $\partial\xi/\partial x = 0$ at a free end of the bar, where the stress $Y \partial\xi/\partial x$ of Section 3.3 vanishes, show that the speed of sound in the air is $c = s\sqrt{Y/\rho}/L_b$.
4. Find the four lowest angular frequencies of the modes of a cubic cavity of side L with rigid walls, and the number of modes that have each of them.
5. A flat cavity with rigid walls has L_z much smaller than L_x and L_y . Show that the modes with angular frequency below $\pi c/L_z$ all have $n_z = 0$, and that when ω/c is also large compared with π/L_x and π/L_y the number of them with angular frequency below ω is approximately $L_x L_y \omega^2 / 4\pi c^2$.

§4.6 Interference, Coherence, and Diffraction

1. The wires to one of the two loudspeakers of the text are reversed, so that the loudspeakers vibrate in antiphase. Show that the intensity far away is $4I_1 \sin^2(\Delta\phi/2)$, and find the directions in which it is zero and those in which it is greatest. What happens at low frequencies, when d is much smaller than a wavelength?
2. By writing $|\Sigma|^2 = \Sigma\Sigma^*$ as a double sum over the loudspeakers, show that the intensity of a row of N loudspeakers, averaged over a period 2π of $\Delta\phi$, is NI_1 .
3. A row of N sources with spacing d emits two wavelengths, λ and $\lambda + \Delta\lambda$, which are said to be resolved when the principal maximum of order q of one falls on the first zero beside the principal maximum of the same order of the other. Show that the smallest $\Delta\lambda$ that can be resolved satisfies $\lambda/\Delta\lambda = nN$.

4. Two long parallel openings, each of width w , have their centres a distance d apart. By integrating the contributions of the strips of width dx' across each opening, show that the intensity is $I_0 \cos^2(\Delta\phi/2)(\sin\beta/\beta)^2$, with $\Delta\phi = kd \sin\theta$ and β as in the text, where I_0 is the intensity at $\theta = 0$. Show that when d/w is a whole number, every principal maximum whose order is a multiple of d/w is missing.
5. Two loudspeakers are fed from one signal whose band of frequencies has the width Δf about the frequency f . Show that the number of maxima of the pattern that can be seen on either side of $\theta = 0$ is about $f/\Delta f$.

Waves in Two and Three Dimensions, Dispersion, and Wave Packets

5.1 Waves in Two and Three Dimensions

In this part we turn from waves that travel along a line to waves that spread out in two and three dimensions. This section first writes the wave equation in three dimensions and finds its plane and spherical waves, then derives the laws of reflection and refraction at a boundary, and finally describes the polarisation of a transverse wave. The two sections that follow return to one dimension and take up media in which waves of different wavelengths travel at different speeds: Section 5.2 defines the phase and group velocities, and Section 5.3 builds a pulse from harmonic waves and follows it as it travels through such a medium.

Refraction and Polarisation

At school, refraction is usually met for light and for water waves. When a wave crosses the boundary between two media in which it travels at different speeds, its frequency stays the same, its wavelength changes in proportion to its speed, and its direction of travel changes. The angles that the directions of travel make with the nor-

mal to the boundary, θ_1 in the first medium and θ_2 in the second, are related by Snell's law,

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2}, \quad (5.1)$$

where v_1 and v_2 are the speeds of the wave in the two media. For light the ratio of speeds is written as the ratio of refractive indices n_2/n_1 , so that Snell's law takes the form $n_1 \sin \theta_1 = n_2 \sin \theta_2$. A wave that passes into a medium where it travels more slowly is bent towards the normal. A wave that passes into a medium where it travels faster is bent away from the normal, and when the angle of incidence exceeds the critical angle θ_c , given by $\sin \theta_c = v_1/v_2$, no wave is transmitted and the whole of the wave is reflected, which is called total internal reflection. Part of a wave is also reflected at every boundary, with the angle of reflection equal to the angle of incidence.

A transverse wave travelling in a given direction can vibrate in any direction perpendicular to it. A wave whose vibration is confined to one such direction is said to be polarised, as light is after passing through a polarising filter, and a longitudinal wave, whose vibration is along its direction of travel, cannot be polarised.

The Wave Equation in Three Dimensions

Let $\psi(\mathbf{r}, t)$ be a disturbance that spreads through a uniform medium in three dimensions, such as the excess pressure in the air or the displacement of a stretched membrane from its plane, which depends on two coordinates only. In one dimension the wave equation is $\partial^2 \psi / \partial t^2 = v^2 \partial^2 \psi / \partial x^2$. A medium that is the same in every direction picks out no axis, so the second derivative with respect to x must be replaced by a combination of second derivatives that does not depend on how the axes are oriented. The Laplacian $\nabla^2 = \partial^2 / \partial x^2 + \partial^2 / \partial y^2 + \partial^2 / \partial z^2$ of Section 3.1 of Vol. 4 of the Fermi Notes on Physics is such a combination, since it is the divergence of the gradient, and each of those two derivatives is defined without reference to the axes. The wave equation in three dimensions is taken to be

$$\frac{\partial^2 \psi}{\partial t^2} = v^2 \nabla^2 \psi. \quad (5.2)$$

You will show in Problem 1 that the small transverse displacement of a stretched membrane obeys the two-dimensional form of this equation, in which ∇^2 has only its first two terms.

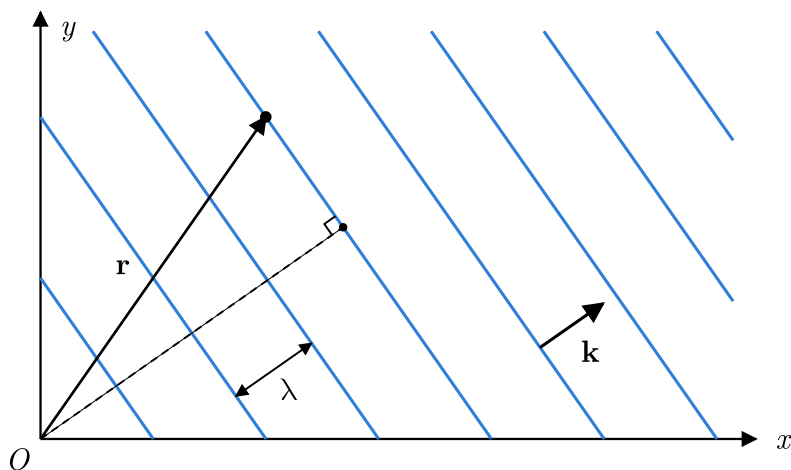


Figure 75. The crests of a plane wave at one instant, drawn where they cut the xy -plane. The wave vector $\mathbf{k}$ is perpendicular to the crests, which are a wavelength λ apart, and every point on a crest, such as the one marked, has a position vector $\mathbf{r}$ with the same projection on the direction of $\mathbf{k}$.

Let $\mathbf{k}$ be a constant vector, and let us try a solution of the form

$$\psi(\mathbf{r}, t) = \text{Re} [\mathcal{A} e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})}] . \quad (5.3)$$

Since $\mathbf{k} \cdot \mathbf{r} = k_x x + k_y y + k_z z$, differentiating the exponential twice with respect to x multiplies it by $-k_x^2$, and similarly for y and z , so that ∇^2 multiplies it by $-(k_x^2 + k_y^2 + k_z^2) = -k^2$, where $k = |\mathbf{k}|$. Differentiating twice with respect to time multiplies it by $-\omega^2$, and the wave equation is satisfied provided that $\omega = vk$. At a given instant the phase $\omega t - \mathbf{k} \cdot \mathbf{r}$ is the same at every point for which $\mathbf{k} \cdot \mathbf{r}$ has a given value. These points form a plane perpendicular to $\mathbf{k}$,

since $\mathbf{k} \cdot \mathbf{r}$ is k multiplied by the projection of $\mathbf{r}$ on the direction of $\mathbf{k}$, as Figure 75 shows. The crests are therefore planes perpendicular to $\mathbf{k}$, a distance $\lambda = 2\pi/k$ apart, and they move in the direction of $\mathbf{k}$ at the speed $\omega/k = v$. This solution is called a plane wave, and $\mathbf{k}$ is called its wave vector. A plane wave travelling along the x -axis has $\mathbf{k} = k \hat{\mathbf{x}}$, and it is the harmonic travelling wave of Section 3.2.

Spherical Waves

A small source in a uniform medium, such as a small sphere whose radius oscillates, sends out waves that are the same in every direction, and the disturbance then depends only on the distance r from the source and on the time. For a function of r alone, the Laplacian in spherical coordinates, worked out in Section 3.4 of Vol. 4, has only its first term,

$$\nabla^2 \psi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) = \frac{\partial^2 \psi}{\partial r^2} + \frac{2}{r} \frac{\partial \psi}{\partial r}, \quad (5.4)$$

where the second form follows from the product rule. The second derivative of the product $r\psi$ is, again by the product rule, $\partial^2(r\psi)/\partial r^2 = r \partial^2 \psi / \partial r^2 + 2 \partial \psi / \partial r$, which is r multiplied by the Laplacian. The wave equation can therefore be rewritten as

$$\frac{\partial^2(r\psi)}{\partial t^2} = v^2 \frac{\partial^2(r\psi)}{\partial r^2}, \quad (5.5)$$

which is the wave equation in one dimension for the product $r\psi$, with r in place of x . Its general solution is d'Alembert's solution, which Section 3.2 took from Vol. 4, and therefore

$$\psi(r, t) = \frac{f(r - vt)}{r} + \frac{f_{\text{in}}(r + vt)}{r}, \quad (5.6)$$

where f_1 and f_2 are arbitrary functions. The first term is a wave travelling outwards from the source and the second a wave travelling inwards towards it, and a wave of either kind is called a spherical wave. A small source in an unbounded medium produces only the

outgoing wave, and its harmonic form is $\psi = \text{Re} [(\mathcal{A}/r) e^{i(\omega t - kr)}]$ with $\omega = vk$.

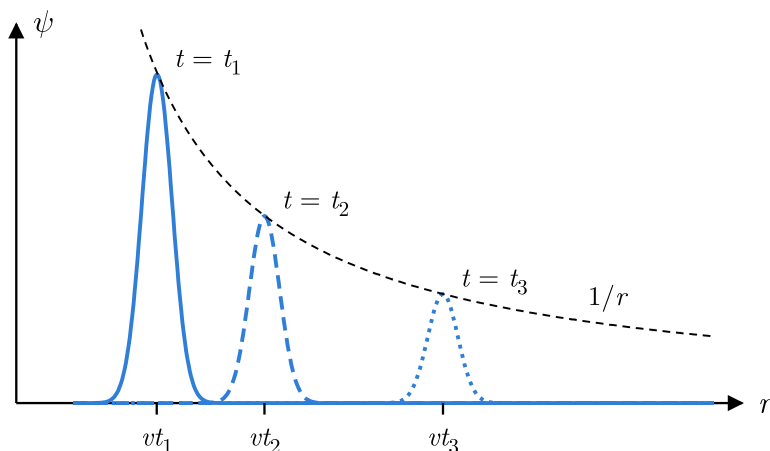


Figure 76. An outgoing spherical pulse $f_1(r - vt)/r$ drawn against r at three times. The pulse keeps its shape, and its height falls as $1/r$, following the dashed curve.

The outgoing pulse, drawn in Figure 76, travels at the speed v and keeps its shape, and its height falls as $1/r$. This fall can also be understood from the conservation of energy. Section 3.4 found the energy per unit length of a string, and per unit volume of a longitudinal wave, to be proportional to the square of the amplitude, so far from the source, where the pulse is thin compared with its distance from the source, the energy per unit volume in the pulse is $1/r^2$ multiplied by a function of $r - vt$, since the energy density depends on the derivatives of ψ , and $\partial\psi/\partial t$ and $\partial\psi/\partial r$ are both $1/r$ multiplied by a function of $r - vt$ far from the source. The pulse fills a spherical shell of area $4\pi r^2$ and of fixed thickness, since it keeps its shape, so the energy in the shell is proportional to $4\pi r^2 \times 1/r^2$, which does not depend on r . The same argument applied to a harmonic wave says that the mean power crossing a sphere of radius r centred on the source, which is $4\pi r^2 I$, is the same for every sphere, so that the intensity I goes as $1/r^2$ and the amplitude as $1/r$. This result is the inverse-square law reviewed in Section 3.1.

Cylindrical Waves

A long, straight source, such as a long thin cylinder whose radius oscillates, sends out waves that depend on the distance r from the source and on the time. The same argument now applies to cylinders instead of spheres. The mean power that crosses unit length of a cylinder of radius r around the source, $2\pi r I$, must be the same for every r , so the intensity goes as $1/r$ and the amplitude as $1/\sqrt{r}$.

The argument rests on the conservation of energy, and it holds at distances from the source of many wavelengths. You will show in Problem 3, with the Laplacian in cylindrical coordinates of Section 3.4 of Vol. 4, that at such distances the wave equation has the outgoing solutions $f_1(r - vt)/\sqrt{r}$, and that close to the source a cylindrical pulse, unlike a spherical one, does not keep its shape exactly.

Snell's Law

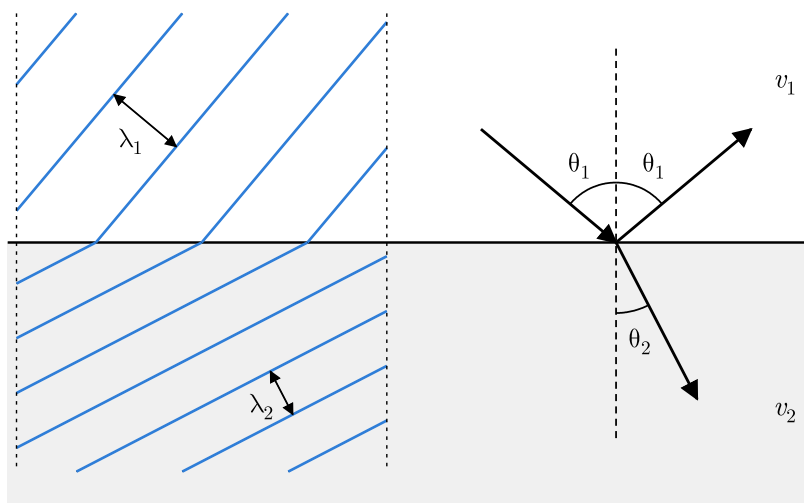


Figure 77. A plane wave crossing a plane boundary into a medium where it travels more slowly. On the left, the crests of the incident and transmitted waves, which meet along the boundary; on the right, the directions of the incident, reflected, and transmitted waves, with the normal to the boundary drawn dashed. The medium in which the wave travels more slowly, the one of speed v_2 , is shaded.

Let the plane $z = 0$ be the boundary between two media, with the wave speed v_1 for $z < 0$ and v_2 for $z > 0$, and let a plane wave of angular frequency ω travel in the first medium towards the boundary, with its wave vector in the xz -plane at an angle θ_1 to the z -axis, which is normal to the boundary. The incident wave is proportional to $e^{i(\omega t - k_x x - k_z z)}$, where $k_x = k_1 \sin \theta_1$ and $k_z = k_1 \cos \theta_1$ are the components of its wave vector, and $k_1 = \omega/v_1$. It produces a reflected plane wave in the first medium and a transmitted plane wave in the second, travelling at angles θ'_1 and θ_2 to the normal. The conditions at the boundary, such as the continuity of the displacement and of the transverse force required at the junction of two strings in Section 4.3, relate the three waves at $z = 0$, where the incident wave goes as $e^{i(\omega t - k_x x)}$, and they must hold for all x and t . This is possible only if every wave has the same dependence on x and on t at the boundary, since exponentials with different exponents are not constant multiples of one another. The reflected and the transmitted waves therefore have the frequency ω and the component k_x of the wave vector along the boundary.

For the reflected wave, $k_1 \sin \theta'_1 = k_1 \sin \theta_1$, so the angle of reflection equals the angle of incidence. For the transmitted wave, whose wave number is $k_2 = \omega/v_2$, the same condition gives $k_2 \sin \theta_2 = k_1 \sin \theta_1$, and dividing by ω gives

$$\frac{\sin \theta_1}{v_1} = \frac{\sin \theta_2}{v_2}, \quad (5.7)$$

which is Snell's law. The condition has a simple geometrical meaning, shown in Figure 77. The crests of the incident wave meet the boundary along lines a distance $\lambda_1/\sin \theta_1$ apart, where $\lambda_1 = 2\pi/k_1$, and the crests of the transmitted wave meet it along lines a distance $\lambda_2/\sin \theta_2$ apart. Since the transmitted wave is driven by the incident wave along the whole boundary, the two sets of crests must meet there, and the two spacings are equal.

Total Internal Reflection

Let us now suppose that the wave travels faster in the second medium, so that $v_2 > v_1$, and that $\sin \theta_1 > v_1/v_2$. Snell's law would then require $\sin \theta_2 > 1$, and no real angle θ_2 satisfies it. The

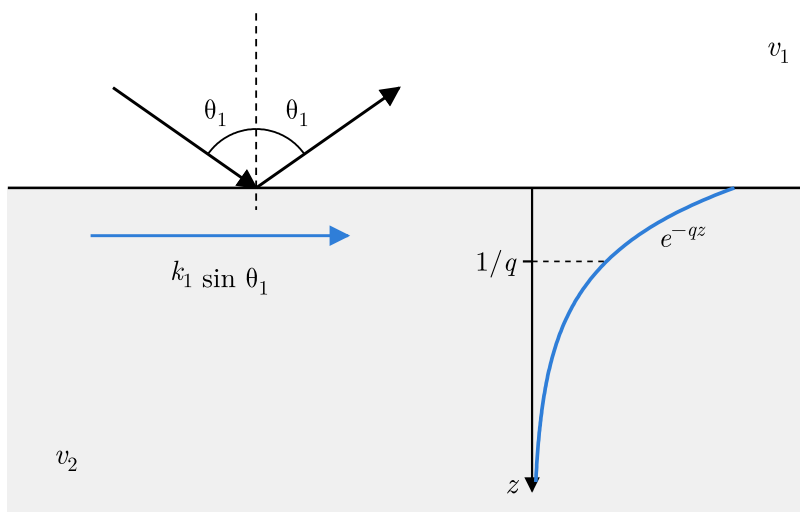


Figure 78. Total internal reflection. The wave in the upper medium is reflected with the angle of reflection equal to the angle of incidence. In the lower medium the disturbance travels along the boundary with wave number $k_1 \sin \theta_1$, and its amplitude, drawn against the depth z on the right, falls by a factor of e in a distance $1/q$. The medium in which the wave travels faster, the one of speed v_2 , is shaded.

argument that every wave must have the same dependence on x and on t at the boundary still holds, however, and the disturbance in the second medium must still go as $e^{i(\omega t - k_x x)}$ along the boundary. Let it be proportional to $e^{i(\omega t - k_x x)}\zeta(z)$, where ζ is a function to be determined. Substituting this into the wave equation of the second medium, the derivatives with respect to t and x multiply it by $-\omega^2$ and $-k_x^2$, and we find

$$\frac{d^2\zeta}{dz^2} = \left(k_x^2 - \frac{\omega^2}{v_2^2}\right)\zeta = q^2\zeta, \quad q = \sqrt{k_1^2 \sin^2 \theta_1 - k_2^2}, \quad (5.8)$$

where q is real and positive because $k_1 \sin \theta_1 > k_1 v_1 / v_2 = k_2$. The solutions of this equation are e^{-qz} and e^{qz} , and the second grows without limit as z increases, so it is excluded. The disturbance in the second medium is therefore

$$\psi \propto e^{-qz} \cos(\omega t - k_1 x \sin \theta_1 + \phi), \quad (5.9)$$

where ϕ is a phase constant. It travels along the boundary and dies away exponentially with distance from it, as Figure 78 shows, and it is an evanescent wave of the kind met on the string on an elastic foundation in Section 4.4. The distance $1/q$ over which it falls by a factor of e is of the order of a wavelength, except close to the critical angle, where q tends to zero. In the steady state the energy in the second medium does not change, and none is carried away to large z , where the disturbance vanishes, so the mean power crossing the boundary is zero and all the incident power is reflected.

Polarisation of Waves on a String

A string stretched along the x -axis can be displaced in the y -direction and in the z -direction. Let $y(x, t)$ and $z(x, t)$ be the displacements of the string in these two directions. For small slopes the tension produces a force $\tau \partial^2 y / \partial x^2$ per unit length in the y -direction and a force $\tau \partial^2 z / \partial x^2$ per unit length in the z -direction, as in Section 3.2, and each displacement obeys the wave equation separately, with the same speed v . A harmonic wave travelling in the positive x -direction therefore has

$$y = A_y \cos(\omega t - kx), \quad z = A_z \cos(\omega t - kx + \Delta\phi), \quad (5.10)$$

where A_y and A_z are the amplitudes of the two components and $\Delta\phi$ is the phase difference between them. At a fixed point of the string the two displacements are two perpendicular oscillations of the same frequency, and Section 4.2 showed that the point then moves on an ellipse. There are three cases of particular interest, drawn in Figure 79.

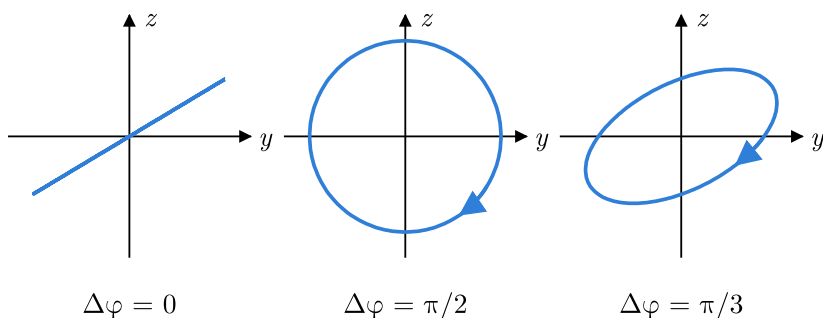


Figure 79. The path followed by one point of a string carrying a harmonic wave, drawn in the plane of the displacements y and z , for three values of the phase difference between the components. The amplitudes are unequal in the first and third panels and equal in the second, and in the second and third panels the arrows show the sense in which the point goes round.

When $\Delta\phi = 0$ or π , the ratio z/y is the constant $\pm A_z/A_y$, and every point of the string moves back and forth along the same line, which makes the angle $\tan^{-1}(\pm A_z/A_y)$ with the y -axis. The wave is then said to be linearly polarised. When $A_y = A_z$ and $\Delta\phi = \pm\pi/2$, we have $z = \mp A_y \sin(\omega t - kx)$, and therefore $y^2 + z^2 = A_y^2$, since $\cos^2(\omega t - kx) + \sin^2(\omega t - kx) = 1$, and every point of the string moves round a circle at the angular frequency ω . The wave is then said to be circularly polarised, and at a given instant the string has the shape of a helix. In every other case each point moves round an ellipse, and the wave is said to be elliptically polarised. You will show in Problem 5 that a linearly polarised wave is the sum of two circularly polarised waves turning in opposite senses.

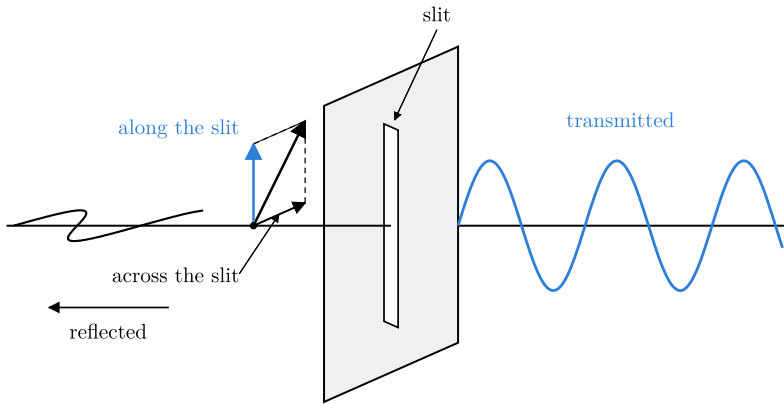


Figure 80. A polariser for waves on a string: a board with a narrow slit through which the string passes. At the point marked on the near side the displacement of the string, the longer arrow, is resolved into a component along the slit (vertical) and a component across it. The sides of the slit press on the string only across the slit, so the component along the slit passes through and travels on beyond the board, while the component across the slit meets what is in effect a fixed end and is reflected back along the string.

A polariser for waves on a string can be made by passing the string through a narrow slit in a board, as in Figure 80. The sides of the slit exert forces only perpendicular to the slit, so the component of the displacement along the slit passes through unchanged, while the component across it meets a fixed end and is reflected, as in Section 4.3.

5.2 Dispersion, Phase Velocity, and Group Velocity

On a stretched string every harmonic wave travels at the same speed v , and a pulse of any shape travels without changing its shape, as d'Alembert's solution in Section 3.2 shows. In the chain of Section 2.3 and on the string on an elastic foundation of Section 4.4, waves of different wavelengths travel at different speeds, and a pulse changes its shape as it travels. We describe such media here by the relation between the frequency and the wave number of their harmonic waves, and find the speed at which a group of waves, and the energy it carries, travels through them.

The Dispersion Relation

A harmonic wave $\text{Re} [\mathcal{A}e^{i(\omega t - kx)}]$ satisfies the equation of motion of a medium only when its angular frequency and its wave number are related in a particular way. The relation $\omega(k)$ is called the dispersion relation of the medium. For the stretched string it is $\omega = vk$, and the following examples, which do not exhaust the possibilities, show some of the other forms it can take:

1. the chain of masses and springs of Section 2.3, and the loaded string, for which $\omega = 2\Omega \sin(ka/2)$ for $0 < k \leq \pi/a$;
2. the string on an elastic foundation of Section 4.4, for which $\omega^2 = v^2k^2 + \omega_c^2$;
3. a stiff string, and water waves on deep water, which we treat below.

The crests of the wave are points at which the phase $\omega t - kx$ has a fixed value, so a crest moves according to $x = (\omega t - \text{constant})/k$, at the speed

$$v_p = \frac{\omega}{k}, \quad (5.11)$$

which is called the phase velocity. A medium in which v_p depends on k is said to be dispersive, and the stretched string, with $v_p = v$ for every k , is non-dispersive. In a dispersive medium the harmonic components of a pulse drift apart as they travel, which is why the shape of the pulse changes. The phase velocity is the slope of the straight line drawn from the origin to the point (k, ω) on the graph of the dispersion relation, as Figures 81 and 82 show.

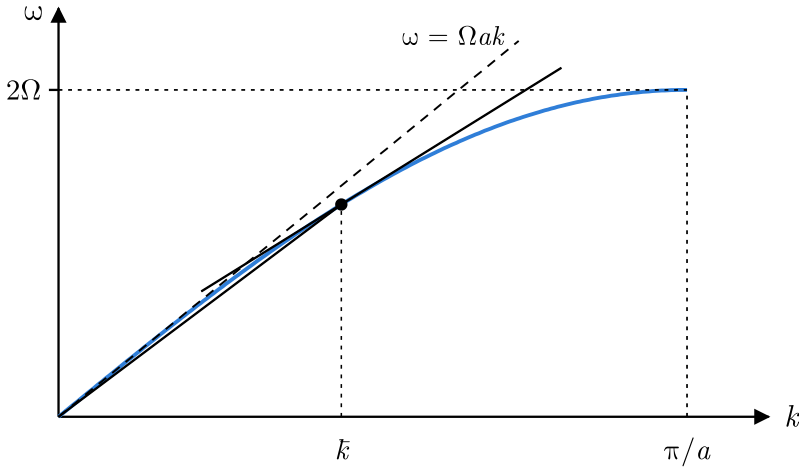


Figure 81. The dispersion relation $\omega = 2\Omega \sin(ka/2)$ of a chain of masses and springs (accent), with the straight line $\omega = \Omega ak$ that it approaches at long wavelengths (dashed). At the wave number $\bar{k}$, the slope of the line from the origin is the phase velocity and the slope of the tangent is the group velocity, which is defined later in this section. The curve is horizontal at the highest frequency 2Ω , where $k = \pi/a$, and the group velocity is zero there.

For the chain, the phase velocity is

$$v_p = \frac{2\Omega \sin(ka/2)}{k}. \quad (5.12)$$

At long wavelengths, where $ka \ll 1$, the sine can be replaced by its argument, and $v_p \rightarrow \Omega a$. For the loaded string, with $\Omega^2 = \tau/ma$ and $\mu = m/a$, this limit is $\sqrt{\tau a/m} = \sqrt{\tau/\mu} = v$, the speed of waves on the continuous string of Section 2.4. As k increases the phase

velocity falls, since $\sin(ka/2)$ grows more slowly than $ka/2$, and at $k = \pi/a$ it is $2\Omega a/\pi$. The graph of the dispersion relation, drawn in Figure 81, bends below the straight line $\omega = \Omega ak$ because the chain is not a continuous medium; when the wavelength is comparable to the spacing a , the masses no longer follow a smooth curve.

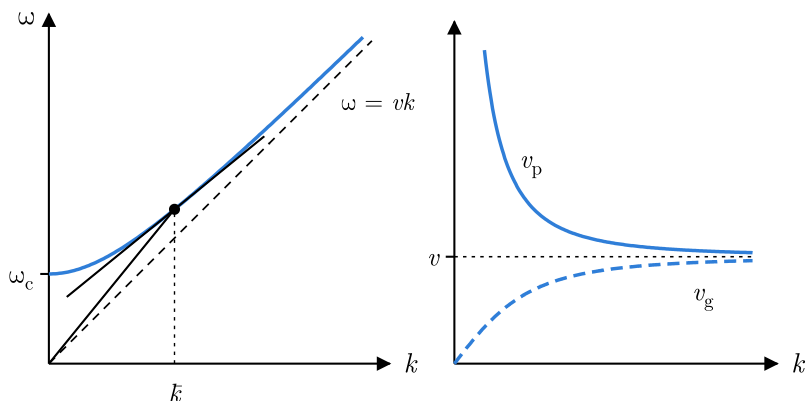


Figure 82. Left: the dispersion relation $\omega^2 = v^2 k^2 + \omega_c^2$ of a string on an elastic foundation (accent), which approaches the dashed line $\omega = vk$ at large k . At the wave number $\bar{k}$ the line from the origin, whose slope is the phase velocity, is steeper than the tangent, whose slope is the group velocity. Right: the phase velocity (solid) and the group velocity (dashed) against k . The phase velocity is always greater than v and the group velocity always less, and the two velocities tend to v at large k .

For the string on an elastic foundation,

$$v_p = \frac{\sqrt{v^2 k^2 + \omega_c^2}}{k} = \sqrt{v^2 + \frac{\omega_c^2}{k^2}}, \quad (5.13)$$

which is always greater than v and grows without limit as $k \rightarrow 0$. At $k = 0$ the whole string moves up and down in phase at the cutoff frequency ω_c , held only by the foundation, and every point reaches its crest at the same instant. At short wavelengths the tension dominates the restoring force, and $v_p \rightarrow v$.

A real string, such as a steel piano string, resists being bent as well as being stretched, and its equation of motion is

$$\mu \frac{\partial^2 y}{\partial t^2} = \tau \frac{\partial^2 y}{\partial x^2} - \beta \frac{\partial^4 y}{\partial x^4}, \quad (5.14)$$

where D is a constant, called the flexural rigidity (with D replacing β in the equation of motion, the dispersion relation, the sentence on k^2 comparable to τ/D , and Problem 4 and its solution), which for a wire of circular cross-section is proportional to Young's modulus and to the fourth power of the radius of the wire. The derivation of the last term, which rests on the relation between the curvature of a bent wire and the moment needed to bend it, is beyond the scope of this course. Substituting the harmonic wave, each derivative with respect to x multiplies it by $-ik$, and each derivative with respect to t by $i\omega$, so that $\mu\omega^2 = \tau k^2 + \beta k^4$. Dividing by μ and taking the square root gives

$$\omega = vk\sqrt{1 + \frac{\beta k^2}{\tau}}, \quad v_p = v\sqrt{1 + \frac{\beta k^2}{\tau}}. \quad (5.15)$$

Here the phase velocity rises with k , and the stiffness affects the waves appreciably only when k^2 is comparable to τ/β , that is, when the wavelength is short enough that bending the wire takes an appreciable force compared with the tension.

As a last example, waves on the surface of water much deeper than their wavelength, and long enough for the surface tension of the water to be negligible, have the dispersion relation

$$\omega^2 = gk, \quad (5.16)$$

where g is the acceleration due to gravity. Its derivation needs the equations of motion of a fluid with a free surface, and is beyond the scope of this course. The phase velocity $\sqrt{g/k}$ falls as k rises, so long waves on the ocean travel faster than short ones.

Two Waves of Nearby Frequencies

To find how a group of waves travels through a dispersive medium, let us add two harmonic waves of equal amplitude A whose wave

numbers, $\bar{k} + \frac{1}{2}\Delta k$ and $\bar{k} - \frac{1}{2}\Delta k$, are close together, where $\Delta k \ll \bar{k}$. Let their angular frequencies, given by the dispersion relation, be $\bar{\omega} + \frac{1}{2}\Delta\omega$ and $\bar{\omega} - \frac{1}{2}\Delta\omega$. The trigonometric identity $\cos a + \cos b = 2\cos\frac{1}{2}(a-b)\cos\frac{1}{2}(a+b)$, which follows from the formulas for the cosine of a sum and of a difference, gives

$$\begin{aligned}
 & A \cos \left[(\bar{\omega} + \tfrac{1}{2}\Delta\omega)t - (\bar{k} + \tfrac{1}{2}\Delta k)x \right] \\
 & + A \cos \left[(\bar{\omega} - \tfrac{1}{2}\Delta\omega)t - (\bar{k} - \tfrac{1}{2}\Delta k)x \right] \\
 & = 2A \cos \left[\tfrac{1}{2}(\Delta\omega t - \Delta k x) \right] \cos(\bar{\omega}t - \bar{k}x).
 \end{aligned} \tag{5.17}$$

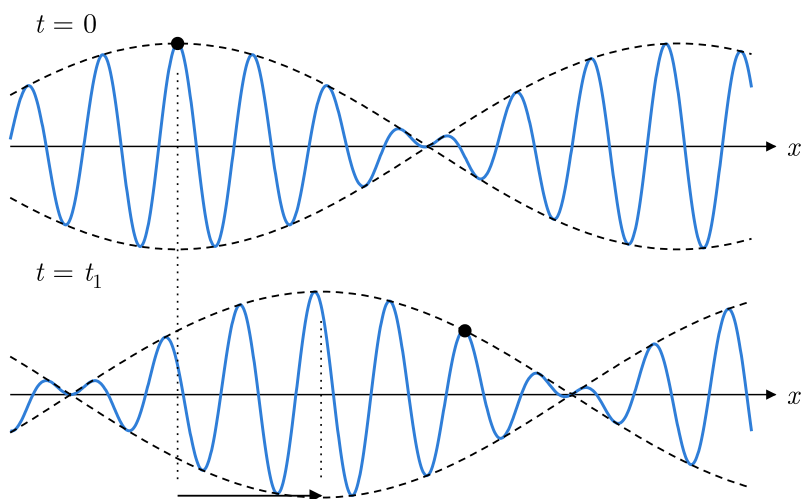


Figure 83. The sum of two waves of equal amplitude on a string on an elastic foundation, with wave numbers $k \pm \frac{1}{2}\Delta k$, at $t = 0$ and at a later time t_1 . The dashed curves are the envelope. The dot marks one crest, which moves at the phase velocity $\bar{\omega}/\bar{k}$, and the arrow shows the distance moved by the envelope, which moves at $\Delta\omega/\Delta k$, about half as fast in this example.

The sum, drawn in Figure 83, is a wave of wave number $\bar{k}$ whose amplitude varies slowly along the string, between 0 and $2A$, over a distance of $2\pi/\Delta k$. It is the counterpart in space of the beats of Section 4.2, and the same sum was formed for two matter waves in Problem 1 of Section 2.4 of Vol. 3 of the Fermi Notes on Physics. The slowly varying factor $2A \cos[\frac{1}{2}(\Delta\omega t - \Delta k x)]$ is called the en-

velope. Its value stays the same at points where $\Delta\omega t - \Delta k x$ is constant, so the envelope moves at the speed $\Delta\omega/\Delta k$. The crests within it move at the phase velocity $\bar{\omega}/\bar{k}$. In a dispersive medium these two speeds differ, and crests appear at one end of each group, move through it, and disappear at the other end, as the dot in Figure 83 does.

Group Velocity

As $\Delta k \rightarrow 0$, the ratio $\Delta\omega/\Delta k$ tends to the derivative of the dispersion relation at $\bar{k}$. The speed

$$v_g = \frac{d\omega}{dk} \quad (5.18)$$

is called the group velocity, and it is the slope of the tangent to the graph of the dispersion relation, as in Figures 81 and 82. For a non-dispersive medium $\omega = vk$, and $v_g = v_p = v$. In general, writing $\omega = kv_p$ and using the product rule gives

$$v_g = v_p + k \frac{dv_p}{dk}. \quad (5.19)$$

When the phase velocity falls as the wave number rises, $v_g < v_p$, and the dispersion is called normal. When the phase velocity rises with the wave number, $v_g > v_p$, and the dispersion is called anomalous. The names were first used in optics, where normal dispersion is the more common.

For the chain, differentiating $2\Omega \sin(ka/2)$ gives

$$v_g = \Omega a \cos \frac{ka}{2}, \quad (5.20)$$

which is Ωa at long wavelengths and falls to zero at $k = \pi/a$, where the frequency takes its highest value 2Ω . The reason is that at $k = \pi/a$ the displacement of mass j is proportional to $e^{i(\omega t - \pi j)} = (-1)^j e^{i\omega t}$, so neighbouring masses move in opposite directions. The wave with $k = -\pi/a$ describes the same motion, since $e^{i\pi j} = e^{-i\pi j}$, and a wave travelling to the right is therefore indistinguishable from

one travelling to the left. The motion is a standing wave, which carries no energy along the chain. The chain has normal dispersion, and you will show in Problem 2 that $v_g < v_p$ for every k between 0 and π/a .

For the string on an elastic foundation, differentiating $\omega^2 = v^2 k^2 + \omega_c^2$ with respect to k gives $2\omega d\omega/dk = 2v^2 k$, and therefore

$$v_g = \frac{v^2 k}{\omega}, \quad v_p v_g = v^2. \quad (5.21)$$

The group velocity is less than v , since $\omega > vk$, and it falls to zero at the cutoff frequency, as Figure 82 shows, while the phase velocity grows without limit there. The dispersion is normal. For the stiff string the phase velocity rises with k , the dispersion is anomalous, and $v_g > v_p$, as you will check in Problem 4. For deep water, $\omega = \sqrt{gk}$ gives $v_g = \frac{1}{2}\sqrt{g/k} = \frac{1}{2}v_p$, so the crests move through a group of waves at twice the speed of the group, and the dispersion is normal.

Energy Travels at the Group Velocity

The envelope marks where the waves are large, and therefore where their energy is, so we would expect the energy to travel at the group velocity. We can check this for the string on an elastic foundation. The equation of motion of Section 4.4 is

$$\mu \frac{\partial^2 y}{\partial t^2} = \tau \frac{\partial^2 y}{\partial x^2} - \kappa_f y, \quad (5.22)$$

and the energy per unit length of the string is the sum of the kinetic energy per unit length $\frac{1}{2}\mu(\partial y/\partial t)^2$, the potential energy of the stretched string $\frac{1}{2}\tau(\partial y/\partial x)^2$ found in Section 3.4, and the potential energy $\frac{1}{2}\kappa_f y^2$ stored in the foundation, which acts on each piece of string as a spring of stiffness κ_f per unit length. Let $y = A \cos(\omega t - kx)$. Then $\partial y/\partial t = -A\omega \sin(\omega t - kx)$ and $\partial y/\partial x = Ak \sin(\omega t - kx)$, and the mean of $\sin^2$ and of $\cos^2$ over a cycle is $\frac{1}{2}$, so the mean energy per unit length is

$$\langle \varepsilon \rangle = \frac{1}{4} A^2 (\mu \omega^2 + \tau k^2 + \kappa_f) = \frac{1}{4} \mu A^2 (\omega^2 + v^2 k^2 + \omega_c^2) = \frac{1}{2} \mu \omega^2 A^2, \quad (5.23)$$

where we have used $\tau = \mu v^2$, $\kappa_f = \mu \omega_c^2$, and the dispersion relation.

The energy is carried along the string by the tension. As in Section 3.4, the part of the string to the left of a point pulls on the part to the right with a transverse force $-\tau \partial y / \partial x$, and the rate at which it does work is

$$\mathcal{P} = -\tau \frac{\partial y}{\partial x} \frac{\partial y}{\partial t} = \tau \omega k A^2 \sin^2(\omega t - kx). \quad (5.24)$$

The foundation acts on each piece of string separately, pushing it towards its equilibrium position, and it carries no energy along the string. The mean power is therefore $\langle \mathcal{P} \rangle = \frac{1}{2} \tau \omega k A^2$, and the ratio of the mean power to the mean energy per unit length is

$$\frac{\langle \mathcal{P} \rangle}{\langle \varepsilon \rangle} = \frac{\tau k}{\mu \omega} = \frac{v^2 k}{\omega} = v_g. \quad (5.25)$$

The power carried past a point is the energy per unit length multiplied by the speed at which the energy moves, so the energy moves at the group velocity. At the cutoff frequency, where $k = 0$, the string moves up and down as a whole and carries no energy along its length, which agrees with $v_g = 0$ there. The phase velocity, which is greater than v , therefore describes only the motion of the crests.

5.3 Wave Packets

A pulse of finite length on a string, or a short burst of sound, can be built from harmonic waves with a continuous range of wave numbers, and the result is called a wave packet. In this section we build such packets with the Fourier transform, and then use the dispersion relation of Section 5.2 to follow a packet as it travels through a dispersive medium. Wave packets of matter waves are treated at length in Section 2.4 of Vol. 3, and the mathematics of the classical packets here is the same.

Building a Packet from Harmonic Waves

Let us add harmonic waves travelling in the positive x -direction, with wave numbers spread over a continuous range, and write the displacement as $y = \text{Re } \psi$, where

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathcal{A}(k) e^{i[\omega(k)t - kx]} dk. \quad (5.26)$$

The quantity $\mathcal{A}(k) dk / \sqrt{2\pi}$ is the complex amplitude contributed by the waves with wave numbers between k and $k + dk$, and $\omega(k)$ is the dispersion relation of the medium. Each harmonic wave in the integral satisfies the equation of motion, and the equation of motion is linear, so the integral satisfies it as well. At $t = 0$ the exponential is e^{-ikx} , and substituting $k = -k'$ turns the integral into the inversion formula of Section 6.2 of Vol. 4, with $\mathcal{A}(-k')$ in place of the Fourier transform. The amplitudes are therefore found from the value of ψ at $t = 0$ as

$$\mathcal{A}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(x, 0) e^{ikx} dx. \quad (5.27)$$

We shall consider packets whose displacement at $t = 0$ has the form $y(x, 0) = h(x) \cos \bar{k}x$, where the envelope $h(x)$ varies slowly compared with $\cos \bar{k}x$, and we take $\psi(x, 0) = h(x) e^{-i\bar{k}x}$, whose real part is $y(x, 0)$. Since each component travels in the positive x -direction, this choice also fixes the velocity of each point of the string at $t = 0$. The amplitudes are then

$$\mathcal{A}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} h(x) e^{i(k - \bar{k})x} dx, \quad (5.28)$$

which is the Fourier transform of h evaluated at the wave number $\bar{k} - k$. The amplitudes are therefore concentrated near $k = \bar{k}$, over a range of wave numbers that is narrow when the envelope is long.

As an example, consider the Gaussian envelope $h(x) = e^{-x^2/4\sigma^2}$, where σ is a constant length. In Section 6.2 of Vol. 4 the Fourier transform of a Gaussian was found to be another Gaussian, and the

result found there gives the Fourier transform of h as $\sqrt{2}\sigma e^{-\sigma^2 k^2}$. Evaluating it at the wave number $\bar{k} - k$ gives

$$\mathcal{A}(k) = \sqrt{2}\sigma e^{-\sigma^2(k-\bar{k})^2}. \quad (5.29)$$

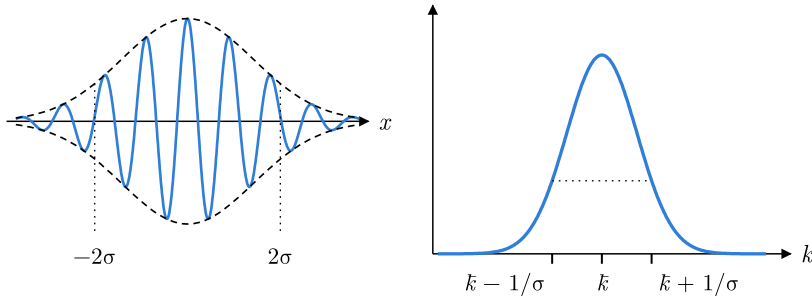


Figure 84. Left: a Gaussian wave packet $e^{-x^2/4\sigma^2} \cos \bar{k}x$ at $t = 0$, with its envelope dashed. The envelope falls to $1/e$ of its greatest value at $x = \pm 2\sigma$. Right: the amplitudes $\mathcal{A}(k)$ of its harmonic components, which fall to $1/e$ of their greatest value at $k = \bar{k} \pm 1/\sigma$.

The packet and its amplitudes are drawn in Figure 84. The envelope falls to $1/e$ of its greatest value at $x = \pm 2\sigma$, and the amplitudes fall to $1/e$ of theirs at $k = \bar{k} \pm 1/\sigma$. For the packet to contain many wavelengths within its envelope, as in the figure, $\bar{k}\sigma$ must be large, and the amplitudes at negative k are then negligible.

The Reciprocal Relation

The Gaussian packet illustrates the reciprocal relation of Section 6.2 of Vol. 4 between the width of a function and the width of its Fourier transform. Let $\Delta x = 4\sigma$ be the length of the packet between the points where the envelope is $1/e$ of its greatest value, and $\Delta k = 2/\sigma$ the corresponding range of wave numbers. The product $\Delta k \Delta x = 8$ does not depend on σ , so a shorter packet needs a wider range of wave numbers.

The same relation holds for packets of every shape, although the value of the product depends on the shape and on how the widths are defined. Consider, for example, a train of harmonic waves of

wave number $\bar{k}$ and length Δx , with a constant envelope along its length and nothing outside it. Its amplitudes have the same form as the spectrum of the finite train of oscillations in Section 2.5, with x in place of t and k in place of ω , and their first zeros lie at $k = \bar{k} \pm 2\pi/\Delta x$. The reason is that the amplitude at the wave number k is proportional to the integral of $e^{i(k-\bar{k})x}$ along the train, and when $k - \bar{k} = \pm 2\pi/\Delta x$ this exponential runs through one whole cycle over the length of the train, so that its integral is zero. In every case, a packet of length Δx contains wave numbers spread over a range Δk whose product with Δx is of the order of 2π ,

$$\Delta k \Delta x \approx 2\pi, \quad (5.30)$$

which is the counterpart in space of the relation $\Delta\omega \Delta t \approx 2\pi$ found in Section 2.5 for a signal in time.

Motion to First Order

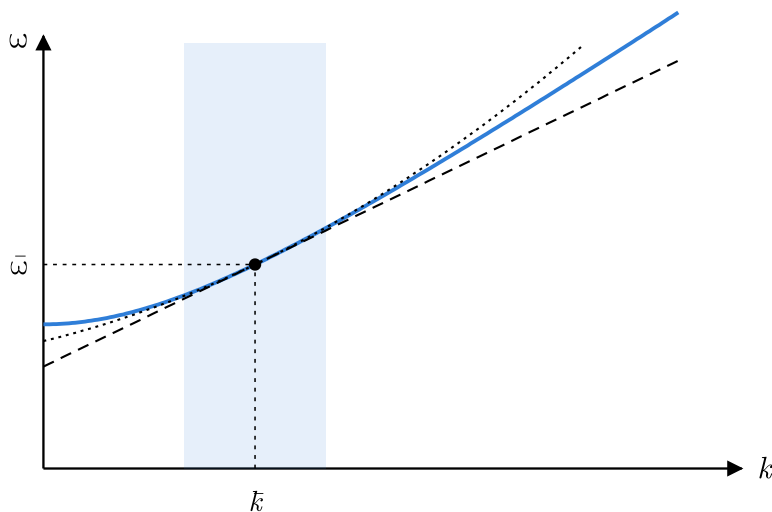


Figure 85. The dispersion relation of a string on an elastic foundation (accent) near the central wave number $\bar{k}$ of a packet, with its approximation to the first order (dashed), which is the tangent at $\bar{k}$, and to the second order (dotted), which is a parabola. The shaded band marks the range of wave numbers that the packet contains, over which the parabola and the curve are hard to tell apart.

When the amplitudes are concentrated near $\bar{k}$, only the values of $\omega(k)$ near $\bar{k}$ matter, and we can expand the dispersion relation about $\bar{k}$ by Taylor's theorem,

$$\omega(k) = \bar{\omega} + v_g(k - \bar{k}) + \frac{1}{2}\omega''(k - \bar{k})^2 + \dots, \quad (5.31)$$

where $\bar{\omega} = \omega(\bar{k})$, and v_g and ω'' are the first and the second derivatives of $\omega(k)$ at $\bar{k}$, so that v_g is the group velocity at $\bar{k}$. Figure 85 compares the first two approximations with the curve itself.

Let us keep only the first two terms. The phase of each component can then be rearranged as

$$\omega(k)t - kx = (\bar{\omega}t - \bar{k}x) - (k - \bar{k})(x - v_gt), \quad (5.32)$$

and the first bracket does not depend on k and comes outside the integral. At $t = 0$ the same rearrangement gives $\psi(x, 0) = e^{-i\bar{k}x}(2\pi)^{-1/2} \int \mathcal{A}(k) e^{-i(k-\bar{k})x} dk$, and since $\psi(x, 0) = h(x) e^{-i\bar{k}x}$, the integral here is $h(x)$. The integral that remains at time t is the same integral with x replaced by $x - v_gt$, and therefore

$$\psi(x, t) = h(x - v_gt) e^{i(\bar{\omega}t - \bar{k}x)}, \quad y(x, t) = h(x - v_gt) \cos(\bar{\omega}t - \bar{k}x). \quad (5.33)$$

To the first order, the envelope moves without change of shape at the group velocity, and the crests within it move at the phase velocity $\bar{\omega}/\bar{k}$. This generalises the result of Section 5.2 for two waves to a packet of any shape. The approximation holds as long as the neglected term $\frac{1}{2}\omega''(k - \bar{k})^2t$ in the phase is small compared with one radian for every wave number in the packet, that is, for times $t \ll 8/(|\omega''|\Delta k^2)$. In a non-dispersive medium ω'' is zero, and the packet keeps its shape for all time, as you will show in Problem 2.

Spreading of the Packet

The second-order term makes the packet spread. Differentiating the expansion of $\omega(k)$ shows that the components with wave numbers near k form a group of their own, which travels at the group velocity

$v_g + \omega''(k - \bar{k})$. The components across the range Δk therefore travel at speeds that differ by about $|\omega''|\Delta k$, and after a time t they have separated by a distance of about $|\omega''|\Delta k t$, which adds to the length of the packet.

For the Gaussian packet the spreading can be found exactly, to the second order. With all three terms of the expansion, the phase of each component is $(\bar{\omega}t - \bar{k}x) - (k - \bar{k})(x - v_g t) + \frac{1}{2}\omega''t(k - \bar{k})^2$, and substituting the amplitudes of the Gaussian packet gives

$$\psi(x, t) = e^{i(\bar{\omega}t - \bar{k}x)} \frac{\sigma}{\sqrt{\pi}} \int_{-\infty}^{\infty} \exp \left[-\Lambda(k - \bar{k})^2 - i(x - v_g t)(k - \bar{k}) \right] dk, \quad \Lambda = \sigma^2 - \frac{1}{2}i\omega''t. \quad (5.34)$$

This integral has the same form as the one that describes the free Gaussian packet of matter waves in Section 2.4 of Vol. 3. The free particle has $\omega = \hbar k^2/2m$, whose expansion about $\bar{k}$ ends at the second-order term, with $\omega'' = \hbar/m$, and the calculation there applies here with $\hbar/m$ replaced by ω'' and, since Vol. 3 writes the phase of each component as $kx - \omega t$, with i replaced by $-i$, which does not affect the modulus. It rests on the Gaussian integral of that section, which gives $\int \exp[-\Lambda(k - \bar{k})^2 - i(x - v_g t)(k - \bar{k})] dk = \sqrt{\pi/\Lambda} e^{-(x - v_g t)^2/4\Lambda}$ for a complex constant Λ with a positive real part, and therefore

$$\psi(x, t) = \frac{\sigma}{\sqrt{\Lambda}} e^{-(x - v_g t)^2/4\Lambda} e^{i(\bar{\omega}t - \bar{k}x)}. \quad (5.35)$$

To find the envelope we need the modulus of ψ . Multiplying the numerator and the denominator of $1/\Lambda$ by the complex conjugate of Λ gives $1/\Lambda = \Lambda^*/|\Lambda|^2$, whose real part is $\sigma^2/|\Lambda|^2$, and $|\Lambda|^2 = \sigma^4 + \frac{1}{4}\omega''^2 t^2$. The modulus of an exponential depends only on the real part of its exponent, so the envelope is

$$|\psi(x, t)| = \sqrt{\frac{\sigma}{\sigma(t)}} e^{-(x - v_g t)^2/4\sigma(t)^2}, \quad \sigma(t) = \sigma \sqrt{1 + \left(\frac{\omega'' t}{2\sigma^2} \right)^2}, \quad (5.36)$$

where we have written $|\Lambda| = \sigma\sigma(t)$, and the factor in front is $\sigma/|\Lambda|^{1/2}$. With ω'' replaced by $\hbar/m$ this is the width found in Section 2.4 of Vol. 3.

The envelope stays a Gaussian and its centre moves at the group velocity, as it did to the first order, while its width grows as $\sigma(t)$ and its height falls as $[\sigma/\sigma(t)]^{1/2}$. The energy per unit length of the packet is proportional to the square of the envelope, as it was for the harmonic wave of Section 5.2, and the integral of $|\psi|^2$ over x is proportional to $\sigma(t) \times \sigma/\sigma(t) = \sigma$, which does not change, so the energy of the packet is conserved as it spreads. For times short compared with $2\sigma^2/|\omega''|$ the width hardly changes. For much longer times $\sigma(t) \approx |\omega''|t/2\sigma$, which grows in proportion to t , and you will show in Problem 5 that this is the spread of the group velocities of the components multiplied by the time. A narrow packet contains a wide range of wave numbers, and it therefore spreads quickly. The sign of ω'' does not affect the width. It decides instead which components gather at the front of the packet; when ω'' is positive, the group velocity rises with k , and the shorter waves move ahead of the longer ones. The result holds as long as the third-order term of the expansion, which was neglected, remains small, which requires the dispersion relation to be close to its parabola across the range of wave numbers in the packet, as it is in Figure 85.

A Pulse on a String on an Elastic Foundation

Let us apply these results to a Gaussian packet on the string on an elastic foundation of Section 4.4, with central wave number $\bar{k}$ and angular frequency $\bar{\omega} = (v^2\bar{k}^2 + \omega_c^2)^{1/2}$. Section 5.2 found the group velocity $v_g = v^2\bar{k}/\bar{\omega}$ from the equation $\omega d\omega/dk = v^2k$. Differentiating that equation once more with respect to k , with the product rule on the left-hand side, gives $(d\omega/dk)^2 + \omega d^2\omega/dk^2 = v^2$, so that

$$\omega'' = \frac{v^2 - v_g^2}{\bar{\omega}} = \frac{v^2}{\bar{\omega}} \left(1 - \frac{v^2\bar{k}^2}{\bar{\omega}^2} \right) = \frac{v^2\omega_c^2}{\bar{\omega}^3}, \quad (5.37)$$

which is positive. The width of the packet doubles when $\sigma(t) = 2\sigma$, that is, when $\omega''t/2\sigma^2 = \sqrt{3}$, at the time

$$t = \frac{2\sqrt{3}\sigma^2\bar{\omega}^3}{v^2\omega_c^2} . \quad (5.38)$$

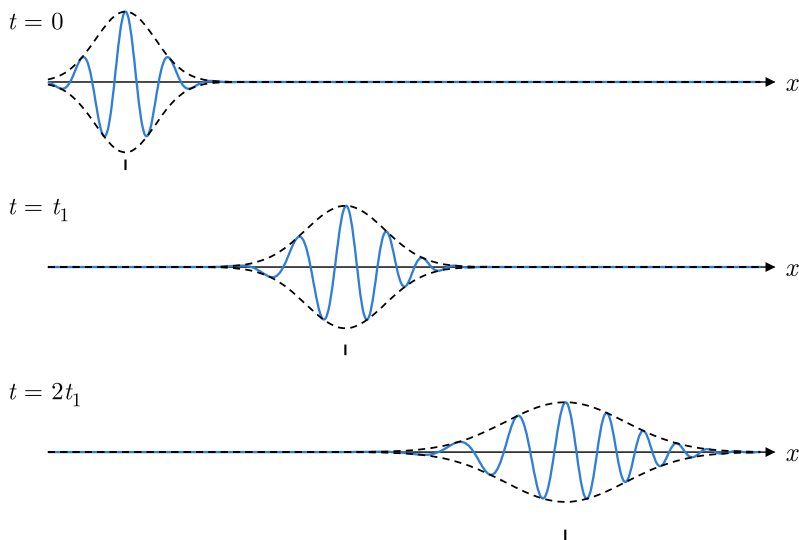


Figure 86. A Gaussian packet on a string on an elastic foundation at the times 0 , t_1 , and $2t_1$, computed from the second-order result, with the envelope $\pm|\psi|$ dashed. The ticks under the axes mark the distance $v_g t$ travelled by the centre, and $2t_1$ is the time at which the width of the envelope has doubled. At the later times the crests are closer together at the front of the packet than at the back.

Figure 86 shows the packet at three times. Without the foundation, $\omega_c = 0$, and the time for the width to double becomes infinite, since the string is then non-dispersive. When the central wavelength is long, so that $v\bar{k} \ll \omega_c$, the central frequency is close to the cutoff, $\bar{\omega} \approx \omega_c$, and the packet travels at the small group velocity $v_g \approx v^2\bar{k}/\omega_c$ while doubling its width in a time of about $2\sqrt{3}\sigma^2\omega_c/v^2$. Since ω'' is positive, the shorter waves in the packet travel faster than the longer ones and gather at its front, as Figure 86 shows. The phase velocity is greater than the group velocity, so the crests move forwards through the packet, appearing at its back and disappearing at its front.

Problems

§5.1 Waves in Two and Three Dimensions

1. A membrane lies in the xy -plane, stretched so that the force exerted across any line drawn in it is Γ per unit length of the line, and its mass per unit area is η . Show that its small transverse displacement $\psi(x, y, t)$ obeys $\partial^2\psi/\partial t^2 = (\Gamma/\eta)(\partial^2\psi/\partial x^2 + \partial^2\psi/\partial y^2)$, and find the speed of its waves.
2. Show that $\psi = (A/r)\sin kr\cos\omega t$, with $\omega = vk$, satisfies the wave equation, that it remains finite at $r = 0$, and that it vanishes at every instant on a set of spheres centred on the origin. Find their radii, and write ψ as the sum of an outgoing and an incoming spherical wave.
3. Let $\psi(r, t)$ depend only on the distance r from a straight line, and let $\psi = r^{-1/2}\chi(r, t)$. Using the Laplacian in cylindrical coordinates of Section 3.4 of Vol. 4, with the distance from the axis written as r (Vol. 4 writes it as s), show that the wave equation becomes $\partial^2\chi/\partial t^2 = v^2(\partial^2\chi/\partial r^2 + \chi/4r^2)$. Hence show that at distances of many wavelengths the outgoing wave is approximately $f_1(r - vt)/\sqrt{r}$, and explain why a cylindrical pulse close to its source does not keep its shape.
4. For the total internal reflection of this section, show that $q = (\omega/v_1)(\sin^2\theta_1 - \sin^2\theta_c)^{1/2}$, and that the evanescent wave travels along the boundary at a speed that lies between v_1 and v_2 .
5. On a string along the x -axis, a wave with $y = A\cos(\omega t - kx)$ and $z = A\sin(\omega t - kx)$ is added to a wave with $y = A\cos(\omega t - kx + \Delta\phi)$ and $z = -A\sin(\omega t - kx + \Delta\phi)$. Show that each of these waves is circularly polarised, the two turning in opposite senses, and that their sum is linearly polarised, along a line at an angle $\Delta\phi/2$ to the y -axis.

§5.2 Dispersion, Phase Velocity, and Group Velocity

1. Show that the group velocity can be written as $v_g = v_p - \lambda dv_p/d\lambda$, where $\lambda = 2\pi/k$ is the wavelength.
2. For the chain of Section 2.3, show that $v_g/v_p =$

$(ka/2)/\tan(ka/2)$, and hence that $v_g < v_p$ for $0 < k \leq \pi/a$. Find v_p and v_g at $k = \pi/2a$.

3. Short waves on the surface of water, for which surface tension dominates over gravity, have a dispersion relation of the form $\omega^2 = Ck^3$, where C is a constant. Show that $v_g = \frac{3}{2}v_p$, and say whether the dispersion is normal or anomalous. Compare the result with that for deep-water waves under gravity.
4. For the stiff string of this section, find v_g and show that $v_g > v_p$. The string has length L and is held at its ends in such a way that y and $\partial^2 y / \partial x^2$ both vanish there. Show that its modes are $\sin(n\pi x/L)$, and that their frequencies are $\omega_n = (n\pi v/L)(1 + \beta n^2 \pi^2 / \tau L^2)^{1/2}$, so that the higher modes are not exact harmonics of the lowest.
5. For a harmonic travelling wave on the string on an elastic foundation, show that the mean kinetic energy per unit length is equal to the sum of the mean potential energies per unit length of the stretched string and of the foundation.

§5.3 Wave Packets

1. The amplitudes of a packet are $\mathcal{A}(k) = \mathcal{A}_0$, a constant, for $|k - \bar{k}| < \frac{1}{2}\Delta k$, and zero for other k . Show that at $t = 0$ its displacement is $y(x, 0) = (2/\pi)^{1/2} \mathcal{A}_0 x^{-1} \sin(\frac{1}{2}\Delta k x) \cos \bar{k}x$, with $\mathcal{A}_0$ taken to be real, and find the distance from the centre of the packet to the first zero of its envelope. Compare the result with the relation $\Delta k \Delta x \approx 2\pi$.
2. Show that in a non-dispersive medium, with $\omega = vk$, the packet satisfies $\psi(x, t) = \psi(x - vt, 0)$ exactly, for any amplitudes $\mathcal{A}(k)$, and relate the result to d'Alembert's solution.
3. A Gaussian packet of central wave number $\bar{k}$ on the string on an elastic foundation is made twice as long, by doubling σ . By what factor does the time for its width to double change? Find the distance the centre of the packet travels in that time, and describe how this distance depends on $\bar{k}$ when $v\bar{k}$ is small and when it is large compared with ω_c .
4. For waves on deep water, with $\omega^2 = gk$, find v_g and ω'' at the

central wave number $\bar{k}$ of a Gaussian packet, and the time for its width to double. A crest appears at the back of the packet and disappears at its front; find the speed at which it moves through the packet.

5. Show that at times long compared with $2\sigma^2/|\omega''|$, the width of a Gaussian packet is $\sigma(t) \approx |\omega''|t/2\sigma$. The quantity $|\mathcal{A}(k)|^2$ is a Gaussian in k with standard deviation $1/2\sigma$. Show that the group velocities of the components have a spread of $|\omega''|/2\sigma$, and hence explain the long-time width.

Acoustics of Fluids: Generation, Propagation, and Attenuation

6.1 The Acoustic Wave Equation and the Speed of Sound

In Section 3.3 we treated a column of fluid in a narrow tube, and found that the longitudinal displacement of the fluid obeys the wave equation with speed $\sqrt{B/\rho_0}$, where ρ_0 is the density of the fluid at rest and the bulk modulus B was left as a parameter to be measured. In this section we first derive the wave equation for a fluid that fills space, then find the bulk modulus of an ideal gas, and finally show why the compressions and rarefactions of a sound wave in a gas take place without any appreciable flow of heat.

The Acoustic Variables

Let a fluid be at rest with a uniform pressure p_0 and a uniform density ρ_0 . We ignore the weight of the fluid, which makes the pressure of a fluid at rest increase with depth, because in air the increase over the wavelength of an audible sound is a very small fraction of p_0 . When a sound wave passes, the pressure at the point $\mathbf{r}$ at time t becomes $p_0 + p(\mathbf{r}, t)$ and the density becomes $\rho_0 + \rho'(\mathbf{r}, t)$, and the fluid there moves with a velocity $\mathbf{u}(\mathbf{r}, t)$. The quantity p is called the excess pressure, or the acoustic pressure, and ρ' is called

the excess density. The velocity $\mathbf{u}$ is called the particle velocity. It is the velocity of the small element of fluid that is at $\mathbf{r}$ at time t , a different quantity from the speed at which the wave travels. The ratio

$$s = \frac{\rho'}{\rho_0} \quad (6.1)$$

is called the condensation, and it is the fractional increase in the density.

The disturbance is taken to be small, in the sense that $|s| \ll 1$, and the condition that this places on $\mathbf{u}$ is found below. The excess pressure in ordinary speech, for example, is a very small fraction of atmospheric pressure. Wherever a product of two of the acoustic variables p , ρ' , and $\mathbf{u}$ appears in an equation beside a term containing only one of them, the product is dropped, and the equations that remain are called the linearised equations. Their solutions obey the principle of superposition, as the waves on a string of Section 3.2 did.

The Equations in One Dimension

Let us take first a disturbance that depends only on x and t , with the particle velocity u directed along the x -axis, and let us consider an imaginary tube of cross-section S whose sides are parallel to the x -axis. Since nothing varies across the tube, no fluid crosses its sides. We apply the conservation of mass and Newton's second law to the fluid in a slab of the tube, fixed in space, between x and $x + \Delta x$, as in Figure 87.

The mass in the slab is $(\rho_0 + \rho')S \Delta x$. It increases at the rate at which mass flows in across the face at x minus the rate at which it flows out across the face at $x + \Delta x$, so that

$$\begin{aligned} \frac{\partial \rho'}{\partial t} S \Delta x &= [(\rho_0 + \rho')uS]_x - [(\rho_0 + \rho')uS]_{x+\Delta x} \\ &\approx -\frac{\partial}{\partial x} [(\rho_0 + \rho')u] S \Delta x, \end{aligned} \quad (6.2)$$

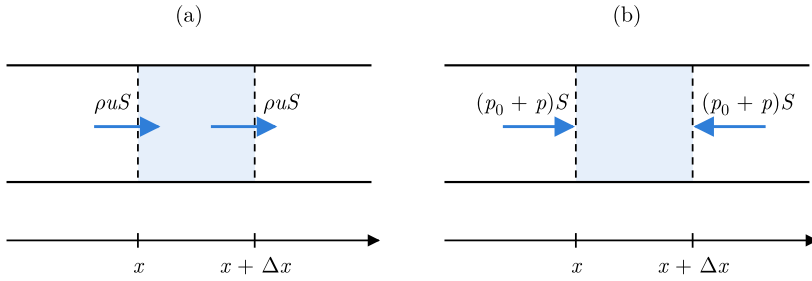


Figure 87. A slab of fluid of cross-section S between the planes x and $x + \Delta x$, which are fixed in space. (a) Fluid carries mass into the slab across the face at x at the rate $\rho u S$ evaluated there, and out across the face at $x + \Delta x$ at the rate $\rho u S$ evaluated there, where $\rho = \rho_0 + \rho'$. (b) The fluid on either side pushes on the fluid in the slab with the forces $(p_0 + p)S$, evaluated at the two faces.

where the last step is Taylor's theorem to the first order in Δx . The product $\rho' u$ is of the second order in the acoustic variables, and dropping it and dividing by $S \Delta x$ gives

$$\frac{\partial \rho'}{\partial t} = -\rho_0 \frac{\partial u}{\partial x}. \quad (6.3)$$

The fluid in the slab is pushed towards increasing x by the force $(p_0 + p)S$ on the face at x and towards decreasing x by the force $(p_0 + p)S$ on the face at $x + \Delta x$. The ambient pressure p_0 contributes equal and opposite forces, and the net force is

$$p(x) S - p(x + \Delta x) S \approx -\frac{\partial p}{\partial x} S \Delta x. \quad (6.4)$$

The force acts on the fluid that is in the slab at the instant considered, and it gives that fluid an acceleration. The fluid does not stay in the slab, and its velocity changes both because time passes and because it moves to places where u is different. An element of fluid that is at x at time t is at $x + u dt$ at time $t + dt$, and by the chain rule for a function of two variables (Section 2.2 of Vol. 4 of the Fermi Notes on Physics) its acceleration is

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x}. \quad (6.5)$$

The second term is of the second order and is dropped. For a wave travelling at speed c , the derivative $\partial u/\partial x$ is of the order of $(1/c) \partial u/\partial t$, so the dropped term is smaller than the first by a factor of the order of u/c , and the approximation holds as long as the particle velocity is small compared with the speed of the wave. Newton's second law for the fluid in the slab, whose mass is $\rho_0 S \Delta x$ once the second-order product $\rho' \partial u/\partial t$ is dropped, then gives

$$\rho_0 \frac{\partial u}{\partial t} = - \frac{\partial p}{\partial x}. \quad (6.6)$$

The frictional forces between layers of fluid that move at different speeds have been left out, since they cause the attenuation of sound, which is not treated in this course.

A third relation is needed, between the pressure and the density. The pressure of a given fluid depends on its density and on its temperature, and in a sound wave the temperature changes along with the density in a way that depends on how much heat flows, which we find later in this section. For the present, we suppose only that the pressure is some function of the density during the passage of the wave, $p_0 + p = \mathcal{F}(\rho_0 + \rho')$, with the same letter (or any letter outside the fixed list) replacing P in the five later places named, so that the adiabat reads $\mathcal{F} \rho^{-C_p/C_v} = \text{constant}$ and the logarithmic derivative $d\mathcal{F}/\mathcal{F}$. By Taylor's theorem, to the first order,

$$p = \left(\frac{dP}{d\rho} \right)_0 \rho' = \frac{B}{\rho_0} \rho' = Bs, \quad (6.7)$$

where the derivative is taken at the density ρ_0 , and $B = \rho_0 (dP/d\rho)_0$ is the bulk modulus of Section 3.3, the increase in pressure divided by the fractional decrease in volume that it produces. The two forms of the bulk modulus agree, since a fixed mass of fluid occupies a volume ΔV inversely proportional to its density, so that the fractional decrease $-d(\Delta V)/\Delta V$ in its volume is equal to the fractional increase $d\rho/\rho$ in its density.

To eliminate the particle velocity, we differentiate the equation of mass conservation with respect to t and the equation of motion with respect to x . The mixed derivative $\partial^2 u / \partial x \partial t$ is the same in either order (Section 2.2 of Vol. 4), so

$$\frac{\partial^2 \rho'}{\partial t^2} = -\rho_0 \frac{\partial^2 u}{\partial x \partial t} = \frac{\partial^2 p}{\partial x^2}. \quad (6.8)$$

Substituting $\rho' = \rho_0 p / B$ on the left gives

$$\frac{\partial^2 p}{\partial t^2} = \frac{B}{\rho_0} \frac{\partial^2 p}{\partial x^2}, \quad (6.9)$$

which is the wave equation with the speed $c = \sqrt{B/\rho_0}$ found in Section 3.3. The same section wrote the excess pressure as $p = -B \partial \xi / \partial x$, where ξ is the displacement of the fluid, and you will show in Problem 2 that the equation of mass conservation gives this relation.

The Equations in Three Dimensions

In three dimensions the argument is the same, with the slab replaced by a small box whose edges Δx , Δy , and Δz are parallel to the axes. The conservation of mass for a fixed volume was expressed in Section 3.3 of Vol. 4 by the continuity equation,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0, \quad (6.10)$$

where $\rho = \rho_0 + \rho'$ is the total density. Dropping the product $\rho' \mathbf{u}$ leaves

$$\frac{\partial \rho'}{\partial t} + \rho_0 \nabla \cdot \mathbf{u} = 0. \quad (6.11)$$

The faces of the box perpendicular to the x -axis give a force $-(\partial p / \partial x) \Delta x \Delta y \Delta z$ along the x -axis, as in one dimension, and the other two pairs of faces give the corresponding forces along the

y - and z -axes, so the net force on the box is $-\nabla p \Delta x \Delta y \Delta z$. The acceleration of the fluid is $\partial \mathbf{u} / \partial t$ together with terms of the second order, found by the chain rule as before, and Newton's second law gives

$$\rho_0 \frac{\partial \mathbf{u}}{\partial t} = -\nabla p. \quad (6.12)$$

The equation of motion of a fluid without friction is called Euler's equation, and this is its linearised form.

Taking the divergence of Euler's equation gives $\rho_0 \partial(\nabla \cdot \mathbf{u}) / \partial t = -\nabla^2 p$, and differentiating the continuity equation with respect to time gives $\partial^2 \rho' / \partial t^2 = -\rho_0 \partial(\nabla \cdot \mathbf{u}) / \partial t$. Eliminating $\nabla \cdot \mathbf{u}$ between these two equations, and substituting $\rho' = p/c^2$, we find

$$\nabla^2 p = \frac{1}{c^2} \frac{\partial^2 p}{\partial t^2}, \quad c^2 = \frac{B}{\rho_0}, \quad (6.13)$$

which is called the acoustic wave equation. It is the wave equation in three dimensions of Section 5.1, with the speed c . Since ρ' and s are constant multiples of p , they obey the same equation.

Euler's equation has a further consequence. The curl of a gradient is zero (Section 3.1 of Vol. 4), so taking the curl of Euler's equation gives $\partial(\nabla \times \mathbf{u}) / \partial t = \mathbf{0}$, and in a fluid that was at rest before the wave arrived, $\nabla \times \mathbf{u}$ remains zero. By the interpretation of the curl in Section 3.1 of Vol. 4, a sound wave sets the small elements of the fluid moving to and fro without setting them rotating. You will use this result in Problem 1 to show that each component of $\mathbf{u}$ also obeys the acoustic wave equation.

Newton's and Laplace's Speeds of Sound

For a gas the bulk modulus can be calculated, once we know how the temperature changes as the gas is compressed. An ideal gas of molar mass M_m has the equation of state $p_0 + p = (\rho_0 + \rho')RT/M_m$, since a mass m of gas is m/M_m moles and its density is m divided by its volume.

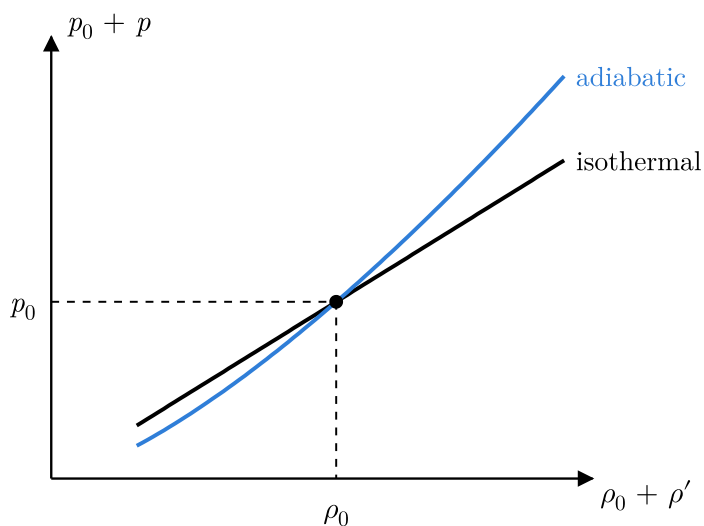


Figure 88. The pressure $p_0 + p$ of an ideal gas against its density $\rho_0 + \rho'$, for a compression at constant temperature (solid) and for a reversible adiabatic compression (accent) from the undisturbed state (ρ_0, p_0) , drawn for a diatomic gas. The slope of each curve at the undisturbed state is the square of the speed of sound that the corresponding assumption predicts, and the adiabat is the steeper of the two curves.

Newton, who gave the first theoretical calculation of the speed of sound in 1687, supposed that the temperature of the gas stays the same as it is compressed and expanded. At constant temperature the pressure is proportional to the density, as the straight line in Figure 88 shows, and $dP/d\rho = RT/M_m = p_0/\rho_0$. The isothermal bulk modulus is therefore p_0 , and Newton's speed of sound is

$$c_N = \sqrt{\frac{p_0}{\rho_0}}. \quad (6.14)$$

For air, this is smaller than the measured speed by about a sixth.

Laplace pointed out in 1816 that the compressions and rarefactions of a sound wave follow one another too quickly for heat to flow between them, so that the compressions are adiabatic. Section 2.2 of Vol. 2 of the Fermi Notes on Physics showed that in a reversible adiabatic change of a fixed amount of ideal gas, the pressure multiplied by the power C_p/C_V of the volume is constant, with the ratio of the heat capacities written γ there. The volume of a fixed amount of gas is inversely proportional to its density, so along the adiabat

$$P\rho^{-C_p/C_V} = \text{constant}. \quad (6.15)$$

Taking logarithms and differentiating, which is called logarithmic differentiation, gives $dP/P - (C_p/C_V) d\rho/\rho = 0$, so that $dP/d\rho = (C_p/C_V)p_0/\rho_0$ at the undisturbed state. The adiabatic bulk modulus is $(C_p/C_V)p_0$, and the speed of sound is

$$c = \sqrt{\frac{C_p}{C_V} \frac{p_0}{\rho_0}} = \sqrt{\frac{C_p}{C_V} \frac{RT}{M_m}}. \quad (6.16)$$

The two assumptions are compared in Figure 88, where the adiabat through the undisturbed state is steeper than the isotherm by the factor C_p/C_V , because a gas compressed without losing heat also grows hotter. As you showed in Problem 4 of Section 5.2 of Vol. 2, C_p/C_V is 5/3 for a monatomic gas and 7/5 for a diatomic gas at ordinary temperatures. Air is mostly nitrogen and oxygen,

whose molecules are diatomic, so Laplace's speed exceeds Newton's by the factor $\sqrt{7/5}$, which accounts for the discrepancy. Measuring the speed of sound in a gas of known molar mass at a known temperature is therefore one way of measuring C_p/C_V .

The adiabat in Figure 88 is drawn as reversible. The compression of an element of gas small compared with the wavelength is quasistatic, since the pressure is nearly uniform across the element and the element passes through a succession of equilibrium states, and the frictional forces and the flow of heat that would make it irreversible are small, as the next subsection shows for the flow of heat.

Why the Compressions Are Adiabatic

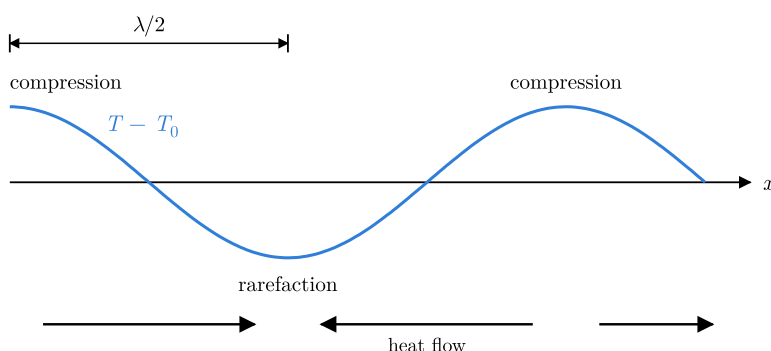


Figure 89. The excess temperature $T - T_0$ along a sound wave in a gas at one instant, which is greatest in the compressions and least in the rarefactions. Heat is conducted from each compression towards the rarefactions on either side of it, over a distance of half a wavelength.

In a sound wave the gas in the compressions is heated and the gas in the rarefactions is cooled, and you will show in Problem 3 that the fractional rise in temperature is $(C_p/C_V - 1)s$. Heat is conducted from each compression to the rarefactions on either side of it, half a wavelength away, as in Figure 89. If the heat flowed fast enough to equalise the temperatures within a small fraction of a period, the compressions would be isothermal, as Newton supposed. If it flows too slowly to have an appreciable effect within a period, the compressions are adiabatic, as Laplace supposed. The question is which of the two times, the period or the time for heat to be conducted over half a wavelength, is the shorter.

To estimate the time needed for the heat to be conducted, we use the diffusion equation of Section 3.3 of Vol. 4, $\partial T/\partial t = D\nabla^2 T$, where the thermal diffusivity D is the thermal conductivity divided by the heat capacity per unit volume. Let us suppose that the gas is held still with a temperature that varies along the x -axis as $T = T_0 + \Theta(t) \cos kx$, with the same wave number as the sound wave. Substituting this temperature into the diffusion equation gives $d\Theta/dt = -Dk^2\Theta$, whose solution is

$$\Theta(t) = \Theta(0) e^{-Dk^2 t}, \quad (6.17)$$

so the variation of the temperature decays in a time of the order of $1/Dk^2$. The corresponding time for the sound wave is $1/\omega = 1/c k$, the time in which its phase changes by one radian. The ratio of the decay rate Dk^2 to the angular frequency of the wave is

$$\frac{Dk^2}{\omega} = \frac{Dk}{c}, \quad (6.18)$$

and the compressions are adiabatic if this ratio is small compared with one.

Section 1.4 of Vol. 2 found the thermal conductivity of a gas, which is written κ there, to be one third of the product of the heat capacity per unit volume, the mean speed of the molecules, and the mean free path. Dividing by the heat capacity per unit volume gives $D \approx \frac{1}{3} \bar{v}_m \lambda_f$, where $\bar{v}_m$ is the mean speed of the molecules and λ_f is the mean free path, which are written $\bar{v}$ and λ in Vol. 2. Here we have ignored the difference between the heat capacities per unit volume at constant volume and at constant pressure, which changes D only by a factor of order one. With $k = 2\pi/\lambda$,

$$\frac{Dk^2}{\omega} \approx \frac{2\pi}{3} \frac{\bar{v}_m}{c} \frac{\lambda_f}{\lambda}. \quad (6.19)$$

You will show in Problem 4 that $c/\bar{v}_m$ is a little less than one. Sound is carried from one part of the gas to the next by the motion of the molecules, so it is to be expected that it travels at a speed

comparable with theirs. The ratio Dk^2/ω is therefore of the order of λ_f/λ . The mean free path in air at atmospheric pressure is a small fraction of a micrometre, while the wavelengths of audible sounds range from centimetres to metres, so the heat conducted in a period is negligible and the compressions are adiabatic, as Laplace supposed.

The result runs against what one might first expect, since one might think that a slow oscillation gives the heat more time to flow, and so that sound of low frequency should be isothermal. The period does grow in proportion to the wavelength, but the time for heat to be conducted over half a wavelength grows as the square of the wavelength, because a temperature variation of wave number k decays at the rate Dk^2 . Longer waves are therefore further from isothermal. The approximation would fail only for wavelengths close to the mean free path, where the gas can no longer be treated as a continuous fluid.

How the Speed Depends on the Temperature

Laplace's formula shows that the speed of sound in an ideal gas depends on its temperature, on its molar mass, and on the ratio of its heat capacities, and not on its pressure. At a fixed temperature, raising the pressure raises the density in the same proportion, since $p_0/\rho_0 = RT/M_m$, and the stiffness of the gas and its inertia increase together. The speed is proportional to $\sqrt{T}$, so sound travels faster in warm air than in cold air, which is one reason the pitch of a wind instrument rises a little as the air inside it warms up, as you will find in Problem 5. At the same temperature, sound travels faster in helium than in air, since the molar mass of helium is several times smaller and its ratio C_p/C_V is larger.

6.2 Plane and Spherical Sound Waves

The acoustic wave equation of Section 6.1 is an equation for the excess pressure alone. Once the pressure in a wave is known, the linearised Euler equation gives the particle velocity, and we now find the particle velocity first for a plane wave and then for a spherical wave. The plane and spherical waves are compared through the ratio of the pressure to the particle velocity, which is called the

specific acoustic impedance.

Plane Waves

As in Section 1.1, we write each oscillating quantity as the real part of a complex expression with the time dependence $e^{i\omega t}$. The linearised equations of Section 6.1 are linear and have real coefficients, so the real parts of complex solutions are solutions, and in this section p and $\mathbf{u}$ stand for the complex expressions. A harmonic plane wave travelling towards increasing x has the excess pressure

$$p = \mathcal{A}e^{i(\omega t - kx)}, \quad (6.20)$$

which satisfies the acoustic wave equation if $\omega = ck$. Every derivative $\partial/\partial t$ of this expression multiplies it by $i\omega$ and every derivative $\partial/\partial x$ multiplies it by $-ik$. The x -component of Euler's equation, $\rho_0 \partial u/\partial t = -\partial p/\partial x$, is satisfied by a particle velocity of the same form, and it gives $i\omega\rho_0 u = ikp$, or

$$u = \frac{p}{\rho_0 c}. \quad (6.21)$$

The particle velocity is in phase with the excess pressure at every point, so the fluid moves forwards in the compressions, in the direction in which the wave travels, and backwards in the rarefactions. For a wave travelling towards decreasing x the relation is $u = -p/\rho_0 c$, as you will show in Problem 1.

The displacement ξ of the fluid satisfies $u = \partial\xi/\partial t = i\omega\xi$, so that $\xi = -iu/\omega$. Since $-i = e^{-i\pi/2}$, the displacement is a quarter of a period behind the particle velocity, and therefore a quarter of a period behind the excess pressure. The wave at one instant is drawn in Figure 90. The compressions lie where the layers of fluid ahead have been pushed back and the layers behind have been pushed forward, so that the displacement is zero and decreasing with x , in agreement with $p = -B \partial\xi/\partial x$ from Section 3.3.

The condensation is $s = p/B = p/\rho_0 c^2$, and with $u = p/\rho_0 c$ this gives $s = u/c$. The condition $|s| \ll 1$ of Section 6.1 is therefore the same as the condition that the particle velocity be small compared

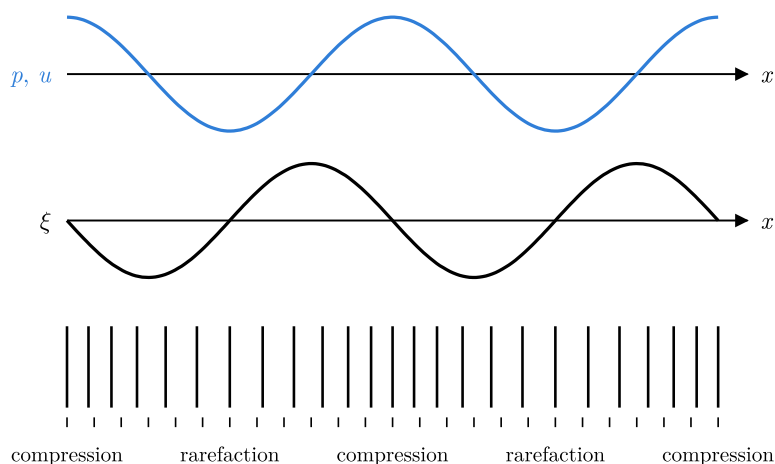


Figure 90. A plane harmonic sound wave travelling towards increasing x , at one instant. Top: the excess pressure and the particle velocity, which are in phase (accent). Middle: the displacement of the fluid, which is a quarter of a period behind them. Bottom: layers of fluid, originally equally spaced at the positions of the short ticks, displaced by the wave, with the displacement exaggerated. The layers crowd together where the pressure is greatest.

with the speed of sound, which was needed to drop the second-order term in the acceleration, and the same as the condition $|p| \ll \rho_0 c^2 = B$. For a gas the adiabatic bulk modulus is $(C_p/C_V) p_0$, so the linearised equations hold as long as the excess pressure is small compared with the ambient pressure.

A plane wave travelling in the direction of a unit vector $\hat{\mathbf{n}}$ has $p = \mathcal{A}e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})}$ with $\mathbf{k} = k\hat{\mathbf{n}}$, and its gradient is $\nabla p = -i\mathbf{k}p$. Euler's equation gives $i\omega\rho_0\mathbf{u} = i\mathbf{k}p$, so that $\mathbf{u} = \hat{\mathbf{n}}p/\rho_0c$. The particle velocity is along the direction in which the wave travels, and sound in a fluid is therefore a longitudinal wave. A fluid at rest exerts no force opposing the sliding of one layer over another, so it has no restoring force that could carry a transverse wave.

The Specific Acoustic Impedance

For a harmonic wave, the ratio

$$z = \frac{p}{u} \quad (6.22)$$

is called the specific acoustic impedance, where u is the component of the particle velocity along a chosen direction, usually the direction in which the wave travels. The word specific indicates that the pressure is a force per unit area, so that z is an impedance per unit area. For a plane wave travelling along the chosen direction, the specific acoustic impedance is

$$z = \rho_0 c = \sqrt{\rho_0 B}, \quad (6.23)$$

which depends only on the fluid and is called the characteristic impedance of the fluid. It is the counterpart for a fluid of the characteristic impedance $Z = \sqrt{\tau\mu} = \mu v$ of a string (Section 3.4), which is the ratio of the transverse force at a point of the string to the transverse velocity there. The excess pressure takes the place of the transverse force, now per unit area, and the particle velocity takes the place of the transverse velocity. In each case the impedance is the square root of the product of a stiffness, the tension or the bulk modulus, and an inertia, the mass per unit length or the density.

The characteristic impedance of water is several thousand times that of air, because water is both far denser and far stiffer.

The analogy can be carried further. A piston at the end of a long tube of cross-section S filled with fluid, moving with velocity u , sends a plane wave down the tube, and the fluid presses back on the piston with the force $\rho_0 c S u$, opposing its motion. The piston therefore does work on the fluid as it would against a resistive force $-\rho_0 c S u$ of the kind described in Section 1.2, as the end of a long string did in Section 3.4.

Spherical Waves

A wave from a small source in an open space spreads out in every direction, and if the source is spherically symmetric its wavefronts are spheres. Let the excess pressure depend only on the distance r from the source and on t . With the Laplacian in spherical coordinates (Section 3.4 of Vol. 4),

$$\nabla^2 p = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial p}{\partial r} \right) = \frac{1}{r} \frac{\partial^2 (rp)}{\partial r^2}, \quad (6.24)$$

where the second form is checked by carrying out the differentiations, since each is equal to $\partial^2 p / \partial r^2 + (2/r) \partial p / \partial r$. The acoustic wave equation therefore says that rp obeys the wave equation in one dimension, and, as we found in Section 5.1, a wave travelling outwards has the form $p = f_1(t - r/c)/r$, which is the form $f_1(r - vt)/r$ of Section 5.1 with the argument rescaled. A harmonic wave travelling outwards has the excess pressure

$$p = \frac{\mathcal{A}}{r} e^{i(\omega t - kr)}. \quad (6.25)$$

By symmetry the particle velocity is radial, and its radial component u obeys the radial component of Euler's equation, $\rho_0 \partial u / \partial t = -\partial p / \partial r$, with the gradient in spherical coordinates from Section 3.4 of Vol. 4. By the product rule,

$$\frac{\partial p}{\partial r} = - \left(ik + \frac{1}{r} \right) p, \quad (6.26)$$

and therefore $i\omega\rho_0 u = (ik + 1/r)p$. Dividing by $i\omega\rho_0$ and using $\omega = ck$ gives

$$u = \frac{p}{\rho_0 c} \left(1 - \frac{i}{kr}\right). \quad (6.27)$$

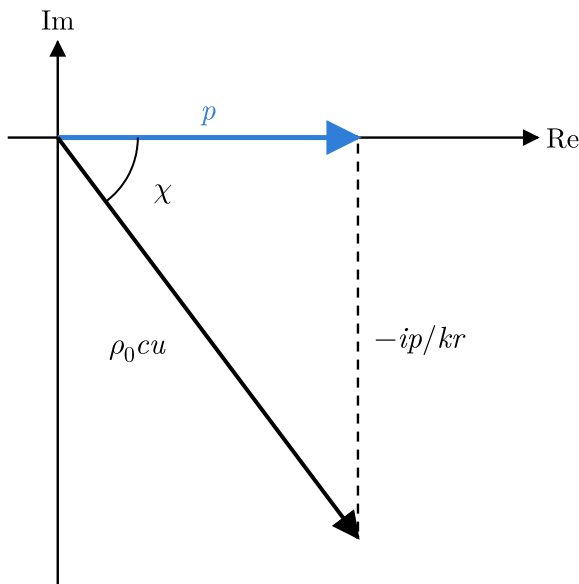


Figure 91. Phasors for the excess pressure p (accent) and for $\rho_0 cu$ in a spherical wave, at a distance from the source less than $1/k$. The phasor $\rho_0 cu$ is the sum of p and the part $-ip/kr$, which is a quarter of a period behind p , and it lags behind p by the angle χ , with $\tan \chi = 1/kr$.

The particle velocity has a part $p/\rho_0 c$ in phase with the excess pressure, as in a plane wave, and a part $-ip/\rho_0 ckr$, which is a quarter of a period behind the pressure, since $-i = e^{-i\pi/2}$. The second part is small far from the source and large near it. The two parts are drawn as phasors in Figure 91, where the particle velocity lags behind the pressure by the angle χ with $\tan \chi = 1/kr$.

The specific acoustic impedance of the spherical wave is

$$z = \frac{p}{u} = \frac{\rho_0 c}{1 - i/kr} = \rho_0 c \frac{ikr}{1 + ikr}. \quad (6.28)$$

Multiplying the numerator and the denominator of the last form by $1 - ikr$ separates the real and imaginary parts,

$$z = \rho_0 c \frac{k^2 r^2}{1 + k^2 r^2} + i \rho_0 c \frac{kr}{1 + k^2 r^2}, \quad (6.29)$$

which are drawn in Figure 92. The impedance is complex, as the mechanical impedance of Section 1.3 is, and its imaginary part is positive, as the term $i m \omega$ of a mass is in $Z_m = b + i(m\omega - \kappa/\omega)$.

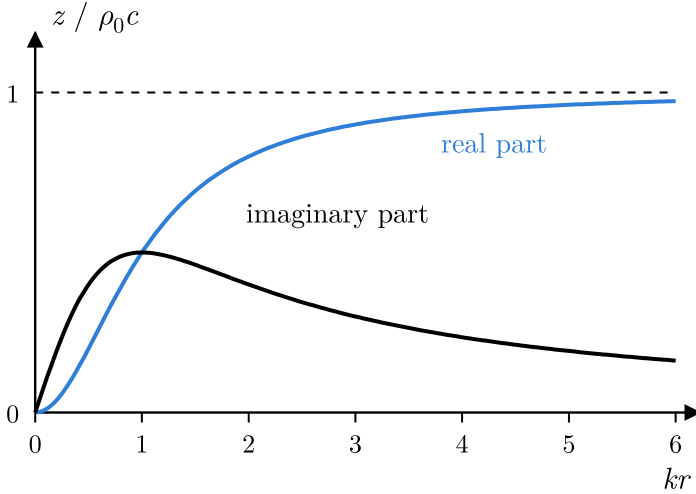


Figure 92. The real part (accent) and the imaginary part (solid) of the specific acoustic impedance of a spherical wave, divided by the characteristic impedance $\rho_0 c$, against kr . Far from the source the impedance tends to $\rho_0 c$, and near the source it is small and almost purely imaginary.

Far from the source, where $kr \gg 1$, the impedance tends to $\rho_0 c$, and the wave is locally a plane wave, since the curvature of a wavefront of radius r is small over a distance of a wavelength. Near the source, where $kr \ll 1$, the impedance is approximately

$$z \approx i \rho_0 c k r = i \omega \rho_0 r, \quad (6.30)$$

which is the impedance per unit area of a mass $\rho_0 r$ per unit area. Near the source, most of the excess pressure goes into accelerating

the fluid to and fro, and the particle velocity is much larger than the value $p/\rho_0 c$ it would have in a plane wave of the same pressure.

The Small Pulsating Sphere

As an example, consider a sphere of radius r_0 whose radius oscillates with a small amplitude, so that its surface moves radially with the velocity $u_0 e^{i\omega t}$, where u_0 is real. The amplitude of the oscillation of the radius is taken to be small compared with r_0 , so that the surface can be treated as staying at $r = r_0$. The fluid neither separates from the surface nor passes into it, so the particle velocity at r_0 is equal to the velocity of the surface. Setting $r = r_0$ in the particle velocity found above gives

$$\frac{\mathcal{A}}{\rho_0 c r_0} e^{-ikr_0} \left(1 - \frac{i}{kr_0}\right) = u_0, \quad (6.31)$$

and solving for the amplitude,

$$\mathcal{A} = \rho_0 c u_0 r_0 e^{ikr_0} \frac{ikr_0}{1 + ikr_0}. \quad (6.32)$$

Let the sphere be small compared with the wavelength, so that $kr_0 \ll 1$. Then e^{ikr_0} and $1 + ikr_0$ are both close to one, and $\mathcal{A} \approx i\rho_0 c k r_0^2 u_0 = i\omega \rho_0 r_0^2 u_0$. Let $q = 4\pi r_0^2 u_0$ be the amplitude of the rate at which the surface of the sphere sweeps out volume, which is called the source strength. The excess pressure at a distance r is

$$p = \frac{i\omega \rho_0 q}{4\pi r} e^{i(\omega t - kr)}, \quad (6.33)$$

and it depends on the source only through q . The waves from the sphere are drawn in Figure 93.

Near the source, the fluid moves almost as it would if it could not be compressed. To see this, suppose that the fluid is incompressible. The volume $q e^{i\omega t}$ per unit time pushed out by the sphere must then cross every sphere of radius r about the source, so that $4\pi r^2 u = q e^{i\omega t}$ and

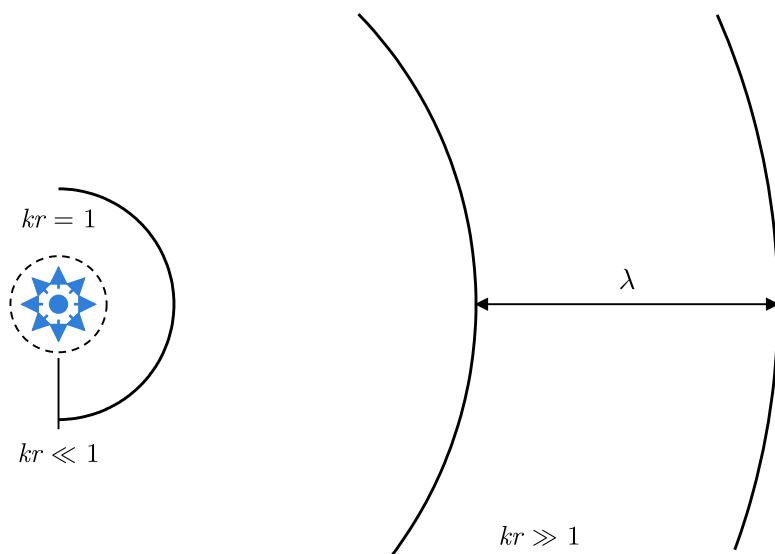


Figure 93. A small sphere (accent) pulsating in a fluid, with its surface moving radially, and three wavefronts of the outgoing spherical wave, a wavelength apart. The dashed circle has the radius $\lambda/2\pi$, at which $kr = 1$. Well inside it the fluid moves almost as an incompressible fluid would, and far outside it the wave is locally a plane wave.

$$u = \frac{q}{4\pi r^2} e^{i\omega t}. \quad (6.34)$$

Euler's equation gives $\partial p/\partial r = -\rho_0 \partial u/\partial t = -i\omega\rho_0 q e^{i\omega t}/4\pi r^2$, and integrating with respect to r , with p tending to zero far from the source, gives

$$p = \frac{i\omega\rho_0 q}{4\pi r} e^{i\omega t}. \quad (6.35)$$

For $kr \ll 1$ the factor e^{-ikr} in the acoustic result is close to one, and the two expressions for the pressure agree. The particle velocities agree as well, since for $kr \ll 1$ the acoustic particle velocity is $p(-i/kr)/\rho_0 c = q e^{i(\omega t - kr)}/4\pi r^2$. The reason is that the region $kr \ll 1$ is small compared with the distance c/ω that sound travels in a time $1/\omega$, so a change of pressure at the source is felt across the region in a time short compared with a period, and within that time the fluid has not moved far enough for its density to change appreciably. The incompressible flow near the source is the part of the motion that is a quarter of a period behind the pressure. The part in phase with the pressure, which is small near the source, becomes the plane-like wave that travels outwards.

6.3 Sound Energy, Intensity, and Decibel Scales

Section 3.4 found the energy carried by a wave on a string, and carried the result over to the longitudinal waves of Section 3.3, where the intensity of a wave was defined as the mean power that it carries across unit area. Here we first find the energy per unit volume of a sound wave in terms of the excess pressure and the particle velocity, then find the intensity of plane and spherical waves, and finally define the decibel scales on which sound levels are measured, use them to combine two sources and to follow the level away from a source, and treat the reflection of sound at the boundary between two fluids.

The Energy Density of a Sound Wave

The kinetic energy per unit volume of the moving fluid is $\frac{1}{2}(\rho_0 + \rho')u^2$, where $u = |\mathbf{u}|$, and to the second order in the acoustic variables it is $\frac{1}{2}\rho_0 u^2$.

The potential energy is the work done in compressing the fluid. Let us follow a small element of fluid of fixed mass, whose volume is ΔV_0 before the wave arrives and ΔV when the condensation has risen to s . The mass of the element, $(\rho_0 + \rho')\Delta V$, does not change, so that $d(\Delta V)/\Delta V = -d\rho'/(\rho_0 + \rho')$, which to the first order is $-ds$. The excess pressure does the work $-p d(\Delta V) = p \Delta V_0 ds$ on the element as its condensation rises by ds , and with $p = Bs$ from Section 6.1 the work done as the condensation rises from zero to s is

$$\Delta V_0 \int_0^s Bs' ds' = \frac{1}{2}Bs^2 \Delta V_0, \quad (6.36)$$

where s' is the variable of integration. The ambient pressure p_0 also does the work $-p_0 d(\Delta V)$ on the element, but this work is done by the neighbouring elements, which expand by the same total volume as the element is compressed. The total volume of the fluid is unchanged, whether it fills a closed vessel or extends beyond the wave into fluid that is at rest, so the work done by the ambient pressure, summed over all the elements, is zero, and it passes energy from one element to another without storing any. The potential energy per unit volume is therefore $\frac{1}{2}Bs^2 = p^2/2\rho_0 c^2$, and the energy per unit volume of the sound wave is

$$\varepsilon = \frac{1}{2}\rho_0 u^2 + \frac{p^2}{2\rho_0 c^2}. \quad (6.37)$$

In a plane wave travelling in one direction, $u = p/\rho_0 c$ at every point and instant, as you showed in Problem 2 of Section 6.2, and the kinetic and potential energies per unit volume are equal, as the kinetic and potential energies per unit length of a string were for a travelling wave in Section 3.4. The energy per unit volume is then $\varepsilon = p^2/\rho_0 c^2$.

The Flow of Energy and the Intensity

To find how the energy moves, we differentiate ε with respect to time,

$$\frac{\partial \varepsilon}{\partial t} = \rho_0 \mathbf{u} \cdot \frac{\partial \mathbf{u}}{\partial t} + \frac{p}{\rho_0 c^2} \frac{\partial p}{\partial t}. \quad (6.38)$$

Euler's equation replaces $\rho_0 \partial \mathbf{u} / \partial t$ by $-\nabla p$, and the continuity equation with $\rho' = p/c^2$ replaces $\partial p / \partial t$ by $-\rho_0 c^2 \nabla \cdot \mathbf{u}$. The right-hand side becomes $-(\mathbf{u} \cdot \nabla p + p \nabla \cdot \mathbf{u})$, which by the product rule for the divergence (Section 3.1 of Vol. 4) is $-\nabla \cdot (p\mathbf{u})$. Therefore

$$\frac{\partial \varepsilon}{\partial t} + \nabla \cdot (p\mathbf{u}) = 0. \quad (6.39)$$

Integrating this equation over a fixed volume and applying the divergence theorem (Section 3.3 of Vol. 4) shows that the energy inside the volume increases at the rate $-\oint p\mathbf{u} \cdot d\mathbf{S}$ over its surface. The vector $p\mathbf{u}$ is therefore the rate at which energy flows across unit area, and it is the rate at which the fluid on one side of a small surface does work on the fluid on the other side, since the excess pressure pushes on a surface of area dS with the force $p dS$, and the fluid there moves with the velocity $\mathbf{u}$. The ambient pressure does the work $p_0 \mathbf{u} \cdot d\mathbf{S}$ per unit time as well, but its mean over a cycle is zero, since $\mathbf{u}$ oscillates about zero. The intensity I is the mean over a cycle of the component of $p\mathbf{u}$ along the direction in which the wave travels, in agreement with the definition of Section 3.4.

For a plane harmonic wave with the pressure amplitude $|\mathcal{A}|$, the excess pressure is $|\mathcal{A}| \cos(\omega t - kx + \phi)$ and its mean square over a cycle is $\frac{1}{2}|\mathcal{A}|^2$, since the mean of $\cos^2$ over a cycle is one half. The quantity $p_{\text{rms}} = |\mathcal{A}|/\sqrt{2}$ is called the root mean square pressure, and the intensity is

$$I = \frac{\langle p^2 \rangle}{\rho_0 c} = \frac{p_{\text{rms}}^2}{\rho_0 c}. \quad (6.40)$$

The mean energy per unit volume is $\langle \varepsilon \rangle = p_{\text{rms}}^2 / \rho_0 c^2$, so $I = c \langle \varepsilon \rangle$,

and the energy of a plane sound wave travels at the speed of sound, as the energy of a wave on a string travels at the speed of the wave. You will show in Problem 1 that the intensity, written with the amplitude of the displacement, has the same form as the mean power carried by a string.

The Intensity of a Spherical Wave

In a spherical wave the pressure and the particle velocity are not in phase, and we need the mean of the product of two harmonic quantities with different phases. Let $p = \text{Re}(\hat{p}e^{i\omega t})$ and $u = \text{Re}(\hat{u}e^{i\omega t})$, where $\hat{p} = |\hat{p}|e^{i\phi_p}$ and $\hat{u} = |\hat{u}|e^{i\phi_u}$ are the complex amplitudes, so that $p = |\hat{p}|\cos(\omega t + \phi_p)$ and $u = |\hat{u}|\cos(\omega t + \phi_u)$. By the trigonometric identity that the product of two cosines is half the sum of the cosine of the difference of their arguments and the cosine of the sum,

$$pu = \frac{1}{2}|\hat{p}||\hat{u}| [\cos(\phi_p - \phi_u) + \cos(2\omega t + \phi_p + \phi_u)] . \quad (6.41)$$

The second term oscillates at the angular frequency 2ω and its mean over a cycle is zero, so

$$\langle pu \rangle = \frac{1}{2}|\hat{p}||\hat{u}| \cos(\phi_p - \phi_u) = \frac{1}{2}\text{Re}(\hat{p}\hat{u}^*) . \quad (6.42)$$

For the spherical wave of Section 6.2, $\hat{p} = \mathcal{A}e^{-ikr}/r$ and $\hat{u} = \hat{p}(1 - i/kr)/\rho_0 c$, so that $\hat{p}\hat{u}^* = |\hat{p}|^2(1 + i/kr)/\rho_0 c$, whose real part is $|\hat{p}|^2/\rho_0 c$. The intensity is

$$I = \frac{|\hat{p}|^2}{2\rho_0 c} = \frac{|\mathcal{A}|^2}{2\rho_0 cr^2} , \quad (6.43)$$

and the relation $I = p_{\text{rms}}^2/\rho_0 c$ of a plane wave holds at every distance from the source, even close to it, where the pressure and the particle velocity are far from being in phase. The part of the particle velocity a quarter of a period behind the pressure contributes nothing to the mean, since $\cos(\pi/2) = 0$. It carries energy outwards during one half of each cycle and back towards the source during

the other half, and near the source it holds a store of kinetic energy that stays with the source, as you will find in Problem 5.

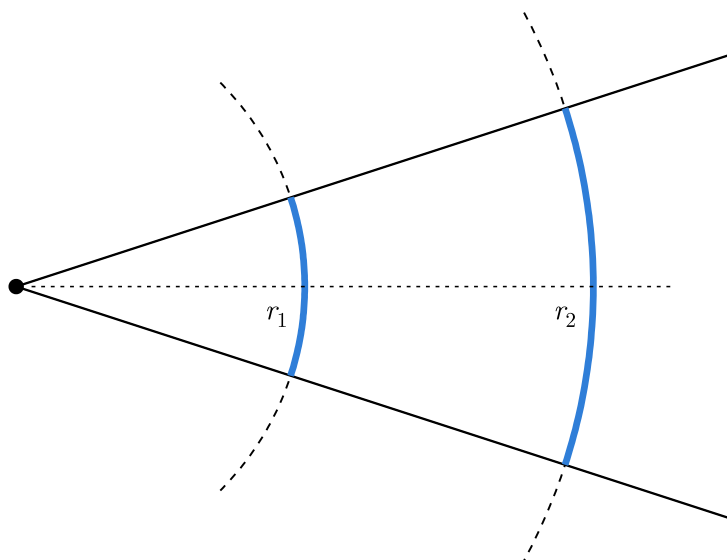


Figure 94. The mean power emitted by a point source into a narrow cone crosses the patches (accent) of the spheres of radius r_1 and r_2 cut out by the same cone. The areas of the two patches are in the ratio r_2^2/r_1^2 , so the intensity falls in the inverse ratio.

The mean power crossing a sphere of radius r about the source is

$$\langle \mathcal{P} \rangle = 4\pi r^2 I = \frac{2\pi |\mathcal{A}|^2}{\rho_0 c}, \quad (6.44)$$

which does not depend on r , as the conservation of energy requires in a fluid that absorbs no energy. The intensity is therefore $I = \langle \mathcal{P} \rangle / 4\pi r^2$, which is the inverse-square law for a point source quoted in Section 3.1, and Figure 94 shows the geometry behind it. Section 5.1 obtained the fall of the amplitude of a spherical wave as $1/r$ from the same requirement, and here the intensity has been calculated from the pressure and the particle velocity at every distance.

Decibel Scales

The human ear responds to sound over a range of intensities of about twelve powers of ten, and the loudness that a listener perceives grows roughly as the logarithm of the intensity. Sound levels are therefore measured on logarithmic scales. The intensity level of a sound of intensity I is defined as

$$L_I = 10 \log_{10} \frac{I}{I_{\text{ref}}} \text{ dB}, \quad I_{\text{ref}} = 10^{-12} \text{ W m}^{-2}, \quad (6.45)$$

where the unit dB is called the decibel. It is one tenth of a bel, which is the unit of the common logarithm of a ratio of powers. The reference intensity is roughly the intensity of the faintest sound that a young listener can hear at frequencies near one kilohertz. The sound pressure level of a sound whose root mean square pressure is p_{rms} is defined as

$$L_p = 20 \log_{10} \frac{p_{\text{rms}}}{p_{\text{ref}}} \text{ dB} = 10 \log_{10} \frac{p_{\text{rms}}^2}{p_{\text{ref}}^2} \text{ dB}, \quad (6.46)$$

with the reference pressure $p_{\text{ref}} = 2 \times 10^{-5} \text{ Pa}$ in air. Underwater acoustics commonly uses 10^{-6} Pa instead, so the reference has to be stated with any level quoted for sound in water. The factor 20 appears because the intensity is proportional to the square of the pressure, so that a given ratio of intensities corresponds to the same number of decibels on either scale. The sound pressure level is more often quoted, since most microphones respond to the pressure.

For a plane wave, $I = p_{\text{rms}}^2 / \rho_0 c$, and the intensity level and the sound pressure level differ by

$$L_I - L_p = 10 \log_{10} \frac{p_{\text{ref}}^2}{\rho_0 c I_{\text{ref}}} \text{ dB}. \quad (6.47)$$

The reference values were chosen so that $p_{\text{ref}}^2 / I_{\text{ref}}$ is $400 \text{ kg m}^{-2} \text{ s}^{-1}$, which is close to the characteristic impedance of air under ordinary conditions, and L_I and L_p for a plane wave in air differ by a small fraction of a decibel. A ratio of ten in intensity is a difference of 10

dB, and a ratio of two is a difference of $10 \log_{10} 2$, which is close to 3 dB.

Combining the Levels of Two Sources

Let two sources give, at some point, the excess pressures p_1 and p_2 , with the sound pressure levels L_{p1} and L_{p2} when each sounds alone. The total pressure is $p_1 + p_2$, and its mean square is

$$\langle (p_1 + p_2)^2 \rangle = \langle p_1^2 \rangle + \langle p_2^2 \rangle + 2\langle p_1 p_2 \rangle. \quad (6.48)$$

The result depends on the last term, and there are two cases to consider. If the sources are incoherent in the sense of Section 4.6, as two musicians playing the same note are, the phase difference between p_1 and p_2 wanders, and the mean of $p_1 p_2$ over a long time is zero. The mean squares, and therefore the intensities, add, and the combined level is

$$L_p = 10 \log_{10} (10^{L_{p1}/10} + 10^{L_{p2}/10}) \text{ dB}. \quad (6.49)$$

For two incoherent sources of equal level, the combined level is $10 \log_{10} 2$, about 3 dB, higher than that of either source alone.

If the sources are coherent, as two loudspeakers driven from the same signal are in Section 4.6, the pressures have the same frequency and a fixed phase difference $\Delta\phi$ at the point, which depends on the difference between the distances from the point to the two sources. The mean of the product, found above, is $\langle p_1 p_2 \rangle = p_{1,\text{rms}} p_{2,\text{rms}} \cos \Delta\phi$, and

$$\langle (p_1 + p_2)^2 \rangle = p_{1,\text{rms}}^2 + p_{2,\text{rms}}^2 + 2p_{1,\text{rms}} p_{2,\text{rms}} \cos \Delta\phi. \quad (6.50)$$

Where two sources of equal level arrive in phase, the mean square pressure is four times that of one source, and the level is $20 \log_{10} 2$, about 6 dB, higher. Where they arrive in antiphase, the pressures cancel. Averaging over the phase difference, whose cosine has the mean zero, gives the incoherent result, so the mean square pressure of two incoherent sources is the average, over the points of an

interference pattern, of the mean square pressure of two coherent sources.

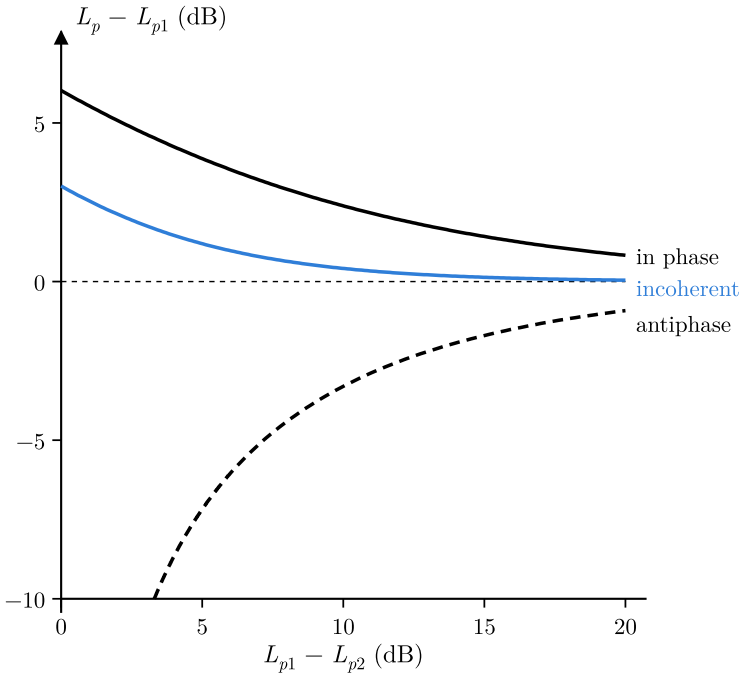


Figure 95. The combined sound pressure level L_p of two sources, less the level L_{p1} of the louder source, against the difference $L_{p1} - L_{p2}$ between the levels of the two sources alone. The curves are for incoherent sources (accident), and for coherent sources arriving in phase (solid) and in antiphase (dashed).

The increase of the combined level above the larger of the two levels is drawn in Figure 95, where the ratio of the root mean square pressures is $p_{2,\text{rms}}/p_{1,\text{rms}} = 10^{-(L_{p1}-L_{p2})/20}$. When one source is louder than the other by ten decibels, the weaker source raises the combined level by less than half a decibel if the two sources are incoherent, while for coherent sources it can still raise or lower the level by two or three decibels, according to the phase difference.

The Fall of the Level with Distance

The intensity at a distance r from a point source is $\langle \mathcal{P} \rangle / 4\pi r^2$, so the levels at the distances r_1 and r_2 differ by

$$L_I(r_1) - L_I(r_2) = 10 \log_{10} \frac{r_2^2}{r_1^2} \text{ dB} = 20 \log_{10} \frac{r_2}{r_1} \text{ dB}, \quad (6.51)$$

and the level falls by $20 \log_{10} 2$, about 6 dB, each time the distance from the source is doubled. The result holds for a point source in an open space, far from any surface that reflects sound, and in a fluid that absorbs no energy. In a room, the sound reflected from the walls adds to the sound arriving directly, and the level falls more slowly with distance; room acoustics is not treated in this course. For a long line of incoherent sources, such as the traffic on a long straight road, the intensities of the sources add, and summing the inverse-square contributions of the sources along the line gives a total intensity that falls as $1/r$, as the intensity of the cylindrical waves of Section 5.1 does and the level falls by about 3 dB for each doubling of the distance.

Reflection at the Boundary between Two Fluids

As a last application, consider a plane wave travelling towards increasing x in a fluid of density ρ_1 and speed of sound c_1 , which meets at normal incidence a plane boundary at $x = 0$ with a second fluid, of density ρ_2 and speed of sound c_2 , as in Figure 96. Let $z_1 = \rho_1 c_1$ and $z_2 = \rho_2 c_2$ be the characteristic impedances of the two fluids. The frequency is the same on both sides of the boundary, since the conditions at the boundary have to hold at every instant, and the wave numbers are $k_1 = \omega/c_1$ and $k_2 = \omega/c_2$. The excess pressure is

$$\begin{aligned} p &= \mathcal{A} [e^{i(\omega t - k_1 x)} + \mathcal{R} e^{i(\omega t + k_1 x)}] \quad (x < 0), \\ p &= \mathcal{T} \mathcal{A} e^{i(\omega t - k_2 x)} \quad (x > 0), \end{aligned} \quad (6.52)$$

where $\mathcal{R}$ and $\mathcal{T}$ are the reflection and transmission coefficients for the pressure. By the results of Section 6.2, the particle velocity of each wave is its pressure divided by the characteristic impedance

of its fluid, with a minus sign for the reflected wave, which travels towards decreasing x , as you showed in Problem 1 of that section.

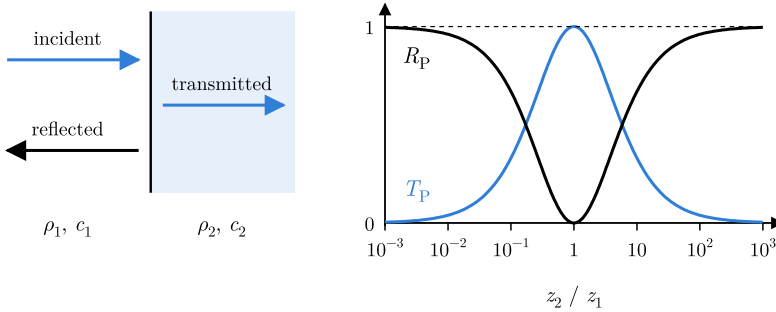


Figure 96. Left: a plane sound wave meets at normal incidence the boundary between two fluids, and is partly reflected and partly transmitted. Right: the fractions of the incident power reflected, $\mathcal{R}_P$ (solid), and transmitted, $\mathcal{T}_P$ (accent), against the ratio z_2/z_1 of the characteristic impedances, on a logarithmic scale.

There are two conditions at the boundary. The first is that the pressure is continuous across it, since the boundary has no mass, and a net force on a small piece of it would give it an unlimited acceleration. The second is that the particle velocity normal to the boundary is continuous, since the two fluids stay in contact and neither passes into the other. At $x = 0$ the two conditions give

$$1 + \mathcal{R} = \mathcal{T}, \quad \frac{1 - \mathcal{R}}{z_1} = \frac{\mathcal{T}}{z_2}. \quad (6.53)$$

Substituting the first equation into the second and solving for $\mathcal{R}$, and then for $\mathcal{T}$, gives

$$\mathcal{R} = \frac{z_2 - z_1}{z_2 + z_1}, \quad \mathcal{T} = \frac{2z_2}{z_1 + z_2}. \quad (6.54)$$

These have the form found for two strings in Section 4.3, where the reflection coefficient for the displacement is $(Z_1 - Z_2)/(Z_1 + Z_2)$. For sound, the reflection coefficient for the particle velocity, and therefore for the displacement, is $-\mathcal{R} = (z_1 - z_2)/(z_1 + z_2)$, of the

same form, while the pressure, which corresponds to the transverse force on the string, is reflected with the opposite sign.

A plane wave of pressure amplitude $|\hat{p}|$ in a fluid of characteristic impedance z has the intensity $|\hat{p}|^2/2z$, so the fractions of the incident power that are reflected and transmitted are

$$\mathcal{R}_P = \mathcal{R}^2 = \left(\frac{z_2 - z_1}{z_2 + z_1} \right)^2, \quad \mathcal{T}_P = \frac{z_1}{z_2} \mathcal{T}^2 = \frac{4z_1 z_2}{(z_1 + z_2)^2}, \quad (6.55)$$

and they add to one, since $(z_2 - z_1)^2 + 4z_1 z_2 = (z_1 + z_2)^2$. The fractions $\mathcal{R}_P$ and $\mathcal{T}_P$ depend only on the ratio z_2/z_1 , as Figure 96 shows. All the power is transmitted when the impedances are equal, even if the densities and the speeds of sound in the two fluids differ. The argument of Section 4.4 for a section of string a quarter of a wavelength long carries over to sound, so that a layer of a third fluid a quarter of a wavelength thick, whose characteristic impedance is $\sqrt{z_1 z_2}$, transmits all the power from one fluid to the other at one frequency.

When $z_2 \gg z_1$, as for sound in air meeting a rigid wall, $\mathcal{R}$ tends to one. The reflected pressure then adds to the incident pressure at the boundary, so that the pressure there is twice the incident pressure, and the particle velocity is zero, as at the closed end of a pipe in Section 4.5. When $z_2 \ll z_1$, as for sound in water meeting the surface of the water from below, $\mathcal{R}$ tends to minus one and the pressure at the boundary is zero, which is the condition Section 4.5 used as an approximation at the open end of a pipe.

Sound travelling from air into water has z_2/z_1 of several thousand. The pressure transmission coefficient is then close to two, and the pressure amplitude in the water is almost twice that of the incident wave. This seems to suggest that sound passes easily into the water, but the transmitted power is small, because the particle velocity in the water is the pressure divided by the large impedance z_2 . The fraction transmitted is approximately $4z_1/z_2$, of the order of a thousandth, which is a loss of about 30 dB and one reason why sounds made in air are hard to hear under water. Since $\mathcal{T}_P$ does not change when z_1 and z_2 are exchanged, sound made under water loses the same fraction of its power in passing into the air.

Problems

§6.1 The Acoustic Wave Equation and the Speed of Sound

1. In a fluid that was at rest before a sound wave arrived, $\nabla \times \mathbf{u} = \mathbf{0}$. Differentiate the linearised Euler equation with respect to time, and use the continuity equation, the relation $\rho' = p/c^2$, and the identity $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$, which you proved in Problem 9 of Section 3.1 of Vol. 4, to show that each Cartesian component of $\mathbf{u}$ obeys the acoustic wave equation.
2. Let $\xi(\mathbf{r}, t)$ be the displacement of the element of fluid whose undisturbed position is $\mathbf{r}$, so that $\mathbf{u} = \partial \xi / \partial t$ to the first order. Show from the linearised continuity equation that $\rho' = -\rho_0 \nabla \cdot \xi$ and $p = -B \nabla \cdot \xi$, and check that the second relation reduces to $p = -B \partial \xi / \partial x$ in one dimension.
3. In a reversible adiabatic change of a fixed amount of ideal gas, the temperature multiplied by the power $C_p/C_V - 1$ of the volume is constant (Section 2.2 of Vol. 2). Use this result to show that the excess temperature T' in a sound wave satisfies $T'/T_0 = (C_p/C_V - 1)s$, where T_0 is the temperature of the undisturbed gas, and express T'/T_0 in terms of p/p_0 .
4. The mean speed of the molecules of an ideal gas, found in Section 1.2 of Vol. 2, is $\bar{v}_m = \sqrt{8RT/\pi M_m}$ when written with the molar mass. Show that $c/\bar{v}_m = \sqrt{\pi C_p/8C_V}$, and evaluate the ratio for a monatomic gas and for a diatomic gas.
5. The modes of a pipe of length L open at both ends have the frequencies $nc/2L$ (Section 4.5). Show that a small rise ΔT in the temperature of the air in the pipe raises each frequency by the fraction $\Delta T/2T$. Find the ratio of the frequencies of the same mode when the pipe is filled with helium and with air at the same temperature, in terms of the molar masses and the ratios of the heat capacities of the two gases.

§6.2 Plane and Spherical Sound Waves

1. Show that for a plane harmonic wave travelling towards decreasing x the particle velocity is $u = -p/\rho_0 c$.

2. Let $p = f_1(t - x/c)$ be a plane wave of any shape travelling towards increasing x into fluid at rest. Integrate Euler's equation with respect to time to show that $u = p/\rho_0 c$ at every point and at every instant.
3. Let two plane waves of equal real amplitude $\mathcal{A}$ travel in opposite directions along the x -axis, so that the excess pressure is $p = 2\mathcal{A} \cos(kx) e^{i\omega t}$. Find the particle velocity, and show that the specific acoustic impedance p/u is purely imaginary everywhere. Show that at $x = 0$ the particle velocity is always zero and the pressure oscillates with its largest amplitude, and relate this to the closed end of a pipe in Section 4.5.
4. Find the magnitude of the particle velocity, in units of $|p|/\rho_0 c$, and its phase lag behind the excess pressure, in a spherical wave at the distances $r = \lambda/2\pi$ and $r = \lambda$ from the source.
5. A spherical wave travelling inwards towards the centre has the excess pressure $p = (\mathcal{A}/r) e^{i(\omega t + kr)}$. Find the radial particle velocity and the specific acoustic impedance, and explain the sign of the particle velocity when $kr \gg 1$.

§6.3 Sound Energy, Intensity, and Decibel Scales

1. Show that the intensity $I = p_{\text{rms}}^2/\rho_0 c$ of a plane harmonic sound wave whose displacement has the amplitude ξ_A reduces to the result $I = \frac{1}{2} \rho_0 c \omega^2 \xi_A^2$ of Section 3.4, and compare this with the mean power $\frac{1}{2} \mu v \omega^2 A^2$ carried by a wave on a string (Section 3.4).
2. For the standing wave $p = 2\mathcal{A} \cos(kx) \cos(\omega t)$, with $\mathcal{A}$ real, whose particle velocity you found in Problem 3 of Section 6.2, find the mean kinetic and potential energies per unit volume as functions of x . Show that their sum does not depend on x , and that the intensity is zero everywhere.
3. Show that N incoherent sources of equal level at a point give a level higher than one of them by $10 \log_{10} N$ dB, and that N coherent sources of equal level arriving in phase give a level higher by $20 \log_{10} N$ dB.
4. Show that the fraction of the incident power transmitted across

the boundary between two fluids is the same whichever way the wave crosses it. When $z_2 \gg z_1$, show that $\mathcal{T}_P \approx 4z_1/z_2$, and express the loss of level in the transmitted wave in decibels.

5. For the outgoing spherical wave of Section 6.2, find the mean energy per unit volume $\langle \varepsilon \rangle$ as a function of kr , and show that $I/\langle \varepsilon \rangle = c [2k^2 r^2 / (1 + 2k^2 r^2)]$, which is less than c and tends to c far from the source.

Vibrating Bodies, Musical and Architectural Acoustics

7.1 Pipes, Horns, and Acoustic Waveguides

This part applies the acoustics of the previous part to sound confined by walls. Section 7.1 treats pipes, horns, and ducts as channels that guide sound, and Section 7.2 treats the Helmholtz resonator and the filters that can be made by attaching cavities and chambers to a pipe.

The pipes of Section 4.5 were treated by finding their standing waves. In this section we first describe the same pipes with impedances, which lets a pipe be ended by anything whose response to the pressure is known, then allow the area of the pipe to change along its length, which leads to the theory of horns, and finally allow the pressure to vary across a duct that is wide compared with a wavelength.

The Input Impedance of a Pipe

Consider a uniform pipe of cross-section S lying along the x -axis, as shown in Figure 97. We take the pipe to be narrow compared with a wavelength, so that the excess pressure p and the particle velocity u are nearly the same at every point of a cross-section; the

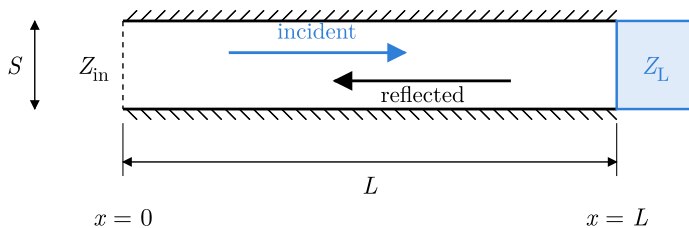


Figure 97. A uniform pipe of cross-section S runs from $x = 0$ to $x = L$, where it is ended by a load of acoustic impedance Z_L . A wave travelling towards the load (accent) is partly reflected by it, and the ratio of the pressure to the volume velocity at $x = 0$ is the input impedance Z_{in} .

condition under which this holds is found at the end of this section. The product uS is the volume of fluid that crosses a section in unit time, and it is called the volume velocity. The ratio

$$Z_A = \frac{p}{uS} \quad (7.1)$$

is called the acoustic impedance at the section. As in Section 1.3, p and u here stand for complex quantities with the time dependence $e^{i\omega t}$, whose real parts are the pressure and the velocity. The real part of Z_A is called the resistance and its imaginary part the reactance, as for the mechanical impedance of Section 1.3.

A harmonic sound wave in the pipe is the sum of a plane wave travelling towards increasing x and one travelling back,

$$p = (\mathcal{A}_+ e^{-ikx} + \mathcal{A}_- e^{ikx}) e^{i\omega t}, \quad (7.2)$$

with $k = \omega/c$. In Section 6.2 the particle velocity of a plane wave travelling towards increasing x was found to be $p/\rho_0 c$. For the wave travelling back, the equation of motion $\rho_0 \partial u / \partial t = -\partial p / \partial x$ of Section 6.1 gives $i\omega \rho_0 u = -ikp$, so its velocity is $-p/\rho_0 c$, and therefore

$$uS = \frac{1}{Z_p} (\mathcal{A}_+ e^{-ikx} - \mathcal{A}_- e^{ikx}) e^{i\omega t}, \quad Z_p = \frac{\rho_0 c}{S}. \quad (7.3)$$

For a single wave travelling towards increasing x the acoustic impedance is Z_p at every section, and Z_p is called the characteristic acoustic impedance of the pipe.

The end of the pipe at $x = L$ may be a rigid cap, an open end, another pipe, or a resonator, among other things. It is called the load, and it is described by the acoustic impedance Z_L that it presents to the fluid in the pipe, so that $p/uS = Z_L$ at $x = L$. Where the load is another pipe, the pressure and the volume velocity are continuous across the junction, since a junction short compared with a wavelength can neither sustain a difference of pressure nor store fluid, so the load is the acoustic impedance at the input of that pipe. Let $\mathcal{R} = \mathcal{A}_- e^{2ikL} / \mathcal{A}_+$ be the ratio of the reflected to the incident pressure at the load. The condition at $x = L$ then reads $Z_L = Z_p(1 + \mathcal{R})/(1 - \mathcal{R})$, and solving for $\mathcal{R}$ gives

$$\mathcal{R} = \frac{Z_L - Z_p}{Z_L + Z_p}. \quad (7.4)$$

A rigid cap, through which no fluid can pass, has $Z_L \rightarrow \infty$ and reflects the pressure without a change of sign. An ideal open end, at which the pressure is held at the ambient value, has $Z_L = 0$ and gives $\mathcal{R} = -1$. A load equal to Z_p reflects nothing, as the string ended by a piston in oil of resistance Z did in Section 4.4.

The acoustic impedance at the input end $x = 0$ is called the input impedance of the pipe, Z_{in} . Setting $x = 0$ in the ratio of the expressions for p and uS gives $Z_{\text{in}} = Z_p(\mathcal{A}_+ + \mathcal{A}_-)/(\mathcal{A}_+ - \mathcal{A}_-)$, and the definition of $\mathcal{R}$ gives $\mathcal{A}_-/\mathcal{A}_+ = \mathcal{R}e^{-2ikL}$. Substituting the expression for $\mathcal{R}$ and multiplying the numerator and the denominator by $(Z_L + Z_p)e^{ikL}$, the numerator becomes $(Z_L + Z_p)e^{ikL} + (Z_L - Z_p)e^{-ikL}$, which by Euler's formula is $2Z_L \cos kL + 2iZ_p \sin kL$, and the denominator becomes $2Z_p \cos kL + 2iZ_L \sin kL$ in the same way. Therefore

$$Z_{\text{in}} = Z_{\text{p}} \frac{Z_{\text{L}} \cos kL + iZ_{\text{p}} \sin kL}{Z_{\text{p}} \cos kL + iZ_{\text{L}} \sin kL}. \quad (7.5)$$

A pipe whose length is a whole number of half wavelengths has $\sin kL = 0$ and presents the load unchanged at its input, and a pipe a quarter of a wavelength long has $\cos kL = 0$ and presents $Z_{\text{p}}^2/Z_{\text{L}}$, which is the property that the quarter-wave section of Section 4.4 relies on.

Resonances of Open and Closed Pipes

Letting $Z_{\text{L}} \rightarrow \infty$, for a pipe closed at $x = L$ by a rigid cap, and setting $Z_{\text{L}} = 0$, for a pipe open at $x = L$, gives

$$\begin{aligned} Z_{\text{in}} &= -iZ_{\text{p}} \cot kL & (\text{closed far end}), \\ Z_{\text{in}} &= iZ_{\text{p}} \tan kL & (\text{open far end}). \end{aligned} \quad (7.6)$$

Both input impedances are purely imaginary, since neither end absorbs energy, and Figure 98 shows them as functions of kL .

The extremes of the input impedance are the resonances of the pipe. A mode of a pipe whose end at $x = 0$ is closed has $u = 0$ and $p \neq 0$ there, so the rest of the pipe must present an infinite input impedance at the frequency of the mode, and a mode of a pipe whose end at $x = 0$ is open has $p = 0$ and $u \neq 0$ there, so the input impedance must be zero. For the pipe open at its far end, the zeros of $\tan kL$ at $kL = n\pi$ give the modes of a pipe open at both ends, with $\omega_n = n\pi c/L$, and its infinities at $kL = (n - \frac{1}{2})\pi$ give the modes of a pipe closed at one end, with $\omega_n = (2n - 1)\pi c/2L$. For the pipe closed at its far end, the infinities of $\cot kL$ give the modes of a pipe closed at both ends, and its zeros give the pipe closed at one end once more. These are the frequencies found in Section 4.5.

The same extremes govern a pipe that is driven. A piston at $x = 0$ moving with a given velocity produces the pressure $Z_{\text{in}}uS$ at its face, which is largest at the maxima of $|Z_{\text{in}}|$, while a given pressure at an open input produces the volume velocity p/Z_{in} , which is largest at the zeros. A clarinet, for example, is often modelled as a cylinder closed at the reed, and it sounds near the maxima of its input

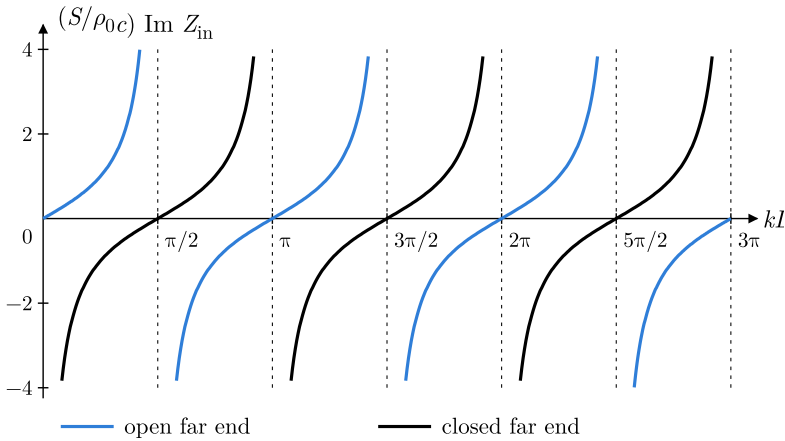


Figure 98. The reactance of the input impedance of a pipe, in units of $Z_p = \rho_0 c / S$, for a pipe open at its far end ($\tan kL$, accent) and for one closed at its far end ($-\cot kL$). Each zero of one curve lies at an infinity of the other, and both curves repeat whenever kL increases by π .

impedance, while a flute, whose embouchure hole is open to the air, sounds near the minima. In a real pipe the friction of the air at the walls and the sound radiated from an open end remove energy, and the peaks of $|Z_{\text{in}}|$ have finite heights and widths, as the resonance of a damped oscillator has in Section 1.3.

The End Correction

The condition $p = 0$ at an open end is an approximation. The air just beyond the end moves with the air inside the pipe, and since this air has mass, a pressure is needed at the end to accelerate it. Suppose that the moving air outside behaves like a plug of cross-section S and length ΔL (and ΔL in place of δL throughout the end-correction passage, including the display for Z_L , the display for Z_{in} , and "an ideal open pipe of length $L + \Delta L$, and ΔL is called the end correction"), that is, a short column of air in which every part has the same displacement. The force pS needed to give its mass $\rho_0 S \delta L$ the acceleration $i\omega u$ satisfies $pS = \rho_0 S \delta L i\omega u$, so the load it presents is

$$Z_L = \frac{i\omega\rho_0\delta L}{S} = iZ_p k\delta L. \quad (7.7)$$

When $k\delta L$ is small it is close to $\tan k\delta L$. Substituting $Z_L = iZ_p \tan k\delta L$ into the input impedance and dividing the numerator and the denominator by $Z_p \cos kL$ gives

$$Z_{\text{in}} = iZ_p \frac{\tan k\delta L + \tan kL}{1 - \tan k\delta L \tan kL} = iZ_p \tan k(L + \delta L), \quad (7.8)$$

by the addition formula $\tan(\theta_1 + \theta_2) = (\tan\theta_1 + \tan\theta_2)/(1 - \tan\theta_1 \tan\theta_2)$. The pipe therefore behaves as an ideal open pipe of length $L + \delta L$, and δL is called the end correction. Its value is about 0.61 times the radius of the pipe for an end that stands free in the air, and about 0.82 times the radius for an end set in a large flat wall. Finding these values requires the theory of the radiation of sound from the end of the pipe, which, as Section 4.5 noted, is beyond the scope of this course. The open end also has a small resistance, through which the pipe loses energy by radiating sound.

Webster's Horn Equation

A horn is a pipe whose area $S(x)$ changes along its length; its narrow end is called the throat and its wide end the mouth. We take the area to change little over a distance equal to the width of the horn, and the width to be small compared with a wavelength, so that the pressure and the velocity are again nearly uniform over each cross-section. The equation they obey follows from the laws of Section 6.1 applied to the thin slice between x and $x + dx$ shown in Figure 99.

The excess mass in the slice is $\rho'S dx$, and by the conservation of mass it increases at the rate at which fluid flows in at x less the rate at which it flows out at $x + dx$. To the first order in small quantities the mass crossing a section in unit time is $\rho_0 u S$, so

$$S \frac{\partial \rho'}{\partial t} = -\rho_0 \frac{\partial (uS)}{\partial x}. \quad (7.9)$$

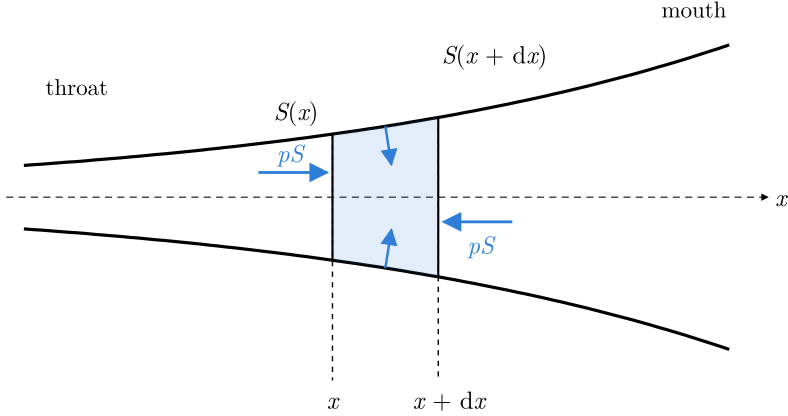


Figure 99. A thin slice of a horn between x and $x + dx$ (shaded). The fluid in the slice is pushed forwards by the pressure on its face of area $S(x)$ and backwards by the pressure on its face of area $S(x + dx)$, and the flaring wall pushes on it along its inward normal, which has a component along the axis.

For the equation of motion, the pressure pushes the slice forwards with the force $pS(x)$ on its face at x and backwards with the force $pS(x + dx)$ on its face at $x + dx$. The wall pushes on the fluid along its inward normal, which in a flaring horn has a component pointing forwards along the axis. The part of the wall between the faces, projected onto a plane across the axis, is a ring of area $(dS/dx) dx$, so the wall adds the force $p (dS/dx) dx$ along the axis. By the product rule the total, $-\partial(pS)/\partial x dx + p (dS/dx) dx$, is $-S (\partial p/\partial x) dx$, and dividing by the mass $\rho_0 S dx$ of the slice gives the equation of motion of Section 6.1, $\rho_0 \partial u/\partial t = -\partial p/\partial x$, unaffected by the flare.

To combine the two equations, we differentiate the equation of mass conservation with respect to time, replace ρ' by p/c^2 as in Section 6.1, and replace $\rho_0 \partial u/\partial t$ by $-\partial p/\partial x$. This gives $(S/c^2) \partial^2 p/\partial t^2 = \partial(S \partial p/\partial x)/\partial x$, or

$$\frac{\partial^2 p}{\partial x^2} + \frac{1}{S} \frac{dS}{dx} \frac{\partial p}{\partial x} = \frac{1}{c^2} \frac{\partial^2 p}{\partial t^2}, \quad (7.10)$$

which is called Webster's equation, after A. G. Webster, who used it in 1919 to study horns. For a uniform pipe the second term on

the left is zero, and the equation is the wave equation of Section 6.1 in one dimension.

The Exponential Horn

As a first example, consider a horn whose area grows exponentially with distance from the throat, $S = S_0 e^{2\alpha x}$, so that its radius grows as $e^{\alpha x}$. The constant α is called the flare constant. Here $(1/S) dS/dx = 2\alpha$, and for a harmonic wave $p = \mathcal{A}(x)e^{i\omega t}$ Webster's equation becomes

$$\frac{d^2 \mathcal{A}}{dx^2} + 2\alpha \frac{d\mathcal{A}}{dx} + k^2 \mathcal{A} = 0. \quad (7.11)$$

To remove the first derivative, let $\mathcal{A} = e^{-\alpha x} \psi(x)$. Then $\mathcal{A}' = (\psi' - \alpha\psi)e^{-\alpha x}$ and $\mathcal{A}'' = (\psi'' - 2\alpha\psi' + \alpha^2\psi)e^{-\alpha x}$, where the primes denote derivatives with respect to x , and substituting these gives

$$\frac{d^2 \psi}{dx^2} + (k^2 - \alpha^2) \psi = 0. \quad (7.12)$$

When $k > \alpha$, let $k'^2 = k^2 - \alpha^2$. The solutions for ψ are $e^{\mp ik'x}$, and a wave travelling away from the throat is

$$p = \mathcal{A}_0 e^{-\alpha x} e^{i(\omega t - k'x)}, \quad \omega^2 = c^2 k'^2 + \omega_c^2, \quad \omega_c = \alpha c. \quad (7.13)$$

This dispersion relation has the form of the relation $\omega^2 = v^2 k^2 + \omega_c^2$ for the string on an elastic foundation of Section 4.4, with the flare of the horn in the place of the foundation. The amplitude of the pressure goes as $e^{-\alpha x} = \sqrt{S_0/S}$, so the product $|p|^2 S$ is the same at every section.

To find the power, we need the particle velocity. The equation of motion gives $i\omega\rho_0 u = -\partial p/\partial x = (\alpha + ik')p$, so the specific acoustic impedance $z = p/u$ is $i\omega\rho_0/(\alpha + ik')$. Multiplying the numerator and the denominator by $\alpha - ik'$, and using $\alpha^2 + k'^2 = k^2$ and $\omega/k = c$, gives

$$z = \rho_0 c \left[\sqrt{1 - \frac{\omega_c^2}{\omega^2}} + i \frac{\omega_c}{\omega} \right]. \quad (7.14)$$

The power carried through a section is the force pS multiplied by the velocity u , and the mean over a cycle of the product of two harmonic quantities is half the real part of the product of one complex amplitude with the complex conjugate of the other, as stated in Section 1.1. With $u = p/z$ and $|z| = \rho_0 c$, the mean power is

$$\langle \mathcal{P} \rangle = \frac{1}{2} \operatorname{Re} (p u^*) S = \frac{|p|^2 S}{2\rho_0 c} \sqrt{1 - \frac{\omega_c^2}{\omega^2}}, \quad (7.15)$$

which is the same at every section, since $|p|^2 S$ is, so the energy entering at the throat is carried along the horn without loss.

When $k < \alpha$, the equation for ψ has the real solutions $e^{\pm\beta x}$, with $\beta = \sqrt{\alpha^2 - k^2}$. In a long horn driven at its throat, the quantity $|p|^2 S$, which is proportional to $|\psi|^2$, cannot grow without limit along the horn, so $\psi = e^{-\beta x}$ and p goes as $e^{-(\alpha+\beta)x}$. The specific acoustic impedance $i\omega\rho_0/(\alpha + \beta)$ is then purely imaginary, so the mean power is zero, and the disturbance is evanescent, as on the string on an elastic foundation below its cutoff in Section 4.4. For this reason $\omega_c = \alpha c$ is called the cutoff angular frequency of the horn. Well below it, β is close to α , and the impedance is close to $i\omega\rho_0/2\alpha$, which is the impedance of a plug of air of length $1/2\alpha$ by the calculation of the end correction, so the source at the throat only accelerates the air near it back and forth.

Figure 100 shows the impedance at the throat. Well above the cutoff the resistance is close to $\rho_0 c$, the value for a plane wave in a uniform pipe, so a small source at the throat delivers power as it would to a plane wave, which is one reason horns have been used with loudspeakers and on brass instruments. A horn that works at low frequencies needs a small flare constant, and it must then be long before its mouth is wide.

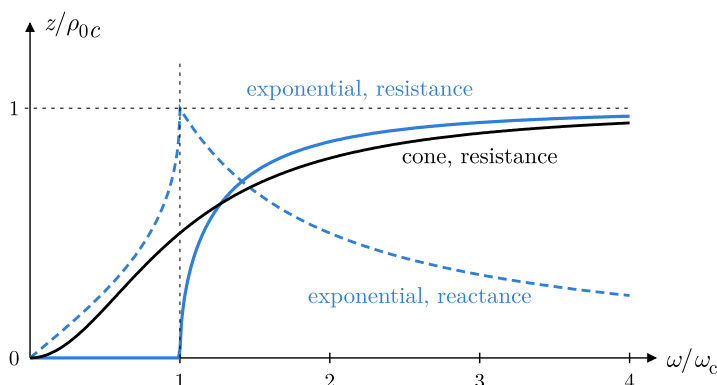


Figure 100. The specific acoustic impedance at the throat of an exponential horn (accent), in units of $\rho_0 c$, against ω/ω_c : the resistance (solid) is zero below the cutoff and rises towards $\rho_0 c$ above it, and the reactance (dashed) is largest at the cutoff. The black curve is the resistance at the throat of a conical horn whose throat lies a distance $1/\alpha$ from its apex, which rises gradually with no cutoff.

The Conical Horn

As a second example, consider a cone, whose area grows as the square of the distance x from its apex, $S = S_0(x/x_0)^2$, with the throat at $x = x_0$. Here $(1/S)dS/dx = 2/x$, and by the product rule $\partial^2(xp)/\partial x^2 = x \partial^2 p/\partial x^2 + 2 \partial p/\partial x$, so multiplying Webster's equation by x gives

$$\frac{\partial^2(xp)}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2(xp)}{\partial t^2}. \quad (7.16)$$

The product xp obeys the wave equation in one dimension, and a wave travelling away from the apex is $p = (\mathcal{A}/x) e^{i(\omega t - kx)}$. This is the spherical wave of Section 6.2, with the distance x from the apex in the place of the distance r from the source. The particle velocity of the spherical wave is along the radial lines, which the walls of the cone follow, so the cone guides a part of a spherical wave without disturbing it, and there is no cutoff. The equation of motion gives $i\omega\rho_0 u = (1/x + ik)p$, so

$$z = \rho_0 c \frac{ikx}{1 + ikx}, \quad (7.17)$$

whose real part at the throat, $\rho_0 c (kx_0)^2 / [1 + (kx_0)^2]$, rises gradually from zero as the frequency rises. Figure 100 compares it with the exponential horn. The resistance of the exponential horn rises more steeply above its cutoff, and a designer who needs a horn to present nearly $\rho_0 c$ over a given band of frequencies might choose the exponential shape for this reason.

Modes of a Rectangular Duct

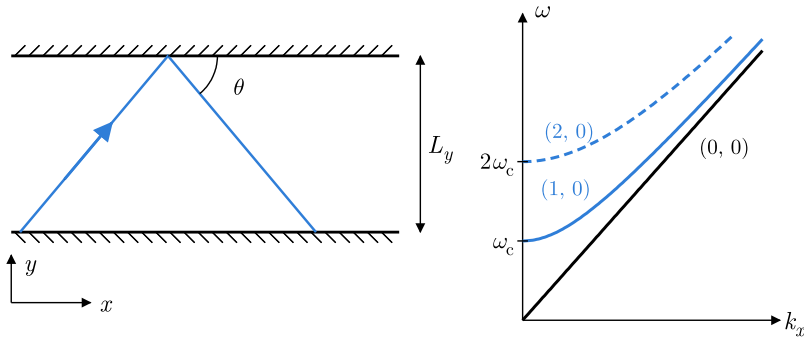


Figure 101. Left: in the mode $(1,0)$ of a rectangular duct, the sound travels as a plane wave that is reflected back and forth between the walls $y = 0$ and $y = L_y$, making an angle θ with the axis. Right: the dispersion relations of the modes $(0,0)$, $(1,0)$, and $(2,0)$, each of which travels only above its cutoff angular frequency, with $\omega_c = \pi c/L_y$ for the mode $(1,0)$.

We now drop the assumption that the pipe is narrow. Consider a duct with rigid walls along the x -axis, whose cross-section is a rectangle with walls at $y = 0$ and $y = L_y$ and at $z = 0$ and $z = L_z$. The excess pressure obeys the wave equation $\nabla^2 p = (1/c^2) \partial^2 p / \partial t^2$ of Section 6.1. At a rigid wall the normal component of the particle velocity is zero, and since $\rho_0 \partial \mathbf{u} / \partial t = -\nabla p$, the normal derivative of the pressure is zero there. The functions $\cos(n_y \pi y / L_y)$ and $\cos(n_z \pi z / L_z)$, with n_y and n_z whole numbers starting from zero, have zero derivatives at the walls, so we try

$$p = \mathcal{A} \cos \frac{n_y \pi y}{L_y} \cos \frac{n_z \pi z}{L_z} e^{i(\omega t - k_x x)}. \quad (7.18)$$

Substituting it into the wave equation gives $k_x^2 + (n_y \pi / L_y)^2 + (n_z \pi / L_z)^2 = \omega^2 / c^2$, so each pair (n_y, n_z) labels a mode of the duct with the dispersion relation

$$\omega^2 = c^2 k_x^2 + \omega_c^2, \quad \omega_c = \pi c \sqrt{\frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2}}, \quad (7.19)$$

which has the same form as that of the exponential horn. The mode $(0, 0)$ is a plane wave with $\omega = ck_x$. Every other mode has a cutoff angular frequency ω_c , below which k_x is imaginary and the mode is evanescent. Above the cutoff, differentiating the dispersion relation gives $\omega d\omega = c^2 k_x dk_x$, so the group velocity of Section 5.2 and the phase velocity are

$$v_g = \frac{d\omega}{dk_x} = c \sqrt{1 - \frac{\omega_c^2}{\omega^2}}, \quad v_p = \frac{\omega}{k_x} = \frac{c}{\sqrt{1 - \omega_c^2/\omega^2}}. \quad (7.20)$$

These modes are dispersive, with $v_p v_g = c^2$, and Figure 101 shows the dispersion relations of the first few modes that vary across the side L_y .

By Euler's formula, $\cos(\pi y / L_y) = \frac{1}{2}(e^{i\pi y / L_y} + e^{-i\pi y / L_y})$, so the mode $(1, 0)$ is the sum of two plane waves whose wave vectors have the components $(k_x, \pm \pi / L_y, 0)$. Each wave vector has magnitude ω / c and makes an angle θ with the axis, where $\cos \theta = ck_x / \omega$, and reflection at a wall turns each plane wave into the other. The sound therefore travels along the zigzag path of Figure 101 at the speed c , and advances along the duct at $c \cos \theta$, which is the group velocity found above. As the frequency falls to the cutoff, θ rises to a right angle, and the waves are reflected straight across the duct without advancing along it.

Let L_y be the larger side. The lowest cutoff other than that of the plane wave is then $\pi c / L_y$, at which the wavelength is $2L_y$. At lower frequencies only the plane wave travels along the duct, and

any other mode set up by a source or a bend dies away within a distance of about $1/|k_x|$, which is comparable with L_y unless the frequency is close to that cutoff. This is the condition assumed earlier in this section: a pipe can be treated as one-dimensional when its width is less than half a wavelength. The modes of a pipe of circular cross-section involve functions that are not treated in this course, but their lowest cutoff also falls at a wavelength comparable with the width of the pipe.

7.2 Helmholtz Resonators and Acoustic Filters

A bottle sounds a note when air is blown across its mouth, and for a bottle with a narrow neck the note is lower than a pipe of the same length closed at one end would give. The air in the bottle oscillates as a single mass on a spring, with a wavelength much longer than the bottle, whereas the modes of a pipe found in Section 4.5 have wavelengths comparable with its length. We begin by treating such a cavity as an oscillator of the kind studied in Sections 1.1 to 1.3, and then use it, together with other pieces of pipe, to make filters that pass sound in some bands of frequency and reflect it in others.

The Helmholtz Resonator

Consider a cavity of volume V with a short neck of length ℓ and cross-section S , as shown in Figure 102, and suppose that every dimension of the cavity and the neck is small compared with a wavelength. The pressure in the cavity is then nearly uniform, since a difference of pressure between two parts of it would be evened out by sound in a time much shorter than a period. The air in the neck is hardly compressed, since the neck is short compared with a wavelength, and it moves back and forth as a plug. The air just beyond each end of the neck moves with it, as the air beyond the open end of a pipe did in Section 7.1, and this is included by replacing ℓ with ℓ' , the length of the neck increased by an end correction at each end. The mass of the plug is then $m = \rho_0 S \ell'$.

Let x be the displacement of the plug into the cavity. The plug pushes a volume Sx of air into the cavity, so the density there rises by the fraction Sx/V , which is the condensation, and the excess pressure is $BSx/V = \rho_0 c^2 Sx/V$, with the adiabatic bulk modulus

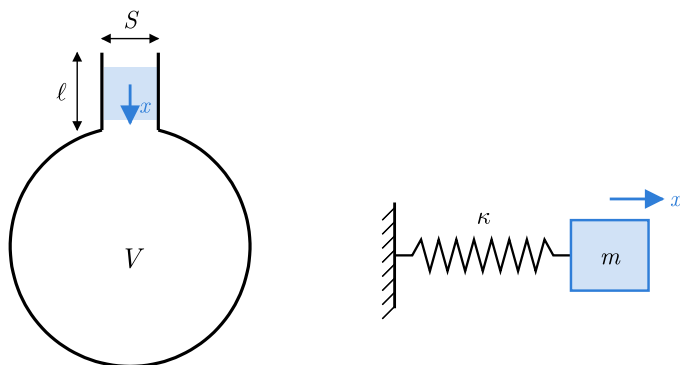


Figure 102. Left: a Helmholtz resonator, a cavity of volume V with a neck of length ℓ and cross-section S . The air in the neck (shaded) moves as a plug, a column of air in which every part has the same displacement, displaced a distance x into the cavity, and the air in the cavity acts as a spring. Right: the oscillator of Section 1.1 with the same equation of motion, a mass m on a spring of constant κ .

for the reason given in Section 6.1. The air outside is at the ambient pressure, so the net force on the plug is $-\rho_0 c^2 S^2 x / V$, and Newton's second law for the plug gives

$$m\ddot{x} = -\kappa x, \quad \kappa = \frac{\rho_0 c^2 S^2}{V}. \quad (7.21)$$

The plug oscillates with simple harmonic motion at the natural angular frequency

$$\omega_0 = \sqrt{\frac{\kappa}{m}} = c\sqrt{\frac{S}{V\ell}}, \quad (7.22)$$

and the arrangement is called a Helmholtz resonator, after Helmholtz, who used such resonators in the 1860s to study the harmonics of musical notes. The frequency depends on the volume of the cavity but not on its shape, provided that all its dimensions are small compared with a wavelength. A larger cavity lowers the frequency because it is a softer spring, and a longer or narrower neck lowers it because the air in the neck then has a larger inertia

for a given volume velocity. The wavelength at the natural frequency is $2\pi\sqrt{V\ell'/S}$, which is long compared with the size of the cavity when the neck is narrow, so that the assumptions above are consistent.

The same result follows from the impedances of Section 7.1. The acoustic impedance at the mouth of the neck is the sum of the impedance $i\omega\rho_0\ell'/S$ of the plug, found for the end correction in Section 7.1, and the impedance $\rho_0c^2/i\omega V$ of the air in the cavity, which you found for a short closed pipe in Problem 1 of Section 7.1,

$$Z_A = i\omega\frac{\rho_0\ell'}{S} + \frac{\rho_0c^2}{i\omega V}. \quad (7.23)$$

The impedances of the plug and of the cavity add because the same volume velocity passes through both, and the pressure at the mouth is the sum of the pressure that accelerates the plug and the pressure in the cavity. The sum is zero at ω_0 , so at this frequency the air can oscillate with no pressure applied at the mouth. The acoustic impedance Z_A is also Z_m/S^2 , where $Z_m = i(m\omega - \kappa/\omega)$ is the mechanical impedance of Section 1.3 without damping, since a force F on the plug is a pressure F/S at the mouth and a velocity $\dot{x}$ is a volume velocity $S\dot{x}$.

Losses and the Driven Resonator

A real resonator loses energy in several ways, among them the friction between the moving air and the walls of the neck and the radiation of sound from the mouth. We describe their combined effect by a resistive force $-b\dot{x}$ on the plug, where b is a constant to be measured, for example from the number of cycles of a free decay, as in Section 1.2. The damping rate is $\gamma = b/2m$ and the quality factor is $Q = \omega_0/2\gamma$.

Suppose now that a sound wave outside the resonator has the excess pressure $p_e \cos \omega t$ at the mouth. The mouth is small compared with a wavelength, so this pressure is nearly uniform over it, and it pushes on the plug with the force $F_0 \cos \omega t$, where $F_0 = Sp_e$. The equation of motion $m\ddot{x} + b\dot{x} + \kappa x = F_0 \cos \omega t$ is that of the driven oscillator of Section 1.3, whose steady amplitude is

$$A = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}}. \quad (7.24)$$

The excess pressure in the cavity is $\kappa x/S$, so its amplitude is $\kappa A/S$. Dividing by $p_e = F_0/S$ and using $\kappa/m = \omega_0^2$ gives the ratio of the pressure inside to the pressure outside,

$$\frac{\kappa A}{Sp_e} = \frac{\omega_0^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}}. \quad (7.25)$$

At low frequencies the ratio tends to one, since the air in the cavity then has time to follow the outside pressure. At ω_0 it is $\omega_0/2\gamma = Q$, so a lightly damped resonator raises the pressure inside it by the factor Q at its natural frequency, over a band of width ω_0/Q as in Section 1.3. A lightly damped oscillator driven by a periodic force picks out the harmonic nearest its natural frequency, as Section 2.5 showed, and Helmholtz used this property by fitting each of his resonators with a second, small opening opposite the mouth, which he placed in his ear, so that he could hear whether a given harmonic was present in a musical note.

A Resonator on the Side of a Pipe

As an example of an acoustic filter, consider a pipe of cross-section S_p and characteristic acoustic impedance $Z_p = \rho_0 c/S_p$, with a resonator attached to its side at $x = 0$, as shown in Figure 103; a resonator attached in this way is called a side branch. Let Z_b be the acoustic impedance of the resonator, the ratio of the pressure at its mouth to the volume velocity flowing into it. A wave with pressure amplitude $\mathcal{A}_i$ arrives from the left, a wave with amplitude $\mathcal{A}_r$ is reflected, and a wave with amplitude $\mathcal{A}_t$ is transmitted.

The junction is small compared with a wavelength, and the following conditions hold there. The pressure is the same on both sides of the junction and at the mouth of the resonator, since a difference of pressure across so short a distance would give the small mass of air there an unbounded acceleration. The volume velocity arriving from the left equals the sum of the volume velocities leaving to the

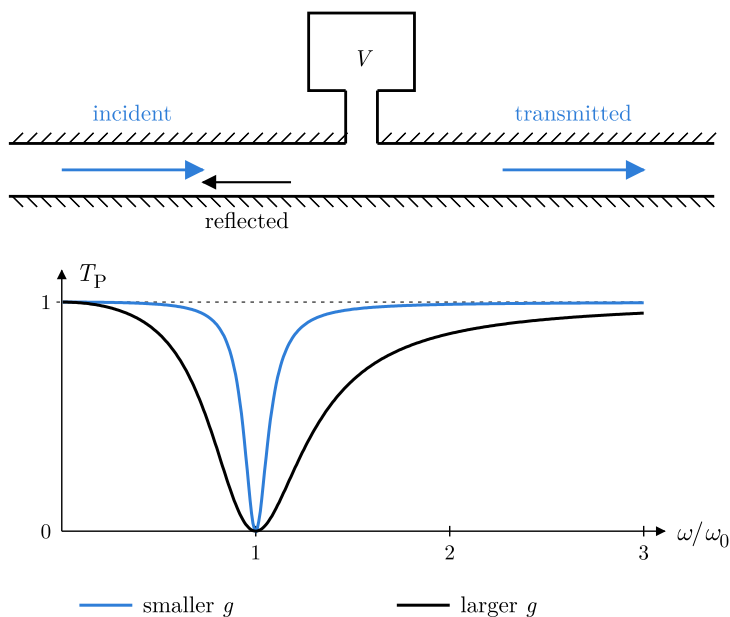


Figure 103. Top: a Helmholtz resonator attached to the side of a pipe. A wave travelling along the pipe (accent) is partly transmitted past the junction and partly reflected. Bottom: the fraction $\mathcal{T}_P$ of the incident power transmitted, against ω/ω_0 , for a smaller (accent) and a larger (black) value of $g = \sqrt{SV/\ell'}/2S_p$. No power is transmitted at the natural frequency of the resonator.

right and entering the resonator, since the junction holds too little air to store any appreciable amount. With the velocities of the waves from Section 7.1, these conditions read

$$\mathcal{A}_i + \mathcal{A}_r = \mathcal{A}_t, \quad \frac{\mathcal{A}_i - \mathcal{A}_r}{Z_p} = \frac{\mathcal{A}_t}{Z_p} + \frac{\mathcal{A}_t}{Z_b}. \quad (7.26)$$

Dividing the first equation by Z_p and adding it to the second gives $2\mathcal{A}_i/Z_p = \mathcal{A}_t(2/Z_p + 1/Z_b)$, so the amplitude transmission and reflection coefficients are

$$\mathcal{T} = \frac{\mathcal{A}_t}{\mathcal{A}_i} = \frac{2Z_b}{2Z_b + Z_p}, \quad \mathcal{R} = \frac{\mathcal{A}_r}{\mathcal{A}_i} = \mathcal{T} - 1 = -\frac{Z_p}{2Z_b + Z_p}. \quad (7.27)$$

The pipe has the same area on both sides, and by Section 6.3 the intensity of a plane wave is proportional to the square of its pressure amplitude, so the fraction of the incident power transmitted is $\mathcal{T}_P = |\mathcal{T}|^2$. For a resonator without losses, Z_b is the impedance Z_A above, which can be written as $i(\rho_0 \ell' / S)(\omega - \omega_0^2 / \omega)$. It is purely imaginary, so $|2Z_b + Z_p|^2 = 4|Z_b|^2 + Z_p^2$, and

$$\begin{aligned} \mathcal{T}_P &= \frac{1}{1 + Z_p^2 / 4|Z_b|^2} = \frac{1}{1 + g^2 / (\omega / \omega_0 - \omega_0 / \omega)^2}, \\ g &= \frac{cS}{2S_p \ell' \omega_0} = \frac{1}{2S_p} \sqrt{\frac{SV}{\ell'}}, \end{aligned} \quad (7.28)$$

where the last form of g uses $\omega_0^2 = c^2 S / V \ell'$. At ω_0 the resonator presents zero impedance and holds the pressure at the junction at the ambient value, as an open end does, so no power is transmitted and the wave is reflected with $\mathcal{R} = -1$. A resonator without losses absorbs nothing, and it stops the transmission by reflecting the sound back towards its source. The transmitted fraction is less than one half when $|\omega / \omega_0 - \omega_0 / \omega| < g$, and when g is small, $\omega / \omega_0 - \omega_0 / \omega$ is close to $2(\omega - \omega_0) / \omega_0$ near the resonance, so the band that is stopped has a width of about $g\omega_0$. A large cavity and a wide neck, compared with the pipe, therefore stop a wide band, as Figure 103 shows. Resonators of this kind are sometimes attached

to ventilation ducts and to the intake pipes of engines to reduce the sound at one troublesome frequency.

The Expansion Chamber

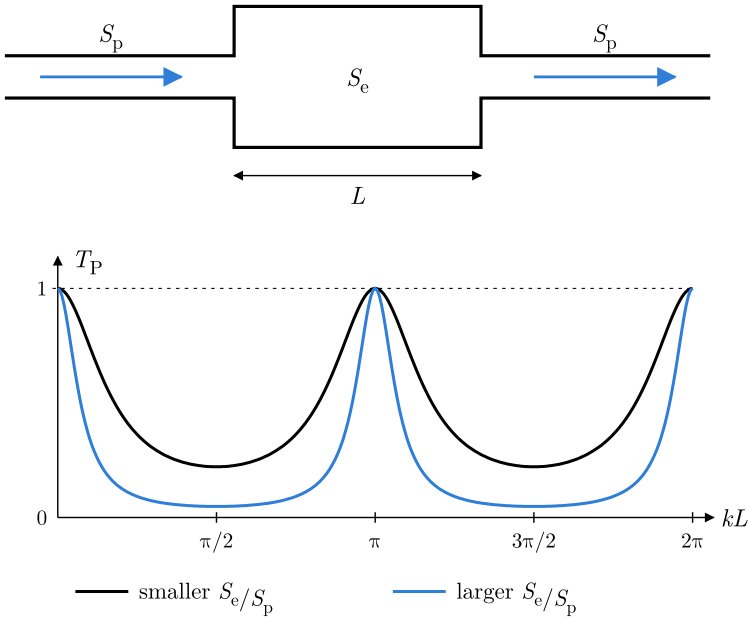


Figure 104. Top: an expansion chamber of length L and cross-section S_e in a pipe of cross-section S_p . Bottom: the fraction $\mathcal{T}_p$ of the incident power transmitted, against kL , for a smaller (black) and a larger (accent) ratio S_e/S_p . The transmission is complete when the chamber is a whole number of half wavelengths long and least when it is an odd number of quarter wavelengths long.

As another example, consider a pipe of cross-section S_p in which a length L is replaced by a wider chamber of cross-section S_e , as shown in Figure 104, an arrangement that is called an expansion chamber, with the pipe beyond the chamber long enough that no wave returns from its far end. At each change of area the pressure and the volume velocity are continuous, for the reasons given for the side branch, so the acoustic impedance is continuous. The outlet pipe carries a single wave travelling away from the chamber, so

it presents the load Z_p at the far end of the chamber. Let $Z_e = \rho_0 c / S_e$ and $\varsigma = S_e / S_p = Z_p / Z_e$. The formula of Section 7.1 with $Z_L = Z_p = \varsigma Z_e$ gives the input impedance of the chamber,

$$Z_{\text{in}} = Z_p \frac{\cos kL + i\varsigma^{-1} \sin kL}{\cos kL + i\varsigma \sin kL}. \quad (7.29)$$

The inlet pipe is loaded by Z_{in} , so by Section 7.1 its pressure reflection coefficient at the chamber is $\mathcal{R} = (Z_{\text{in}} - Z_p) / (Z_{\text{in}} + Z_p)$. Substituting Z_{in} and multiplying the numerator and the denominator by $\cos kL + i\varsigma \sin kL$ gives

$$\mathcal{R} = \frac{i(\varsigma^{-1} - \varsigma) \sin kL}{2 \cos kL + i(\varsigma + \varsigma^{-1}) \sin kL}. \quad (7.30)$$

No energy is lost in the chamber, and the outlet has the same area as the inlet, so the fraction of the power transmitted is $\mathcal{T}_P = 1 - |\mathcal{R}|^2$. The squared modulus of the denominator is $4 \cos^2 kL + (\varsigma + \varsigma^{-1})^2 \sin^2 kL$, which, since $(\varsigma + \varsigma^{-1})^2 = (\varsigma - \varsigma^{-1})^2 + 4$, can be rewritten as $4 + (\varsigma - \varsigma^{-1})^2 \sin^2 kL$. Subtracting the squared modulus of the numerator leaves 4, and therefore

$$\mathcal{T}_P = \frac{1}{1 + \frac{1}{4} (\varsigma - \varsigma^{-1})^2 \sin^2 kL}. \quad (7.31)$$

The transmission is complete when the chamber is a whole number of half wavelengths long, since it then presents the load Z_p unchanged at its input, as Section 7.1 showed. It is least, with $\mathcal{T}_P = 4 / (\varsigma + \varsigma^{-1})^2$, when the chamber is an odd number of quarter wavelengths long. The chamber then acts as the quarter-wave section of Section 4.4 and presents $Z_e^2 / Z_p = Z_p / \varsigma^2$ at its input, and since its impedance is far from the geometric mean of the impedances on either side, which would transmit all of the incident power, the mismatch is as large as it can be. Unlike the side-branch resonator, the expansion chamber stops a series of bands, as Figure 104 shows, and passes low frequencies. Since $\mathcal{T}_P$ is unchanged when ς is replaced by $1/\varsigma$, a narrow constriction of the same length

has the same effect. Chambers of this kind are one of the elements often used in the silencers of engine exhausts.

A Chain of Necks and Cavities

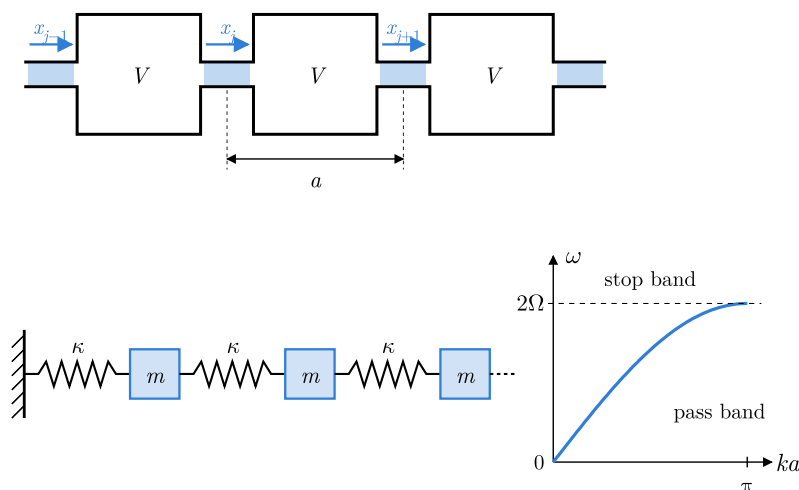


Figure 105. Top: a row of cavities of volume V joined by necks a distance a apart, with the displacements x_j of the plugs of air in the necks. Bottom left: the chain of masses and springs of Section 2.3, which has the same equations of motion. Bottom right: the dispersion relation $\omega = 2\Omega \sin(ka/2)$ of the chain, which passes waves below 2Ω and stops them above.

As a final example, consider a row of identical cavities of volume V joined in line by identical necks of cross-section S and effective length ℓ' , with neighbouring necks a distance a apart, as shown in Figure 105. Each neck holds a plug of air of mass $m = \rho_0 S \ell'$, and each cavity has a uniform pressure. Let x_j be the displacement of the plug in neck j towards increasing x . The cavity between necks j and $j + 1$ receives the volume Sx_j and loses Sx_{j+1} , so by the argument for the Helmholtz resonator its excess pressure is $\kappa(x_j - x_{j+1})/S$, with $\kappa = \rho_0 c^2 S^2/V$. The plug in neck j is pushed forwards by the pressure in the cavity behind it and backwards by the pressure in the cavity ahead of it, so

$$m\ddot{x}_j = \kappa [(x_{j-1} - x_j) - (x_j - x_{j+1})] = \kappa (x_{j+1} - 2x_j + x_{j-1}) . \quad (7.32)$$

This is the equation of motion of the chain of masses and springs of Section 2.3, with the frequency scale

$$\Omega^2 = \frac{\kappa}{m} = \frac{c^2 S}{V \ell'} , \quad (7.33)$$

which is the square of the natural angular frequency of a Helmholtz resonator made from one cavity and one neck. By Section 2.3, a wave $x_j = \mathcal{A}e^{i(\omega t - kja)}$ travels along the chain with the dispersion relation $\omega = 2\Omega|\sin(ka/2)|$, and the highest angular frequency at which the chain can carry a wave is 2Ω . Above it the disturbance is evanescent, as it was for the chain of masses and springs driven above its highest frequency in Section 4.4. The row of cavities is therefore what is called a low-pass filter: it passes sound at angular frequencies below 2Ω , which make up its pass band, and reflects sound at angular frequencies above 2Ω , which make up its stop band.

At low frequencies ω is close to $\Omega a k$, so sound travels along the chain at the speed $\Omega a = c\sqrt{Sa^2/V\ell'}$. The treatment of the necks and cavities as plugs and springs requires the wavelength at the cutoff, $\pi\sqrt{V\ell'}/S$, to be long compared with the spacing a , which holds when Sa^2 is small compared with $V\ell'$. The speed Ωa is then less than c by the small factor $\sqrt{Sa^2/V\ell'}$.

Problems

§7.1 Pipes, Horns, and Acoustic Waveguides

1. A uniform pipe of length L and cross-section S is closed at $x = L$ by a rigid cap. Show that its input impedance is $-iZ_p \cot kL$. Show that when kL is small it is close to $\rho_0 c^2 / i\omega V$, where $V = SL$ is the volume of the pipe, and show that this is the ratio of pressure to volume velocity found by treating the air in the pipe as a spring that is compressed uniformly by a piston at $x = 0$.

2. A pipe of cross-section S_1 is joined to a long pipe of cross-section S_3 by a section of length L and cross-section S_2 . Show that no sound is reflected back along the first pipe at the frequency for which L is a quarter of a wavelength, provided that $S_2 = \sqrt{S_1 S_3}$.
3. For the exponential horn, find the group velocity and show that it equals $c\sqrt{1 - \omega_c^2/\omega^2}$. Below the cutoff, find the distance over which the amplitude of the pressure falls by a factor e , and find its limit at low frequencies.
4. For the conical horn, show that the mean power carried through a section at distance x from the apex does not depend on x , and find it in terms of $\mathcal{A}$, S_0 , and x_0 .
5. A rectangular duct has $L_y = 2L_z$. List the modes in order of their cutoff angular frequencies, as far as the fifth mode, and find the band of angular frequencies in which exactly two modes can travel along the duct.

§7.2 Helmholtz Resonators and Acoustic Filters

1. A cavity of volume V has N identical necks, each of cross-section S and effective length ℓ' . Show that in the mode in which the plugs in all the necks move together, the natural angular frequency is $\sqrt{N}$ times that of the same cavity with one neck.
2. For the driven Helmholtz resonator, show that at frequencies well above ω_0 the ratio of the pressure in the cavity to the pressure outside goes as ω_0^2/ω^2 , and find the phase by which the pressure in the cavity lags behind the pressure outside at ω_0 .
3. A side-branch resonator with losses presents the impedance b/S^2 at its natural frequency. Find the fractions of the incident power that are transmitted along the pipe, reflected, and absorbed by the resonator at this frequency. Show that the absorbed fraction is greatest when $b/S^2 = Z_p/2$, and find its value.
4. A short open tube of cross-section S and effective length ℓ' , set in the wall of a pipe of cross-section S_p , presents the impedance $i\omega\rho_0\ell'/S$. Show that it transmits the fraction $[1 + (cS/2S_p\ell'\omega)^2]^{-1}$ of the incident power along the pipe, and explain why such a hole in the wall of a pipe reflects low-frequency

sound and acts approximately as the end of the pipe.

5. For the expansion chamber, find the ratio $\varsigma = S_e/S_p$, greater than one, for which the least transmitted fraction is a given value $\mathcal{T}_{P,\min}$.

Solutions

Chapter 1. Simple, Damped, and Driven Oscillations

§1.1 Simple Harmonic Motion Revisited

1. The potential energy is $E \cos^2 \theta$ and the kinetic energy is $E \sin^2 \theta$, where $\theta = \omega_0 t + \phi$ is the angle of the representative point below the x -axis in Figure 1. The potential and kinetic energies are equal when $\cos^2 \theta = \sin^2 \theta = \frac{1}{2}$, that is at $x = \pm A/\sqrt{2}$. The potential energy is the larger when $|\cos \theta| > |\sin \theta|$, which holds within an angle $\pi/4$ of the horizontal axis, that is on two arcs of angle $\pi/2$ each, one on either side of the origin, making half the circle. Since the point goes round at a constant rate, it spends half of each period on these arcs.
2. The effective mass is still $\rho_L SH$. A displacement x along the tube raises the surface in the first arm by $x \sin \theta_1$ and lowers the surface in the second by $x \sin \theta_2$. The column of liquid of length x moved from one arm to the other has its centre of mass raised by $\frac{1}{2}x \sin \theta_1 + \frac{1}{2}x \sin \theta_2$, so $U = \frac{1}{2}\rho_L g S(\sin \theta_1 + \sin \theta_2)x^2$ and $\omega_0^2 = U''(0)/(\rho_L SH) = g(\sin \theta_1 + \sin \theta_2)/H$. With $\theta_1 = \theta_2 = \pi/2$ this is $2g/H$.
3. When the body is displaced sideways by x , each half of the string makes an angle with the line joining the fixed points whose sine is approximately $x/(L/2) = 2x/L$, for $x \ll L$. The two halves each pull on the body with tension τ , and their transverse components

add to $-2\tau(2x/L) = -4\tau x/L$. The equation of motion is $m\ddot{x} = -(4\tau/L)x$, so $\omega_0^2 = 4\tau/(mL)$.

4. The complex amplitude of $\dot{x}$ is $i\omega_0\mathcal{A}$, so $\langle x\dot{x} \rangle = \frac{1}{2}\text{Re}[\mathcal{A}(i\omega_0\mathcal{A})^*] = \frac{1}{2}\text{Re}(-i\omega_0|\mathcal{A}|^2) = 0$. By the chain rule, $x\dot{x} = \frac{1}{2}d(x^2)/dt$, and the mean of the derivative of a periodic function over a period is the change in the function over the period divided by P , which is zero. In the phase plane, $x\dot{x}$ is positive in the first and third quadrants and negative in the second and fourth, and the point spends equal times in each.
5. At equilibrium the weight $\rho_b g S_b H_c$ of the cylinder, where S_b is its cross-sectional area, equals the weight $\rho_L g S_b h$ of the liquid displaced, so $h = (\rho_b/\rho_L)H_c$. From the text, $\omega_0^2 = g/h = \rho_L g/(\rho_b H_c)$.

§1.2 Damped Oscillations and the Quality Factor

1. From $\Lambda = 2\pi\gamma/\omega_d$ with $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$, dividing the numerator and the denominator by γ gives $\Lambda = 2\pi/\sqrt{\omega_0^2/\gamma^2 - 1} = 2\pi/\sqrt{4Q^2 - 1}$. Then $\pi/\Lambda = \frac{1}{2}\sqrt{4Q^2 - 1} = Q\sqrt{1 - 1/4Q^2}$, and by the binomial expansion this is approximately $Q(1 - 1/8Q^2)$.
2. The amplitude falls by $e^{-n_{1/2}\Lambda} = \frac{1}{2}$, so $\Lambda = \ln 2/n_{1/2}$, and $Q \approx \pi/\Lambda = \pi n_{1/2}/\ln 2$. The energy goes as the square of the amplitude and falls by $e^{-2n\Lambda}$ in n cycles, so it falls to half in $n_{1/2}/2$ cycles.
3. The initial conditions give $D_+ + D_- = A$ and $\beta_+ D_+ + \beta_- D_- = \dot{x}_i$, so $D_+ = (\dot{x}_i - \beta_- A)/(\beta_+ - \beta_-)$ and $D_- = (\beta_+ A - \dot{x}_i)/(\beta_+ - \beta_-)$. A zero at $t > 0$ needs $e^{(\beta_+ - \beta_-)t} = -D_-/D_+ > 1$. If $\dot{x}_i < \beta_- A$, then $D_+ < 0 < D_-$ and $-D_-/D_+ = (\beta_+ A - \dot{x}_i)/(\beta_- A - \dot{x}_i)$, which exceeds 1 because $\beta_+ > \beta_-$. If $\beta_- A \leq \dot{x}_i \leq \beta_+ A$, the two constants have the same sign or one is zero, and if $\dot{x}_i > \beta_+ A$, then $D_- < 0 < D_+$ and $-D_-/D_+ < 1$; in neither case is there a zero at a positive time. In the phase plane, the starting point lies below the straight line $\dot{x} = \beta_- x$ of the faster exponential, so the body is moving towards equilibrium faster than that exponential would carry it.
4. From $x = A(1 + \omega_0 t)e^{-\omega_0 t}$, differentiating by the product rule gives $\dot{x} = -A\omega_0^2 t e^{-\omega_0 t}$. Its derivative with respect to t vanishes

at $t = 1/\omega_0$, where the speed is $A\omega_0 e^{-1}$. The trajectory leaves $(A, 0)$ downwards, reaches its lowest point $\dot{x}/\omega_0 = -A/e$ at $x = 2A/e$, and returns towards the origin. It meets the line $\dot{x} = -\omega_0 x$ only in the limit, since $\dot{x}/x = -\omega_0^2 t/(1 + \omega_0 t)$ approaches $-\omega_0$ as t grows without reaching it.

5. At a maximum $\dot{x} = 0$, so the energy there is $\frac{1}{2}\kappa X^2$, where X is the maximum. Successive maxima are in the ratio e^Λ , so the energies are in the ratio $e^{2\Lambda}$. The fraction lost in a cycle is $1 - e^{-2\Lambda} \approx 2\Lambda \approx 2\pi/Q$, by the expansion of the exponential for small Λ and the result $\Lambda \approx \pi/Q$.

§1.3 Forced Oscillation, Resonance, and Impedance

1. The complex amplitude of the velocity is F_0/Z_m , so by the formula for the mean of a product $\langle b\dot{x}^2 \rangle = \frac{1}{2}b|F_0/Z_m|^2 = \frac{1}{2}F_0^2 b/|Z_m|^2$, which is the expression for $\langle \mathcal{P} \rangle$ in the text.
2. From $\omega_\pm = \sqrt{\omega_0^2 + \gamma^2} \pm \gamma$, the product is $(\omega_0^2 + \gamma^2) - \gamma^2 = \omega_0^2$. At ω_- the reactance is $-b$, so $Z_m = b(1 - i)$ and the velocity amplitude $F_0/Z_m = F_0(1 + i)/2b$ is $\pi/4$ ahead of the force. The displacement amplitude is the velocity amplitude divided by $i\omega$, a quarter of a period behind the velocity, so $\delta = \pi/4$ at ω_- . At ω_+ the velocity is $\pi/4$ behind the force and $\delta = 3\pi/4$.
3. At ω_0 , $x = A \sin \omega_0 t$ with $A = F_0/b\omega_0$, and the energy is $\frac{1}{2}m\omega_0^2 A^2 \cos^2 \omega_0 t + \frac{1}{2}\kappa A^2 \sin^2 \omega_0 t = \frac{1}{2}\kappa A^2$, since $m\omega_0^2 = \kappa$, which is constant and equal to $\frac{1}{2}mF_0^2/b^2$. The energy dissipated in one cycle is $(F_0^2/2b)(2\pi/\omega_0) = \pi F_0^2/b\omega_0$. The ratio is $m\omega_0/2\pi b = Q/2\pi$, since $Q = \omega_0/2\gamma = m\omega_0/b$.
4. When $\omega \ll \omega_0$ the denominator $\omega_0^2 - \omega^2 + 2i\gamma\omega$ is approximately ω_0^2 , provided that γ is not much larger than ω_0 , so $\mathcal{A} - \zeta_0 \approx \omega^2 \zeta_0/\omega_0^2$. The acceleration of the ground has complex amplitude $-\omega^2 \zeta_0$, so the relative displacement is $-\ddot{\zeta}/\omega_0^2$.
5. At resonance $A = F_0/b\omega_0 = F_0/(2m\gamma\omega_0)$, and $1 - e^{-\gamma t} \approx \gamma t$ when $\gamma t \ll 1$, so $A(1 - e^{-\gamma t}) \sin \omega_0 t$ tends to $(F_0 t/2m\omega_0) \sin \omega_0 t$ as $\gamma \rightarrow 0$. Differentiating twice by the product rule gives $\ddot{x} = (F_0/2m\omega_0)(2\omega_0 \cos \omega_0 t - \omega_0^2 t \sin \omega_0 t)$, so $m\ddot{x} + \kappa x = F_0 \cos \omega_0 t$, since the terms in $t \sin \omega_0 t$ cancel when $\kappa = m\omega_0^2$. The amplitude grows in proportion to the time without limit.

Chapter 2. Coupled Oscillators, Normal Modes, and Fourier Analysis

§2.1 Two Coupled Oscillators

1. From the text,

$$\begin{aligned} x_1 - x_2 &= A [\cos(\omega_b t/2) \cos \bar{\omega} t - \sin(\omega_b t/2) \sin \bar{\omega} t] \\ &= A \cos(\bar{\omega} t + \omega_b t/2) \end{aligned} \quad (\text{S.1})$$

by the compound-angle formula, and $\bar{\omega} + \omega_b/2 = \omega_2$, so $x_2 - x_1 = -A \cos \omega_2 t$. This is $-2q_2$, and $q_2 = \frac{1}{2}A \cos \omega_2 t$ exactly. The energy stored is $\frac{1}{2}\kappa_c A^2 \cos^2 \omega_2 t$, whose mean is $\frac{1}{4}\kappa_c A^2$. Dividing by $E = \frac{1}{2}m\omega_0^2 A^2$ gives $\kappa_c/2m\omega_0^2$.

2. The upper component of the solution in the text is $A \cos \psi \cos \psi \cos \omega_1 t - A \sin \psi (-\sin \psi) \cos \omega_2 t$, which is the stated result. At $t = \pi/\omega_b$, $\omega_2 t = \omega_1 t + \pi$, so $\cos \omega_2 t = -\cos \omega_1 t$ and $x_I = A(\cos^2 \psi - \sin^2 \psi) \cos \omega_1 t = A \cos 2\psi \cos \omega_1 t$. The fraction $\sin^2 2\psi$ is one half when $|\omega_I^2 - \omega_{II}^2| = 2\kappa_c/m$, and since $\omega_I^2 - \omega_{II}^2 = g(1/L_I - 1/L_{II})$, this is $g|L_{II} - L_I|/L_I L_{II} = 2\kappa_c/m$.
3. Let x and x_a be the displacements of the machine and the small mass. The equations are $m\ddot{x} = -\kappa x + \kappa_a(x_a - x) + F_0 \cos \omega t$ and $m_a\ddot{x}_a = -\kappa_a(x_a - x)$. With complex amplitudes $\mathcal{A}$ and $\mathcal{A}_a$, the second gives $\mathcal{A}_a = \kappa_a \mathcal{A}/(\kappa_a - m_a \omega^2)$, and substituting into the first gives $\mathcal{A} [\kappa + \kappa_a - m\omega^2 - \kappa_a^2/(\kappa_a - m_a \omega^2)] = F_0$, which on multiplying through by $\kappa_a - m_a \omega^2$ becomes $\mathcal{A} = F_0(\kappa_a - m_a \omega^2)/[(\kappa + \kappa_a - m\omega^2)(\kappa_a - m_a \omega^2) - \kappa_a^2]$. This vanishes when $\kappa_a = m_a \omega^2$, where the denominator is $-\kappa_a^2$. Then $\mathcal{A}_a$ is found from the first equation with $\mathcal{A} = 0$, $\kappa_a \mathcal{A}_a + F_0 = 0$, so $\mathcal{A}_a = -F_0/\kappa_a$.
4. Adding and subtracting the equations gives $m\ddot{q}_n + b\dot{q}_n + m\omega_n^2 q_n = \frac{1}{2}F_0 \cos \omega t$ for $n = 1, 2$, so each amplitude is $F_0/[2m(\omega_n^2 - \omega^2 + 2i\gamma\omega)]$, as in Section 1.3, and $\mathcal{A}_1$ is their sum, $\mathcal{A}_1 = (F_0/2m)(\omega_1^2 + \omega_2^2 - 2\omega^2 + 4i\gamma\omega)/[(\omega_1^2 - \omega^2 + 2i\gamma\omega)(\omega_2^2 - \omega^2 + 2i\gamma\omega)]$. At $\omega^2 = \frac{1}{2}(\omega_1^2 + \omega_2^2)$ the numerator is $4i\gamma\omega$, which is not zero, and the denominator is $(-\kappa_c/m + 2i\gamma\omega)(\kappa_c/m + 2i\gamma\omega)$, whose modulus is $\kappa_c^2/m^2 + 4\gamma^2\omega^2$. Hence $|\mathcal{A}_1| = 2F_0\gamma\omega m/(\kappa_c^2 + 4m^2\gamma^2\omega^2) \approx 2F_0m\gamma/\kappa_c^2$ when $2\gamma\omega \ll \kappa_c/m$.

§2.2 Many Coupled Oscillators and the Loaded String

1. The springs on either side of mass j are stretched by $y_{j+1} - y_j$ and $y_j - y_{j-1}$, so $m\ddot{y}_j = \kappa(y_{j+1} - y_j) - \kappa(y_j - y_{j-1})$, which is the equation of the loaded string with τ/a replaced by κ , and the walls give $y_0 = y_{N+1} = 0$. For $N = 2$ the frequencies are Ω and $\sqrt{3}\Omega$, which are $\sqrt{\kappa/m}$ and $\sqrt{(\kappa + 2\kappa_c)/m}$ with $\kappa_c = \kappa$.
2. The ring has no mass, so the net transverse force on it must be zero, and the only transverse force is from the last segment, $-\tau(y_{N+1} - y_N)/a$, which gives $y_{N+1} = y_N$. The trial amplitudes $\sin(kja)$ then require $\sin[(N+1)ka] = \sin(Nka)$. Using $\sin\alpha - \sin\beta = 2\cos\frac{1}{2}(\alpha + \beta)\sin\frac{1}{2}(\alpha - \beta)$ from the compound-angle formulas, this is $2\cos[(2N+1)ka/2]\sin(ka/2) = 0$, so $ka = (2n-1)\pi/(2N+1)$, and $\omega = 2\Omega\sin(ka/2)$ gives the stated frequencies. For $N = 1$ the formula gives $2\Omega\sin(\pi/6) = \Omega$, and the bead feels only the force $-\tau y_1/a$ from the left segment, since the right segment stays horizontal, so $\omega^2 = \tau/ma = \Omega^2$.
3. By the binomial expansion, the segment between beads j and $j+1$ is longer than a by $(y_{j+1} - y_j)^2/2a$, and stretching it by this amount at constant tension τ takes the work $\tau(y_{j+1} - y_j)^2/2a$; summing over the $N+1$ segments gives the result. In mode n , when every bead passes through equilibrium the potential energy is zero, and the kinetic energy is $\frac{1}{2}m\omega_n^2 A_n^2 \sum_j \sin^2[jn\pi/(N+1)] = \frac{1}{4}(N+1)m\omega_n^2 A_n^2$ by the normalisation sum of the text. The total energy is constant, so this is its value at every instant.
4. Only $j = 1$ contributes, so $A_n = \frac{1}{2}\eta\sin(n\pi/4)$, which gives $A_1 = A_3 = \eta/2\sqrt{2}$ and $A_2 = \eta/2$, with every $\phi_n = 0$. The middle bead has pattern entries 1, 0, and -1 in the three modes, so $y_2 = (\eta/2\sqrt{2})(\cos\omega_1 t - \cos\omega_3 t)$.

§2.3 Symmetry and Normal Modes

1. The exchange matrix has rows $(0, 1)$ and $(1, 0)$, and its square is the identity, so its eigenvalues are ± 1 , each with a single eigenvector, $(1, 1)$ and $(1, -1)$. Each is therefore a mode, and multiplying by D gives $\omega^2 = \kappa/m$ for $(1, 1)$ and $(\kappa + 2\kappa_c)/m$ for $(1, -1)$.
2. Exchanging y_j and y_{N+1-j} carries the equation for bead j into

the equation for bead $N+1-j$, since each bead is joined in the same way to its neighbours and the conditions $y_0 = y_{N+1} = 0$ are exchanged with each other, so $\text{PD} = \text{DP}$. In mode n , $\sin[(N+1-j)n\pi/(N+1)] = \sin(n\pi - jn\pi/(N+1)) = -\cos(n\pi)\sin[jn\pi/(N+1)] = (-1)^{n+1}\sin[jn\pi/(N+1)]$ by the compound-angle formula, so odd n gives even modes and even n gives odd modes.

3. With $N = 3$, $n = 0, 1, 2$ give $e_j = 1$, $e^{-2\pi ij/3}$, and $e^{-4\pi ij/3}$, with frequencies 0 and $2\Omega \sin(\pi/3) = \sqrt{3}\Omega$ twice. The mode of zero frequency is a rotation of the ring at constant speed. A real mode of frequency $\sqrt{3}\Omega$ is $x_j \propto \cos(2\pi j/3) \cos \omega t$, in which mass 3 has twice the displacement of masses 1 and 2 and moves opposite to them, so the centre of mass along the track stays fixed. The same frequency follows from the forces, since mass 3 is displaced by x_3 and its neighbours by $-x_3/2$, so each spring on it changes length by $\frac{3}{2}x_3$, and the restoring force is $-3\kappa x_3$.
4. The equations for masses 1 and 3 are unchanged, and the equation for mass 2 becomes $m_2 \ddot{x}_2 = -\kappa_c(x_2 - x_1) - \kappa_c(x_2 - x_3)$, which is unchanged by the exchange. For $(e_1, 0, -e_1)$ the middle equation gives $0 = 0$ and the outer equations give $m\omega^2 = \kappa + \kappa_c$, as before. For (e_1, e_2, e_1) the equations are $(\kappa + \kappa_c - m\omega^2)e_1 - \kappa_c e_2 = 0$ and $-2\kappa_c e_1 + (2\kappa_c - m_2\omega^2)e_2 = 0$, whose determinant gives $(\kappa + \kappa_c - m\omega^2)(2\kappa_c - m_2\omega^2) - 2\kappa_c^2 = 0$.
5. The waves with k and $-k$ have the same frequency, so their sum divided by two and their difference divided by $2i$, which are $\mathcal{A}e^{i\omega t} \cos(jka)$ and $\mathcal{A}e^{i\omega t} \sin(jka)$, are motions, and with $\mathcal{A} = A$ their real parts are the stated displacements. The scalar product of the patterns is $\sum_j \cos(jka) \sin(jka) = \frac{1}{2} \sum_j \sin(2jka)$, which is the imaginary part of $\sum_{j=1}^N e^{2ijka}$. With $2ka = 4\pi n/N$, this is a geometric series whose ratio $e^{4\pi in/N}$ is not one for $0 < n < N/2$ and whose N th power is one, so it sums to zero, as the sum in Section 2.2 did.

§2.4 The Continuum Limit

1. With $\theta = n\pi a/2L$, the formula $\omega_n = (2v/a)\sin\theta$ gives $\omega_n \approx (2v/a)(\theta - \theta^3/6) = (n\pi v/L)(1 - \theta^2/6) = n^2\pi^2 a^2/(24L^2)$. The fractional difference is less than β when

$n^2\pi^2a^2/(24L^2) < \beta$, that is when $n < (L/\pi a)\sqrt{24\beta}$. With $L/a = N + 1$ the largest such mode number is the largest whole number below $(N + 1)\sqrt{24\beta}/\pi$.

2. The slope is y_p/x_p on the first segment and $-y_p/(L - x_p)$ on the second, so $\frac{1}{2}\tau \int_0^L (\partial y/\partial x)^2 dx = \frac{1}{2}\tau y_p^2[1/x_p + 1/(L - x_p)] = \frac{1}{2}\tau y_p^2 L/[x_p(L - x_p)]$. The string starts at rest, so this is its total energy. Setting it equal to $\sum_n E_n$ with the E_n of the text, and writing $\theta = \pi x_p/L$, gives $\sum_n \sin^2(n\theta)/n^2 = \pi^2 x_p(L - x_p)/(2L^2) = \frac{1}{2}\theta(\pi - \theta)$. The result holds for $\theta = 0$ and $\theta = \pi$ as well, where both sides are zero.
3. With $x_p = L/2$ the coefficients are $a_n = (8y_p/n^2\pi^2) \sin(n\pi/2)$, and at $x = L/2$ each term carries a further factor $\sin(n\pi/2)$. The product $\sin^2(n\pi/2)$ is one for odd n and zero for even n , which gives the stated series. Section 6.1 of Vol. 4 gives $|x| = L/2 - (4L/\pi^2) \sum_{n \text{ odd}} \cos(n\pi x/L)/n^2$ for $-L \leq x \leq L$, so with $x = vt$ the sum is $(\pi^2/4L)(L/2 - vt)$, and the displacement of the midpoint is $y_p(1 - 2vt/L)$. The corners of Figure 26 explain this motion. The two corners leave the midpoint at $t = 0$, reach the ends at $t = L/2v$, are reflected, and meet again at the midpoint at $t = L/v$. Throughout this time the midpoint lies on the flat part of the string between the corners, whose height falls at the constant rate $2y_p v/L$.
4. The kinetic energy of mode n is $\frac{1}{4}\mu L \dot{q}_n^2 = \frac{1}{4}\mu L \omega_n^2 (a_n \sin \omega_n t - b_n \cos \omega_n t)^2$, and the mean values of $\sin^2 \omega_n t$ and $\cos^2 \omega_n t$ over a period are each one half while that of $\sin \omega_n t \cos \omega_n t$ is zero, so the mean kinetic energy is $\frac{1}{8}\mu L \omega_n^2 (a_n^2 + b_n^2) = \frac{1}{2}E_n$. The same calculation gives the mean potential energy $\frac{1}{4}\mu L \omega_n^2 \langle q_n^2 \rangle = \frac{1}{2}E_n$. The time $2L/v$ is a whole number of periods of every mode, so over that time each mode has mean kinetic energy $\frac{1}{2}E_n$, and summing over the modes gives $\frac{1}{2}E$.

§2.5 Fourier Analysis of Signals

1. For $n \neq 0$, $C_n = (F_0/P) \int_0^{t_1} e^{-in\omega t} dt = (F_0/P)(1 - e^{-in\omega t_1})/(in\omega)$. Taking out the factor $e^{-in\omega t_1/2}$ leaves $e^{in\omega t_1/2} - e^{-in\omega t_1/2} = 2i \sin(n\omega t_1/2)$, and with $P\omega = 2\pi$ this gives the stated result. The mean force is $C_0 = F_0 t_1/P$. The

amplitude of harmonic n is $2|C_n| = (2F_0/n\pi)|\sin(n\omega t_1/2)|$, and when $n\omega t_1/2 = n\pi t_1/P \ll 1$ the sine can be replaced by its argument, which gives $2|C_n| \approx 2F_0 t_1/P = 2C_0$ for every such n . The sine vanishes when $n\omega t_1/2 = n\pi t_1/P$ is a whole multiple of π , that is when nt_1/P is a whole number.

2. The square wave contains only the odd harmonics, so none of its harmonics has the angular frequency $\omega_0 = 2\omega$. For every odd n , $|\omega_0^2 - n^2\omega^2| = |4 - n^2|\omega^2 \geq 3\omega^2$, while $2\gamma n\omega = n\omega_0\omega/Q$ is much smaller, so the damping term can be neglected. The response at ω has the amplitude $(4F_0/\pi)/3m\omega^2$, and the response at 3ω has the amplitude $(4F_0/3\pi)/5m\omega^2$, whose ratio to the first is $1/5$.
3. The peak near ω_d contributes $(A^2/8\pi) \int d\omega/[(\omega - \omega_d)^2 + \gamma^2] = (A^2/8\pi)(1/\gamma)[\arctan((\omega - \omega_d)/\gamma)]_{-\infty}^{\infty} = A^2/8\gamma$, and the peak near $-\omega_d$, which comes from the second term in $\tilde{x}$, contributes the same, which gives $A^2/4\gamma$. For the time integral, $\cos^2 \omega_d t = \frac{1}{2}(1 + \cos 2\omega_d t)$, and $\int_0^\infty e^{-2\gamma t} \cos 2\omega_d t dt = \operatorname{Re}[1/(2\gamma - 2i\omega_d)] = \gamma/2(\gamma^2 + \omega_d^2)$, so $\int_0^\infty x^2 dt = (A^2/2)[1/2\gamma + \gamma/2(\gamma^2 + \omega_d^2)]$, which is the stated result. The second term is smaller than the first by a factor of order γ^2/ω_d^2 , so the two integrals agree when $\gamma \ll \omega_d$.
4. Near the positive frequencies the spectrum of each train is $(1/\sqrt{2\pi}) \sin[(\omega - \omega_{si})\Delta t/2]/(\omega - \omega_{si})$, with $i = 1$ or 2 , whose first zeros lie at $\omega_{si} \pm 2\pi/\Delta t$. When $\omega_{s2} - \omega_{s1} = 2\pi/\Delta t$, the peak at ω_{s2} is the first zero on the right of the first spectrum and the peak at ω_{s1} is the first zero on the left of the second. If the angular frequencies are much closer than this, each peak lies well inside the central peak of the other, and the sum has a single peak whose width is set by Δt ; if they are much further apart, the sum has two separate peaks. The two trains can therefore be told apart only if their angular frequencies differ by at least about $2\pi/\Delta t$.

Chapter 3. Wave Motion, the Wave Equation, and Wave Energy

§3.1 Waves at School

1. The speed goes as $1/\sqrt{\mu}$ at fixed tension, so it doubles. The frequency is fixed by the source and is unchanged. Since $\lambda = v/f$,

the wavelength doubles.

2. The intensity goes as $1/r^2$, so it falls by a factor of nine at the distance $3r_1$. The amplitude goes as $1/r$, so the amplitude at $3r_1$ is a third of the amplitude at r_1 .
3. The listener ahead hears $f_a = cf/(c - V_s)$ and the listener behind hears $f_b = cf/(c + V_s)$. Putting these two fractions over the common denominator $c^2 - V_s^2$ gives $f_a - f_b = 2cfV_s/(c^2 - V_s^2)$ and $f_a + f_b = 2c^2f/(c^2 - V_s^2)$, and the ratio of the first to the second is V_s/c .
4. Before passing the source the listener hears $(c + V_o)f/c$, and afterwards $(c - V_o)f/c$, so the ratio is $(c + V_o)/(c - V_o)$.

§3.2 Travelling Waves on a String

1. The wave is $y = A \cos(\omega t + kx)$, with $k = \omega/v$, which reduces to $A \cos \omega t$ at $x = 0$. In complex form it is $y = \text{Re} [Ae^{i(\omega t + kx)}]$. The transverse velocity is $\partial y / \partial t = -A\omega \sin(\omega t + kx)$, and at $x = \lambda/4$ we have $kx = \pi/2$, so at $t = 0$ the velocity is $-A\omega$.
2. From the solution for a string released from rest, $y(0, t) = \frac{1}{2}y_0(-vt) + \frac{1}{2}y_0(vt) = y_0(vt)$, since y_0 is even. The displacement at the origin is therefore $y_m(1 - vt/x_0)$ for $t < x_0/v$, falling steadily to zero, and zero afterwards.
3. With $q = vy'_0$ the integral is $v^{-1} \int_0^x q \, dx' = y_0(x) - y_0(0)$, so $f_2(x) = y_0(x) - \frac{1}{2}y_0(0)$ and $f_1(x) = \frac{1}{2}y_0(0)$. The constants cancel in $f_1(x - vt) + f_2(x + vt)$, which is therefore $y_0(x + vt)$.
4. D'Alembert's formula is $y = \frac{1}{2}[y_0(x - vt) + y_0(x + vt)] + (2v)^{-1} \int_{x-vt}^{x+vt} q(x') \, dx'$. For $x > x_0$ the argument $x + vt$ exceeds x_0 , so $y_0(x + vt) = 0$. While $t < (x - x_0)/v$, the argument $x - vt$ also exceeds x_0 , so $y_0(x - vt) = 0$, and the interval of integration lies wholly in $x' > x_0$, where $q = 0$, so the point is undisturbed. When $t > (x + x_0)/v$, the argument $x - vt$ is less than $-x_0$, so $y_0(x - vt) = 0$, and the interval of integration contains the whole of $-x_0 < x' < x_0$, which gives the stated constant displacement.
5. By the product rule and the chain rule, $\partial y / \partial t = (2v\eta/\sigma^2) y_m e^{-\eta^2/\sigma^2}$, where $\eta = x - vt$. (And η for s in the rest of the solution.) Setting the derivative of se^{-s^2/σ^2} with

respect to s to zero gives $1 - 2s^2/\sigma^2 = 0$, so the greatest speed occurs at $s = \sigma/\sqrt{2}$ and is $\sqrt{2}e^{-1/2}vy_m/\sigma$. This is small compared with v provided $y_m \ll \sigma$, which is the condition that the slope of the pulse be small.

§3.3 Longitudinal Waves in Bars and Fluids

1. For $\xi = f(x - vt)$ the chain rule gives $\partial\xi/\partial t = -v\partial\xi/\partial x$, so the strain is $\partial\xi/\partial x = -v^{-1}\partial\xi/\partial t$. Then $-F = -YS\partial\xi/\partial x = (YS/v)\partial\xi/\partial t$, and since $Y = \rho v^2$, this is $\rho v S \partial\xi/\partial t$.
2. The slice has mass $\rho_0 S \delta x$ at rest and $(\rho_0 + \rho')S(1 + \partial\xi/\partial x)\delta x$ when disturbed. Equating the two masses and dropping the product of the small quantities ρ' and $\partial\xi/\partial x$ gives $\rho' = -\rho_0 \partial\xi/\partial x$. Then $p = -B \partial\xi/\partial x = (B/\rho_0)\rho' = c^2\rho'$.
3. The time taken by a longitudinal pulse is $L/v_l = L/(L\sqrt{\kappa/m}) = \sqrt{m/\kappa}$. A transverse pulse takes $L/v_t = \sqrt{m/\kappa} \sqrt{L/(L - L_0)}$, which is longer, and which approaches the time for a longitudinal pulse when $L \gg L_0$.
4. The strain is $\partial\xi/\partial x = kA \sin(\omega t - kx)$, whose greatest value is $kA = \omega A/c$. The condition $|\partial\xi/\partial x| \ll 1$ is therefore $kA \ll 1$, or $A \ll \lambda/2\pi$, since $k = 2\pi/\lambda$.

§3.4 Energy and Power in Waves

1. Differentiating ε with respect to t gives $\mu \dot{y}\ddot{y} + \tau y'\dot{y}'$, where dots denote derivatives with respect to t and primes derivatives with respect to x . By the product rule, $\partial\mathcal{P}/\partial x = -\tau(y''\dot{y} + y'\dot{y}')$. The sum is $\dot{y}(\mu\ddot{y} - \tau y'')$, which is zero by the wave equation, since $\tau/\mu = v^2$.
2. With $\chi_1 = \omega t - kx$ and $\chi_2 = \omega t + kx$, the derivatives are $\partial y/\partial x = kA_1 \sin \chi_1 - kA_2 \sin \chi_2$ and $\partial y/\partial t = -\omega A_1 \sin \chi_1 - \omega A_2 \sin \chi_2$. Multiplying them, the cross terms $A_1 A_2 \sin \chi_1 \sin \chi_2$ cancel, and $\mathcal{P} = \tau k \omega (A_1^2 \sin^2 \chi_1 - A_2^2 \sin^2 \chi_2)$. The mean of each $\sin^2$ is one half, and $\tau k = \tau \omega / v = Z\omega$, which gives $\frac{1}{2}Z\omega^2(A_1^2 - A_2^2)$. The waves in opposite directions carry their powers independently.
3. The forces on the mass are $-\kappa y$ from the spring and $\tau \partial y/\partial x$ at $x = 0$ from the string. With only an outgoing wave on the string,

$\tau \partial y / \partial x = -Z \partial y / \partial t$, as in the text, so $m\ddot{y} = -\kappa y - Z\dot{y}$. This is the damped oscillator of Section 1.2 with $b = Z$, so the damping rate is $\gamma = Z/2m$ and $Q = \omega_0/2\gamma = m\omega_0/Z = \sqrt{\kappa m}/Z$.

4. With $\eta = x - vt$, the energy per unit length is $\tau f'^2$, and $f' = -(2\eta/\sigma^2) y_m e^{-\eta^2/\sigma^2}$, so ε is proportional to $\eta^2 e^{-2\eta^2/\sigma^2}$. (And η for s in the rest of the solution.) This vanishes at $s = 0$, and setting its derivative $2s(1 - 2s^2/\sigma^2)e^{-2s^2/\sigma^2}$ to zero gives its greatest values at $s = \pm\sigma/\sqrt{2}$.
5. As for the string, the only wave is $\xi = f(x - vt)$. The end has no mass, so the applied force balances the force $F = YS \partial \xi / \partial x$ exerted on it by the bar, and $F_e = -YS \partial \xi / \partial x = (YS/v) \partial \xi / \partial t = \rho v S \partial \xi / \partial t$ at $x = 0$.

§3.5 The Doppler Effect and Shock Waves

1. In the frame of the air, the source and the observer both move with speed V_a in the direction from the observer towards the source. The source is therefore moving away from the observer, with $V_s = -V_a$, and the observer is moving towards the source, with $V_o = V_a$, and the general formula gives $f' = f(c + V_a)/(c + V_a) = f$. In the frame of the ground the crests travel at $c + V_a$ and arrive at the frequency f , so the wavelength measured on the ground is $(c + V_a)/f$.
2. The wall is an observer at rest approached by the source, so it receives $f_1 = cf/(c - V_s)$ and reflects sound of this frequency, acting as a source at rest. The driver is an observer approaching this source at V_s and hears $f_1(c + V_s)/c = f(c + V_s)/(c - V_s)$. The beat frequency is the difference between this frequency and f , which is $2fV_s/(c - V_s)$.
3. The boom reaches the observer when the cone, which trails the aircraft at the half-angle θ_M , passes through the observer's position. The aircraft is then a horizontal distance $H/\tan \theta_M$ beyond the point overhead, which it reached a time $H/(V_s \tan \theta_M)$ earlier. With $\sin \theta_M = c/V_s$, we have $\tan \theta_M = (c/V_s)/\sqrt{1 - c^2/V_s^2}$, and the delay is $(H/c)\sqrt{1 - c^2/V_s^2}$.
4. The crests from a distant source are planes that travel at c along the direction from the source to the observer, spaced by

$\lambda = c/f$. The component of the observer's velocity towards the source is $V_o \cos \theta$, so the observer meets the crests at the rate $(c + V_o \cos \theta)/\lambda = f(1 + V_o \cos \theta/c)$.

- From $\sin \theta_M = \sqrt{gh}/V_s$, the speed is $V_s = \sqrt{gh}/\sin \theta_M$. A boat moving more slowly than $\sqrt{gh}$ is always inside the crests it has already made, which spread ahead of it, and the construction of Figure 47 has no cone, since $\sin \theta_M$ would exceed one.

Chapter 4. Superposition, Boundaries, and Interference

§4.1 Superposition at School

- A standing wave with n half-wavelengths in the string has nodes at the whole-number multiples of L/n . The point $L/4$ is one of them when $n/4$ is a whole number, so $n = 4, 8, 12, \dots$, and the frequencies are $f_n = nv/2L = 2v/L, 4v/L, 6v/L, \dots$
- The fundamentals are $c/2L_o$ for the open pipe of length L_o and $c/4L_c$ for the closed pipe of length L_c . They are equal when $L_o = 2L_c$. The second-lowest frequency of the open pipe is twice the fundamental, and that of the closed pipe is three times the fundamental, so the ratio is $2 : 3$.
- At a small angle θ from the perpendicular bisector of the loudspeakers, the path difference is $d \sin \theta \approx d\theta$, and the quiet positions have $d\theta = (q + \frac{1}{2})\lambda$. Neighbouring quiet positions differ in angle by λ/d , and at a distance D this is a distance $\lambda D/d$ along the line.
- The maximum of order 2 for λ is at $\sin \theta = 2\lambda/d$, and the maximum of order 3 for λ' is at $\sin \theta = 3\lambda'/d$. These are equal when $2\lambda = 3\lambda'$. The order q can be seen only if $\sin \theta = q\lambda/d$ is less than one, so $q < d/\lambda$.

§4.2 Superposition of Oscillations

- The sum of the complex amplitudes is $A(1 + e^{i\Delta\phi} + e^{2i\Delta\phi}) = Ae^{i\Delta\phi}(e^{-i\Delta\phi} + 1 + e^{i\Delta\phi}) = Ae^{i\Delta\phi}(1 + 2\cos \Delta\phi)$, using the exponential form of the cosine, and its modulus is $A|1 + 2\cos \Delta\phi|$. It vanishes when $\cos \Delta\phi = -\frac{1}{2}$, that is at $\Delta\phi = 2\pi/3$. The three

phasors then form an equilateral triangle, each turned by $2\pi/3$ from the one before, and the third ends at the tail of the first.

2. The slowly varying complex amplitude is $A_a + A_b e^{-i(\omega_a - \omega_b)t}$, and its argument is the phase of the sum relative to $\omega_a t$. Its tip moves on a circle of radius A_b centred on the point A_a of the real axis. The argument is greatest in magnitude when the line from the origin touches this circle, and the radius to the point of contact is then perpendicular to that line, so the sine of the angle is A_b/A_a .
3. The equation of the path has the form $\sigma_y y^2 + 2\sigma_{yz}yz + \sigma_z z^2 = \sin^2 \Delta\phi$, with $\sigma_y = 1/A_y^2$, $\sigma_z = 1/A_z^2$, and $\sigma_{yz} = -\cos \Delta\phi/A_y A_z$. In coordinates y' and z' turned through the angle ψ , with $y = y' \cos \psi - z' \sin \psi$ and $z = y' \sin \psi + z' \cos \psi$, the coefficient of $y' z'$ (and in the next sentence "The axes lie along y' and z' ") is $2\sigma_{yz} \cos 2\psi - (\sigma_y - \sigma_z) \sin 2\psi$, using the double-angle formulas. The axes lie along Y and Z when this coefficient vanishes, which gives $\tan 2\psi = 2\sigma_{yz}/(\sigma_y - \sigma_z) = 2A_y A_z \cos \Delta\phi/(A_y^2 - A_z^2)$. In Figure 52 $A_y > A_z$, and for $\Delta\phi = \pi/4$ the angle ψ is positive, so the major axis rises to the right, while for $\Delta\phi = 3\pi/4$ it falls to the right.
4. The double-angle formula gives $\cos 2\omega t = 2\cos^2 \omega t - 1 = 2y^2/A_y^2 - 1$, so every point of the path lies on the parabola. As ωt runs from 0 to π , y runs from A_y to $-A_y$, and as it runs from π to 2π , y runs back again, so the arc is traced back and forth.
5. Squaring the expression for x as the sum of three oscillations gives the squares of the three terms and their cross products. Each cross product is a product of cosines of different frequencies, which, by the product formula of this section, is a sum of cosines that average to zero over a long time. The square of each term averages to half the square of its amplitude, so the mean of x^2 is $\frac{1}{2}A^2 + 2 \cdot \frac{1}{2}(hA/2)^2 = \frac{1}{2}A^2(1 + \frac{1}{2}h^2)$. The two sidebands contribute $\frac{1}{4}h^2 A^2$, and the fraction is $\frac{1}{4}h^2/[\frac{1}{2}(1 + \frac{1}{2}h^2)] = h^2/(2 + h^2)$. It is largest at $h = 1$, where it is $1/3$.

§4.3 Reflection and Transmission at a Boundary

1. The ring has negligible mass and the rod exerts no force along it, so the transverse force $-\tau \partial y / \partial x$ of the string on the ring must vanish. With $y = \text{Re}[(\mathcal{A}e^{-ikx} + \mathcal{A}_r e^{ikx})e^{i\omega t}]$, the condition at $x = 0$ gives $-ik(\mathcal{A} - \mathcal{A}_r) = 0$, so $\mathcal{A}_r = \mathcal{A}$ and $\mathcal{R} = 1$. The displacement of the ring is $\text{Re}[(\mathcal{A} + \mathcal{A}_r)e^{i\omega t}]$, whose amplitude is $2|\mathcal{A}|$. This is the limit $Z_2 \rightarrow 0$ of the junction.
2. The calculation of the text applies with the two strings interchanged, so $\mathcal{R}' = (Z_2 - Z_1)/(Z_1 + Z_2) = -\mathcal{R}$ and $\mathcal{T}' = 2Z_2/(Z_1 + Z_2)$. Then $\mathcal{T}\mathcal{T}' = 4Z_1Z_2/(Z_1 + Z_2)^2$, and $1 - \mathcal{R}^2 = [(Z_1 + Z_2)^2 - (Z_1 - Z_2)^2]/(Z_1 + Z_2)^2$ has the same value.
3. The spring and the bead have the mechanical impedance $Z_m = i(m\omega - \kappa/\omega)$ of Section 1.3 with $b = 0$, and Z_2 is replaced by $Z + Z_m$. Then $\mathcal{T} = 2Z/[2Z + i(m\omega - \kappa/\omega)]$, and since the strings are identical $\mathcal{T}_P = |\mathcal{T}|^2$, which is the stated result. It equals one when $m\omega = \kappa/\omega$, that is at $\omega = \sqrt{\kappa/m} = \omega_0$, where the forces needed to accelerate the bead and to stretch the spring cancel. Half the power is transmitted when $|m\omega - \kappa/\omega| = 2Z$, that is when $m\omega^2 \mp 2Z\omega - \kappa = 0$, whose positive roots are $\omega = \pm Z/m + \sqrt{Z^2/m^2 + \omega_0^2}$.
4. For the incident pulse $\partial y / \partial t = -v_1 \eta'_i$, so its energy is $\mu_1 v_1^2 \int \eta_i'^2(s)^2 ds = \tau \int \eta_i'^2 ds$, using $\mu_1 v_1^2 = \tau$. The reflected pulse has the energy $\mathcal{R}^2 \tau \int \eta_i'^2 ds$ by the same calculation. For the transmitted pulse, $\partial y / \partial t = -v_1 \mathcal{T} \eta'_i$ evaluated at the argument $s = (v_1/v_2)(x - v_2 t)$, so its energy is $\mu_2 v_1^2 \mathcal{T}^2 \int \eta_i'^2(s)^2 dx = \mu_2 v_1 v_2 \mathcal{T}^2 \int \eta_i'^2 ds$, since $dx = (v_2/v_1) ds$. Dividing by the incident energy and using $\mu_2 v_2 = Z_2$ and $\tau/v_1 = Z_1$ gives $(Z_2/Z_1)\mathcal{T}^2 = \mathcal{T}_P$, and the two fractions add to one, as in the text.
5. With $\xi = \eta_i(x - vt) + \eta_r(x + vt)$, the condition $\partial \xi / \partial x = 0$ at $x = 0$ gives $\eta'_i(-vt) + \eta'_r(vt) = 0$ for all t , so $\eta'_r(s) = -\eta'_i(-s)$ for every argument s . Integrating, with both functions zero before the pulse arrives, gives $\eta_r(s) = \eta_i(-s)$, which is the limit $\mathcal{R} = 1$ of the junction. The part of the incident pulse labelled by the value s of its argument has the strain $\eta'_i(s)$. In the reflected pulse $\xi = \eta_i(-x - vt)$, the same part is where $-x - vt = s$, and by the chain rule its strain there is $-\eta'_i(s)$. A compression, with

negative strain, is therefore reflected as a tension, with positive strain.

§4.4 Impedance Matching and Evanescent Waves

1. The fraction reflected is $\mathcal{R}^2 = (Z-b)^2/(Z+b)^2$, and the fraction absorbed is $1-\mathcal{R}^2 = [(Z+b)^2-(Z-b)^2]/(Z+b)^2 = 4bZ/(b+Z)^2$. Its derivative with respect to b is $4Z(Z-b)/(Z+b)^3$, which is positive for $b < Z$ and negative for $b > Z$, so the fraction is largest, and equal to one, at $b = Z$.
2. The spring has the mechanical impedance $Z_m = -i\kappa/\omega$, so $\mathcal{R} = (Z+i\kappa/\omega)/(Z-i\kappa/\omega)$. The numerator and the denominator are complex conjugates, so $|\mathcal{R}| = 1$, and the argument of $\mathcal{R}$ is twice the argument of the numerator, $2 \tan^{-1}(\kappa/\omega Z)$. As $\omega \rightarrow 0$ this tends to π , so $\mathcal{R} \rightarrow -1$, as at a fixed end, and as $\omega \rightarrow \infty$ it tends to zero, so $\mathcal{R} \rightarrow 1$, as at a free end.
3. With $k_2 L = \pi$ we have $e^{-2ik_2 L} = 1$, and the formula for the input impedance gives $Z_{\text{in}} = Z_2(1 - \mathcal{R}_{23})/(1 + \mathcal{R}_{23}) = Z_2 \cdot 2Z_3/2Z_2 = Z_3$. At ω_q , $Z_{\text{in}} = Z_2^2/Z_3$ for any Z_2 , so $\mathcal{R} = (Z_1 - Z_2^2/Z_3)/(Z_1 + Z_2^2/Z_3)$, and multiplying the numerator and the denominator by Z_3 and squaring gives the stated fraction.
4. At a point of the stretch the force exerted on the string to its right is $F = -\tau \partial y / \partial x = \text{Re}[\tau \beta (\mathcal{B}_+ e^{-\beta x} - \mathcal{B}_- e^{\beta x}) e^{i\omega t}]$, and the velocity is $\text{Re}[i\omega (\mathcal{B}_+ e^{-\beta x} + \mathcal{B}_- e^{\beta x}) e^{i\omega t}]$. By the result of Vol. 4, the mean of their product is half the real part of the product of the first complex amplitude and the conjugate of the second, $\frac{1}{2} \tau \beta \omega \text{Re}[-i(|\mathcal{B}_+|^2 e^{-2\beta x} - |\mathcal{B}_-|^2 e^{2\beta x} + \mathcal{B}_+ \mathcal{B}_-^* - \mathcal{B}_+^* \mathcal{B}_-)]$. The first two terms in the round brackets are real, so after multiplication by $-i$ they contribute nothing to the real part. The last two are the difference between a complex number and its conjugate, $2i \text{Im}(\mathcal{B}_+ \mathcal{B}_-^*)$, and multiplying by $-i$ gives $2 \text{Im}(\mathcal{B}_+ \mathcal{B}_-^*)$. The mean power is therefore $\tau \beta \omega \text{Im}(\mathcal{B}_+ \mathcal{B}_-^*)$, which does not depend on x .
5. At rest the equation of motion becomes $\tau d^2 y / dx^2 = \kappa_f y$, whose solutions are $e^{\pm x \sqrt{\kappa_f / \tau}}$. The growing solution is excluded on a long string, and the condition at $x = 0$ gives $y = y_0 e^{-x \sqrt{\kappa_f / \tau}}$. As $\omega \rightarrow 0$, $1/\beta \rightarrow v/\omega_c = \sqrt{\tau/\mu} \sqrt{\mu/\kappa_f} = \sqrt{\tau/\kappa_f}$, so the evanescent

wave at low frequencies has the shape of the static deflection, and every point moves in phase with the end as the end is slowly moved up and down.

§4.5 Standing Waves in Strings, Pipes, and Cavities

1. The kinetic energy per unit length is $2\mu\omega^2 A^2 \cos^2 kx \sin^2 \omega t$ and, since $\tau k^2 = \mu\omega^2$, the potential energy per unit length is $2\mu\omega^2 A^2 \sin^2 kx \cos^2 \omega t$. Between neighbouring nodes, a length $\lambda/2$, the integrals of $\cos^2 kx$ and of $\sin^2 kx$ are both $\lambda/4$, so the total energy is $\frac{1}{2}\mu\omega^2 A^2 \lambda (\sin^2 \omega t + \cos^2 \omega t) = \frac{1}{2}\mu\omega^2 A^2 \lambda$. A travelling wave of amplitude A has the mean energy per unit length $\frac{1}{2}\mu\omega^2 A^2$, by Section 3.4, so two of them carry $2 \times \frac{1}{2}\mu\omega^2 A^2 \times \lambda/2$ in half a wavelength, which is the same.
2. The ring has negligible mass, so the applied force equals the force F exerted on the string, and the complex amplitude of the velocity is $F_0/(-iZ \cot kL)$, whose magnitude is $F_0/(Z|\cot kL|)$. It grows without limit where $\cot kL = 0$, that is where $kL = (n - \frac{1}{2})\pi$, which are the frequencies of the string of Figure 66 with its ends exchanged. For $kL \ll 1$, $\cot kL \approx 1/kL$ and $F/\dot{y} \approx -iZ/kL = -i\tau/\omega L$, which is the impedance $-i\kappa/\omega$ of a spring in Section 1.3 with $\kappa = \tau/L$. This is the transverse force per unit displacement of a string of length L pulled sideways at its end, since the string then makes the angle y/L with the axis.
3. The bar has free ends, where $\partial\xi/\partial x = 0$, and a node at its midpoint. Measuring x from the midpoint, $\xi \propto \sin kx$, and $\cos(kL_b/2) = 0$ in the lowest mode, so $k = \pi/L_b$, the wavelength in the bar is $2L_b$, and the frequency is $f = \sqrt{Y/\rho}/2L_b$. The nodes of displacement in the air are half a wavelength apart, so the wavelength in the air is $2s$ and $c = 2sf = s\sqrt{Y/\rho}/L_b$.
4. The angular frequency is $(\pi c/L)\sqrt{n_x^2 + n_y^2 + n_z^2}$. The four smallest values of the sum of squares, with the integers not all zero, are 1, from the arrangements of $(1, 0, 0)$, giving three modes; 2, from the arrangements of $(1, 1, 0)$, giving three modes; 3, from $(1, 1, 1)$, giving one mode; and 4, from the arrangements of $(2, 0, 0)$, giving three modes. The angular frequencies are $\pi c/L$, $\sqrt{2}\pi c/L$, $\sqrt{3}\pi c/L$, and $2\pi c/L$.

5. A mode with $n_z \geq 1$ has $\omega \geq \pi c/L_z$, so below that frequency $n_z = 0$. The modes are then points of a plane lattice with cells of area $\pi^2/L_x L_y$, lying inside a quarter of the circle of radius ω/c , as in Figure 68. Their number is approximately $\frac{1}{4}\pi(\omega/c)^2 L_x L_y / \pi^2 = L_x L_y \omega^2 / 4\pi c^2$.

§4.6 Interference, Coherence, and Diffraction

1. The waves now differ in phase by $\Delta\phi + \pi$, so the intensity is $4I_1 \cos^2[(\Delta\phi + \pi)/2] = 4I_1 \sin^2(\Delta\phi/2)$. It is zero where $d \sin \theta = q\lambda$, including the direction $\theta = 0$, and greatest where $d \sin \theta = (q + \frac{1}{2})\lambda$. When $d \ll \lambda$, $\Delta\phi$ is small in every direction and $I \approx I_1 (kd \sin \theta)^2$, which is much less than I_1 , so two loudspeakers in antiphase and close together radiate little sound at low frequencies.
2. We have $|\Sigma|^2 = \sum_j \sum_{j'} e^{-i(j-j')\Delta\phi}$, with j and j' running from 0 to $N-1$. The mean of $e^{-i(j-j')\Delta\phi}$ over a period of $\Delta\phi$ is zero unless $j = j'$, when it is one. The N terms with $j = j'$ give a mean of N , so the mean intensity is NI_1 .
3. The principal maximum of order q for $\lambda + \Delta\lambda$ is at $d \sin \theta = q(\lambda + \Delta\lambda)$. The first zero beside the principal maximum of order q for λ is where $\Delta\phi = 2\pi q + 2\pi/N$, that is at $d \sin \theta = q\lambda + \lambda/N$. Equating the two expressions for $d \sin \theta$ gives $q\Delta\lambda = \lambda/N$, so $\lambda/\Delta\lambda = qN$.
4. Let the openings be centred at $x' = 0$ and $x' = d$. The strip at the distance x' from the centre contributes a phasor proportional to $e^{-ikx' \sin \theta} dx'$, and the integral over the opening centred at x'_c is $e^{-ikx'_c \sin \theta} [e^{-ikx' \sin \theta} / (-ik \sin \theta)]$ taken between $-w/2$ and $w/2$ relative to x'_c , which is $e^{-ikx'_c \sin \theta} w \sin \beta / \beta$. Adding the two openings gives $w(\sin \beta / \beta)(1 + e^{-i\Delta\phi})$, whose squared modulus is $4w^2 \cos^2(\Delta\phi/2)(\sin \beta / \beta)^2$, and dividing by its value at $\theta = 0$ gives the result. The principal maxima of order q are at $\sin \theta = q\lambda/d$, and when q is the multiple $j(d/w)$ of d/w this is $j\lambda/w$, where $\sin \beta = 0$, so those maxima are missing.
5. The maximum of order q has the path difference $q\lambda$, and the pattern is visible while the path difference is less than $\Lambda_{\text{coh}} = c/\Delta f$. The largest order is therefore about $c/(\lambda\Delta f) = f/\Delta f$,

since $c = f\lambda$.

Chapter 5. Waves in Two and Three Dimensions, Dispersion, and Wave Packets

§5.1 Waves in Two and Three Dimensions

1. Consider a small rectangle of the membrane with sides Δx and Δy . Its edges of length Δy , at x and at $x + \Delta x$, are pulled along the membrane with forces $\Gamma \Delta y$, whose components along z are, for small slopes, $\Gamma \Delta y \partial\psi/\partial x$ at $x + \Delta x$ and $-\Gamma \Delta y \partial\psi/\partial x$ at x . By Taylor's theorem their sum is $\Gamma \Delta x \Delta y \partial^2\psi/\partial x^2$, and the other two edges give $\Gamma \Delta x \Delta y \partial^2\psi/\partial y^2$. Newton's second law for the mass $\eta \Delta x \Delta y$ gives the stated equation, and the speed is $v = \sqrt{\Gamma/\eta}$.
2. The product $r\psi = A \sin kr \cos \omega t$ is a standing wave of the one-dimensional wave equation in r when $\omega = vk$, so ψ satisfies the wave equation, as shown in this section. As $r \rightarrow 0$, $\sin kr \approx kr$ and $\psi \rightarrow Ak \cos \omega t$, which is finite. The function vanishes where $\sin kr = 0$, on the spheres $r = n\pi/k = n\lambda/2$ with $n = 1, 2, 3$, and so on. Writing $\sin kr \cos \omega t = \frac{1}{2}[\sin(kr - \omega t) + \sin(kr + \omega t)]$ gives $\psi = (A/2r) \sin k(r - vt) + (A/2r) \sin k(r + vt)$.
3. For a function of r alone the Laplacian is $r^{-1} \partial(r \partial\psi/\partial r)/\partial r$. By the product rule, $r \partial\psi/\partial r = r^{1/2} \partial\chi/\partial r - \frac{1}{2}r^{-1/2}\chi$, and differentiating again gives $r^{1/2} \partial^2\chi/\partial r^2 + \frac{1}{4}r^{-3/2}\chi$, since the two terms in $\partial\chi/\partial r$ cancel. Dividing by r gives $\nabla^2\psi = r^{-1/2}(\partial^2\chi/\partial r^2 + \chi/4r^2)$, and the stated equation follows. For a wave of wave number k , $\partial^2\chi/\partial r^2$ is of order $k^2\chi$, so the last term is smaller by a factor of order $1/k^2r^2$ and can be neglected many wavelengths from the line. Then χ obeys the one-dimensional wave equation, whose outgoing solution is $f_1(r - vt)$. Close to the line the last term is not negligible, and it changes the shape of χ , and therefore of the pulse, as the pulse travels.
4. From the definition,

$$q^2 = k_1^2 \sin^2 \theta_1 - k_2^2 = (\omega/v_1)^2 [\sin^2 \theta_1 - (v_1/v_2)^2] \quad (\text{S.2})$$

, and $v_1/v_2 = \sin \theta_c$. The phase $\omega t - k_1 x \sin \theta_1$ is constant on points moving along the boundary at the speed $\omega/(k_1 \sin \theta_1) =$

$v_1/\sin\theta_1$. This speed is greater than v_1 because $\sin\theta_1 < 1$, and less than v_2 because $\sin\theta_1 > v_1/v_2$.

- For the first wave $y^2 + z^2 = A^2$, and at $x = 0$ the point $(y, z) = (A \cos \omega t, A \sin \omega t)$ goes round the circle anticlockwise in the yz -plane; for the second, $(y, z) = (A \cos(\omega t + \Delta\phi), -A \sin(\omega t + \Delta\phi))$ goes round clockwise. Let $\Theta = \omega t - kx + \Delta\phi/2$. By the sum-to-product identities $\cos a + \cos b = 2 \cos \frac{1}{2}(a + b) \cos \frac{1}{2}(a - b)$ and $\sin a - \sin b = 2 \cos \frac{1}{2}(a + b) \sin \frac{1}{2}(a - b)$, the sum has $y = 2A \cos \Theta \cos(\Delta\phi/2)$ and $z = -2A \cos \Theta \sin(\Delta\phi/2)$. The ratio $z/y = -\tan(\Delta\phi/2)$ is constant, so the sum is linearly polarised along a line at an angle $\Delta\phi/2$ to the y -axis, measured from the y -axis towards the negative z -axis.

§5.2 Dispersion, Phase Velocity, and Group Velocity

- Since $k = 2\pi/\lambda$, the chain rule gives $dv_p/dk = (dv_p/d\lambda)(d\lambda/dk) = -(2\pi/k^2) dv_p/d\lambda$, so $k dv_p/dk = -\lambda dv_p/d\lambda$. Substituting this into $v_g = v_p + k dv_p/dk$ gives the result.
- Dividing $v_g = \Omega a \cos(ka/2)$ by $v_p = 2\Omega \sin(ka/2)/k$ gives $(ka/2) \cos(ka/2)/\sin(ka/2)$. For $0 < \theta < \pi/2$, $\tan \theta > \theta$, and at $\theta = \pi/2$ the ratio is zero, so the ratio is less than one throughout. At $k = \pi/2a$, $v_p = 2\Omega \sin(\pi/4) \cdot 2a/\pi = 2\sqrt{2}\Omega a/\pi$ and $v_g = \Omega a \cos(\pi/4) = \Omega a/\sqrt{2}$.
- With $\omega = C^{1/2}k^{3/2}$, $v_p = C^{1/2}k^{1/2}$ and $v_g = \frac{3}{2}C^{1/2}k^{1/2} = \frac{3}{2}v_p$. The phase velocity rises with k , so the dispersion is anomalous, and the crests move backwards through a group; for deep-water waves under gravity $v_g = \frac{1}{2}v_p$, the dispersion is normal, and the crests move forwards through a group.
- Differentiating $\omega = v(k^2 + \beta k^4/\tau)^{1/2}$ with the chain rule gives $v_g = v(k + 2\beta k^3/\tau)(k^2 + \beta k^4/\tau)^{-1/2} = v(1 + 2\beta k^2/\tau)(1 + \beta k^2/\tau)^{-1/2}$. The ratio $v_g/v_p = (1 + 2\beta k^2/\tau)/(1 + \beta k^2/\tau)$ is greater than one. The function $\sin(n\pi x/L)$ and its second derivative, $-(n\pi/L)^2 \sin(n\pi x/L)$, both vanish at $x = 0$ and $x = L$. A standing wave $\sin(n\pi x/L) \cos \omega t$ satisfies the equation of motion provided that $\mu\omega^2 = \tau k^2 + \beta k^4$ with $k = n\pi/L$, which gives the stated frequencies. The factor in brackets grows with n , so the frequency of mode n is slightly greater than n times the

frequency of the lowest mode.

- From the calculation in this section, the mean kinetic energy per unit length is $\frac{1}{4}\mu\omega^2 A^2$, and the mean potential energies of the string and the foundation add to $\frac{1}{4}A^2(\tau k^2 + \kappa_f) = \frac{1}{4}\mu A^2(v^2 k^2 + \omega_c^2)$, which equals $\frac{1}{4}\mu\omega^2 A^2$ by the dispersion relation.

§5.3 Wave Packets

- The integral is $(2\pi)^{-1/2} \mathcal{A}_0 \int e^{-ikx} dk$ over $\bar{k} - \frac{1}{2}\Delta k < k < \bar{k} + \frac{1}{2}\Delta k$, which is $(2\pi)^{-1/2} \mathcal{A}_0 e^{-i\bar{k}x} (e^{i\Delta k x/2} - e^{-i\Delta k x/2})/ix = (2/\pi)^{1/2} \mathcal{A}_0 e^{-i\bar{k}x} \sin(\frac{1}{2}\Delta k x)/x$, by Euler's formula, and its real part is the stated displacement. The envelope first vanishes at $x = \pm 2\pi/\Delta k$, so the product of Δk and the distance from the centre to the first zero is 2π .
- With $\omega = vk$ the exponential is $e^{i(vkt-kx)} = e^{-ik(x-vt)}$, so the integral for $\psi(x, t)$ is the integral for $\psi(x, 0)$ with x replaced by $x - vt$. Taking the real part, $y(x, t) = y(x - vt, 0)$, which is the wave travelling in the positive x -direction of d'Alembert's solution, keeping its shape.
- The doubling time $2\sqrt{3}\sigma^2\bar{\omega}^3/v^2\omega_c^2$ is proportional to σ^2 , so it grows by a factor of four. The centre travels $v_g t = 2\sqrt{3}\sigma^2\bar{k}\bar{\omega}^2/\omega_c^2 = 2\sqrt{3}\sigma^2\bar{k}(1 + v^2\bar{k}^2/\omega_c^2)$, which grows with $\bar{k}$, in proportion to $\bar{k}$ while $v\bar{k} \ll \omega_c$ and as $\bar{k}^3$ when $v\bar{k} \gg \omega_c$.
- With $\omega = g^{1/2}k^{1/2}$, $v_g = \frac{1}{2}(g/\bar{k})^{1/2}$ and $\omega'' = -\frac{1}{4}g^{1/2}\bar{k}^{-3/2}$. The width doubles when $|\omega''|t/2\sigma^2 = \sqrt{3}$, at $t = 8\sqrt{3}\sigma^2\bar{k}^{3/2}/g^{1/2}$. A crest moves at $v_p = (g/\bar{k})^{1/2}$ and the packet at $v_g = \frac{1}{2}v_p$, so the crest moves through the packet at $\frac{1}{2}(g/\bar{k})^{1/2}$.
- When $\omega''t/2\sigma^2 \gg 1$, the square root in $\sigma(t)$ is close to $|\omega''|t/2\sigma^2$, and $\sigma(t) \approx |\omega''|t/2\sigma$. Squaring the amplitudes gives $|\mathcal{A}|^2 = 2\sigma^2 e^{-2\sigma^2(k-\bar{k})^2}$, a Gaussian of standard deviation $1/2\sigma$ in k . The group velocity of the components near k is $v_g + \omega''(k - \bar{k})$, so the group velocities have a spread of $|\omega''|/2\sigma$. After a long time the components have separated by this spread multiplied by t , which is $|\omega''|t/2\sigma$, and the packet has become as wide as this separation, since its initial width σ is by then negligible. The standard deviation of $|\psi|^2$ in x is $\sigma(t)$, as the envelope found in

this section shows.

Chapter 6. Acoustics of Fluids: Generation, Propagation, and Attenuation

§6.1 The Acoustic Wave Equation and the Speed of Sound

1. Differentiating $\rho_0 \partial \mathbf{u} / \partial t = -\nabla p$ with respect to time gives $\rho_0 \partial^2 \mathbf{u} / \partial t^2 = -\nabla (\partial p / \partial t)$. Since $\partial p / \partial t = c^2 \partial \rho' / \partial t = -\rho_0 c^2 \nabla \cdot \mathbf{u}$ by the continuity equation, this becomes $\partial^2 \mathbf{u} / \partial t^2 = c^2 \nabla (\nabla \cdot \mathbf{u})$. The identity with $\mathbf{A} = \mathbf{u}$ and $\nabla \times \mathbf{u} = \mathbf{0}$ gives $\nabla (\nabla \cdot \mathbf{u}) = \nabla^2 \mathbf{u}$, so $\partial^2 \mathbf{u} / \partial t^2 = c^2 \nabla^2 \mathbf{u}$, which holds for each Cartesian component.
2. The continuity equation gives $\partial \rho' / \partial t = -\rho_0 \nabla \cdot (\partial \xi / \partial t) = -\partial (\rho_0 \nabla \cdot \xi) / \partial t$. Integrating with respect to time, with $\rho' = 0$ and $\xi = \mathbf{0}$ before the wave arrives, gives $\rho' = -\rho_0 \nabla \cdot \xi$, and $p = (B / \rho_0) \rho' = -B \nabla \cdot \xi$. In one dimension $\nabla \cdot \xi = \partial \xi / \partial x$.
3. The volume of a fixed amount of gas is inversely proportional to ρ , so $T \rho^{1-C_p/C_V}$ is constant. Logarithmic differentiation gives $dT/T = (C_p/C_V - 1) d\rho/\rho$, and to the first order $T'/T_0 = (C_p/C_V - 1) \rho'/\rho_0 = (C_p/C_V - 1)s$. Since $s = p/B = (C_V/C_p) p/p_0$, this is $T'/T_0 = (1 - C_V/C_p) p/p_0$.
4. The ratio of the squares is $[(C_p/C_V)RT/M_m]/[8RT/\pi M_m] = \pi C_p/8C_V$. For a monatomic gas it gives $c/\bar{v}_m = \sqrt{5\pi/24}$, and for a diatomic gas $\sqrt{7\pi/40}$, both a little less than one.
5. The frequency is proportional to c , which is proportional to $T^{1/2}$. By the binomial expansion, $(T + \Delta T)^{1/2} \approx T^{1/2}(1 + \Delta T/2T)$, so the fractional rise is $\Delta T/2T$. With the subscripts He and air for the two gases, the ratio of the frequencies is $c_{\text{He}}/c_{\text{air}} = \sqrt{(C_p/C_V)_{\text{He}} M_{\text{m,air}} / [(C_p/C_V)_{\text{air}} M_{\text{m,He}}]}$, since the length of the pipe is the same.

§6.2 Plane and Spherical Sound Waves

1. For $p = \mathcal{A}e^{i(\omega t + kx)}$ each derivative $\partial/\partial x$ multiplies p by ik , and Euler's equation gives $i\omega\rho_0 u = -ikp$, so $u = -p/\rho_0 c$. The fluid

moves backwards, in the direction in which the wave travels, in the compressions.

2. Euler's equation gives $\partial u / \partial t = -(1/\rho_0) \partial p / \partial x = (1/\rho_0 c) f_1'(t - x/c)$, where f_1' is the derivative of f_1 with respect to its argument, by the chain rule. Integrating with respect to time gives $u = f_1(t - x/c)/\rho_0 c$, with no added constant, since both u and p are zero before the wave arrives.
3. The pressure is the sum of $\mathcal{A}e^{i(\omega t - kx)}$ and $\mathcal{A}e^{i(\omega t + kx)}$, whose particle velocities are $\pm p/\rho_0 c$ for the two waves, so $u = (\mathcal{A}/\rho_0 c)(e^{-ikx} - e^{ikx})e^{i\omega t} = -(2i\mathcal{A}/\rho_0 c) \sin(kx) e^{i\omega t}$. Then $p/u = i\rho_0 c \cot kx$, which is imaginary. At $x = 0$, $\sin kx = 0$ and $\cos kx = 1$, so the fluid does not move and the pressure has its largest amplitude $2|\mathcal{A}|$. A rigid wall at $x = 0$ imposes $u = 0$ there, and this is the closed end of a pipe in Section 4.5, where the displacement has a node and the pressure an antinode.
4. The magnitude is $\sqrt{1 + 1/k^2 r^2}$ and the lag is $\chi = \tan^{-1}(1/kr)$. At $r = \lambda/2\pi$, $kr = 1$, the magnitude is $\sqrt{2}$ and the lag is an eighth of a period. At $r = \lambda$, $kr = 2\pi$, the magnitude is $\sqrt{1 + 1/4\pi^2}$, only a little more than one, and the lag is $\tan^{-1}(1/2\pi)$, a small fraction of a period.
5. By the product rule, $\partial p / \partial r = (ik - 1/r)p$, and Euler's equation gives $i\omega\rho_0 u = (1/r - ik)p$, so $u = -(p/\rho_0 c)(1 + i/kr)$. The impedance is $z = -\rho_0 c / (1 + i/kr) = -\rho_0 c (k^2 r^2 - ikr) / (1 + k^2 r^2)$. For $kr \gg 1$, $u \approx -p/\rho_0 c$, and the fluid in the compressions moves inwards, which is the direction in which the wave travels.

§6.3 Sound Energy, Intensity, and Decibel Scales

1. The particle velocity has the amplitude $\omega\xi_A$, since $u = i\omega\xi$, and the pressure amplitude is $\rho_0 c \omega\xi_A$. Then $I = \frac{1}{2}|\mathcal{A}|^2/\rho_0 c = \frac{1}{2}\rho_0 c \omega^2 \xi_A^2$. The string gives the same form with the density ρ_0 replaced by the mass per unit length μ and the speed c by v , the intensity being a power per unit area where the string carries a power.
2. From Problem 3 of Section 6.2, taking real parts, $u = (2\mathcal{A}/\rho_0 c) \sin(kx) \sin(\omega t)$. The mean kinetic energy per unit volume is $\frac{1}{2}\rho_0 (4\mathcal{A}^2/\rho_0^2 c^2) \sin^2(kx) \cdot \frac{1}{2} = (\mathcal{A}^2/\rho_0 c^2) \sin^2 kx$, and

the mean potential energy per unit volume is $(4\mathcal{A}^2 \cos^2 kx \cdot \frac{1}{2})/2\rho_0 c^2 = (\mathcal{A}^2/\rho_0 c^2) \cos^2 kx$. Their sum is $\mathcal{A}^2/\rho_0 c^2$, independent of x . The product pu is proportional to $\cos(\omega t) \sin(\omega t) = \frac{1}{2} \sin 2\omega t$, whose mean is zero.

3. For incoherent sources the mean squares add, giving N times the mean square of one source, and $10 \log_{10} N$ dB more. For coherent sources in phase the pressures add, giving N times the pressure and N^2 times the mean square, which is $10 \log_{10} N^2 = 20 \log_{10} N$ dB more.
4. The expression $4z_1 z_2 / (z_1 + z_2)^2$ is unchanged when z_1 and z_2 are exchanged. When $z_2 \gg z_1$, the denominator is approximately z_2^2 , so $\mathcal{T}_P \approx 4z_1/z_2$. The transmitted level is lower than the incident level by $10 \log_{10}(z_2/4z_1)$ dB.
5. With $|\hat{u}|^2 = (|\hat{p}|^2/\rho_0^2 c^2)(1 + 1/k^2 r^2)$, the mean kinetic energy per unit volume is $\frac{1}{4}\rho_0 |\hat{u}|^2 = (|\hat{p}|^2/4\rho_0 c^2)(1 + 1/k^2 r^2)$, and the mean potential energy per unit volume is $|\hat{p}|^2/4\rho_0 c^2$. The sum is $\langle \varepsilon \rangle = (|\hat{p}|^2/4\rho_0 c^2)(2 + 1/k^2 r^2)$. Dividing $I = |\hat{p}|^2/2\rho_0 c$ by it gives $c \cdot 2/(2 + 1/k^2 r^2) = c [2k^2 r^2/(1 + 2k^2 r^2)]$, which tends to c as kr grows. The extra kinetic energy near the source is carried by the part of the particle velocity that is a quarter of a period behind the pressure, which transports no energy on average.

Chapter 7. Vibrating Bodies, Musical and Architectural Acoustics

§7.1 Pipes, Horns, and Acoustic Waveguides

1. For $Z_L \rightarrow \infty$ the input impedance tends to

$$Z_p \cos kL / (i \sin kL) = -iZ_p \cot kL \quad (\text{S.3})$$

. For small kL , $\cot kL \approx 1/kL$, so

$$Z_{\text{in}} \approx -i\rho_0 c / (SkL) = -i\rho_0 c^2 / (\omega SL) = \rho_0 c^2 / i\omega V \quad (\text{S.4})$$

. A piston displaced a distance x_1 into the pipe reduces the volume of the air by Sx_1 , which raises the density by the fraction Sx_1/V and the pressure by $p = BSx_1/V = \rho_0 c^2 Sx_1/V$. With $u = i\omega x_1$, the ratio p/uS is $\rho_0 c^2 / i\omega V$.

2. The long pipe presents $Z_3 = \rho_0 c / S_3$ to the end of the middle section, whose characteristic acoustic impedance is $Z_2 = \rho_0 c / S_2$. With $\cos kL = 0$ the input impedance is Z_2^2 / Z_3 . The reflection coefficient at the junction with the first pipe is $(Z_{\text{in}} - Z_1) / (Z_{\text{in}} + Z_1)$, with $Z_1 = \rho_0 c / S_1$, and it is zero when $Z_2^2 = Z_1 Z_3$, which is $S_2^2 = S_1 S_3$.
3. Differentiating $\omega^2 = c^2 k'^2 + \omega_c^2$ gives $\omega d\omega = c^2 k' dk'$, so $v_g = c^2 k' / \omega = c \sqrt{1 - \omega_c^2 / \omega^2}$. Below the cutoff p goes as $e^{-(\alpha + \beta)x}$, so the distance is $1 / (\alpha + \sqrt{\alpha^2 - \omega^2 / c^2})$, which tends to $1 / 2\alpha$ as $\omega \rightarrow 0$ and rises to $1 / \alpha$ at the cutoff.
4. With $z = \rho_0 c i k x / (1 + i k x)$, $1/z^* = (1 - i k x) / (-i \rho_0 c k x)$, whose real part is $1 / \rho_0 c$. The mean power is $\frac{1}{2} \text{Re}(p u^*) S = \frac{1}{2} |p|^2 S \text{Re}(1/z^*) = |A|^2 S / (2 \rho_0 c x^2)$, and since $S = S_0 x^2 / x_0^2$ this is $|A|^2 S_0 / (2 \rho_0 c x_0^2)$, independent of x .
5. With $L_z = L_y / 2$, the cutoffs are $(\pi c / L_y) \sqrt{n_y^2 + 4n_z^2}$. In order: (0,0) with no cutoff, (1,0) at $\pi c / L_y$, then (2,0) and (0,1) together at $2\pi c / L_y$, and then (1,1) at $\sqrt{5} \pi c / L_y$. Exactly two modes travel when $\pi c / L_y < \omega < 2\pi c / L_y$.

§7.2 Helmholtz Resonators and Acoustic Filters

1. When every plug moves into the cavity by x , the cavity receives the volume NSx , and the excess pressure is $\rho_0 c^2 NSx / V$. The force on each plug is $-\rho_0 c^2 NS^2 x / V$, so each plug has the spring constant $N\kappa$ and the same mass m , and ω_0 is multiplied by $\sqrt{N}$.
2. When $\omega \gg \omega_0$ and $\omega \gg \gamma$, the square root is close to ω^2 , and the ratio goes as ω_0^2 / ω^2 . The pressure in the cavity is proportional to the displacement x , which by Section 1.3 lags behind the driving force by the phase δ with $\tan \delta = 2\gamma\omega / (\omega_0^2 - \omega^2)$. At ω_0 this gives $\delta = \pi/2$, so the pressure in the cavity lags the outside pressure by a quarter of a period.
3. Let $\zeta = b / S^2$. Then $\mathcal{T} = 2\zeta / (2\zeta + Z_p)$ and $\mathcal{R} = -Z_p / (2\zeta + Z_p)$, so $\mathcal{T}_p = 4\zeta^2 / (2\zeta + Z_p)^2$ and $\mathcal{R}_p = Z_p^2 / (2\zeta + Z_p)^2$. The absorbed fraction is $1 - \mathcal{T}_p - \mathcal{R}_p = 4\zeta Z_p / (2\zeta + Z_p)^2$. Differentiating with respect to ζ and setting the result to zero gives $2\zeta = Z_p$, where the absorbed fraction is one half.

4. With $Z_b = i\omega\rho_0\ell'/S$, purely imaginary, $\mathcal{T}_P = [1 + Z_p^2/4|Z_b|^2]^{-1}$, and $Z_p/2|Z_b| = (\rho_0 c/S_p)S/(2\omega\rho_0\ell') = cS/2S_p\ell'\omega$. At low frequencies the impedance of the hole is small, the pressure at the junction is held close to the ambient value, as at an open end, and almost all the sound is reflected there; at high frequencies the impedance of the hole is large, because the air in it must be accelerated back and forth more rapidly, and little sound leaves the pipe through it.
5. The least transmitted fraction is $4/(\varsigma + \varsigma^{-1})^2$, so $\varsigma + \varsigma^{-1} = 2/\sqrt{\mathcal{T}_{P,\min}}$. Multiplying by ς gives a quadratic equation for ς , whose root greater than one is $\varsigma = [1 + \sqrt{1 - \mathcal{T}_{P,\min}}]/\sqrt{\mathcal{T}_{P,\min}}$.